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ADVANCES IN AUSTRIAN ECONOMICS VOLUME 13

THE SOCIAL SCIENCE OF HAYEK'S 'THE SENSORY ORDER'

EDITED BY
WILLIAM N. BUTOS
Trinity College



United Kingdom – North America – Japan
India – Malaysia – China

LIST OF CONTRIBUTORS

<i>Peter J. Boettke</i>	Mercatus Center, George Mason University and Department of Economics, George Mason University Fairfax, VA, USA
<i>William N. Butos</i>	Department of Economics Trinity College, Hartford, CT, USA
<i>Jean-Paul Carvalho</i>	Department of Economics, Oxford University, Oxford, UK
<i>Daniel J. D'Amico</i>	Joseph A. Butt S. J. College of Business, Loyola University New Orleans, New Orleans, LA, USA
<i>Francesco Di Iorio</i>	EHESS – CREA, École Polytechnique, Paris, France
<i>Peter E. Earl</i>	School of Economics, University of Queensland, St. Lucia, Brisbane, Australia
<i>Evelyn Gick</i>	The Master of Arts in Liberal Studies Program, Dartmouth College, Hanover, NH, USA
<i>Steven Horwitz</i>	Department of Economics, St. Lawrence University, Canton, NY, USA
<i>Lorenzo Infantino</i>	Department of History and Political Science, Luiss Guido Carli University, Rome, Italy
<i>Roger Koppl</i>	Institute for Forensic Science Administration, Fairleigh Dickinson University, Madison, NJ, USA

<i>Mark Koyama</i>	Political Theory Project, Brown University, Providence, RI, USA
<i>Samuli Leppälä</i>	Department of Economics, Turku School of Economics, Turku, Finland
<i>Leslie Marsh</i>	New England Institute for Cognitive Science and Evolutionary Studies, University of New England, Portland, ME, USA
<i>Thomas J. McQuade</i>	Department of Economics, New York University, Brooklyn, NY, USA
<i>Robert F. Mulligan</i>	Department of Accountancy, Finance, Information Systems, & Economics, College of Business, Western Carolina University, Cullowhee, NC, USA
<i>G. R. Steele</i>	Economics Department, Lancaster University Management School, Lancaster, Lancashire, UK
<i>Nikolai G. Wenzel</i>	Wallace and Marion Reemelin Chair in Free-Market Economics, Department of Economics and Business Administration, Hillsdale College, Hillsdale, Michigan, USA

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INVESTOR IN PEOPLE

THE UNEXPECTED FERTILITY OF HAYEK'S COGNITIVE THEORY: AN INTRODUCTION TO *THE SOCIAL SCIENCE OF HAYEK'S “THE SENSORY ORDER”*

William N. Butos

ABSTRACT

Purpose – Overview of Hayek's cognitive theory and the contributions of chapters.

Methodology/approach – Perspective on significance of Hayek's cognitive theory for the social sciences.

Findings – Hayek's cognitive theory provides insight into his oeuvre; more importantly, it is relevant for social theory in its own right.

Research limitations/implications – Hayek's cognitive theory warrants further attention by economists and social theorists interested in evolutionary social processes.

Originality/value of paper – To counter a widespread view that the contribution to economics and social science of Hayek's cognitive theory

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is largely confined to methodology. Hayek's cognitive theory also provides a useful framework for furthering the understanding of evolution within the social realm.

INTRODUCTION

Friedrich Hayek (1899–1999), one of the great intellectuals of the 20th century, is widely known for his contributions to economics and to the social sciences in general. For several decades, his ideas have attracted widespread interest and have wielded a strong and enduring influence on other scholars in areas that comfortably fit under the rubric of “economics, politics, and philosophy.” Hayek was a prolific scholar (29 volumes are planned for his *Collected Works* by the University of Chicago Press) and the secondary literature on Hayek is vast.¹ His work in technical economics, beginning in the late 1920s, and his extensive writings in broader social theory still cast a robust and relevant light on the many issues centering on the operation of the market economy and social institutions and processes.

It is especially useful for the purpose at hand to reaffirm that much of Hayek's work in the social sciences typically addresses, directly or indirectly, questions and problems of the nature, transmission, and use of knowledge. Curiously, Hayek's work in cognitive psychology, which falls within the province of such questions, has not been as widely studied by Hayek scholars. The aim of this volume is to explore possible connections between Hayek's cognitive psychology and social theories, hence the title *The Social Science of Hayek's “The Sensory Order.”* It will show that Hayek's cognitive theory has, unexpectedly and despite its being virtually ignored for many years, begun to inspire a wide range of endeavors to mine its insights to better understand social phenomena.

HISTORICAL COMMENTS ON *THE SENSORY ORDER*

Published in 1952 (but based on an exploratory essay written as a student in 1920), *The Sensory Order* remained generally unnoticed by psychologists, philosophers, social scientists, and Hayek scholars for the next few decades. Hayek's Nobel Prize in 1974 helped induce a resurgence of interest around the world in his economics and broader social theory, but it was chiefly

cognitive psychologist Walter B. Weimer who should be credited for bringing Hayek's cognitive work before a larger audience and making the case for its wider significance for economics and social theories generally. Beginning in the mid-1970s, Weimer (1979) started using and developing the ideas in *The Sensory Order* for his work in cognitive science and the methodology of scientific research. Coming from a background in the philosophy of science, he was especially influenced by the late W.W. Bartley's *The Retreat to Commitment* (1962) and Kuhn's *The Structure of Scientific Revolutions* (1970) in seeing Hayek's work in cognitive science, constraints on knowledge, and spontaneous order, as compatible to Bartley's epistemological stance on "comprehensively critical rationalism" and Kuhn's evolutionary theory of science. In 1981, Weimer organized a cognitive science conference at Penn State University that featured Hayek as the principal speaker (Hayek, 1982), while Weimer's own address (Weimer, 1982) presented the case for the significance of Hayek's cognitive psychology for the social sciences; this paper became the preeminent, and for several years the only useful entrée to *The Sensory Order*.

Interest in Hayek's cognitive theory, however, was not completely moribund in the period after 1980. Aside from Weimer's continued work,² we can take note that several notable neurophysiologists such as Gerald Edelman (1985) and Joaquin Fuster (1995) specifically highlighted the prescient ideas put forth by Hayek, particularly with respect to evolutionary neuromechanisms. In the social sciences, Yeager (1984) seems to have been the first economist to seriously apply Hayek's writings on cognitive psychology to economics. In his short paper, Yeager sought to establish the proposition that "appreciating Hayek's work in economics and political science requires knowing of his interest in psychology" (p. 2). To make his point, Yeager centered his attention on *The Road to Serfdom*. The connections Yeager forged are based on the "insights of 'literary' psychology" exhibited in *The Road to Serfdom* regarding "the influence of beliefs and attitudes" as opposed to the more technical treatment of cognitive processes found in *The Sensory Order* (p. 3).

While Yeager's discussion of the *The Sensory Order* is an excellent introduction to Hayek's cognitive theory and all the more remarkable given its brevity, the overall cast of the paper illustrates an especially vexing question dogging the literature on *The Sensory Order* by social scientists: Do we need *The Sensory Order* at all to do our social science and how does it make a difference in providing insight toward understanding the social realm? Yeager's paper, which sadly made no identifiable impact on other economists until about 2003,³ plus three other papers to be noted

immediately below, illustrate the several directions researchers subsequent to Weimer have pursued in attempting to understand and develop the implications of *The Sensory Order*.

First, Manfred Streit (1993)⁴ broke new ground in a regrettably neglected paper by giving an extended overview of Hayek's work in epistemology as presented in *The Sensory Order* and an argument for its foundational significance for his economic and social theories. The relevant connections Streit suggested include cognitively grounded subjectivity, limitations on individual knowledge, and the significance of tacit (or unconscious) knowledge. These elements from Hayek's cognitive theory, Streit maintained, provide central ideas upon which it is possible to construct a theoretically coherent Hayekian social theory.

In the second paper, Butos and Koppl (1993) used *The Sensory Order* to generate a theory of "Hayekian economic expectations" that are coherent, competitive, and self-correcting, and which are the result of cognitive processes that embody learning and the capacity to formulate conjectures and self-corrective routines that allow the individual to adapt to the external environment. The thrust of this and related subsequent research (see, e.g., Butos & Koppl, 1997) was to demonstrate that a judicious application of insights gleaned from Hayek's cognitive theory could be useful in addressing questions and issues in economics.

A third and related direction is reflected in the attempt to explore the systemic implications of Hayek's work, incorporating the ideas of evolution and self-organization. McQuade (2007), for example, saw *The Sensory Order* as providing a "prototypical description of an adaptive classifying system" (p. 58) useful for understanding the social orders of science and market. The key point of intersection is that both the brain and other emergent (or spontaneous) orders, such as science and the market, "implement a classifying process" as a byproduct of the "interactions between their components" that induce "adaptive reactions of the system as a whole" (p. 59).

These papers of Yeager, Streit, Butos and Koppl, and McQuade illustrate the breadth and variety of the secondary literature on Hayek's cognitive theory in economics and broader social theory. This literature, which has now grown quite large, can be sorted into two distinct yet often overlapping paths. One path has followed the work of Yeager and Streit in the sense of using *The Sensory Order* to better understand Hayek's oeuvre. This approach, amply demonstrated by scholarship through Caldwell (2004), provides an assessment and appreciation of the deep and wide-ranging insights of one of the 20th century's intellectual giants. An important part of

this literature, as evidenced, for example, in Streit's paper, may be seen as reflecting the broader interests of political economy in which connections are drawn for Hayek's own thinking across various domains of inquiry in some rather fundamental ways. An important contribution of this literature has been to establish the continuity in Hayek's own thinking (see, e.g., Gray, 1984; Birner, 1994; Butos & McQuade, 2002; Steele, 2007). But a significant aspect of actually establishing the coherence of Hayek's ideas has quite often involved the relevance of his cognitive work for methodological concerns (see Birner, 1999; Caldwell, 2004). In this area, there is little disagreement among researchers of the significance for social science methodology of Hayek's cognitive work regarding subjectivism (or at least some variant of it), constraints on knowledge, and the nature and limits of prediction.

The second, and as yet less traveled, path, illustrated by the work of Butos, Koppl, and McQuade, seeks to stand on Hayek's shoulders in applying insight gleaned from *The Sensory Order* to push forward the understanding of social phenomena.

These two paths of inquiry, both emanating from ideas in *The Sensory Order*, have, since the early 1990s, generated a small but nonetheless fairly regular stream of papers and books by economists and others who have been at pains to explain Hayek's ideas in this area and in several cases have pursued the application of those ideas to significant questions in social science. Slowly, Hayek's ideas have been made more familiar and accessible to a wider audience. Although I will briefly highlight the basic gist of Hayek's cognitive theory below, I will do so in a way that gears the discussion to the application of those ideas from the standpoint of social theory and economics. Hopefully, this will direct the reader's attention toward questions inspired by *The Sensory Order* that bear on both the theory of agency (at the individual level) and market and, more generally, social theory (at the level of the emergent phenomena generated by groups of individuals).

AN OVERVIEW OF HAYEK'S COGNITIVE THEORY

The objective of this volume is not to rehash or assess Hayek's cognitive theory but to provide some insight into whether and in what ways that theory may be of value to social scientists. This approach, it is hoped, will redirect attention toward questions that matter for social scientists and away from questions that concern the neurophysiology of the brain. The aim is to address the question posed above – to show that Hayek's cognitive theory

assists in deepening our understanding of Hayek as a social thinker and that it offers a body of ideas having relevance for social science in and of themselves.

Weimer (1982, p. 263) noted that throughout the expanse of his work, Hayek's "constant concern is how knowledge is manifested in phenomena of organized complexity." Hayek's investigations into such questions first took form in the area of cognitive psychology while he was still a student. In 1920, he wrote a paper on "Contributions to a Theory of How Consciousness Develops," a paper not yet generally available.⁵ The central point of this paper was to argue that cognitive phenomena – such as consciousness and emotions – are byproducts of physiological processes, in contrast to nonscientifically established "basic psychic processes." This allows scientific understanding of the brain to shift "to a determination of the relationships within which interventions into the organism take place," and "hence the genesis of the qualitatively different sensations can be adduced from experience" (Hayek, 1920, p. 1).⁶ Three decades later Hayek would develop these ideas into a considerably more detailed treatment of cognitive psychology, including their philosophical and epistemological consequences, which were not addressed in the 1920 essay. Yet, the 21-year-old Hayek's emphasis on a physiological explanation for cognitive activity and his claim that such mental phenomena rely on prior experience formed the basis for his more mature thinking.⁷ As Hayek himself indicates, the principal thesis of *The Sensory Order* is that

[we] do not first have sensations which are then preserved by memory, but it is the result of physiological memory that the physiological are converted into sensations. The connexions between the physiological elements are thus the primary phenomenon which creates the mental phenomena" (1952a, p. 53).⁸

The cognitive theory that Hayek builds from this starting point sees the mental realm as a complex but orderly system having the capacity to transform sensory inputs, which even precede sensations, through a complex chain of classification into an emergent interpretation of reality. In this way, Hayek's theory provides an explanation of the generation of individual knowledge about reality geared toward the adjustment of individual actions to that external reality. As discussed below, this positions Hayek's theory firmly within a rubric in which individual cognitive functioning is analyzed as an emergent adaptive system, a perspective McQuade (2007) and others have used that establishes not only the stunning modernity of *The Sensory Order* but also the possible generalizability of a *Sensory Order*-inspired

analysis applied to other emergent social orders, such as the market, science, and cities (see, e.g., McQuade & Butos, 2005; Butos & Koppl, 2007).

For Hayek, there is no presumption that a given individual's knowledge corresponds to what really exists or is true in some absolute sense. Rather, Hayek's cognitive theory provides an explanation for the relationship between two orders – the subjective order produced by the individual and the objective order external to the individual. Mental processes in Hayek's account produce a classification between flows of sensory inputs and the backdrop of a preexisting classification that partly has been learnt and partly has been made possible by the evolved genetic makeup of the species. For Hayek, classification is the primary activity of cognitive functioning. Thus, any sensory input is itself problematic in the special sense that what it is and why it may be significant to the individual cannot be distilled in the absence of the mental process by which that stimulus is classified. What an individual knows is actually something that is happening to the mind. As Hayek says in *The Sensory Order*, “the qualitative differences between experiences” refer to “mental and not physical events” and that what “we believe we know about the external world is, in fact knowledge about ourselves” (1952a, pp. 6–7). This idea reappears in the “The Primacy of the Abstract” where Hayek argues characterizes the “formation of abstractions ... as something which happens to the mind” (1978, p. 43). And what happens to the mind according to Hayek is the emergence of a classification based on the arrangement of sensory data based on perceived “equivalences or differences” (Hayek, 1952a, p. 41) in the qualities attributed to experienced objects that enables “the placing of something into one or several classes of objects.” Thus, “if sensory perception must be regarded as an act of classification, what we perceive can never be unique properties of individual objects but always only properties which the objects have in common with other objects” (Hayek 1952a, p. 142).

This supports a view of cognitive functioning that understands the subjective order of perceived events as a relational structure governed by rules by which sensory inputs are classified. As Hayek notes, “all the ‘knowledge’ ... an organism possesses consists in action patterns which the stimuli tend to evoke, or ... what we call knowledge is primarily a system of rules of action” (1978, p. 41). These rules, whether innate or learnt, must be prior to what is experienced, since without them classification in Hayek's theory (and hence, perception) could not occur. In this way, Hayek claims that “the richness of the sensory world in which we live is not the starting point from which the mind derives abstractions, but the product ... of

abstractions which the mind must possess” to experience that richness (Hayek, 1978, p. 44).

But Hayek also identifies an epistemological constraint in terms of what any given classificatory system can know. Here, the constraint refers to any classificatory system’s inability to fully explain itself. As Hayek says in *The Sensory Order*:

The proposition which we shall attempt to establish is that any apparatus of classification must possess a structure of higher complexity than is possessed by the objects which it classifies ... If this is correct, it means ... that the human brain can never fully explain its own operations (Hayek, 1952a, p. 185).

This places a limit on what any single classifying system can know about itself and implies that for each individual there is some level beyond which his mental processes are effectively “closed” to him. Some portion of what we think, according to Hayek, is governed by higher-level mental processes that allow an individual to generate knowledge that cannot necessarily be articulated. It is the sort of knowledge that riding a bicycle or any other skilled activity requires. Polanyi (1967) refers to this kind of knowledge as “tacit knowledge.”

In Hayek’s theory, an individual’s classificatory mental processes have the capacity to adapt on their own accord as new classifications are created and others fade away, whether induced by reclassification by the individual or as a consequence of social intercourse by which conventions, cultural mores, and other forms of social knowledge are conveyed. As a result, Hayek’s theory is quite consistent in drawing attention to both the inherent limits on individual knowledge and also the capacity of the individual to generate knowledge. Hayek’s theory of the mutability of mental processes, and the resulting capacity of individuals to create and act on such knowledge – that is, to exhibit emergent outcomes – places it squarely within the class of biological, psychological, and social theories emphasizing adaptive and evolutionary processes.

A central implication of *The Sensory Order* is Hayek’s argument, as noted earlier, is that the “capacity of any explaining agent must be limited to objects with a structure possessing a degree of complexity lower than its own” (p. 185). McQuade (2010) points out that Hayek’s complexity argument is order-specific and that while it certainly applies to the mind, it is not correct to assume that the constraints to which an individual’s mind is subject also apply to knowledge-generating orders composed of many individuals, such as science. While it is true that an individual mind cannot explain itself, McQuade’s point is that the interactions of many individuals

are capable of producing a kind of knowledge that is not subject to the classificatory constraints that apply to an individual. Thus, for example, we cannot assume that the knowledge produced by the scientific order is in principle circumscribed by what an individual scientist knows. The problem, then, for studying social orders requires not only a fuller understanding of individual action, but also what happens when individuals interact and in so doing generate outcomes qualitatively different from those an individual could have produced. This nonadditive linking of individual interactions to emergent systemic outcomes is the characteristic of a Hayek-inspired approach to social theory.

GENERALIZING HAYEK'S COGNITIVE THEORY

The relevance of Hayek's cognitive account for social theory at the level of the individual centers on a richer picture of human agency it affords than available from standard neoclassical economics. It is a picture that abstractly connects action with the knowledge available to the individual. That knowledge is not complete, but it may ordinarily be sufficient to usefully guide action. The actor's capacity to learn, implying a capacity for error correction and the generation of novelty, provides a window for seeing the actor in a more dynamic context. When applied to economics, Hayek's cognitive theory would see action reflecting a cognitively complex individual functioning within a complex environment that provided broad scope for action from routine behaviors to the discovery and the creation of knowledge. It is a cognitive theory that provides a view of agency capable of supporting robust behaviors for which modern neoclassical theory, by construction or disposition, is but a starting point. Hayek's cognitive theory invites us to understand human action stemming from complex, tacit, and mutable mental processes, inherently incomplete knowledge, learning, and knowledge generation at the individual level. This view of agency describes, in short, an adaptive classifying system, in which all manifestations of entrepreneurial activity, from arbitrage to innovation, preference change, and the generation of knowledge are not so much analytical complications as necessary vistas of inquiry for understanding human action from an evolutionary perspective.

Hayek's cognitive theory establishes that individual knowledge is necessarily decentralized and incomplete but capable of change. While such knowledge will certainly have value to the individual, it may potentially be useful to others. But if so, for social theory this raises the question of the

processes, mechanisms, and form by which such knowledge becomes available at the social level. Within a market order, human action, in all the ways it is entrepreneurially instantiated and the resulting exchanges by which plans are implemented, produces market-level phenomena. From the standpoint of economics, these constitute unintended byproducts principally in the form of prices and outputs. Thus, individual knowledge is modified, transformed, and transmuted into an explicit social metric or form useful to others. Feedback channels connecting individuals and the social structures in which they participate give full play to the production of emergent phenomena at the individual level and at all levels of social interaction. Notwithstanding the various ways economists might use to understand the social world, it is not possible to escape the fact that the objects of our inquiries require a perspective attuned to the sorts of considerations inhering in Hayek's work.

Of significant importance to social theory is the recognition that adaptive classifying systems differ with respect to their capacities for using and generating knowledge. While this is made explicit at the individual level in Hayek's cognitive theory, it also has clear counterparts at the social level. In systems subject to changes in rules and institutions, whether they arise endogenously or by deliberate design, emergent market-level outcomes are affected because the constraints governing the kind of knowledge produced, how it flows, and the feedback it produces within the system cannot be detached from the process by which the system and its components adapt to a changed environment. Such considerations furnish demonstrable explanatory value for understanding the adaptive and emergent characteristics of systems functioning under alternate institutional arrangements. When Austrian economists argue, for example, that centrally planned systems suffer from inherent flaws by which resources are allocated, those arguments refer to a system whose adaptive capacities and feedback mechanisms have been hindered or rendered inoperative altogether. From an evolutionary perspective, the weakened agility and robustness of the centrally planned system in using and generating knowledge means that its adaptive and self-correcting processes are also compromised.

The value added of Hayek's work in cognitive psychology to social theory, and especially in furthering our understanding of Hayek, has interested researchers since the late 1970s. At the same time, however, the relevance of Hayek's cognitive theory for social theory remains controversial and unsettled. These doubts are not without foundation, after all, if economics concerns social interaction in the context of scarcity, delving into the workings of the brain might be seen as pushing economics far beyond its

legitimate disciplinary boundaries. And it is true that Hayek advised that “it is a mistake” for social scientists “to believe their aim is to *explain* conscious action” because that “is the task of psychology” (Hayek, 1952b, p. 39), a sentiment also shared by Mises (1957, chap. 12).⁹ For Mises, economics deals with purposeful action, not with the underlying psychology of motivations and valuations that give rise to action:

The field of our science is human action, not the psychological events which result in an action. ... The theme of psychology is the internal events that result or can result in a definite action. The theme of praxeology is action as such (Mises, 1966, pp. 10–11).

Later in his career, however, Hayek did acknowledge that his cognitive theory perhaps had a wider significance than he had earlier thought. In the third volume of *Law, Legislation and Liberty*, he noted that *The Sensory Order* “helped me greatly to clear my mind on much that is very relevant to social theory” (Hayek, 1979, p. 199 n. 26). He explicitly refers here to methodological issues regarding complexly organized social phenomena, a connection more than just implicit with respect to Chapter III of *The Counter-Revolution of Science* (Hayek, 1952b) and Chapter VIII of *The Sensory Order* dealing with “Philosophical Consequences.” But he also mentions its significance for his “conception of evolution” and “a spontaneous order” (Hayek, 1979, p. 199). These sentiments, though not elaborated in detail, do seem to capture Hayek’s recognition of a field of application for his cognitive theory that extends well beyond methodological concerns. And, indeed, from an Austrian perspective once we enter into areas that go beyond Misesian praxeological theorems, questions concerning such issues as how agents learn or form economic expectations seem necessary to be broached on *some* level, if only to understand unstated or implicit theories of learning and expectations upon which the claims of economic theory rest. As Hayek said in “Economics and Knowledge,” the coordination of plans is based on an empirical claim that the “expectations of the people and particularly of the entrepreneurs will become more and more correct” but that “we are still pretty much in the dark about ... the nature of the process by which knowledge is changed” (Hayek, 1948, p. 45).

We have seen that McQuade, Koppl, and others, while following a Hayekian research agenda that makes use of his cognitive theory, have pursued inquiries that are less concerned with “explaining Hayek” than with using Hayekian insights to develop new approaches and new applications useful for understanding the social realm. Thus, more fully understanding Hayek’s oeuvre has enriched our understanding of his ideas; however, it has also opened up related and new lines of inquiry as we try to analyze the

social realm. That so many scholars have found Hayek's ideas intriguing, engaging, and seminal to their own research in so many areas of social science is perhaps the truest measure of his contributions.

OVERVIEW OF THIS VOLUME

The papers published here are original to this volume and present the ideas of economists and social theorists from around the world. These 15 essays constitute the first such collection by social scientists whose aim is to examine the social science of Hayek's cognitive theory. More importantly, as a group they testify to the allure and seminal quality of Hayek's cognitive theory. They view Hayek's cognitive theory not as a body of doctrine to be dissected or as fodder for the maws of hagiographers, but as providing a new and interesting perspective for formulating questions about the social realm. The papers, in short, are forward-looking in suggesting that Hayek's cognitive theory is a tool, a prod, a window for furthering our understanding of the social realm. While each of the papers are distinct and address a variety of different questions, from an *ex post* perspective as a group they constitute a thematic collection in their pursuit of questions broadly consistent with evolutionary and dynamic approaches to social phenomena.

Identifying connections and insights linking Hayek's cognitive theory and his work in many areas of social theory has provided ample scope for many researchers. But more than that, such identifications may be suggestive of the possibilities for further research that seeks to build upon Hayek's ideas and apply them to problems in the social sciences. The papers assembled here argue for the relevance and significance of Hayek's cognitive theory for the social sciences by examining how it informs questions in the philosophy of science, economic methodology, agency theory, the market pricing process, and our understanding of institutions, cultural phenomena, and a broader social theory.

The papers have been organized into four general (and certainly overlapping) groupings: (I) Thomas McQuade, G. R. Steele, Robert Mulligan, and Leslie Marsh on evolutionary and philosophical questions raised by Hayek's cognitive theory; (II) Lorenzo Infantino, Francesco Di Iorio, Peter Earl, and Samuli Leppälä on the usefulness for economics of Hayek's cognitive theory, in particular with respect to economic methodology, the theory of agency, and the theory of price formation; (III) Steven Horwitz, Jean-Paul Carvalho and Mark Koyama, Nikolai G. Wenzel, and

Evelyn Gick on applying Hayek's cognitive theory to social institutions and broader social theory; and (IV) a mini symposium assessing the value added of Hayek's cognitive theory with a paper by Daniel D'Amico and Peter Boettke followed by comments and rebuttals by Steven Horwitz and Roger Koppl, and a final reply by Daniel D'Amico and Peter Boettke.

The papers by Thomas J. McQuade ("Science and *The Sensory Order*") and G. R. Steele ("Reflecting upon Knowledge: Hayek's Psychology and Social Science"), delve into a "sensory order-inspired" examination of science. McQuade uses Hayek's cognitive theory as a bridge for developing a social theory of science, both of which he places in the broader category of "adaptive systems theory," while Steele draws on Hayek's writings to address the necessary distinctions that must be made between scientific knowledge and psychological and social knowledge.

McQuade argues that *The Sensory Order* can be generalized to provide a basis for understanding the generation of scientific knowledge, an undertaking Hayek himself did not pursue or even endorse. In McQuade's view, science is a system that instantiates a particular social process through which the interactions of scientists produce scientific knowledge as an emergent byproduct of the process itself. The key distinction is between the knowledge of a single scientist and the emergent knowledge generated by many scientists whose individual activities are governed by evolved institutional mechanisms, norms, negative feedback, and other system-level constraints. An important consequence of adaptive systems theory is recognizing that emergent effects are specific to the systems generating those effects. McQuade applies this result to recast Hayek's own "limits to knowledge" argument regarding the impossibility of a system, such as the human brain, to fully explain itself. In particular, McQuade pointedly emphasizes that this claim surely applies to an individual's knowledge of himself, but that it does not necessarily apply to the kind of knowledge science is capable of generating – a polite hint suggesting that Hayek's cognitive theory must be applied at the level of the social realm if it is to be useful to social science.

G. R. Steele uses Hayek's "connectionist paradigm" as a framework for understanding the subjectivity of individual knowledge and its implications for social science. Drawing on Hayek's *The Sensory Order* and *The Counter-Revolution of Science*, Steele argues that the methods appropriate for understanding the highly complex orders studied by psychology and social science are distinct from those methods suitable for the less complex phenomena of the physical sciences. He attributes such distinctions to the increasing distancing of subjective experience from the kind of knowledge obtainable in the physical sciences; in contrast, for psychology and social

science the ties between subjective experience and knowledge are considerably tighter. Steele argues that Hayek's insights about gaining knowledge of social phenomena are firmly rooted within a framework emphasizing adaptive and evolutionary processes.

In "*The Sensory Order: Operational Epistemology as an Evolutionary Adaptation*," Robert Mulligan approaches *The Sensory Order* as an epistemological work concerning the human mind as a naturally evolved adaptive order. Hayek's description of the classificatory process by which a mind generates knowledge represents for Mulligan an actual account of mechanisms that operationalize the epistemology of the species. This, in turn, serves to integrate a number of recent developments in neurobiology and cognitive science. In Mulligan's evolutionary account, this epistemology evolves differentially depending on the pace of structural changes in the brain and the transmission of knowledge when individuals interact; this allows Mulligan to use Hayek's cognitive theory as a tool to better understand connections between the evolution of brain structure and social institutions.

Leslie Marsh, "Hayek: Cognitive Scientist *Avant la Lettre*," in surveying a broad swath of literature dealing with the interface between cognitive and social science, shows that Hayek's *oeuvre* does not conform to either strictly individualist or holistic precepts. Instead, Marsh argues that Hayek, in socializing the mind and cognitivizing social theory, presents a considerably more sophisticated foundation for understanding the social realm. Marsh establishes compatibility between the society as an extra-neural memory store and the "enactive mind," both of which Marsh sees as adaptive emergent systems.

The second set of papers directly uses Hayek's cognitive theory to illuminate and understand various issues in economics. In "Hayek and the Evolutionary Tradition Against *Homo Oeconomicus*" Lorenzo Infantino sees *The Sensory Order* as completing Hayek's methodological critique of rationalistic constructivism and scientism and in so doing establishing a firm basis for an evolutionary account of social phenomena. Infantino argues that Hayek's cognitive theory establishes the groundwork for both methodological individualism and a socially and institutionally embedded view of the individual. It is a view, Infantino suggests, that rejects rationalistic enthusiasms from central economic planning to asserting that the human mind can plan its own growth; it is also a view that requires a rejection of *homo oeconomicus* and its supporting psychological claim that human agency is prior to society.

Francesco Di Iorio argues in "The Sensory Order and the Neurophysiological Basics of Methodological Individualism" that Hayek's cognitive

theory constitutes a defense of methodological individualism in that it sustains a nondeterministic theory of decision-making and supports the interpretative approach of *Verstehen*. Whereas individual action cannot be dislodged from the meanings the individual imparts to knowledge, the analysis of individual action must therefore take account of interpretations generated by the individual. The “sensory order-inspired” approach of Di Iorio allows him to reject deterministic models of individual action and social orders and to support subjectivism, the interpretative approach, and mind as a complexly organized adaptive system. According to Di Iorio, Hayek’s cognitive theory extends the reach of methodological individualism as a broadly applicable approach that deepens our understanding of agency and intentionality.

Peter Earl approaches Hayek’s cognitive theory from the perspective of evolutionary and behavioral economics in “The Sensory Order, the Economic Imagination and the Tacit Dimension.” Earl finds that Hayek provides the foundation for a theory of choice more robust than found in standard rational choice theory. When choice, Earl argues, is nondeliberative, reactive, rule-governed, and based on tacit knowledge, optimization with known alternatives and constraints and fixed preferences cannot be fully explanatory. As Earl notes, from an evolutionary standpoint behavioral responses require only “good enough” responses, not ones that are globally optimal. Earl argues that Hayek’s cognitive theory is central to providing the kind of support necessary for a more realistic theory of agency.

The Sensory Order upholds the contingency of interpretation on past experience and social context. In “Hayek on Prices and Knowledge: Supplementing *The Use of Knowledge in Society* with *The Sensory Order*,” Samuli Leppala extends this point to market interactions and the theory of price formation. Here, Leppala shows that an individual’s interpretation of the meaning or significance of market prices is contingent on specific past experiences of the individual which necessarily give rise to specialized local and personal knowledge. Thus, dispersed local knowledge, which is unique to each individual, provides the relevant context for individuals to appraise current market prices and engage in actions that instantiate the market process based on such interpretations.

The fertility of *The Sensory Order* can be seen in many areas of application. The third group of papers uses Hayek’s cognitive insights to better understand social institutions and broader social theory. Hayek’s cognitive theory describes, for example, an adaptive system whose success hinges on the individual’s ability to adjust to the external environment.

In short, it provides an account of learning. In “*The Sensory Order and Organizational Learning*,” Steven Horwitz uses Hayek’s theory of learning as expressed by Hayek’s “map” and “model” to understand how firms behave adaptively through positive and negative feedback. Firms are conceived by Horwitz as complex and emergent institutions that use routines to adjust the activities that occur within the firm to available resources. Horwitz links firm success to organizational learning of better routines and suggests that the speed of such learning and its successful deployment is likely to be more effective in a competitive environment.

The paper by Jean-Paul Carvalho and Mark Koyama, “Instincts and Institutions: The Rise of the Market,” uses *The Sensory Order* to demonstrate that the emergence of the market economy required the coevolution of human psychology and social institutions to foment cooperative behaviors. Their discussion corroborates the kind of dependence between the evolution of society and mind, a point Hayek emphasizes in arguing that reason is both a cause and effect of the growth of civilization (Hayek, 1962, chap. 2). More than that, though, they offer an original explanation of institutional emergence based on humans’ evolved psychology that actually promoted institutions essential for the rise and persistence of cooperative market arrangements. The particular psychological characteristic Carvalho and Koyama deploy is the idea that “revenge is sweet”; consequently, as a byproduct of people’s willingness to pay for revenge, various institutions arise that extend and sustain trade.

The papers by Nikolai Wenzel and Evelyn Gick identify points of analogy between Hayek’s cognitive theory at the level of the individual and ways of understanding social phenomena at the institutional and cultural levels. In “An Institutional Solution for a Cognitive Problem: Hayek’s Sensory Order as Foundation for Hayek’s Institutional Order,” Wenzel establishes that important analogies exist between Hayek’s cognitive theory and the institutional order with respect to constraints on knowledge, learning and knowledge generation, and the role of mental filters for understanding the world. In “The Role of Dispositions in Hayek’s Cognitive Theory,” Evelyn Gick highlights Hayek’s discussion of dispositions operating at both the level of perception and at the level of action for interacting individuals. Hayek (1982, p. 290) acknowledged that he “could have greatly simplified [his] exposition” had he used the term *disposition* in *The Sensory Order*. At the perceptual level Hayek understood dispositions in the sense of inducing further interpretation of stimuli in contrast to *dispositions to act* as “coming in at a very late stage.” According to Gick, the classificatory rules governing dispositions at the perceptual level provide a basis for Hayek’s “dispersion

of knowledge” argument. Gick applies Hayekian dispositions to act to the social realm by arguing that Hayek’s theory of cultural evolution can be interpreted as the evolution of dispositions that result in the formation of “systems of rules of conduct” (see Hayek, 1967, p. 66).

The fourth grouping is a mini-symposium organized around the paper by Dan D’Amico and Peter Boettke on “Making Sense of *The Sensory Order*.” D’Amico and Boettke are interested in examining the relationship of Hayek’s work in cognitive theory to his larger corpus. Their principal aim is not to evaluate the usefulness of Hayek’s theory as such, but to take on the considerably more knotty question of prioritizing Hayek’s economics and his cognitive theory. According to D’Amico and Boettke, the existing Hayek-inspired secondary literature in neuroscience claims that his psychology precedes his economics; alternatively, they argue that, while Hayek’s cognitive theory may be of some interest, it is not at all crucial for an understanding of his economics. In fact, they directly call into question the value-added of Hayek’s cognitive theory for understanding economics.

D’Amico and Boettke concentrate their critique on the work of Steven Horwitz, Roger Koppl, and the present author. In light of that degree of specificity, Horwitz and Koppl were invited to submit replies to “Making Sense of *The Sensory Order*” and their comments appear, respectively, as “I Am Not a ‘Neuro-Hayekian’: I’m a Subjectivist” and “Confessions of a Neuro-Hayekian,” followed by a reply by D’Amico and Boettke.

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NOTES

1. See Caldwell’s (2004) intellectual biography and Boettke’s (2000) three-volume collection of journal articles on Hayek. Both testify to the breadth and depth of Hayek’s oeuvre. Other useful monographs and collections specifically on Hayek include Gray (1984), Birner (1994), and Steele (2007).

2. See for example Weimer (1982). During the mid-1970s through the early 1980s Hayek’s cognitive theory was central to the research of Weimer and two doctoral students in economics (James Wible and the present author) who came under

Weimer's influence at Penn State. The influence of Hayek's cognitive theory is clearly seen, for example, in Wible (1980) and some of his subsequent published work. Also, early on, the political scientist Eugene Miller (1974) used *The Sensory Order* to mount a critique of Hayek's attack on constructive rationalism.

3. To date, five published economics papers cite Yeager's paper, with the earliest at 2003 and three by the same coauthors.

4. The 1993 paper by Streit was an English version of a paper he published in German in 1992 at the first Hayek Symposium in Bleibach, Germany.

5. I thank the estate of F. A. Hayek for granting permission to quote from his 1920 paper and Bruce Caldwell for making this manuscript available. This and other essays, including *The Sensory Order*, are scheduled to be published in Volume XVIII of the *Collected Works of F. A. Hayek*.

6. Hayek's 1920 paper can be seen as a reaction against Mach's widely influential psychology. For a useful discussion of this point, see De Vries (1994). Hayek (1952a, pp. 175–176) noted, however, that “although the theory developed here was suggested in the first instance by the psychological views [of] Mach, its systematic development leads to a refutation of his and similar phenomenalist philosophies.” I thank Roger Koppl for this point.

7. Also see Hayek, “The Primacy of the Abstract” (1978).

8. Hayek (1952a, p. 55) notes that this quotation is a translation from his 1920 student paper.

9. Mises distinguishes between psychology considered as a natural (experimental) science and psychology that applies to cognition, motivations, ideas, and judgments of value. Mises (1957, pp. 264–274) calls the latter *thymology* and considers it a branch of history bearing “no special relation to praxeology and economics” (p. 271). As Koppl (1997) points out, Mises claimed that thymological knowledge is gained from “understanding” while praxeological knowledge is learned through discursive reasoning or “conception.”

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SCIENCE AND *THE SENSORY ORDER*

Thomas J. McQuade

ABSTRACT

Purpose – First, to look closely and critically at Hayek’s treatment of science in The Sensory Order. This provides hints as to the difficulties in maintaining a theory of scientific knowledge as a selective sum of the identifiable contributions of individual scientists. Second, to generalize from Hayek’s theory of how the brain generates an individual’s knowledge to a theory of how science generates scientific knowledge, knowledge that is not a simple sum of individual contributions. Third, to apply this picture of science to understanding developments in postpositivist philosophy and post-Mertonian sociology of science.

Approach – We provide a short survey of the conventional understanding of science and scientific knowledge, including that of Hayek in The Sensory Order. We examine in more depth the ways in which developments in postpositivist philosophy and sociology have transformed our understanding of science. We describe how, by analogy with Hayek’s theory of the brain, science can be seen as an adaptive system that adjusts to its environment by classifying the phenomena in that environment to which it is sensitive, and we apply this systemic picture of science with a view to integrating much of the more moderate content of recent philosophy and sociology of science.

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INTRODUCTION

The Sensory Order is a book about a theory of cognitive psychology, not about science in general. But science in general does play an important role in the book – it is cast in the role of providing a definitive classification of the phenomena of the world external to the brain, against which the sensory classification generated by an individual brain is contrasted. These two classifications are different: the one generated by science is in terms of the relationships of physical phenomena to each other, while that generated by the brain in sensory contact with the physical world is in terms of the interrelationships of sensory qualities. Hayek (1952, pp. 3–4) points out that, while both are classifications of the phenomena of the world, there is no simple one-to-one correspondence between them – for example, objects that to our senses resemble each other need not be physically similar, and objects that appear to be altogether different may be physically very similar. In the face of this difference, Hayek takes it as his task to explain by what kind of process a given physical situation as represented by science is classified by a brain in terms of sensory qualities.

At least for the purpose of developing a theory of cognition, then, science is assumed, unproblematically, to present a reliable and true characterization of the world outside the brain. While Hayek himself was far from being a naïve realist, this pragmatic realism was, given the state of the philosophy of science at the time the book was written, not at all a controversial position to take. But all Hayek actually required was that science be taken as providing a “different” and “more consistent” (1952, p. 173) characterization of the world than the senses provide by themselves – a position that, ignoring some of the more recent postmodernist descents into comprehensive and reflexive skepticism, is not likely to provoke much argument even today. Nonetheless, it is worth looking closely and critically at Hayek’s treatment of science in *The Sensory Order* for two reasons: first, one can use Hayek’s observations about science to illustrate the difficulties in maintaining a theory of scientific knowledge as a selective sum of the identifiable contributions of individual scientists, and second, one can generalize from Hayek’s theory of how the brain generates an individual’s knowledge to how science generates scientific knowledge in a way that not only accommodates and illuminates some of the major developments in postpositivist philosophy and post-Mertonian sociology of science but also integrates them into a coherent picture.

To provide the necessary background, we start with a short discussion of what the conventional understanding (both lay and academic) of science and

scientific knowledge seems to be, and we use Hayek's brief comments on science to illustrate some of the problems with a conventional and steadfastly individualistic treatment. We then focus on the ways in which developments in postpositivist philosophy and sociology have transformed our understanding of science by adopting an empirically informed approach to the scientific process and by introducing social factors into the epistemological mix. We describe how, by analogy with Hayek's theory of the brain as an adaptive system, science can be seen as a system that adapts to its environment by classifying the phenomena in that environment to which it is sensitive, and we argue that much of the more moderate content of postpositivist philosophy and sociology of science can be seen to follow naturally from this systemic picture of science. The aim is to show how a picture of science as an "adaptive classifying system," by recognizing the mutability of scientific theories subjected to the social process of integration into existing knowledge, can accommodate both individual and social factors into a naturalistic philosophy of science without stumbling into the dead ends of extreme relativism and nihilistic reflexivity.

WHAT IS SCIENCE?

Science, in dictionary definitions, encyclopedia explanations, newspaper articles, and pronouncements of practicing scientists, is generally characterized as the investigation of the physical world by individuals seeking to understand the phenomena they observe, employing procedures that have been found to be effective for the production of reliable knowledge. These individuals present data and explanations – theories – as potential contributions to the body of scientific knowledge. Some contributions might be judged (by other scientists) more acceptable than others on the basis of various criteria, including logical coherency and repeatability of results and usefulness for prediction, and it is the acceptable ones that add, brick by brick, to the growing edifice of scientific knowledge.

Philosophers of science, at least prior to 1960 or so, have not, by and large, found fundamental fault with these general statements as far as they go, but have worked to explore the detailed ramifications of each of them. In traditional philosophy of science – which, for the purposes of discussion, we take to be that characterized by Suppe (1977, pp. 50–52) as the logical empiricist "Received View," a development and refinement of logical positivism that was generally accepted prior to the "sociological turn" initiated by Kuhn (1962) and others – attention was directed to the

propositions generated as candidate knowledge by individual scientists, especially to a consideration of what it meant for a scientific proposition to be true and justified.

The object was to articulate an objective procedure or set of criteria by which such candidate knowledge could be justified as being acceptable. Such criteria were applied to the candidate propositions themselves, without regard to the context in which they were produced. This distinction between “the context of justification” and “the context of discovery” was articulated by Reichenbach (1938, pp. 6–7) and was accepted, either explicitly or tacitly, by all of the major proponents of the Received View. Under such an assumption, theories could be formalized and examined as static, final products subject to confirmation or falsification. To the extent that scientific process was considered, it was also in terms of the propositions generated by individual scientists and not the behavior or interactions of those scientists. The emphasis was on the specification of a proper scientific method for individual scientists (or research teams) to generate potentially true and justifiable statements, and on the application of such a specification in the demarcation between science and nonscience. Justification can be seen in terms of arguments that show a proposition or theory to be true, or at least probably true, or more likely true than competing propositions. Or, as Popper (1982, pp. 19–20) has pointed out, one could justify a *preference* for a theory rather than justifying the theory itself. But if the grounds for preference are some form of relative net truth content, or “versimilitude,” this distinction is rather subtle, although it can be construed to favor a process of critical assessment rather than objective proof.

While epistemologists claimed the context of justification as their territory, sociologists of science (prior to the 1960s) tended to accept a division of labor and restrict themselves to concern with the context of discovery. Within that circumscribed domain, however, they did introduce two ideas whose implications have had an impact on postpositivist philosophy as well as sociology. First, they pointed out that scientists did not have to be selfless seekers of truth for science to generate knowledge. According to Merton (1942, p. 276):

A passion for knowledge, idle curiosity, altruistic concern with the benefit of humanity, and a host of other special motives have been attributed to the scientist. The quest for distinctive motives appears to have been misdirected. It is rather a distinctive pattern of institutional control of a wide range of motives which characterizes the behavior of scientists.

Second, as hinted in the above quote, Merton claimed that it is the institutional environment in which scientists work which conditions them to

pursue their work in a way consistent with scientific knowledge generation. A social community that is structured such that certain norms of scientific behavior are inculcated, supported, and enforced is the specification of a functioning context of discovery. Whatever the individual motives are for engaging in scientific work, when played out in the context of the proper institutional constraints they will (with allowance for human error and attempts to deviate from the scientific norms) result in the production of properly scientific output. But whether or not this output is valid scientific knowledge is not a sociological issue – *that* is a matter for the context of justification.

To summarize: science is the product of individual contributions to the understanding of the physical world, contributions that arise from within a particular social context but are assessed as logical units according to objective criteria (with the capacity for successful empirical prediction being very high on the list) and only add to scientific knowledge if they pass this test (and continue to pass it as new – and acceptable – data emerges). This general characterization of science would, more or less, have been taken for granted in the early 1950s, when Hayek produced *The Sensory Order*.

SCIENCE IN *THE SENSORY ORDER*

Hayek opens *The Sensory Order* (1952, p. 3) with a discussion of the fact of the difference between “the physical order” (the classification of features of the world generated by science) and “the phenomenal, or sensory, order” (the classification of features of the world generated by an individual brain):

There exist now, in fact, at least two different orders in which we arrange or classify the objects of the world around us: one is the order of our sense experiences in which events are classified according to their sensory properties such as colours, sounds, odours, feeling of touch, etc.; the other is an order which includes both these same and other events but which treats them as similar or different according as, in conjunction with other events, they produce similar or different other external events.

He notes that (p. 2) “the progress of the physical sciences has all but eliminated [the sensory properties or ‘qualities’] from our scientific picture of the external world.” Science systematically disregards the way in which objects appear in human experience (in which the classification is tuned to the implications of the environmental components for individual survival) and substitutes instead a relational classification of events. But, there is no one-to-one correspondence between these two classifications, and so,

as he points out, (pp. 4–5) “a question like ‘what is x ?’ has meaning only within a given order, and ... must always refer to the relation of one particular event to other events belonging to the same order.”

The question Hayek sets himself to answer is the puzzle of the relationship between the physical and phenomenal orders, between the order of science and the sensory order. He answers it by providing, in general terms, a physical description of the generation and operation of the sensory order. Hayek’s posing (p. 7) of “the central problem of psychology” as the elucidation of “the kind of process by which a given physical situation is transformed into a certain phenomenal picture” is a sensible and productive approach. His interest is psychology, not physics; he takes the order generated by physics and the other sciences as given. The classifications generated by science are taken to be *the* description of the environment to which the brain adapts.

His theory of the sensory order leads him (p. 174) to the following conception of science:

Science consists ... in a constant search for new classes so defined that propositions about the behavior of their elements are universally and necessarily true. For this purpose these classes cannot be defined in terms of sensory properties of the particular individual events perceived by the individual person; they must be defined in terms of their relations to other individual events.

Hayek does not (nor, for his purposes does he need to) say how these classes come to be defined. He describes science in terms of a development within the individual mind, synonymous with the development of conceptual thought. Experience shows that the sensory classification is not always adequate for the prediction of events, and so (pp. 145–146):

a process of reclassification [on the results of sensory experience] is forced on us because we find that the classification of objects and events which our senses effect is only a rough and imperfect approximation to a reproduction of the differences between the physical objects which would enable us correctly to predict their behavior.

Hence, the experience of failed prediction and the desire for better is the motive force for the evolution, within the individual, of an augmented classification in which the objects and events of the environment are classified – not in sensory terms but in terms of their relationships to each other. Hayek (p. 171) equates this development with science:

when we complete the process of defining all objects by explicit relations instead of by the implicit relations inherent in our sensory distinctions, those sense data disappear completely from the system. ... Science thus tends necessarily towards an ultimate state

in which all knowledge is embodied in the definitions of the objects with which it is concerned.

That is as far as Hayek goes. But, while the development of conceptual thought is most certainly a prerequisite for the development of science, it is far from being the whole story – in fact, it is only the first step. The major claim of this paper is that the story of science is more than the story of the development of conceptual thought. Science is a social phenomenon, a social system whose operation produces a relational classification of objects and events in the physical world. The form and content of this classification depends not only on the conceptually aware participants in the system but also on the sort of transactions in which they engage in the process of generating this classification and on the structure of interaction in which these transactions take place and that, in turn, they modify. Knowing the properties of neurons is necessary but not sufficient in explaining the nature and content of the classifications generated by the brain; knowing the properties of individual humans is necessary but not sufficient in explaining the nature and content of the classifications generated by science. While it is clear that for Hayek's purposes in *The Sensory Order* it is sufficient to pass from the individual to the social, from personal knowledge to scientific knowledge, simply by the use of plural pronouns such as “we,” “us,” and “our,” such a move is insufficient for the purpose of understanding science itself.

Directing our attention to science itself, we might ask: *why* does science generate the sort of relational classification that it does? It is not enough to say (p. 3, emphasis added) that “the physical sciences have been *forced* to define the objects of which this world exists increasingly in terms of the observed relations between these objects, and at the same time more and more disregard the way in which these objects appear to us.” As an observation of a strong tendency within science, this is perfectly accurate, but invoking an unnamed “force” does not go far by way of explanation. In fact, by suggesting an effect outside of science itself, it is quite misleading, for there really is no candidate for the source of such a force. In another passage (p. 6), Hayek says that:

in order to build up a science capable of predicting events, [it has been necessary] to replace the classification of objects or events which our senses effect by a new classification which corresponds more perfectly to the manner in which those objects or events resemble or differ from each other in the effects which they have upon each other.

The “force,” then, appears to be a general drive for prediction (or, more accurately, better prediction, for the individual brain unaided by science

certainly shows significant anticipatory ability). However, there are two problems with this. The first is the fact that the classifications of science *do* provide for better prediction is not in itself an explanation for the particular knowledge-generating structure of science. The second is that an *individual* drive for better prediction (no doubt because of its survival-enhancement possibilities) is in itself insufficient for explaining the characteristics of a *social* system that generates a predictive classification. But this latter problem requires an excursion into postpositivist philosophy and sociology of science for its full force to be appreciated.

SCIENCE IN POSTPOSITIVIST PHILOSOPHY AND SOCIOLOGY

Although the Received View was never the monolithic structure that the name implies, internalist criticism of it began to pick up in the 1950s, most notably with Quine's (1951) attacks on the analytic-synthetic distinction and on the claim that individual propositions were the natural units of meaning and reference. A more radical, externalist challenge, however, was the view that science cannot be rationally justified in the positivist manner and that full epistemic understanding requires attention to the values and methods of the social group out of which the science arises. This position was first broached by Fleck (1935 in Redman, 1991), but at the time of its publication received very little attention. However, it was a crucial input for Kuhn's (1962) influential exposition of a sociological epistemology of science. And the challenge to Reichenbach's thesis was not limited to Fleck and Kuhn only, but was pressed by Polanyi (1952, 1958), Toulmin (1953, 1961), Hanson (1958), and Feyerabend (1965) also. Together with the more direct internal criticisms initiated by Quine and others, it led to the abandonment of the Received View by most philosophers of science.

The rejection of the Received View has not, however, resulted in the wide acceptance of any of the alternatives, for these in turn have been subjected to critical scrutiny, especially by Shapere (1964, 1966, 1971), Scheffler (1967), and Suppe (1977). The charge thought to be the most telling, according to Suppe (p. 648), is that the introduction of social factors into the context of justification is tantamount to reducing scientific knowledge to the mere collective beliefs of scientists as a social group, and necessarily leads to an extreme subjective idealism that is incompatible with any objective,

progressive view of science as a rational attempt to discover how the world really is. Suppe (pp. 633–634) even goes so far as to declare the sociological philosophies of science *passé*, interesting only with regard to their historical role in contributing to the demise of the Received View. Some of the proponents of the postpositivist epistemologies, notably Kuhn (1977), have sought to adjust their views to shore them up against the charge of relativism, but the adjustments have not satisfied the critics.

The Naturalistic Movement

In spite of the severity of the criticisms leveled at them, the expositions of Quine, Kuhn, and the other postpositivist philosophers have left the philosophy of science permanently changed. Although the appearance of consensus that was achieved during the heyday of the Received View is conspicuously lacking, there are still broad areas of general agreement. The importance of the history of knowledge acquisition to epistemology is admitted, at least as a constraint on, or a testing ground for, philosophical speculations. The move away from a timeless logic of justification to an historicization of reason was one of Kuhn's (1962) major contributions, and he was followed in this by a long line of "historicist" philosophers (which included, among others, Toulmin, Shapere, Lakatos, and Laudan) who also challenged the long-held notion of a "first philosophy" as an authoritative epistemology formulating timeless logical rules for assessing scientific validity. They followed a "naturalistic" line in which the observed methods and results of actual scientific practice are taken to be significant inputs.

Laudan, in an attempt to elucidate a normative naturalism, saw progress in the solution of scientific problems as defining rationality (rather than the other way around). He noted (1977, p. 124) that "any appraisal of the rationality of accepting a particular theory is trebly relative: it is relative to its contemporaneous competitors, it is relative to the prevailing doctrines of theory assessment, and it is relative to the previous theories within the research tradition." In a similar vein, Lakatos (1978) sought to ascertain, based on the history of science, a cogent description of the scientific method and then to take this to be the methodology according to which science ought to be conducted. But Lakatos's "rational reconstructions" of various scientific episodes are notorious for their flouting of factual accuracy, as Koertge (1976) has pointed out.

An approach which is naturalistic in the sense of adopting a natural science analogy is that of Toulmin (1981) and Popper (1984) which makes use of the concept of Darwinian natural selection to describe the development of knowledge – competing theories are intellectual variants that are subject to selection or extinction. The issue then becomes the description of the nature of the selection process – for Popper it is an objectively specifiable process of rational criticism and for Toulmin it includes in addition social factors, such as the “intellectual politics” of the discipline.

Yet a third naturalistic approach is one that takes Quine’s (1969a) program of “naturalizing” epistemology literally by placing epistemology as a branch of psychology and examining the causal formation of beliefs with an eye to exposing limitations to knowledge implied by findings in psychology, discovering prescriptions for enhancing the reliability of human cognitive processes, or even reducing such processes to empirical correlations between external conditions and neural states. These three approaches cover a wide range of territory – the naturalistic movement is far from being a monolithic enterprise.

Some other very basic issues have been raised in the context of naturalism. One is the extent to which knowledge has a fundamentally nonpropositional component – a concern raised by Polanyi (1958). If psychological studies of cognitive performance indicate that people possess and use forms of knowledge not expressible as statements, then perhaps the standard epistemological concepts of belief and justification should be adjusted to accommodate such phenomena. A second issue, which has remained very much alive despite the heavy criticism from traditional philosophers, is the extent to which the emphasis on knowledge as an individualistic phenomenon is misplaced. Since what people know is dependent not only on their own direct experience, but also on what their contemporaries know (and on what their predecessors knew), and since all knowledge is generated by people with particular interests in particular social contexts and through interaction with other people, it seems artificial to be epistemologically concerned only with individuals in isolation. In addition, the impact of the surrounding society on science has been a topic of discussion and research that precedes by far the “sociological turn” in the philosophy of science (Mirowski, 2004). So it is not too surprising that a recurrent strand in several variants of naturalistic philosophy has been the idea of “societal consensus” as an important component of justification, and even more traditional epistemologists, such as Goldman (1987, 1999), have argued persuasively for the recognition of social factors in the justification of beliefs and practices.

The Sociology of Scientific Knowledge

Admittedly on the fringes of philosophy, the sociologists Barnes (1974) and Bloor (1976), working in the context of the “sociology of scientific knowledge” and very strongly influenced by Kuhn, developed what amounts to a nonnormative naturalized epistemology. This is something of a departure from traditional Mertonian sociology of science, in which (as we have alluded to above) science is described as a community in which certain ideals have moral force: contributions judged solely on merit, no individual property in ideas once contributed, absence of emotional or financial attachments to any particular result, and maintenance of a rationally skeptical attitude toward any contribution. Adherence to these ideals is encouraged by institutional arrangements that provide for individual recognition and reward. The ideals have a methodological function in that adherence to them is presumed to be the most efficient means to furthering the goal of science – the uncovering of truth.

Barnes and Bloor (1982) did not follow Merton in this normative stance, and they proposed a form of constructivism in which social factors both internal and external to the scientific community not only affect the direction of scientific inquiry (as Merton would agree), but also play a part in the determination within the community of what counts as knowledge. Rather than assuming true beliefs to be the natural outcome of proper scientific method and trying to account for false beliefs as products of social and psychological influences, as the Mertonians had done, the new sociologists saw true beliefs as well as false ones as the products of social forces and perfectly legitimate subjects for causal explanation. The stance was both skeptical (in denying that there can be arguments that would establish a particular epistemology or ontology as ultimately correct) and relativist (in claiming that belief systems cannot be objectively ranked in terms of their proximity to reality or their rationality).

But their relativism is a constrained one. They are clear that, in any well-developed system of knowledge generation, canons of rationality, relevance, good procedure, and established truth are “hard facts” that confront anyone working within that system. And they are not idealists in the sense of denying that the perceived phenomena of experience play any part in the formation of the conventions of science – although they follow Quine in claiming that the classifications of science are *underdetermined* by the phenomena. But their “Strong Programme” has been portrayed as a variant of extreme idealism and relativism that seeks to undermine science by challenging the basis for the general acceptance of the validity of scientific

knowledge (Koertge, 1998; Sokal & Bricmont, 1998) – a criticism that may be more applicable to some constructivist sociologists downstream of the Strong Programme (Latour, 1987; Woolgar, 1988a) than to Bloor and Barnes themselves. More subtly, however, the Achilles heel of the Strong Programme was its vagueness as to how societal factors entered (and hence were reflected in) particular knowledge claims. The problem was that many empirically plausible interests could be invoked and related to beliefs in various ways, and there was no guidance for deciding on any definitive ascription. In the end, then, the Strong Programme offered no compelling reason for the centrality of social causes in knowledge generation.

The Impact of Postpositivism

To summarize the lasting impact of postpositivist philosophy and sociology of science, one can emphasize the major issues discussed above: the trend to naturalism, the undermining of the concept of a theory as simply a formal axiomatic system, the move away from a timeless logic of justification to the historicization of reason, and the attempt to recognize social elements in the construction of knowledge. Often cited specifically as hallmarks of postpositivism are the philosophical claims of “underdetermination” (of theory by data), “incommensurability” (of competing scientific theories), and “theory-ladenness” (of data and methods). Each of these claims needs a little elaboration:

- Underdetermination had its origin in Duhem’s (1914, p. 4) thesis that “an experiment in physics can never condemn an isolated hypothesis.” If the hypothesis together with its auxiliary background assumptions predicts some particular observation, a negative empirical result indicates a problem with either the hypothesis or one or more of the auxiliaries but cannot determine which. There is always the possibility that the empirical adequacy of the theory can be saved by adjustments elsewhere than in the hypothesis under scrutiny. This idea was expanded greatly by Quine (1951, p. 38) to what has come to be known as the Duhem–Quine thesis – “our statements about the external world face the tribunal of sense experience not individually but only as a corporate body” – from which he argued (p. 40) that “any statement can be held true come what may, if we make drastic enough adjustments elsewhere in the system.” Quine took his enlarged thesis to be equivalent to Duhem’s. But there is no reason to go quite so far. As Zammito (2004, p. 33) points out, the Duhem

thesis and the limited underdetermination it implies can stand apart from Quine's more radical extensions, and certainly does not imply a radical egalitarianism with respect to competing theories. The fact that logic by itself cannot compel theory choice does not imply extreme relativism, for even if two different theories were equally empirically successful, they could be evaluated on other criteria such as, for example, simplicity.

- Kuhn (1962) and Feyerabend (1962) introduced the notion of incommensurability between scientific theories. The claim, in Kuhnian terms, was that the proponents of competing paradigms, adhering to different classificatory schemes and following different exemplars, see the world in different terms, and it is not easy, to say the least, for one to properly understand the other. This type of incommensurability was taken by critics to mean conceptual relativism in that it seemed to deny the translatability of the content of past scientific theory into modern terms, but Kuhn (1982) protested that his point was considerably more modest (and more subtle) than that. While acknowledging that it is possible in many cases for there to be sufficient overlap in terms and in the way they function so that theories can be rationally compared, he proposed a constrained variant of incommensurability, which he called "local incommensurability," identifying the problem of translation as a problem of aligning two different classificatory schemes and thus associating it with the issue of theory-ladenness.
- Quine's (1969b) thesis of the "inscrutability of reference" is the claim that there is indeterminacy in the linking of any term or statement with phenomena of the world – that there are no "natural joints" at which nature can be cut. This implies that there is no theory-neutral observation language; in other words, all observations are theory-laden. Hanson (1958) pointed out, with reference to ambiguous drawings that provoke Gestalt switches, that the theory-ladenness of observation was not simply a matter of observers having different interpretations, but an essential feature of it. Observation is "seeing that ...," and the "that" is necessarily expressed in the terms of the theory being invoked. Tycho observes the sun to rise; Kepler observes the horizon to dip. Kuhn (1962, p. 91) elaborated on Hanson's insight by describing how the scientist's environment is different when viewed through the lens of different theory:

Led by a new paradigm, scientists adopt new instruments and look in new places. Even more important, during revolutions scientists see new and different things when looking with familiar instruments in places they have looked before ... paradigm changes do cause scientists to see the world of their research-engagement differently.

Kuhn (p. 122) also argued for the theory-ladenness of scientific methodology, and related it to incommensurability, by citing examples of differences between the scientific practices and standards of acceptance seen as adequate within different scientific schools at different times, each scientific tradition using different methods and standards and differing as to what problems are regarded as important. These observations highlight an element of contingency in scientific explanation in the face of changing norms of practice and theory acceptance, and represent a further challenge to the whole concept of theory choice based on purely logical reasoning.

While these contributions of postpositivism have come to be widely accepted as capturing characteristics of science as it is actually practiced and scientific knowledge as it is actually generated, they are, at best, loosely related findings – insightful about the nature of science, to be sure, but not adequately connected together and not shown to be the natural consequences of any overarching theory of science. However, Hayek’s work in *The Sensory Order* can point the way to integrating them into a more coherent picture, a picture in which the fundamental distinction between an individual’s classification of phenomena and the classification generated as an emergent effect of specific types of social interactions is made clear.

SCIENCE AS AN ADAPTIVE SYSTEM

Hayek has given an explanation for the characteristics of *the sensory order* in terms of the structure of the system that generates that order, without invoking any outside force and without taking the product of that system as sufficient reason for it being the way it is. A consistent Hayekian might well wish to take these insights one step further – to give an explanation for the characteristics of *the order of science* in terms of the structure of the system that generates that order, without invoking any outside force and without taking the product of that system as sufficient reason for it being the way it is. In Hayekian terms, elaborated by Buto and Koppl (2003), science can be described as a complex spontaneous order.

Adaptive Classifying Systems

Hayek’s central idea in *The Sensory Order* is that the brain’s mutable network of interconnected neurons functions as a “map” of the previously

experienced environment by effectively classifying the stimuli that have impinged on the system from its environment, and implements a continually updated “model” of the current environment in the activation pattern in the map’s network paths. As has been described in more detail elsewhere (McQuade & Butos, 2005; McQuade, 2007), these basic processes of classification described by Hayek as operating in the brain have their analogs in specific social arrangements. Quoting from McQuade (2007, p. 59):

Social systems are brain-like in a limited but important respect – specifically, the interactions between their components implement a classifying process on stimuli impinging on the system, and this process can induce real changes in component behavior and interaction that, in turn, engender adaptive reactions of the system as a whole to changes in its environment. In short, they are adaptive classifying systems.

Science (i.e., the network of interacting scientists) can be characterized as one of these adaptive social complexes – a system that reacts to its environment by modifying its internal structure so as to maintain a coherent map of that environment, supporting a relational classification of the environmental phenomena in terms of the system’s own potential activation patterns.

The basic structure in modern science is built from the institutions of publication and citation through which the scientists themselves, the active components of the system, interact by reporting observations, suggesting explanations, criticizing the techniques and hypotheses of others, and incorporating some aspects of their interpretations of the contributions of others into their own work. Such incorporation is not simply the ultimate compliment, but is also the process by which individual contributions are transmuted into scientific knowledge. These fundamental and long-lasting institutions are augmented by more mutable forms of interaction, which include the personal professional habits and routines of individual scientists and their organization into schools and groups. The scientists themselves are not fixed entities, for their tastes and preferences in what they find interesting and relevant and who they take seriously and find it profitable to interact with can be changed as a direct result of their experiences within the system, and their accumulations of reputation affect not only their capabilities for interaction but those tastes and preferences as well. A systemic side effect of such ongoing interactions, and one that is visible at the individual level, is the emergence of widely accepted procedures and taken-for-granted concepts and assumptions, whose visibility is enhanced by incorporation into textbooks.

What emerges as accepted scientific knowledge from the recursive process of contribution, evaluation, criticism, interpretation, selective use, and assimilation is to be clearly distinguished from the individual contributions themselves. Scientific knowledge is far from being the simple sum of individual contributions; it is, rather, the emergent result of individual contributions processed through the particular institutions of science, and altered, selectively abstracted, and reinterpreted in the process. Consider, for example, Keynes's (1936) *General Theory* – a major contribution to economics by any measure. There is a considerable difference between the theory articulated in the 1936 book (and in his reply (Keynes, 1937) to critics of the book) and the “Keynesianism” that emerged shortly thereafter as his work was assimilated to become mainstream orthodoxy. Hicks (1937) effectively transformed Keynes's predominately process-oriented reasoning about an economy whose institutions do not cope well with the effects of uncertainty into the form of an equilibrium model in which all of the unknowns are determined together and no institutional considerations are involved, and it was this Hicksian interpretation, substantially augmented by Modigliani, Hansen, and others into an uneasy synthesis of both Keynesian and classical ideas, that found its way into the textbooks.

Hayek himself (1952, pp. 10–12) provides another example of the transformation of an individual contribution in his discussion of J. Müller's “principle of the specific energy of nerves.” Here, the original point was that perceived sensory qualities depended on the particular nerve fibers stimulated and not on the physical attributes of the stimulus itself, but what became conventional wisdom was the subtly different proposition that the sensory qualities depended on some attributes of the impulse and not on the physical attributes of the stimulus.

The usual concentration on individual contributions to science serves to obscure the systemic nature of the enterprise. As a result, it may seem counterintuitive to picture the community of scientists as a networked system of interacting components that reacts to its environment while maintaining a large measure of stability. But if one is prepared to see the interactions among scientists in the process of proposing, evaluating, promoting, and making use of knowledge claims, all done through transactions of conventional form, as the connections that make up such a system, the analogy is not so far-fetched. Individual scientists have repetitive and constrained patterns of communication and association; they read and publish only in a small subset of journals, and they group themselves not only by specialty but by school within specialty. The usual effect of a scientist's interactions is to reinforce these patterns, not to

alter them. Most publications do not break new ground; they adhere to the consensus of their group, a consensus of doctrine and practice that has emerged from these repetitive publication and citation transactions in a way analogous to that in which market prices emerge from repetitive exchange transactions.

The consensus is stabilized by negative feedback in that scientists who deviate from accepted findings face considerable risk to their reputations within their group. And yet there is the possibility for structural change in response to perceived changes in the environment. Just as perceived shortages or surpluses, or the perceptive recognition of unmet demand, given incentive by the perceived potential for profit, can induce exchange transactions in markets whose terms differ from current market prices and that (if they persist) may result in changes in the spectrum of market prices (initially of local scope, but potentially eventually affecting a wide range of related goods), so the reiteration of anomaly in science, or an attractive explanation of it, given incentive by the perceived potential for enhanced reputation and credibility, can affect both patterns of communication and the content of the prevailing consensus (initially of local scope, but potentially affecting scientific understanding and practice on a wider scale).

Adaptive systems, such as science, work because their internal reactions to environmental inputs provide feedback effects that are in the interests, not merely of the participants in the system who are directly engaged in the reaction, but of many others as well. While recognition, reputation, and credibility directly reward those who make contributions to understanding environmental inputs that other participants find useful in furthering their own work, exposure to contributions recognized as useful (and their downstream modifications and reinterpretations) can be of value to others not yet directly involved in the publication or usage of the contribution. Induced structural changes, the realignments of interaction patterns that are the system's adjustment to new realities, can, like changes in market prices and buying patterns, both further the interests of some participants and represent negative feedback for others with investments in the old consensus who, were they to continue their normal interaction patterns, would suffer losses in credibility. But while these incentive effects are vital in driving adaptation at the system level (the system's building within its own structure a classification of its environmental inputs), this adaptation does not depend on any desire of the participants to consciously aim for it, for the transactions and their feedback effects are such that self-interested behavior on the participants' part in the context of these transaction forms is all that is required.

Scientific Knowledge as a Systemic Phenomenon

In this picture, scientific knowledge is not inherently propositional (although aspects of it can be usefully expressed in such terms), and is not any simple summation of individual contributions chosen for inclusion by participating scientists (although individual contributions are obviously essential elements in the system's classification process). In a very general sense, and from a systemic viewpoint, scientific knowledge is the complex of the focal interaction patterns that the current configuration of the system of science's map supports. These particular patterns of activation, learned from past experience, are the "attractors" that form the basis of the system's classifications of future inputs. To the extent that such patterns depend on the deployment in scientific interactions of widely accepted techniques and hypotheses, these techniques and hypotheses can be codified as established theory and practice and called "scientific knowledge" in accord with the common understanding of that term as denoting a logically organized presentation of propositions.

To give an illustration, if one observes the interaction patterns of Austrian economists – in papers, conference speeches, and conversations – one sees that the notion of capital as a structured entity is deeply embedded in these interactions. One could say that the scientific subsystem of Austrian economists knows that capital is a structure (or is fruitfully modeled as such). Confrontation with empirical phenomena involving capital generates interactions in which structured capital is a feature. Interactions characterized by structured capital tend to get follow-on notice; those with an unstructured, "capital stock" characteristic tend to die out rather quickly. Structured capital is an "attractor" in the interaction space of Austrian economists.

Particular theories and practices may be widely accepted among a one subgroup of scientists but not among their peers, and any progression to a more general and deeper acceptance may sometimes be a long, drawn-out process – involving further input, innovation, argumentation, and change of active personnel – that alters the system's map and its spectrum of focal interaction patterns. But the outcome cannot be predicted, for the system is both subject to unanticipated input and self-modifying in reaction to it.

While this characterization of scientific knowledge in terms of the social system of science's internal states may seem obtuse on first encounter, it is consistent with the relational, self-contained nature of scientific knowledge described by Hayek – and analogous to his theory of the representation of individual knowledge by brain states – in that it is precisely this set of states taken in relation to each other that constitutes the system's classification of

the phenomena of its experience. It also captures quite well the mutable mixture of agreement and dispute typical of actual science. And it would seem to be quite at home with the postpositivist insistence on the historical contingency of scientific truth and rationality. All of this suggests that adaptive systems theory, in the form described here, may be a useful basis on which to interpret, assess, and perhaps integrate the various strands of modern philosophy of science.

**POSTPOSITIVISM IN THE LIGHT OF
ADAPTIVE SYSTEMS THEORY**

One of the characteristics of postpositivist philosophy of science is an appreciation for the relevance of actual scientific concepts and findings (especially about human beings) and for the history of actual scientific discoveries and the methods by which they were produced. (As Kitcher (1992) perceptively puts it, this characteristic is less of a novel adoption than a return to an earlier sensitivity.) A second feature is the appreciation for the relevance of social factors in the generation of knowledge, a concern that lends itself to a naturalistic approach. Entangled with both of these developments is the question of whether or not a normative stance, the traditional stance of epistemology, is still feasible. And further stirring the postpositivist pot is the question of whether, in the absence of any a priori foundation for a normative stance, the way is left open for a relativism in which any sort of belief formation is warranted.

Categories of Postpositivism

For the purpose of bringing adaptive systems theory to bear on this wide field of philosophical activity, it is useful to categorize the varieties of postpositivist naturalism as individually or socially oriented, and to divide each of these further according to whether or not their thrust is normative.

	Individualist	Social
Nonnormative	Quine, Churchland	Barnes, Bloor
Normative	Laudan	Goldman

Here is some elaboration on each of the four possibilities:

- The original nonnormative individualist naturalism is that of Quine (1969a), who suggested that the traditional concerns about justification of beliefs confused two separate issues: explanation and legitimization. The latter is a matter for cultural studies; the former, focusing on a causal understanding of an individual's belief formation, is the province of psychology. This either eliminates any role for traditional epistemology, or reduces and reorients it as a branch of psychology. It is not a popular stand among postpositivist philosophers, although it has had its downstream influence in the work of Rorty (1979), Stich (1990), and Kornblith (1999). From the point of view of Hayek scholars, a particularly interesting variant of nonnormative naturalism is that of Churchland (1995, 2007), based on a theory of mind as patterns of activation in a recurrent Hebbian neural network,¹ which, although apparently pursued in complete ignorance of *The Sensory Order*, has much in common with Hayek's work – while going beyond it in adding considerable neurobiological detail, experimental grounding, and philosophical sophistication.
- Far more mainstream is normative individualist naturalism, of which the work of Laudan (1984, 1990) is an influential example. Here, the findings of science and the history of science are mined for clues as to what constitutes good reasons for belief and good practice. Instead of seeking to formulate a priori standards against which to evaluate scientific work, scientific activity is taken as paradigmatic of rational behavior. The aim is to pick out, from what scientists have actually done, the strategies that have made possible the emergence of science as a successful enterprise. The naturalistic fallacy – the construction of “ought” from “is” – is skirted by keeping any proposed norms strictly hypothetical.
- A prominent example of nonnormative social naturalism is the philosophical stance of Barnes (1974), Bloor (1976), and their followers in the Edinburgh School of sociology. Their claim (Barnes, Bloor, & Henry, 1996), is that they provide an exclusively descriptive account of scientific knowledge while adhering to their four general tenets: causality (the intent to uncover the causal origins of beliefs), impartiality (the commitment not to privilege particular belief-generating systems), symmetry (the explanations of beliefs considered false invoking the same causal mechanisms as those of true beliefs), and reflexivity (the acknowledgment that the beliefs of the investigator are subject to the same analysis as those of the investigated). Knowledge is socially constituted in that norms of rationality emerge in specific social contexts and subject to

the effect of interests from outside of science as well as inside, and knowledge is taken to be whatever beliefs are collectively endorsed in a particular social context. This relativist conception of knowledge would seem on its face to have ignored the rather important matter of the constraints imposed by the physical world itself, but Bloor (1976, p. 7) does acknowledge, if somewhat vaguely, that “naturally there will be other types of causes apart from social ones which will cooperate in bringing about belief.” However, some sociologists downstream of Bloor and Barnes do in fact espouse a radical constructivism with respect to knowledge.

- Probably the best-known normative social naturalism is Goldman’s (1999) “social epistemics,” although there are other interesting versions of normative social naturalism (Longino, 1990; Solomon, 2001). Goldman proposes to associate a “veritistic value” to states of belief, where belief in a truth rates highest on the scale. He is a strong realist about truth; something is true if it conforms to what is observed to be so. Practices (including norms, rules, and forms of interaction) in a scientific (or any other) community can have positive instrumental veritistic value if they tend to increase the mean veritistic value across the members of the community, and this measure allows for a comparative assessment of existing and proposed practices. The evaluative standards proposed – reliability, power, fecundity, speed, and efficiency (Goldman, 1987, pp. 128–129) – are universal ones, not relative to any particular community, although their respective weights may differ in different communities. And Goldman does not imply that some particular practices are always to be used, for a range of diverse practices operative within a community may turn out to score higher on the veritistic scale than uniformity of practice.

Since adaptive systems theory deals with individuals embedded in a social system, studied in a way that has more in common with anthropology and sociology than epistemology as traditionally understood, it is no surprise that it would be closer in spirit to social naturalism than to individualist naturalism, and more in sympathy with nonnormative naturalism than with normative naturalism. However, there is no argument with nonnormative individualist naturalism as far as it goes, for understanding the behavior and reactions of individuals engaged in knowledge-generating activity is clearly important for understanding the system as a whole. This is particularly true with respect to Churchland’s work, since this is, in effect, carrying further Hayek’s work on which the generalization to social systems pursued here is based. Similarly for its normative counterpart – investigations of individual scientific practice, past

and present, are useful inputs to a theory of science even if the motive for pursuing them is to unearth some normative pronouncements.

Normative Social Naturalism

Goldman's social epistemics, our example of a normative social naturalism, is an interesting development – it is naturalistic (and compatible with adaptive systems theory) especially in its accommodation of both individual cognitive limitations and social interaction, but it mixes this with the concern for justification more typical of traditional analytic epistemology and also asserts a strong form of realism that sets it quite apart from the much less adamant realisms (not to mention constructivism) of other postpositivist philosophers. Its focus on truth is not without its problems – Kuhn (1962, pp. 62–63, 126–127), among others, has described a number of historical episodes (the Copernican revolution, for example) in which scientific paradigm change is not evidently accompanied by an increase in the number of truths. And compatibility with observation is not all that can be said about beliefs – indeed, many beliefs that in their time were quite compatible with observation are now regarded as completely false. Besides, it is not necessary to think that scientific beliefs embody definitive truth in order to realize that they can be reflective of a world independent of individual minds and scientific interactions. But, apart from all that, the core of social epistemics is its normative thrust. It is not at all clear that it succeeds on its own terms – whether Goldman's objective measure of veritistic value is up to the task set it has been questioned, for example, by Kitcher (2002), on the grounds of its vagueness and subjectivity.

On the subject of normativity: it is understandable that general rules for justifying scientific procedures (individual and social) and scientific beliefs should have been sought, but it is instructive to note the lack of agreement on any proposed set of such normative rules and the fact that a great deal of what is taken to be knowledge was not generated with proper adherence to such rules. This is not to say that rules of thumb, teachable to science students with a view to instructing them into the general standards of the community they are about to enter, are not useful. But more global attempts at prescription would seem to be rather beside the point. An implication of adaptive systems theory is that science is a discovery process not simply of knowledge, but also of procedures for interacting in the production of knowledge. The currently best procedures are discovered in systemic action and modified by systemic experience; they cannot be postulated from the

outside as disembodied formulae. And that is the failure of all traditional epistemology (and of traditional artificial intelligence, for that matter) – the supposition that the playing out over time of self-modifying adaptive processes can be captured by a formula or an algorithm.

Unfortunately, workable or not, the proposal of prescriptive injunctions appeals to those with a constructivist bent, and in this guise they can be positively damaging if imposed (even with the best of intentions) on an adaptive system from the outside – introducing elements of *taxis* (deliberately constructed organization) into a *kosmos* (a self-generating or spontaneous order), as Hayek (1973, pp. 36–37, 51) would put it. This is not to say that new procedures should not be thought up, but that the only way to prove their worth, to justify them, would be to implement them in practice in the hope that they would thrive within the system without the coercive force of government behind them. The normative epistemologist, or his downstream converts, would be best advised to eschew policy prescription and become entrepreneurs of a sort.

Policy prescription is implicitly (and often explicitly) a call for a governmental agency to impose a suggested institutional form. Such imposition would risk the “kosmos/taxis” problem and sidestep any mechanism of uncoerced acceptance. But even to call for it makes the mistake of exhorting an adaptive system (the relevant government apparatus) to do something that is probably maladaptive for it compared to the current status quo. If one must propose policy, one should at least take the policy-imposing system’s structure and interplay of internal interests into account and not simply assume that it is analogous to an individual benevolently committed to the greater good.

Nonnormative Social Naturalism

That leaves nonnormative social naturalism as the species of postpositivism closest to adaptive systems theory. But there are considerable differences between the two in both emphasis and substance. The Strong Programme has been taken to task, for example, by Goldman (1987, p. 112), for its concentration on factors influencing knowledge generation external to science, particularly ideological and political influences, and on its neglect of the world itself as a major factor. These objections may be somewhat overstated, at least with respect to Barnes and Bloor, although not with respect to their intellectual descendants, as we will see below. But from the perspective of adaptive systems theory, these objections miss an essential

point – it is not so much the interests and motivations of the individual scientists that determine what emerges as knowledge but the fact that these interests and motivations have to be played out in the context of interactions that incorporate very specific feedback effects on those individuals. They are the “invisible hand” of science. And these interactions, particularly the fundamental ones surrounding publication and citation, do not simply influence scientists’ decisions and choices or constrain them to do the right thing, but are an integral part of the process by which individual contributions are transmuted into scientific knowledge. The adaptive system of science is built from *both* the individual scientists and their knowledge-generating interactions, and these in combination determine which external phenomena the system as a whole is sensitive to and to which it adapts. With an emphasis on usefulness based on empirical success built into the basic interactions, the physical world is certainly at the top of the list, but social factors, including, for example, funding (Butos & McQuade, 2006), can have their effect as well.

Where adaptive systems theory radically departs not only from Bloor and Barnes, but also from all of the above naturalisms, is in its insistence on a categorical difference between individual knowledge and scientific knowledge, between knowledge instantiated in the internal structure of a brain and knowledge instantiated in the internal structure of the social system of science. But there is an explanatory difficulty here – using the word “knowledge” to refer to an attribute of a social system is not a generalization readily accepted. To become comfortable with it, a reader has to jump three hurdles:

- To think of individual knowledge in terms of a set of potential states of activation of a neuronal complex refined by adaptation to an environment, rather than as states of belief in propositions with truth content. Hebb–Hayek-like neural-network theories of mind and their implications for epistemology are still very much on the fringe in philosophical circles, as Churchland (2007) attests.
- To set aside the knee-jerk reaction that associates the attribution of “knowledge” to other adaptive complexes, in particular, complexes of individuals, with authoritarian concepts of “group minds” or “class consciousness” or with attempts to portray such complexes as consciously purposeful.
- To deal with the complication that systemic and individual knowledge interacts. In adaptive social systems, the brains of individual scientists are the active components of the system. Individual scientists can observe and be affected by aspects of the system’s state and can acquire knowledge

from such observations and experiences. In other words, knowledge at the system level feeds back into knowledge at the individual level. But this does not mean that the system's state is represented by the sum of these individual repositories of knowledge – there remains the clear distinction between the states of individual brains composed of interacting neurons and the states of the adaptive system composed of interacting brains.

That this distinction between individual and systemic knowledge is not an idle, pedantic one – that it has consequences for understanding the nature and limits of knowledge – will be illustrated below when we return to *The Sensory Order*.

Underdetermination, Incommensurability, and Theory-Ladenness

What all of the above forms of postpositivism have in common (to widely varying extents and levels of emphasis) are the philosophical claims of “underdetermination” (of theory by data), “incommensurability” (of prior and current or of competing scientific theories), and “theory-ladenness” (of data and methods). But these claims are always given an individualistic interpretation, whether in the context of theory choice (by individual scientists) or the analysis of observation (again, by individual scientists). However, from the perspective of adaptive systems theory, they are also systemic characteristics of scientific knowledge – they are a consequence of the way in which scientific knowledge is generated and the form in which it is instantiated in the structure of the social system. All of them arise from the way in which new inputs, incompatible with the system's current “map,” to the extent that they are not simply ignored, result first in local adjustments, only leading to deeper readjustments of the system's classifications after prolonged exposure of the system to more sustained experience of anomaly.

- Underdetermination of theory by observation is a consequence of the fact that the assimilation of empirical data is only a part of the story in the systemic knowledge-generating process. Even given that empirical adequacy is a major contributor to the perceived usefulness of scientific contributions, and, therefore, an important factor in the assimilation process, the assimilation has to proceed, not simply by scientists choosing to incorporate new empirical results, but also through the progressive rearrangements of transaction patterns that such choices initiate. Since these rearrangements are rearrangements of an existing structure, it is

most likely that piecemeal, local adjustments would take place. The resulting global classifications of the system would show only minor change, giving the appearance of a Duhemian reaction to new data.

- Incommensurability of past and present scientific knowledge is not surprising in the face of the major rearrangements of the classificatory categories that can result, after a prolonged assimilation process, from the incorporation of significant new inputs. The extent of the incommensurability will depend on the extent of the differences between the past and present classifications, a difference that is likely to grow as more time passes and the system undergoes further adaptation. The case of the incommensurability of currently competing theories is also a matter of the difficulty of making comparisons from points of view based on different classifications, only here the classifications are more local – representing different schools of thought within science.
- Theory-ladenness of observation is the science analog of what Butos and Koppl (2007, pp. 29–30) have called the *pons asinorum* for *The Sensory Order* (i.e., Hayek’s insight that memory is antecedent to sensation) applied to science rather than to the brain. An adaptive system’s reactions to input depend on *both* the input and the operative classifications, where the latter are the contingent result of the system’s past experience and determine not only how current observations are to be interpreted, but also which observations are to be considered relevant.

In summary, viewed as systemic attributes of knowledge, underdetermination, incommensurability, and theory-ladenness are all natural consequences of the way in which the differentially mutable structure of an adaptive system reacts to its environment.

THE DEAD END OF RADICAL REFLEXIVITY

For completeness, it is necessary to emphasize that the sociology of science has certainly not rested with the Strong Programme. Its downstream successors have taken a “linguistic turn,” moving it away from the basic scientific orientation of Bloor and Barnes toward a form of literary theory, and certainly distancing it from any compatibility with adaptive systems theory.

In reaction to the inconclusive theoretical generalities of the Strong Programme, the social study of science shifted from conceiving of science in terms of knowledge to seeing it as practice, and turned to microlevel studies of laboratory work and specific scientific controversies. While providing

valuable insight into the complexity of day-to-day scientific activity, these studies were no more successful than the Strong Programme in developing a comprehensive sociology of science. And the reaction to this theoretical impasse was a radicalization of the constrained relativism of the Strong Programme to a thorough-going social constructivism and, in the end, a descent to a self-defeating reflexivity.

The path to the “reflexive revolution” in the sociology of scientific knowledge was paved in the 1980s by Mulkay (1985) who argued that it was impossible for sociologists to provide definitive accounts of scientists’ actions and beliefs. The stumbling block was the diversity of the participating scientists’ accounts of “what actually happened,” which left no firm bedrock on which to anchor historical or sociological analysis. Mulkay recommended that, instead, sociologists concentrate on the nature of scientific discourse; his (short-lived) movement became known as “discourse analysis.” The problem with this is that, if scientists’ claims are to be discounted, why not discount sociologists’ claims – including Mulkay’s?

It was Woolgar (1988a) who took the reflexive turn to its extreme conclusion – effectively, a blanket skepticism about all empirical inquiry. For Woolgar, all attempts at explanation involve the use of descriptions, and all descriptions are inherently unreliable. There are three possible responses to this problem: the naïve realist position that the descriptions fully reflect reality; the social constructivist position that allows for some variation (due to social factors) in the relation between descriptions and their ostensible referents; and Woolgar’s own position that the descriptions themselves *are* the reality. Woolgar (1989, p. xix) readily admits that his own position is open to the same charge of deficiency as the others. His strategy is “to sustain and explore the paradoxes which arise when we attempt to escape the inescapable [i.e., the problem of descriptions], not to attempt their resolution” and (Woolgar, 1988b, p. 30) to “juxtapose textual elements such that no single (comfortable) interpretation is readily available.” The upshot of all this studied reflexivity is more a form of fiction than anything else, made irritating and tedious by a writing style that comes across as frivolous or narcissistic. It luxuriates in an absolute skepticism that makes no sense at all for a resolutely antifoundationalist and naturalist epistemology. And from the perspective of adaptive systems theory, to say that “science is discourse” is to invoke a sadly meager and inapt metaphor for a system in which the relevant discourse (papers, talks, discussions, critiques, etc.) always takes place in the context of a prior classificatory structure that constrains the discourse and where reputational and credibility feedbacks are causally efficacious.

Adaptive systems theory is not contaminated by such self-defeating reflexivity because it does not concern itself with assessing the truth or reliability content of the classifications of science. An investigator employing adaptive systems theory applied to science is certainly engaging in science himself, working to get his own contributions accepted, but that is a benign form of reflexivity. The scientific aim is that of understanding, not of debunking.

KNOWLEDGE, INDIVIDUAL, AND SOCIAL

As noted above, one immediate consequence of adaptive systems theory is a reassessment and refinement of what is meant by “knowledge.” To illustrate this, and to show how the change in outlook matters, we return one last time to *The Sensory Order* to discuss an argument of Hayek’s regarding the limits to knowledge that turns out to depend critically on a standard individualistic concept of knowledge.

Adaptive systems theory requires that a basic distinction be maintained between knowledge at the individual level and knowledge at the system level. An individual’s knowledge is the classification of external phenomena performed by the brain; scientific knowledge is the classification of external phenomena performed by the social system of science. Since these are two completely different adaptive systems, it is not at all surprising that their classifications should be quite different. Of course, as noted above, they are not independent of each other – on the one hand, the active components of the system of science are the brains of scientists so that individual knowledge feeds into the social process of scientific knowledge generation, and on the other, the classifications of science are observable to individual scientists and therefore feedback as inputs to their individual processes of personal knowledge generation.

The argument of Hayek’s that relies on a standard individualistic concept of knowledge is his defense (1952, pp. 184–194) of a “mild and ‘practical’ form” of methodological dualism. The leading premise and its immediate consequence (pp. 188–189) are straightforward:

An apparatus capable of building within itself models of different constellations of elements must be more complex ... than any particular constellation of such elements of which it can form a model ... Applying the same general principle to the human brain as an apparatus of classification it would appear to mean that, even though we may ... possess an explanation of the principle on which it operates, we shall never, by means of the same brain, be able to arrive at a detailed explanation of its working in particular

circumstances, or be able to predict what the results of its operations might be. To achieve this would require a brain of a higher order of complexity

This is the uncontroversial assertion that the individual brain cannot contain a model of itself in complete physical detail. For Hayek, for whom the individual brain is the generator of scientific knowledge, it follows that the goal of fully explaining mental processes in purely physical terms cannot be achieved, and so (p. 191, emphasis added):

it also follows that *we* shall never achieve a complete ‘unification’ of all sciences in the sense that all phenomena of which it treats can be described in physical terms. In the study of human action, in particular, *our* starting point will always have to be *our* direct knowledge of the different kinds of mental events, which to *us* must remain irreducible entities.

But notice the plural pronouns. Hayek starts from a perfectly valid argument about the limits to the intrinsic capability of one brain, but applies it, by way of introducing those plurals, to the system of science. He argues as if the only classificatory system involved here is the human brain. But the classifications of science are not produced by a single human brain; they are the result of the interactions between many human brains in a system whose overall complexity exceeds that of a single brain – a system that, therefore, is not subject to the same limits as Hayek supposes to be binding.

This is not to say that Hayek’s conclusion might not be correct – only that his reason for it is inadequate to make the case. The classificatory capability of the system of science may well still be inadequate to the task of fully reducing the full explanation of mental phenomena to physical terms, but that is a question yet to be resolved. But there is no doubt that there are still Hayekian limits to scientific explanation – the system of science certainly cannot contain a model of itself in complete detail and most likely cannot do so for any complex social arrangement, such as an economy. What is pertinent for the purposes of this paper, however, is to emphasize the difference between knowledge at the individual level and knowledge at the system level. The classification generated in the mind of an individual scientist, even though it includes feedback influences from his observations of the classification produced by science and may contribute inputs to the process of scientific classification, is not the same thing as the emergent classification generated by the system of science that we call “scientific knowledge” and should not be conflated with it.

CONCLUSION

Science is a social process, more than that, it is a *system* in the sense that it is more than the simple sum of the endeavors of the scientists who participate in the process. Scientific knowledge is not just a selection from the best offerings of individual scientists; it is the outcome of an extended, institutionalized, social process of publication, criticism, interpretation, citation, argumentation, promotion, rejection, reinterpretation, assimilation, and even (from some participants' points of view) misinterpretation of those individual contributions. Knowledge generation in science cannot be separated from the procedures through which scientists interact and through which individual efforts are transmuted to become accepted as scientific knowledge, and to attempt to characterize "scientific knowledge" apart from "scientific process and institutions" is a dualism on par with separating mind from matter.

It is not simply that the norms of science, having emerged out of past interactions between scientists, have a determining effect on which individual contributions are acceptable contributions to knowledge – it is that such norms, procedural as well as prescriptive, are themselves part of the process through which individual scientific contributions are transmuted into emergent scientific knowledge. Not all social effects are simply influences on individuals on a par with personal interests; the particular transactional forms involved in the absorption of individual contributions are an integral part of the process of knowledge generation. Individuals and their scientific interactions together form a networked system, and scientific knowledge is an emergent side effect of the activity within that system. To be more specific, scientific knowledge is a classification by the system of science of the environment to which that system is sensitive and to which it adapts – indeed, "knowledge" in general should refer to the classification produced as a side effect of the adaptation to its environment by *any* adaptive system.

These are the claims that are pressed in this paper. The ground for such claims is suggested by (but by no means endorsed by) Hayek in *The Sensory Order*. They follow naturally from a generalization of Hayek's treatment of the mind as classificatory activity occurring in the network of neuronal connections that make up the brain. By characterizing the order of science as an adaptive classifying system similar in some very basic respects (but very different in implementation) to the classifying system that produces the sensory order, we have been able to put into context the major claims of postpositivism – including constrained

relativism, strong antifoundationalism, Duhemian underdetermination, inescapable theory-ladenness, and limited incommensurability – and thereby to contribute to a nonnormative naturalistic philosophy of science.

NOTE

1. A recurrent Hebbian network is a complex of neurons with mutable synaptic weights, incorporating feedback connections that temper inputs with previous experience. Hebb (1949) described a theory of the synapse and in addition accounted for short-term memory in terms of current neural activity (which Hayek, independently but later, would characterize as the brain's "model") and long-term memory in terms of mutable synaptic connections (Hayek's "map").

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REFLECTING UPON KNOWLEDGE: HAYEK'S PSYCHOLOGY AND SOCIAL SCIENCE

G. R. Steele

ABSTRACT

Purpose – To present the argument that the paradigm of spontaneously self-ordering open adaptation is common to Hayek's thesis on the mind (The Sensory Order) and to his presentations of social science (the social order).

Methodology/approach – To show how Hayek's methodological stance for social science interrelates with his theoretical work in neuroscience and psychology, where the 'connectionist' paradigm is relevant to extensive writings upon the human condition.

Findings –

- *close parallels across biological, psychological and social adaptations give a basis for determining which methods are appropriate to gain knowledge about knowledge;*
- *broad confirmation is evident that methods of proven worth to physical science have little relevance for the analysis of psychological and social phenomena, which are more complex than the phenomena of the material world.*

The Social Science of Hayek's 'The Sensory Order'

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Research limitations/implications –

- *that the social order rests upon common beliefs;*
- *that no simple distinction separates subjective and objective knowledge;*
- *that any drive for social science to match the precision of physical science is misguided;*
- *that in seeking an objective focus, behaviourism eliminates crucial introspective insights upon motivation and goals.*

Originality/value of paper – *The presentation is one of exegesis showing the relevance of Hayek's seminal work in theoretical psychology to the broadest themes of human understanding and social adaptation.*

INTRODUCTION

Although Friedrich Hayek's reputation rests in political theory and laissez-faire economics, he describes a neglected thesis in psychology – *The Sensory Order* – as 'the most important thing I have yet done' (Hayek, 1948; cited from Caldwell, 1997, p. 1856) and as one of his 'more important contributions to knowledge' (Hayek, 1994, p. 138). He comments further that 'the insights I gained ... both from the first stage in 1920 or later in the 1940s, were probably the most exciting events that ever occurred to me, and which shaped my thinking' (Hayek, 1994, p. 153). *The Sensory Order* was to prove a seminal contribution. However, its neglect over many years is matched by scant appreciation of commonalities in the structure of, and the processes within, the sensory order and the social order; and of the closely related evolutionary dynamics of the human psyche¹ and of society. Although that neglect is part-remedied by more recent work (notable examples include Smith, 1997; Birner, 1999; Horwitz, 2000; McQuade & Butos, 2005), the interdependence of biological, psychological and social adaptations remains contentious, problematic and under-researched.

Here, the very modest present aims are twofold. The first is to identify common features across Friedrich Hayek's extensive writings upon the human condition, where the 'connectionist' paradigm provides the framework for both theoretical psychology and social science. The sensory order and the social order are complex; that is, each is 'determined by the regularity of the actions towards each other of the elements of a structure' (Hayek, 1967, p. 73). In pursuing detailed work within these two areas, Hayek examines the characteristics of the different research methods that are respectively relevant to physical science and to social science: another

seminal exposition – *The Counter-Revolution of Science* – was published in the same year as *The Sensory Order*. Here is the second aim: to show how Hayek’s methodological stance for social science draws from his work on *The Sensory Order*.²

There is one aspect where subconscious mind processes and conscious processes of scientific inquiry are obviously linked. How the mind functions set the parameters for its scientific operation? In probing further, key features are the capacity and limitations of the mind; and the respective methods that are relevant for the mind to improve upon its understanding of (1) the physical world, (2) other minds with which it interacts within that physical world, and (3) its self.

A background overview of the society of men (Man and Society section) and a review of relevant aspects of psychology (Psychology: Alternative Paradigms and Psychology: Mental and Physical Events sections) are preliminary to the core of the paper, which is about ‘doing science’. As physical science advanced, the use of instrumentation (Scientific Method: From Sensory Order to Physical Order section) and the interchange between theory and fact (Scientific Method: Theory and Fact section) were primary factors in a drive for objectivity. Important considerations relate to inherent difficulties both in the concept of objectivity and in the difference between explanations of physical events and explanations of social events (Scientific Method: Objectivity and Subjectivity section); and to introspection which has no relevance to physical science, but is vital to social science (Scientific Method: Social Phenomena section). All of these issues feed the criticism (Scientific Method: ‘Horses for Courses’ section) of ‘scientism’. The latter describes ‘an attitude which is decidedly unscientific’ (Hayek, 1952a, pp. 15–16); that is, the application of methods which, though proven in the science of simple phenomena, are inappropriately applied to complex phenomena. Scientism ‘impedes the progress of the understanding of society’ (Hayek, 1952a, p. 17), by its attempt to deal only with objective facts. Against this, and the mathematical formalisation which has been particularly predominant in twentieth-century economics, the paradigm of evolutionary adaptation (Social Order: Evolutionary Adaptation and Conclusion sections) is more recent, but increasingly recognised for its rich insights into all aspects of the human condition.

MAN AND SOCIETY

Against the protracted timescale of biological evolution, rapid societal changes invoke particular consideration. A general evolutionary pattern is

characterised in which hierarchical rivalry between *hominids* gave way to the 'egalitarianism, cooperation, and sharing' (Whiten, 1996, p. 140) of hunter-gatherer groups; and, while that culture of booty-sharing shaped 'the human mind for more than two million years' (Leaky & Lewin, 1992, p. 142), the most 'impressive evidence of hunting' (as opposed to scavenging) is found 'with the emergence of *Homo sapiens* as recently as 100,000 years ago' (Whiten, 1996, p. 140). Thereafter, and less controversially, the hierarchical structures of farming communities emerged over a very short period of 10,000 years.

With the vast difference between the biological and social timescales, a consequential thesis is that 'human nature' is determined largely by natural selection that occurred during the Pleistocene period; and that our psychological adaptations are largely determined by problems encountered by our hunter-gatherer ancestors. The implication is that many behavioural characteristics of modern man are atavistic; that is, they are the legacy of evolved behavioural traits that are no longer relevant. In short, there is a 'thesis of ancient provenance' (Plotkin, 1998, p. 73) to which common phobias in regard to snakes, open spaces and dark places lend support. Other phenomena ('road rage', for example) arise when rapid societal developments sit beside comparatively static psychological dispositions. A contrasting view is that the developmental plasticity of the human mind constitutes the relevant evolutionary adaptation; and that the mind, like the immune system, is continuously adapting, over both evolutionary time (phylogenic) and an individual lifetime (ontogenic).³ Although these views are not mutually exclusive, Hayek's particular emphasis is 'that modern man is torn by conflicts which torment him and force him into ever-accelerating further changes'; and that instinctive drives are overlain by 'the remains of the traditions acquired in the successive types of social structure through which he has passed' (Hayek, 1979, p. 159). Innate psychological traits and the requirements of the extended socio-economic order of modern industrialised society create many tensions:

innate structures built into man's organisation in the course of perhaps 50,000 generations were adapted to a wholly different life from that which he has made for himself during the last 500, or for most of us only 100, generations or so. ... And although we still share most of the emotional traits of primitive man, he does not share all ours, or the restraints which made civilisation possible. Instead of the direct pursuit of felt needs or perceived objects, the obedience to learnt rules has become necessary to restrain those natural instincts which do not fit into the order of the open society (Hayek, 1979, p.160).

With his work both in psychology and social science, Hayek sought to gain knowledge about knowledge. How does an individual know

(in both the conscious and subconscious sense) how to set his own actions? How does an individual know how to anticipate and to react to the actions of others? In addressing those questions, psychology and social science are constrained by what is knowable about human understanding. In the study of the acquisition, application and adaptation of knowledge, the thesis is that each of these disciplines deals with structures and processes that are analogous both in broad terms and in regard to many elemental details.

A general thought – that he who does not know himself has known nothing – is familiar from many sources. It is ascribed to various Greek philosophers,⁴ and it is a *cliché* in literature and in psychological and religious discourse. Its relevance is that an understanding of the function of the human mind (psychology) gives the basis for an understanding of man's interaction with others (social science). Herein lies the big problem: as a (first) universal methodological principle, '[a]ny apparatus of classification must possess a structure of higher degree of complexity than is possessed by the objects which it classifies' (Hayek, 1952b, p. 185). In particular regard to psychology, this means that no mind can ever fully understand itself, because 'to 'explain' our own knowledge would require that we should know more than we actually do, which is, of courses, is a contradictory statement' (Hayek, 1952b, p. 86). Nevertheless, the parallel, mutually re-enforcing aspects of psychology and social science are central to Hayek's vision of the evolutionary adaptation of the individual and society that exists as a spontaneous and complex social order. While such linkages await (or defy) comprehensive appraisal, some primary features may be examined.

With its focus upon human understanding in its psychological and social dimensions, Hayek's work rests upon the ontological presumption of a material world and of life forms that evolve from – and are an integral part of – the material world. The function of the mind is to give direction and protection to an individual. The function of the social framework is to give direction and protection to individuals. The respective direction and protection given are very different. The mind decides for an individual, whereas the social framework provides the context within which individuals bring effect, coherence and coordination to their interdependent decision-making. The social framework inhibits decisions that individual minds might otherwise be willing to take; and it shapes (and informs of) the consequences that are likely to follow upon those decisions. In short, it creates a culture of common values.

The mind and the social framework are self-organising, spontaneously evolving, open adaptations that are defined, not in terms of their physical composition, but in terms of their abstract⁵ functions. With the capacity to

process only a small proportion of the sensory stimuli that are signalled as touch, taste, sight, smell and sound, the mind has evolved to know what it needs to know. The neurophysical explanation is that there is a deep-seated cerebral structure (the 'value system') that projects over the entire cortex:

[v]alues reflect events involving the nervous system that have been selected during evolution because they contribute to adaptive behaviour and to phenotype fitness. Examples of low-level values are: 'eating is better than not-eating' or 'seeing is better than not seeing' (Edelman & Tononi, 1995, p. 85).

In like achievement, social institutions that guide individuals' actions are a surrogate for that vast array of information that individuals would otherwise need to know. With neither is there (nor could there be) any store of comprehensive knowledge. Instead, neurological processing achieves a simultaneous and selectively purposeful engagement with a multitude of sensory data; and social exchange allows effective use of that knowledge of 'the particular circumstances of time and place' (Hayek, 1945, p. 83) which is dispersed across a multitude of individuals' minds.

PSYCHOLOGY: ALTERNATIVE PARADIGMS

Competing paradigms exist in regard to the nature of cognitive processes. The orthodox 'symbol-processing' paradigm sites the biological memory-store at definite locations in the brain, as in the location of memory within a computer. Evidence to the contrary – rat behaviour after the surgical destruction of part of the brain – is an inability 'to demonstrate the isolated localisation of a memory trace anywhere within the nervous system' (Lashley, 1950, p. 478). An alternative paradigm – of which *The Sensory Order* is an early statement – is that of 'connectionism'. It rejects the notion of a mind memory-store whose contents either reflect, or are correlated with, characteristics of elements in the material world:

we do not first have sensations which are then preserved by memory, but it is a result of physiological memory that the physiological impulses are converted into sensations. The connections between the physiological elements are thus the primary phenomenon which creates the mental phenomena (Hayek, 1952b, p. 53).

Connectionism regards mind as an ever-active classification system, whose dynamic instrumentation is the neural structure of the brain. As raw sensory impulses are fed continuously to the brain via the nervous system, changing patterns of neural interconnections are the instrumental basis for an emergent sensory order (mind). Understanding is achieved as the mind

reorders itself, by its classification (and reclassification) of sensory impulses within multi-layered neural network configurations.

The significance of a neural impulse derives from its relationship to, and combination with, other impulses in producing sensory qualities (sensations) that are determined by the ‘differentiating’ neurological responses of the system as a whole:

sensory qualities are not ... attached to ... individual physiological impulses, but ... [are] ... determined by the system of connections by which the impulses can be connected from neuron to neuron (Hayek, 1952b, p. 53).

At the highest levels of consciousness, a response to stimuli is modified by the influence of the widest range of impulses from other sources. For simple reflex action, higher centres receive simultaneous reports of both stimulus and response. And, in general, a continuous range of ‘engaged’ connections is hypothesised within which no qualitative distinction is afforded to the highest abstract processes of intellectual thought. Included is a set of learned behaviours embedded as supra-conscious; and that our reason is dependent upon such tacit knowledge of cultural habits, rules and procedures implies that we know more than we can express.

Memory and thought processes are indistinguishable neurological processes – particular neural network configurations – that engage (potentially) the whole brain, by the variable strength of neural impulses. The broad feature of connectionism is that mental properties are ‘determined by the place of the impulse in a system of relations between all the neurons through which impulses were passed’; and it was this ‘clear perception’ that led Hayek ‘to interpret the central nervous system as ... a process of continuous and simultaneous classification and constant reclassification’ (Hayek, 1952b, p. 289). As a very simple example, in confronting a solid object of given shape and weight, a set of sensory impulses might be ‘memorised’ simultaneously in networks that identify aspects of stone, fossil, building material, shape, hardness, projectile and so on; and there may be further associations with networks that identify aspects of (say) architecture, paleontology, design, density, ballistics and so on.

The order that is expressed within the mind by multiple configurations of neural networks is a guide to relationships that exist between external events that have previously stimulated, or are currently stimulating, sensory impulses to the brain. Those multiple neural networks provide the structural ‘map’ for the classification of new impulses in constructing ‘a kind of model of the particular environment in which the organism finds itself at the moment and which will enable it to take account of that environment in all its

movements' (Hayek, 1952b, pp. 114–115).⁶ Although continuous sequences of impulses mould the neural structure, in relation to the 'constantly changing patterns of impulses', the structural map 'can be regarded as semi-permanent' (Hayek, 1952b, p. 116). In this manner, the adjustment of neural networks is the mechanism that enhances understanding.⁷

Differentiating responses of the neurophysiological system are determined by linkages previously created within the brain: a system of connections 'acquired in the course of development of the species and the individual by a kind of "experience" or "learning"' (Hayek, 1952b, p. 53). Such phylogenic development determines pre-sensory neurological linkages; that is, a priori knowledge that 'is not learnt by sensory experience, but is rather implicit in the means through which we can obtain such experience' (Hayek, 1952b, p. 167).

In summary, therefore, sensory qualities (or, in more recent parlance, 'qualia') derive from the 'engagement' of multiple neural networks as sensory stimuli, transmitted as physiological impulses, are classified either by the reinforcement of existing neural networks or by the initiation of new neural networks. The notion of memory cells is defunct. Rather, *individual neurons are merely the foci in the network of relationships and it is the multiple interplay of impulses between neurons ... which forms the recurrent, recognisable and familiar elements of the mental structure.* The description of mind that is set in italics is contrived to parallel Hayek's description of social order –

individuals are merely the foci in the network of relationships and it is the various attitudes of the individuals towards each other ... which form the recurrent, recognizable and familiar elements of the [social] structure (Hayek, 1952a, p. 34)

– so as to re-engage with the central thesis, which is that the mind and the social framework are self-organising, spontaneously evolving, open adaptations that display common characteristics. Mutual adaptations shape both: (1) the categorisations and the associations that are embedded within an individual mind as its response to sensory stimuli; and (2) the categorisations and associations that are embedded within social institutions and which define the rules for social behaviour. The spontaneous orders of the mind and of social affairs give the foundation for implicit understandings, conscious deliberation and formal scientific theory.

PSYCHOLOGY: MENTAL AND PHYSICAL EVENTS

The phylogenic cognitive development of every life form is driven by particular adaptations that enhance the likelihood of its survival within a

material world. Since the evolution of that unique knowledge is fundamentally driven by the presentation of particular survival challenges, it follows that knowledge is different for different life forms whose evolutionary experiences differ radically. Mutual understanding between different life forms is, therefore, problematic: a human is more sensitive to (the perceptions of) another human than to a rat or (less still) to a bat or (less still) to a gnat. The indisputable inference is that there is no definitive 'knowledge' that is approached generally by developmental understanding:

the classes into which different events are placed in the process of perception, are not attributes which are possessed by these events and which are in some manner 'communicated' to the mind; they are regarded as consisting entirely in the 'differentiating' responses of the organism by which the classification or order of events is created ... Every sensation, even the 'purest', must therefore be regarded as an interpretation of an event in the light of past experience of the individual or the species (Hayek, 1952b, p. 166).

It follows that it is impossible to explain mental events upon the basis of the physical events that stimulate the senses; and that any inquiry regarding 'why people as a result of particular material circumstances hold particular views at particular moments, is fundamentally misconceived' (Hayek, 1952b, pp. 192–193). Given the obvious relevance of those 'particular views' to an individual's behaviour, Hayek is insisting that 'particular material circumstances' have no practical relevance in explaining that behaviour. More importantly, he denies the moral relevance: the precept that an individual's personal responsibility might be extenuated upon the basis of 'material circumstances' would destroy 'the chief device which society has developed to assure decent conduct – the pressure of opinion making people observe the rules of the game'. The concept of 'some metaphysical self which stands outside the chain of cause and effect' (Hayek, 1967, p. 232) is inadmissible, because it would leave nothing for which an individual could be held personally responsible. To hold an individual 'responsible for the consequences of an action' is neither an assertion of causation nor of fact (which we cannot know), but 'of the nature of a convention introduced to make people observe certain rules' (Hayek, 1960, pp. 74–75).

Given the widespread practice of allowing defence counsel to plead extenuating circumstances prior to passing sentence in courts of law, it is noteworthy that – in a similar context – Bertrand Russell⁸ questions 'how far human volitions are subject to causal laws' (Russell, 1929, p. 404). While it is 'obvious that there is *some* degree of correlation between the brain and the mind', Russell doubts the existence of 'laws of

correlation of the mental and the physical' (Russell, 1929, p. 405); or, as Hayek states it:

though we may know that mental events of the kind we experience can be caused by the same forces which operate in the rest of nature, we shall never be able to say which are the particular physical events which 'correspond' to a particular mental event (Hayek, 1952b, p. 194).

Undoubtedly past events are relevant both in their phylogenetic and ontogenic dimensions, but (and this is a crucial point) the impact of 'our previous history' is upon 'the whole human personality':

[e]ven though we may know the general principle by which all human action is causally determined by physical processes, this would not mean that to us a particular human action can ever be recognizable as a necessary result of the whole human personality – that means the whole of a person's mind – which ... cannot reduce to something else (Hayek, 1952b, p. 193).

It is not the case, for example, that every individual is necessarily brutalised by brutal circumstances. So, while general tendencies that shape motivation may allow the formulation of precepts for (say) education and penal systems, there is no secure basis for mitigating the social requirement for an individual to be held to account for his actions.

SCIENTIFIC METHOD: FROM SENSORY ORDER TO PHYSICAL ORDER

In progressively distancing itself from the experience of 'immediately given sensory qualities', physical science builds

models of the connexions of the events in the external world by breaking up the classes known to us as sensory qualities and by replacing them by classes explicitly defined by the relations of events to each other ... [and as] ... this model of the physical world becomes more and more perfect, its application to any particular phenomenon in the sensory world becomes more and more uncertain (Hayek, 1952b, p. 194).

By that description, the development of physical science is 'a process of progressive emancipation from our innate classification of the external stimuli till in the end they completely disappear' (Hayek, 1952a, p. 20). Indeed, 'things in the external world show uniformity in their behaviour towards each other only if we group them in a way different from that in which they appear to our senses' (Hayek, 1952a, p. 22). For example, white powders that trigger identical sensory stimuli may be placed into different

scientific categories; and with that move from qualitative to quantitative categorisations, ever-greater precision is

the essence of this process of breaking up our immediate sense data and of substituting for a description in terms of sense qualities one in terms of elements which possess no attributes but ... the relations established by systematic testing and experimenting (Hayek, 1952a, p. 23).

In this manner, sense data are ‘pushed steadily further back’ as objects are reclassified ‘by explicit relations instead of by implicit relations inherent in our sensory distinctions’ until, eventually, there remains only a ‘system of explicit definitions’ that is ‘all-comprehensive and self-contained or circular’ (Hayek, 1952b, p. 171). Russell drives⁹ more explicitly to the same conclusion in arguing that ‘same cause, same effect’ is ‘utterly otiose’ as a scientific principle:

[a]s soon as the antecedents have been given sufficiently fully to enable the consequent to be calculated with some exactitude, the antecedents have become so complicated that it is very unlikely that they will ever recur. Hence, if this were the principle involved, science would remain utterly sterile (Russell, 1929, p. 392).

So, for example, while the general experience is that the impact of a stone causes glass to break, such details as mass, velocity, thickness and other contributing circumstances form a vast number of potential configurations. If those antecedents were sufficiently well defined to enable their consequences to be unambiguously calculated (that is, beyond the vagueness of glass usually breaks when hit by stones), every event would be unique and, therefore, offer no basis from which to generalise upon cause and effect.

Russell’s example illustrates how events become recategorised to the point where there remains a unique ‘system of explicit definitions’ that is ‘all-comprehensive and self-contained or circular’ (Hayek, 1952b, p. 171). Russell’s conclusion is that causality is a mark of scientific infancy: ‘laws of probable sequence, though useful in daily life and the infancy of science, tend to be displaced by quite different laws as soon as a science is successful’; and so, with ‘every advance in science’ there is an ever ‘greater differentiation of antecedent and consequent’ until eventually ‘there is nothing that can be called a cause, and nothing that can be called an effect; there is merely a formula’ (Russell, 1929, pp. 392–395).¹⁰

SCIENTIFIC METHOD: THEORY AND FACT

Physical science improves upon the sensory order (that is, upon the innate classification or 'mental structure' of sensory impulses) with categorisations that achieve greater behavioural uniformity in the relationships between material things. Precise instrumentation and experimentation extend the means to differentiate between otherwise identical sensory phenomena. Formal 'scientific' theories deliver ever-more coherent and extensive categories of classification. As different levels of understanding are consciously 'assumed', new patterns supersede primary categorisations of sensory impulses. Whether they are qualitative or quantitative, patterns are necessarily theory laden. Yet, a naïveté in regard to the possibility of observations detached from theory is disturbingly common among practitioners of science.

There is a widely cited response from Albert Einstein to Werner Heisenberg's assertion (in 1926) that only observable magnitudes should contribute to a theory. It is that '[i]n reality the very opposite happens. It is the theory which decides what we can observe' (Heisenberg, 1971, p. 63). Analogous sentiments are expressed by Hayek in *'The Theory of Complex Phenomena'*, where he notes that

[i]ntimate acquaintance with the facts is certainly important; but systematic observation can only start after problems have arisen. Until we have definite questions to ask we cannot employ our intellect; and questions presuppose that we have formed some provisional hypothesis or theory about the events (Hayek, 1967, p. 22);

and, in that same paper, Hayek cites Karl Popper:

[s]cience ... cannot start with observations, or with the "collection of data", as some students of method believe. Before we can collect data, our interest in *data of a certain kind* must be aroused: the *problem* always comes first (Popper, 1957, p. 121).

It is the essence of understanding that abstract constructs are a prerequisite to shaping order from disorder. Facts are not given; they are created. So, there is a (second) universal methodological principle: whether implicitly or explicitly stated, theory pervades every observation.¹¹ Without theory, we cannot know what is taking our attention. And there is an associated measurement problem. Theory is built upon theory: '[m]easuring instruments are constructed in accordance with laws and their readings are tested under the assumption that these laws are correct' (Feyerabend, 1993, p. 232). So, for example, 'Galileo's telescope provided evidence only for those who could accept Galileo's theory of optics, which was less well

established than the hypothesis it was required to support' (Loasby, 1989, p. 16). More generally, a fundamental methodological conundrum is that there is no rigorous definition of scientific rigor.

SCIENTIFIC METHOD: OBJECTIVITY AND SUBJECTIVITY

To the extent that everything that an individual knows is mediated through a unique brain, everything is known subjectively; or, rather, there is no distinction between objective and subjective phenomena. However, as knowledge progresses beyond the immediate experience of sensory phenomena, a distinction may be drawn:

whenever we have to explain human behaviour towards things; these things must ... not be determined in terms of what we might find out about them by the objective methods of science, but in terms of what the person acting thinks about them (Hayek, 1952a, p. 30).

In Hayek's presentation, 'objective facts' are those that 'can be defined without referring to our knowledge of people's conscious intentions with regard to them', because they are determined 'by the objective methods of science' (Hayek, 1952a, p. 30); and he infers that different people are more likely to have identical beliefs about 'an "objective" fact' than they are to agree upon the nature of 'a subjective phenomenon' (Hayek, 1952a, p. 29). The distinction arises because the 'order in social phenomena ... cannot be stated in physical terms. ... It is an order in which things behave in the same way because they mean the same thing to man' (Hayek, 1952a, p. 40).

In explaining a social phenomenon, the scientist needs to know 'what the people dealing with it think' (Hayek, 1952a, p. 33), rather than that what is known objectively. Potential confusion is introduced when Hayek distinguishes between 'laws of nature' as they are generally acknowledged in any 'contemporary society' and those that 'figure in the works of natural scientists' (Hayek, 1952a, p. 30). He suggests that

[i]f the current "scientific" knowledge of society which we study included the belief that the soil will bear no fruit till certain rites or incantations are performed, this would be quite as important for us as any law of nature which we now believe to be correct (Hayek, 1952a, pp. 30–31).

The illustration shows how the two categories intersect. Established 'rites and incantations' are (by their survival) more likely to have practical (objectively scientific) relevance rather than to be devoid of (objectively

scientific) meaning. So, for example, as crop yields are raised if fish accompany the planting of seed, rival explanations are either ('subjective' belief) that deities are pleased by gifts of fish or ('objective' fact) that fish enhance soil fertility. Irrespective of the explanation, scattering fish is acknowledged to be relevant to crop yields. The difficulty arises from the absence both of an impartially 'objective' position and of definitive scientific rigor: (1) material structures and events are accessed only indirectly via the mind; (2) 'objective' methods are corrigible; (3) data (whether scientifically or casually derived) are theory laden; (4) scientific theories are creations of the mind; and (5) science sets no sharp division between a layman's description and a scientist's explanation:

[t]he scientific way of forming concepts differs from that which we use in our daily life, not basically, but merely in the more precise definition of concepts and conclusions; more painstaking and systematic choice of experimental material; and greater logical economy (Einstein, 1940/1953, p. 253).

SCIENTIFIC METHOD: SOCIAL PHENOMENA

A statement of the material properties of a social phenomenon is unlikely to convey its social meaning. If social phenomena were defined in terms of (the current state of 'objective' knowledge about) their material components, 'we should probably find no recognisable order whatever in social phenomena' (Hayek, 1952a, p. 40). A legal agreement, an obligation, a ritual, a religious or community association, et cetera, are unlikely to be identified by the material properties of their component elements.

Where individuals from diverse backgrounds react in similar manner, it is because some commonality – say, in their cultural conditioning – causes them to regard 'situations' as identical, rather than because situations are alike in any material sense. For example, a gift, a punishment or a prize are defined, not by physical properties, but by subjective values and beliefs; and the relevance of the term 'subjective' is 'that the knowledge and beliefs of different people, while possessing that common structure which makes communication possible, will yet be different and conflicting in many respects' (Hayek, 1952a, p. 29).

It is for social science to examine 'the consequences of the fact that people perceive the world and each other through sensations and concepts which are organized in a mental structure common to all of them' (Hayek, 1952a, p. 23). This commonality allows social scientists to 'understand and explain human action in a way we cannot with physical phenomena, ... [so that] ...

the term “explain” tends to remain charged with meaning not applicable to physical phenomena’ (Hayek, 1952b, p. 18). Upon that basis, social science has an advantage over physical science. It would be to abandon that advantage, if social science were to succumb to ‘attempts to dispense with our subjective knowledge of the working of the human mind ... [that is] ... to do without the knowledge derived from “introspection”’ (Hayek, 1952a, pp. 44–45). Introspection affords unique insights:

in his conscious decisions, man classifies external stimuli in a way which we know solely from our own subjective experience of this kind of classification. We take it for granted that other men treat various things as alike or unlike just as we do ... [unless, for example] ... they are colorblind or mad (Hayek, 1952b, p. 26).

Without introspection, it would be impossible to understand human action. To cite one illustration (see Hayek, 1952a, p. 43), the only means by which an archaeologist decides whether a stone has been shaped by nature or by man, is to apply introspection to understand the mind and the capacities of prehistoric man. There is no other way. Introspection is the only rational basis for the scientific understanding of another individual’s behaviour. The validity of introspection is verified most obviously by our ability to communicate with other people and to achieve mutual understanding. Whether consciously or subconsciously, an individual applies introspection to decide how phenomena are most likely to be categorised by the minds of other individuals.

SCIENTIFIC METHOD: ‘HORSES FOR COURSES’

Although not the first to note that a new science might be ‘hampered by too slavish an imitation of the technique of some older science’ (Russell, 1931, p. 178),¹² Hayek gives it a name. He labels as ‘scientism’ and as ‘“scientistic” prejudice’ any investigation that is motivated ‘not with the general spirit of disinterested inquiry but with slavish imitation of the method and language of Science’ (Hayek, 1952a, p. 15). In particular, scientism denotes an uncritical application to social science of method that is appropriate to physical science:

the devices developed by the natural sciences for the special purpose of replacing a description of the world in sensory or phenomenal terms by one in physical terms lose their *raison d’être* in the study of intelligible human action (Hayek, 1952b, p. 193).

In certain respects, scientism presents as an extension of 'behaviourism', with its aversion to unobservable causes and a desire to establish psychology as a science of human behaviour. Unobservable processes of the mind were viewed as an inappropriate subject for scientific inquiry: '[w]hat behaviourism did was forbid causal explanations if they did not lie within the limits of ordinary everyday experience. Scientifically this was an extraordinary bankrupting stance' (Plotkin, 1998, p. 33).

In its particular application to social science, scientism seeks direct access to 'social phenomena', rather than to identify them indirectly in the 'principles of structural coherence of complex phenomena' (Hayek, 1952a, p. 38); and it treats social phenomena, not 'as something of which the human mind is a part and the principles of whose organisation we can reconstruct from the familiar parts, but as if they were objects directly perceived by us as wholes'. Hayek's counter is that, while it is true that 'the "whole" situation ... will greatly differ from place to place and from time to time', without knowing the 'familiar elements from which the unique situation is made up' (Hayek, 1952a, p. 53), no meaning can be assigned to individual actions.

In their aspiration to match the kind of laws emanating from nineteenth-century physical science, many social theorists had constructed grandiose models – Marxism 'the vehicle through which this result of scientism gained so wide an influence' (Hayek, 1952a, p. 74) – as a basis for understanding social development. Hayek likens the resulting narrative to that of 'the proverbial observer from another planet':

his records would in a sense be history, such as, e.g., the history of an ant heap. Such theory would have to be written in purely objective, physical terms. ... But such history could not help us understand any of the events recorded by it in the sense in which we understand human history (Hayek, 1952a, p. 79).

Motivated by a fear 'of starting from the subjective concepts determining individual actions' and a concern 'to avoid using as data the concepts held by individuals', scientism succumbs to the 'collectivist prejudice' (Hayek, 1952a, p. 38); and, yet,

the wholes as such are never given to our observation but are without exception constructions of our mind. They are not "given facts," objective data of a similar kind which we spontaneously recognize as similar by their common physical attributes. They cannot be perceived at all apart from a mental scheme that shows the connection between some of the many individual facts which we can observe (Hayek, 1952a, pp. 53–55).

The ‘specific theoretical method which corresponds to the systematic subjectivism and individualism of the social sciences’ (Hayek, 1952a, p. 38) is that of methodological individualism, which Hayek endorses in a very particular sense; that is, where a social scientist ‘systematically starts from the concepts which guide individuals in their actions’ as distinct from such ‘pseudo-entities’ as “‘society” or the “economic system,” “capitalism” or “imperialism,” and other such collective entities ... which he must not mistake for facts’ (Hayek, 1952a, pp. 38–39).

Hayek refines the distinction between methods that are appropriate to physical science and those that are appropriate to biological and social sciences, when he refocuses upon ‘simplicity’ and ‘complexity’; that is, upon ‘the relatively simple phenomena with which the natural sciences deal’ as distinct from ‘the more complex phenomena of life, of mind, and of society’ (Hayek, 1967, p. 25).¹³

In the broadest terms, science has two vantage points. The first is appropriate in relation to simple phenomena and it allows components to be discovered within a whole; the second is appropriate in relation to complex phenomena and it allows a whole to be built from components. The former offers a general (holistic) view of the systematic interrelationships of the material world, whose component elements may be accessed using ‘objective’ scientific methods that reveal detail within the whole. In understanding metals, for example, the most rudimentary categorisation is made by the immediate sensory qualities of colour, density and so on. Thereafter, in working metal with fire and water, various mechanical properties are revealed; and with the application of ever-more sophisticated scientific methods, electro-chemical and molecular structures are identified. Although understanding is always incomplete and susceptible to modification, mutually consistent characteristics unfold against a set of universal physical laws. In Hayekian terms, immediate differentiation by sensory qualities is replaced by ‘objective’ categorisations of material differences.

The second vantage point presents an ever-widening perspective upon our primary experience of social relationships. Here the application of methods appropriate to social science may ‘lead to the *discovery* of principles of structural coherence’ of complex spontaneous social orders. The approach is ‘compositive’, a variant of methodological individualism that is too readily conflated with the atomistic approach of neoclassical microeconomics (see Caldwell, 2004, pp. 285–287). Unlike physical matter, which is presumed to conform to universal structures (laws), the presumption for the human sciences is quite different. Psychobiological and socio-historical processes are evolved systems; and memory is critical to all systems that follow a path

of evolutionary adaptation. Although constrained by the laws of physical science, there is a vital ancestral heritage that brings uniquely evolutionary aspects to human science; that is to biology, psychology and social science. By reason of their superior usefulness, a series of evolutionary path-dependent adaptations – modifications of physiological, behavioural and organisational characteristics – are selected and preserved (both genetically and institutionally) by way of trial, selection, retention and replication.

SOCIAL ORDER: EVOLUTIONARY ADAPTATION

In particular regard to the development of Hayek's social theorising, the publication in 1937 of '*Economics and Knowledge*' is regarded as a landmark. The paper

stemmed from a joke about economists speaking about given data just to reassure themselves that what was given really was given. That led me, in part, to ask to whom were the data really given. To us, it was of course [given] to nobody. The economist assumes [the data] are given to him, but that's a fiction. In fact, there's no one who knows all the data or the whole process, and that's what led me in the thirties, to the idea that the whole problem was the utilization of information dispersed among thousands of people and not possessed by anyone (Hayek, 1994, p. 147).

Against the neoclassical assumption of 'given data', Hayek refocuses attention upon 'the idea that the whole problem was the utilization of information dispersed among thousands of people and not possessed by anyone' (Hayek, 1994, p. 147). In its further development, Hayek's socio-economic theory emphasises the efficiency of evolving social institutions in the generation, selection and the use of knowledge that is both (1) tacit and embodied within institutional structures and (2) explicit and dispersed across a multitude of individuals. Although ideas exist only within individual minds, the elements of a social structure are little changed even as one generation of individuals replaces another. Attitudes and relationships can be preserved across a succession of individuals; and this allows social structures to be studied apart from any particular individuals, who just happen to be 'the foci in the network of relationships' (Hayek, 1952a, p. 59). Knowledge

only exists in the dispersed, incomplete, and inconsistent form in which it appears in many individual minds, and the dispersion and imperfection of all knowledge are two of the basic facts from which the social sciences have to start (Hayek, 1952a, p. 50);

but, even if comprehensive information were readily available, it would vastly exceed the capacity of any individual mind (or the combined mind-capacity of any organisationally manageable group of individuals) to absorb: 'the most complicated thing in the universe is the collective of human brains and their psychological processes that make up human culture' (Plotkin, 1998, p. 222).

In its widest application, social science seeks to explain issues that arise from the (unintended) consequences of human action,

such as the problem of the compatibility of intentions and expectations of different people, of the division of knowledge between them, and the process by which the relevant knowledge is acquired and expectations are formed (Hayek, 1952a, p. 33).

Social cohesion requires that individuals' ever-changing and inconsistent views, decisions and actions should become compatible, not only with each other, but also with an objective reality. Social science inquires into the manner whereby these tendencies derive from evolutionary forces acting upon social structures (that is, social institutions in the very broadest sense).

Until the late 1950s, the concept of evolution 'was largely confined to its original explanatory brief, the origin of the species, where it remained a flat, unilevel, conception'. However, by its 'success as a scientific theory', other disciplines began to consider how they might elaborate 'evolutionary theory in ways that would allow it to encompass complex structures' (Plotkin, 1998, p. 226). There is little doubt that Hayek was influenced¹⁴ by developments such as these:

Hayek interacted with natural scientists in his Chicago seminar, and biological headings figured prominently on the only mimeograph sheet (besides the provisional syllabus) that remains from the seminar. ... by the middle of the decade Hayek had come to a startling observation, one fully compatible with his new readings: the complex adaptive orders that had been identified by the classical economists, by philologists, and by others, the sorts of orders that he had encountered again in his research on the brain, were in fact to be found in a variety of other scientific fields. He drew the conclusion that the basic dividing line among all the sciences was between those that studied simple and those that studied complex phenomena (Caldwell, 2004, p. 304).

The mutual adaptations that are spontaneously achieved through evolutionary processes allow extended and complex social systems to maintain the coherence necessary for their most effective function. In elaborating further, it is instructive to consider the antithesis of the evolutionary hypothesis.

'Constructivist rationalism' is used by Hayek to indicate the erroneous belief – a derivation of the philosophy of Descartes and Rousseau – that

man created the institutions of civilisation and is able to reconstruct then according to his will: 'Society appeared to them as a deliberate construction for an intended purpose' (Hayek, 1978, pp. 5–6). Such constructivist social engineering is analogous to a designer's attempt to emulate the performance of a bird's wing (see Dawkins, 1982, p. 46). Since the best design meets specifications at lowest cost, minimum performance criteria are necessary. Unlike those pertaining to evolution, *all* those performance criteria are arbitrary: materials, maximum loading, safety margins, *etc.* Given those standards, the designer seeks an optimal solution. By contrast, all aspects of a bird's natural wing are adaptive solutions to the myriad of problems that arose in real time and which (unlike the designer's solution) proved capable of bringing the bird's particular ancestral line thorough to the present. Those same considerations are relevant to the difference between the *minutiae* of evolved social institutions (the natural wing) and to grandiose designs for socio-economic development (the designer's alternative).¹⁵

Interdependent decisions taken by individuals have the most far-reaching consequences that defy both anticipation and retrospective appraisal; and yet they contribute towards an extensive social order. Whether it is the manner 'in which footpaths are formed in a wild broken country' (Hayek, 1952a, p. 40), or the evolution of money, or the formation of language, or the formation of prices or the direction of production under competition, none is the outcome of conscious design. The diversity across individuals, in terms of their subjectively held and inconsistent beliefs, is given coherence within a spontaneously evolving social order. The evolutionary development of social institutions is directed by minor changes (analogous to genetic mutations) that either aid or exacerbate social cohesion. Even where design and forward planning are present, the 'contrived' nature of their contributions is not of the essence. Rather, success is achieved in consequence of the general characteristics of an evolutionary process:

a mechanism of reduplication with transmittable variations and competitive selection of those which prove to have a better chance of survival will in the course of time produce a great variety of structures adapted to continuous adjustment to the environment and to each other (Hayek, 1967, p. 32).

Where the development is perverse, prosperity (again, in the very broadest sense) is diminished and the community is damaged. By their greater catallactic¹⁶ and/or organisational harmony, evolutionary processes allow some communities to prosper as others decline. In all cases, evolved systems develop as a whole and, if separated, their mutually adapted component elements are unlikely to remain viable. In respect of *any* system, knowledge

of those components is never enough. Knowledge of the 'rules' by which components are connected is also required:

[t]his means that, before we can hope successfully to improve them, we must learn to comprehend much better than we do now in what manner the man-made rules and the spontaneous forces of society interact (Hayek, 1967, p. 92).

Social interaction relies upon individuals' conceptualisations (their categorisations of sensory impulses), in the absence of which purposeful action has no meaning. Social processes comprise the interrelationships that (via rational decisions and purposeful actions) derive from those conceptualisations.

Rationality is the achievement of 'some degree of coherence and consistency in a person's action, some lasting knowledge or insight which, once acquired, will affect his action at a later date and in different circumstances' (Hayek, 1960, p. 77). That same 'coherence and consistency' allows others to build upon their own introspective insights, with the effect of enhancing social order. The basis of all rational human action rests upon the validity of some expectation that individuals' actions 'have a good chance of achieving their aims' (Hayek, 1973b, p. 11). That expectation, in turn, requires a significant degree of stability in regard to social institutions and to conventional patterns of behaviour; that is, to custom and practice. The framework of social institutions gives direction to the actions of individuals. In their turn, individuals' actions give shape to the social framework of institutions. The many and varied components of the social framework include tacit and explicit customs, practices, rules, laws and legislation that affects (and is affected by) the choices and behaviour of interacting (socialising) individuals.

CONCLUSION

The Sensory Order and *The Counter-Revolution of Science* are seminal works. Together, they show how close parallels across biological, psychological and social adaptations are a basis for determining which methods are appropriate to gain knowledge about knowledge. Methods of proven worth to physical science have little, if any, relevance for the analysis of either psychological or social phenomena, because these are more complex than the phenomena of the material world:

invoking historical antecedence as cause is difficult science, and ... those, like chemists or physiologists, who do not have to take in the sweep of time with its changing

conditions are fortunate indeed to have to deal only with relatively easily approached now-and-forever causes (Plotkin, 1998, p. 19).

The mathematical precision by which the laws of physical science can be presented is a siren voice to social theorists:

the blind transfer of the striving for quantitative measurement to a field in which the specific conditions are not present which give it its basic importance in the natural sciences, is the result of an unfounded prejudice (Hayek, 1952a, p. 51).

Against 'scientism', Hayek positions socio-economic theory firmly within the adaptive context of evolutionary processes, the capacities of human understanding and the characteristics of an extended social order. Socio-economic data and qualitative information are dispersed, uncertain and of varying quality. In principle, however, evolutionary processes may be expected to reconcile inconsistencies: (1) in respect of the mind, between diverse sensory impulses, so as to achieve a coherent and safe accommodation of an individual to the world around; and (2) in respect of the social order, between different individuals, and between individuals and an objective reality.

NOTES

1. *Psyche*: the mind functioning as the centre of thought, emotion and behaviour and consciously or unconsciously mediating the body's responses to the social and physical environment.

2. In the Preface to *The Sensory Order*, Hayek states how a 'basic idea' first conceived when he was reading psychology 'without much guidance ... in Vienna in 1919 and 1920' subsequently 'proved helpful in dealing with the problems of the methods of the social sciences' (Hayek, 1952b, pp. v–vi). Even experienced Hayek scholars can be mistaken if tested to identify the source of citations taken from *The Sensory Order* and *The Counter-Revolution of Science*.

3. *Ontogeny*: the evolution of individual organisms through the interaction between genetic and environmental determinants. *Phylogeny*: the evolutionary development of species across successive generations.

4. *Gnothi seauton*: 'know thyself', a precept inscribed in gold letters over the portico of the temple at Delphi. Authorship is variously ascribed to Pythagoras, to several of the wise men of Greece, and to Pheonoe, a mythical Greek poetess.

5. For example, the function of numeracy (from the mind) and of the law (from society) are not identified by the respective material composition of any brain or law court.

6. See McQuade and Buto (2005), for explicit socio-economic applications of the map–mind paradigm.

7. Gerald Edelman continues to extend the map–mind relationship. Thus, for illustration, ‘consciousness is an outcome of comparative memory in which previous self-nonsel self categorizations are continuously related to ongoing present categorizations and their short-term succession, before such categorizations have become part of that memory’ (Edelman, 1989, p. 155).

8. Whereas the mutual influence between Hayek and Karl Popper is recognized (see Caldwell, 2006), the influence of Bertrand Russell awaits research. Hayek’s description of his ‘gain from hearing or reading what other people thought’ may be relevant in regard to the meagre citation of Russell: ‘it changed ... the colours of my own concepts. What I heard or read did not enable me to reproduce their thought but altered my thought. I would not retain their ideas or concepts but modify the relations between my own’ (Hayek, 1978, pp. 52–53).

9. Although Hayek cites Russell – see Hayek (1952a, p. 208, fn 13) – to the effect that ‘[o]rdinary language is totally unsuited for expressing what physics really asserts, since the words of everyday life are not sufficiently abstract’ (Russell, 1931, p. 85), there is no reference to Russell’s essay of 1929.

10. Cause and effect are only meaningful within open systems. The ideal gas law serves to illustrate. For a gas with initial and final pressure (P_1 and P_2 , respectively), initial and final volume (V_1 and V_2 , respectively) and initial and final temperature (T_1 and T_2 , respectively), the law states that $P_1 V_1 / T_1 = P_2 V_2 / T_2$. To introduce cause, at least one variable must be exogenously determined (and, thereby, left unexplained). So, if an experimental scientist were to apply an external force to raise the temperature from T_1 to T_2 , while volume is held constant ($V_1 = V_2$), the rise in temperature would ‘cause’ pressure to increase from P_1 to P_2 . In short, ‘an exogenous element cannot be an effect, it can only be a cause’ (Hicks, 1978, p. 22).

11. There is a clear parallel between the theory–observation scientific relationship and the map–model mental relationship: ‘If we were to seek observations that do not embody theoretical interpretations, we are reduced to the level of ‘sense-data’: not anything like ‘The photon emitted by a distant star has the wavelength of 8,000 angstroms’, not even ‘I see a red star’, but ‘Red here now’, whose desperate incorrigibility is entirely useless for building scientific knowledge’ (Chang, 2005, p. 3). Furthermore, modifications to theory and map alike, necessarily invoke Otto Neurath’s metaphor of sailors on the open sea, who are constrained to rebuild their ship, just one plank at a time.

12. Although Hayek offers no acknowledgement, he makes reference to Russell’s *The Scientific Outlook* (from which the immediately preceding citation is taken) elsewhere in his scientism essay (Hayek, 1952a).

13. See Caldwell (2004, p. 361ff) for an extended discussion of this point.

14. Caldwell comments that ‘even though there was not much evolutionary thought in it, *The Sensory Order* helped move Hayek toward thinking in evolutionary terms’ (Caldwell, 2004, p. 279). Given that the relevant research predates the 1950s, the prominence given to adaptive processes (if generally by inference) is noteworthy. Although connectionism, as a paradigm for neuroscience, artificial intelligence research and psychology has developed far beyond Hayek’s seminal contributions, ‘the essence of his analysis still remains with us’ (Edelman, 1982, p. 24).

15. If constructivist engineering to replace evolved systems were justifiable solely by the argument that ‘evolutionary processes do not necessarily lead to – by any reasonable definition – optimal consequences’ (Hodgson, 1993, p. 25), the case for

the former would be unanswerable. Since an evolutionary outcome is always path dependent, optimality is an inappropriate criterion.

16. The catallaxy is an order wherein there is no common purpose; its essential characteristic is that of free exchange rather than (as within a goal-oriented organisation) that of optimal choice.

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THE SENSORY ORDER: OPERATIONAL EPISTEMOLOGY AS AN EVOLUTIONARY ADAPTATION

Robert F. Mulligan

ABSTRACT

Purpose – Recent findings in neurobiology, cognitive psychology, and brain evolution are interpreted in light of Hayek's construction of the sensory order as a spontaneously emergent evolutionary adaptation. The sensory order is an experimental view of reality ordered by experience.

Approach – Natural selection of behavioral and cognitive adaptations is shown to result in structural change within the brain. Individual brains grow, and species brains evolve, through the construction and evaluation of hypothetical classification schemata. This process both results in the construction of the sensory order, as well as results from the particular models of objective reality that individuals have constructed, their evaluation of these models, and the comparison of our own models with those of others.

Findings – Cognitive adaptations, such as belief in agency, causal reasoning, and theory of mind, are inherited because they enhance survival and reproductive opportunities. In addition, behavioral adaptations including empathy, reciprocity, social hierarchy, and peacemaking are also inherited. Socialization in larger groups required the evolution of

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enhanced brain connectivity permitting a greater degree and sophistication of social intercourse.

Research Implications – Recent findings in neurobiology can be better related to one another in terms of how they contribute to the sensory order. Literary Darwinism, a school of literary theory, can also be understood more fully.

Originality/Value of Paper – Varied developments in modern neurobiology and cognitive psychology are shown to lead to spontaneously emergent institutional structures, such as behavioral regularities and rules of morality, which further enhance the survival benefits of inherited brain structure and the sensory order.

INTRODUCTION

Very early in his career, Friedrich Hayek sought to address the relationship between the subjective reality of perception, valuation, and market interaction, and the presumed objective reality that underlies experience. The problem of relating subjective belief to objective reality is complicated by the fact that our understanding of objective reality can only be subjective. Among the problems with which Hayek grappled was the relationship between the static order the mind constructs at any point in time, and the dynamic order necessary to accommodate and seek out new information. Accumulation of design in biological evolution helps explain the path dependency constraining the dynamic order. The human mind constructs networks of interconnections that evolve over the individual's lifetime, and at any point in time, the mind's model of external reality is thus provisional. Social interaction relies on common areas of belief and overlapping congruencies among potentially varying individual beliefs. Individuals measure their own belief systems against those of others, as well as against external reality.

Economics analyzes social communication through spontaneously emergent institutions including prices and markets. Various kinds of spontaneous evolution occur simultaneously on different timescales. Individuals revise entrepreneurial plans frequently throughout their lifetimes, including changing consumption patterns, introducing new production activities, and accommodating altered business conditions – that is, they alter entrepreneurial plans so as to remain in coordination with the

changing plans of others (Mulligan, 2009). The sensory order evolves in a similar timeframe, though childhood development is distinct from the adult mind, which nevertheless continues to evolve as individuals discover new knowledge and reject previously accepted beliefs. Positive legislation changes expectations of behavioral regularities in society nearly as often. Spontaneously evolved common and customary law evolves much more slowly and political organization evolves more slowly still (Mulligan, 2004, 2005). Caldwell (2000, 2003, 2004) places Hayek's work on psychology and the theory of knowledge within his views on economic methodology, technical economics, political theory, and social development.

Though biological evolution proceeds with literal geologic slowness, brain evolution seems to be the most rapid of all forms of biological evolution. In the brain, the physical structure of cortical connectivity evolves to accommodate behavioral regularities exhibited by individuals within a species. Subtle and apparently random variations in local brain structure occur naturally throughout a species and enable adoption of behavioral rules and institutional adaptations, which necessarily take advantage of available structural capacities – adaptations are constrained by the structure currently available. If a behavioral regularity contributes evolutionary advantage, then the structural adaptation it exploits is passed on throughout the species. The economic value of physical artifacts, institutional structure, and behavioral regularities is not atomistic – each confers meaning on and provides context for all others. For rules and other behavioral regularities to survive, they must first be experimentally adopted by entrepreneurial carriers, and then must enhance the survival and reproduction of the carriers (Dopfer & Potts, 2008).

In *The Sensory Order*, Hayek (1952a, sects. 5.25–5.31) suggests the human mind evolved to provide the organism tentative models of an objective, external reality. These models originate as mappings of sensory impulses onto their external objects, and are topological, not topographical (Hayek, 1952a, sect. 5.29). Continual revision of these models constitutes the sensory order and provides a sophisticated, flexible, and yet effectively simple categorization for the otherwise unintelligible mass of experience and memory. More sophisticated and hierarchical mappings relate abstract categories to the external reality (Hayek, 1952a, sect. 6.47). In Hayek's view, the mind organizes sensory impressions into manageable summaries through categorizing new impressions according to classifications already in place. But the internal classificatory structure is also subject to spontaneous reorganization to accommodate new experience. Thus, to Hayek, the sensory order had to be a naturally occurring, naturally evolved, spontaneous order, which emerged through natural selection to enhance the survival and

reproductive opportunities of the organism. Hayek was concerned with the actual mechanisms through which epistemology evolves and becomes operationalized, as opposed to the more stylized rules of formal logic.

Hayek's construction of the sensory order contributes to integrating varied developments in modern neurobiology and cognitive psychology. Individuals implicitly exploit their sensory orders to achieve evolutionary advantage. Clearly, to the somewhat limited extent we exercise volitional control over our own classificatory apparatus, we intentionally construct it to enhance our survival and reproductive opportunities. In the broader sphere where the sensory order operates autonomously, evolutionary advantage accrues to those individuals who achieve the most useful mappings of reality onto the classificatory structure ordering their sensory and mental experience. As the sensory order expands in sophistication, breadth, and sensitivity, this contributes to the cortical interconnectivity necessary to attain and understand further evolutionary advantages. Recent findings in neurobiology show that nerve cells within the brain are structured to categorize related sensory data in spatial proximity. This juxtaposition of related information is autonomous and occurs without the individual's volitional control or intention. This may be the very physical mechanism through which the sensory order works. Humans clearly economize on physical energy and mental effort by attempting to preserve classificatory schemes already established and verified by experience, suggesting at least a temporary justification for confirmation bias (Wason, 1960, 1966, 1968; Lord, Ross, & Lepper, 1979; Evans, Barston, & Pollard, 1983). This behavioral principle is also consistent with modern theories of the emergence of institutions such as religion and morality (Atran, 1990, 2002; Atran & Norenzayan, 2004a, 2004b; Norenzayan & Atran, 2004; Norenzayan & Hansen, 2006; Norenzayan, Atran, Faulkner, & Schaller, 2006). A number of early brain adaptations seem to have been, and continue to be, widely used to classify human experience, including belief in agency, causal reasoning, and theory of mind. Even when these adaptations are objectively misapplied, their application benefits the individual by economizing on scarce energy, cortical capacity, and connectivity.

The paper is organized as follows: following this brief introduction, the section "The Hayekian Synthesis in Cognitive Psychology" demonstrates how Hayek's model of the sensory order offers a reasonably accurate first approximation of how the mind's understanding evolves; the next section, "Brain Evolution as a Spontaneous Order," develops some implications of Hayek's formulation for contemporary neurobiology; followed by the "Conclusion."

THE HAYEKIAN SYNTHESIS IN COGNITIVE PSYCHOLOGY

In *The Sensory Order*, Hayek synthesized those schools of contemporary psychological thought he felt could be applied most fruitfully to explaining human behavior, especially economizing behavior. The first draft of *The Sensory Order* was written just after World War I while Hayek was a student of Mises and Wieser at the University of Vienna, though not published until 1952, and then in English, extensively enlarged and revised to acknowledge contemporary and intervening developments in psychology. Hayek's long association with Michael Polanyi, as well as his own work on spontaneous orders in government and law still lay in the future when the kernel of *The Sensory Order* was written. Nevertheless, Hayek recognized the spontaneous character of markets and the price mechanism, not to mention market organization. The economic significance of the theory of knowledge can be appreciated by contrasting papers strewn at random around a room, a state of noise or chaos, and the same papers organized neatly in filing cabinets (or on computers) and retrievable through a particular, though nonunique, system of organization. Buto (2003) and McQuade and Buto (2005) see the sensory order as an archetypal model of such knowledge-generating orders, including the entrepreneurial activity of generating market-based knowledge. One key ingredient of knowledge-generating orders including the sensory order is a necessary reduction in entropy or disorder.

In preparing his old notes for publication in the early 1950s, the German manuscript underwent significant revision and refinement, as well as translation by the author into English. At the same time Hayek was working on *The Counterrevolution of Science*, which also appeared in 1952. *The Counterrevolution of Science* is, on the one hand, a reaction to logical positivism and its doctrines of logical atomism, unity of science (*Erkenntnis*), and scientism, particularly as applied to the social sciences, and on the other hand, a memoir of the interaction between two rival schools of philosophical thought, the logical positivists on the one hand, and the economic methodologists of the Austrian school on the other. Hayek's rejection of logical atomism helped focus his attention on particular psychological and cognitive theorists and practitioners, both when he began his enquiries in 1919, and when he prepared the final manuscript of *The Sensory Order*. Logical atomism derived from Ernst Mach's positivism, and attempted to apply rigorous analysis to hypothesized independent primary entities. It was (and is) thought by many mainstream social scientists to warrant, and even require, examination of individuals and

phenomena independently of their environment or interaction. The rejection of logical atomism led Hayek to the principal school of contemporary psychological thought, which also rejected it, the gestalt school.

Hayek appreciated the ambition of his research program, and perhaps influenced by the notable lack of critical approval during his lifetime, remained both modest about the extent of his accomplishment and cautious about extending his work on the sensory order. As Mises (1957, p. 97) suggests, “perhaps it is too bold a venture for the human mind to speculate about its own nature and origin.” Hayek’s subsequent work on spontaneous evolution of the liberal order (Hayek, 1960, 1973, 1976, 1979) displays a notable thematic consistency with this seminal early work, in that both address spontaneous order. *The Sensory Order* presents an even more ambitious and original application of spontaneous order doctrine to the field of psychology. In some way this new application was intermediate between traditional applications of spontaneous order to legal, political, and economic theory (by such figures of the Scottish enlightenment as David Hume, Adam Ferguson, and Adam Smith, and related thinkers Sir Matthew Hale and Bernard de Mandeville (Ratnapala, 2001)), on the one hand, and to biological evolution on the other.

Hayek (1952a, sect. 3.74) followed the gestalt school in rejecting the atomistic conception of sensory qualities. According to gestalt psychology, the importance of sensory data lays less in those purely physical aspects which lend themselves to atomism, than to the way they are acted on and integrated by the organism (Ehrenfels, 1890; Wertheimer, 1912; Perls, 1973). The gestalt school called this integrative facility “organization of the field” (Braly, 1933; Leeper, 1935; Duncker, 1939). Hayek felt this rather vague concept could be reinterpreted with greater precision in terms of subjective “causal connections between physiological impulses” (Hayek, 1952a, sect. 3.74). The impulses might be atomistic in some sense, but the opportunistic and subjective use we make of causal connections we organize cannot be. More importantly, this organization must be tentative – the sensory order is subject to revision as new data are integrated into our experience.

By the early twentieth century, the traditional introspective approach to psychology typified by William James (1890) and Wilhelm Wundt (1910) was being criticized as subjective and nonrigorous by such founders of behaviorism as John B. Watson (1913, 1919). Some behaviorists viewed the stimulus–response as an automatic causal relation (Watson & Rayner, 1920; Pavlov, 1927), while others emphasized its volitional character (Skinner, 1938). Behaviorism explored the atomistic–deterministic aspects of psychology, which could be examined through imitating the empirical approach of

the physical sciences. In contrast, gestalt psychology recognized that insight and intuition could enhance the efficiency of the learning process far beyond that offered by exclusive reliance on deterministic trial and error. Tolman & Honzick (1930) were able to demonstrate that rats' ability to learn mazes was enhanced through recollection over time, and this enhanced learning could be demonstrated long after their initial experience with a particular maze configuration. This study contributed to the belief that recall enhances learning because it presumably reinforces or corrects existing knowledge. It was suggested that the rats constructed an internal conceptual model of the maze, but would not necessarily demonstrate that knowledge objectively until later reinforcement was offered. Tolman (1948) later demonstrated that humans construct comparable conceptual mappings, as Hayek (1952a, sects. 5.17–5.49) had theorized.

Hayek's insight was that learning implies something beyond the passive recording of externally objective information, that the brain organizes sensory impulses, and continually reorganizes them. We are thus free to make mistakes, and our recollections can also be mistaken, however since we cannot avoid the consequences of mistaken understanding, there is always an incentive to correct errors, as well as to construct more encompassing models. This contrasted sharply with the behaviorists' atomistic–positivistic conception of the mind that viewed memory as a passive tape recorder, which unselectively perceives everything, though our access to these memories was seen as flawed and limited. It contrasts even more sharply with the radical empirical behaviorism of B. F. Skinner (1953, 1968; Ferster & Skinner, 1957), which sometimes subordinates the role of the organism to that of the environment (Skinner, 1957). Cognitive psychology emerged subsequently to address the implications of environmental and internal context for different stages of cognitive development, what this implied for how we go about learning within a particular developmental stage, and how we progress to higher levels of understanding (Eysenck & Keane, 1990; Mayer, 1992). Cognitive psychology thus owes a strong debt to Hayek. Along with Hayek, the cognitive psychologists also emphasize path dependency and the role of prior knowledge in supporting the acquisition of new knowledge and its comprehension (Schallert, 1991).

Hayek (1952a, sects. 3.25–3.34) starts with the observation that within the brain, neurons serving adjacent sensory receptors are necessarily organized spatially in juxtaposition with the receptors they serve, and therefore with each other. Even viewing the central nervous system as an inactive physical object, Hayek observes its connections are organized for the efficient processing of information from the receptors, and the efficient coordination

of that information, which enables us to form consistent perceptions of experience, coordinating the senses of taste, touch, smell, etc. This explains why we would be confused if we were shown one kind of food, but exposed to the odor or taste of another. Hayek considers further that an additional level of categorization is imposed by the gross physical need to process the different sensory qualities of sight, taste, hearing, smell, touch, and distinguish among them (Hayek, 1952a, sect. 3.43). Although Hayek's discussion is about visual impulses, he generalizes his observation to the other senses. Experience enables us to improve our ability to coordinate and interpret primary sensory data. As infants we learn first to distinguish between light and dark, then to identify shapes and colors, then distinguish objects, and finally to understand and distinguish among an unlimited variety of abstract qualities that are properties of the objects we perceive, including size, proximity, temperature, vibration, and texture. Depth perception rests on such abstract qualities. Sufficient experience is required before we can imagine, for example, fictional objects that may or may not possibly exist, but similarly, at this early stage we are unable to conceptualize actual objects, or coordinate sensory inputs, in a manner more sophisticated than can be supported by our cognitive development.

The vast majority of sensory impulses are set aside and largely ignored, unless subsequent events focus our attention on particular qualities, or cognitive dissonance triggers recall of something we previously failed to pay attention to (Mayer, 1992). Implicitly, the most important category we create is one of inessential impulses. Impulses that do receive the attention of the brain's classificatory apparatus are first implicitly classified with regard to the timing of reception, the quality of sense engaged (i.e., sight, touch, etc.), and the location of the sensation (i.e., fingertip vs. toe, near vs. far, etc.) Because of the structural organization of the central nervous system, this initial classification is a spatial one, where impulses related through qualities, senses, proximity, etc., are organized spatially through temporary storage on nearby neurons. However, this is only the beginning of the classification process. At each stage, the majority of information seems to be judged useless and is discarded, serving to further focus attention on memories that remain. Those categorizations of primary impulses are combined and aggregated to form more sophisticated and intricate classification systems, which now apply not to primary impulses, but to broader classes identified as having some perceptible concrete or synthetic, imperceptible abstract characteristics in common. Since Hayek, numerous cognitive psychologists have described similar classificatory

structures, for example, frames (Minsky, 1975), schemata (Rumelhart, 1975, 1991), and scripts (Schank & Abelson, 1977; Abelson, 1981).

Hayek (1952a, sects. 3.54–3.62) describes the emergence of higher-order classificatory schemata as being limited by the possible permutations of the primary impulses, but efficiency imposes the much lower threshold of physical feasibility, and more importantly, classification is conducted on an ongoing basis, addressing the organism's current needs or interests. Bartlett (1932) found memory recall to be a reconstructive activity – we construct imperfect recreations of the original constellations of sensory impulses, which are generally incomplete and often less than perfectly accurate in other respects. Classification is always experimental, and always subject to revision. Rather than dramatic paradigm shifts, a classification schema tends to be most often incorporated into more intricate schemata, rather than overturned entirely. Often, a more effective classification will evolve out of an older, more primitive one, which served the organism's needs well enough at one point, but eventually sufficient new experience emerges to support a more subtle and effective appreciation of reality (Hayek, 1952a, sect. 6.45). Rizzello (2004) concludes that the sensory order is highly path-dependent in that it so totally conditions the content and nature of subsequent learning that path dependency dominates the cognitive dimension of perception, individual decision-making, and institutional change. Clearly, Hayek emphasizes the extent to which the sensory order is developed through building on previously established associations, but it is also clear that the sensory order can be superseded when sufficient disconfirmatory sensory evidence or logical dissonance is encountered. Institutional change may be more narrowly bounded by path dependency than an individual's sensory order, due to the complexity imposed by interaction among many cooperative individuals, compared with the organization of one individual's perceptual impulses or even their higher and more abstract classificatory schemata. Some degree of revision and reorganization of the sensory order can occur even in the absence of new experience, through sudden insight.

The sensory order is built on the distinction between two kinds of primary sensory data. *Exteroception* or *exteroceptive impulses* refer to perceptual impulses that originate from outside the organism (Hayek, 1952a, sect. 4.54). This contrasts with *proprioception*, awareness of the organism's internal state that responds to external stimuli (Hayek, 1952a, sect. 4.35). Damasio (1995, 1999, 2003) emphasizes that the human body is part of the environment over which the central nervous system exercises awareness. Thus, we can imagine responses to an external stimulus before it is applied.

In extreme cases, merely thinking about stimuli that can trigger anxiety can trigger an anxiety response. For example, imagining a rapid pulse can elevate one's pulse and blood pressure, as can recalling or imagining an embarrassing experience. Hayek notes that expectations require some vision or model of causal relationships, that is, a mapping of temporally ordered impulses onto external reality. The act of revising a course of action to better realize disappointed expectations clearly calls for an ability to model reality and to revise those models in response to new data (Hayek, 1952a, sect. 4.54). Hayek emphasizes the extent to which this activity of revising against current perceptions constitutes a feedback loop. Educated by experience, the mind forms linkages among both simple sensory data, and complex aggregates of sensory information summarized in concepts used to categorize sensory impulses (Hayek, 1952a, sects. 6.37–6.38). Experience continues as long as the life of the organism, and potentially forces continual revision of the complex of linkages. These complexes of hypothesized causal linkages constitute the sensory order, a working model of reality that supports our perception, conceptualization, and logical reasoning, which Hayek describes with terminology attributed to Karl Popper:

The qualities which we attribute to the experienced objects are strictly speaking not properties of that object at all, but a set of relations by which our nervous system classifies them or, to put it differently, *all* we know about the world is of the nature of theories and all 'experience' can do is to change these theories (Hayek, 1952a, sect. 6.37).

Cognitive psychology emerged subsequently (Broadbent, 1958; Neisser, 1967) as the further development of the gestalt school which originally inspired Hayek. Along with Hayek, cognitive psychology recognizes that prior belief and understanding are critical for determining behavior in general, as well as for how we go about learning in a particular context, and even for how we frame problems for possible solution (Eysenck & Keane, 1990; Schallert, 1991; Mayer, 1992). Behavioral patterns and problem-solving algorithms and heuristics we adopt depend both on the problems we define, and on our stage of cognitive development (Piaget, 1951, 1953), which in turn partially determines our approach to defining problems. Cognitive psychologists identify schemata, "hypothesized mental frameworks that give organization to incoming information" (Bruning, 1994, p. 8), as providing a context which confers meaning on continuing experience, or alternatively, which we use to interpret subsequent experience. As Hayek suggests, the classificatory schemata organizing sensory data accommodates the simplest, unassociated, concrete impulses as well as the most abstract, higher-order generalizations. Rumelhart (1991) proposes several features of

these classificatory schemata: (1) they accommodate a range of information from simple to highly complex; (2) they organize information into one or more nonexclusive hierarchical structures; (3) they drive our interpretations of experience, and influence how we integrate current experience with past memory; and (4) they contain slots or place-holders for the reception of fixed or variable values, which may be qualitative or quantitative, and which may be empty at any point in time. Each of these features can be found in various forms throughout *The Sensory Order*.

Maintaining the sensory order is less an intentional activity, especially as it relates to processing concrete perceptions, than a side-effect of its own operation. Nevertheless, our reliance on classificatory schemata allows us to adapt to reality by enhancing the individual's experience and survival opportunities. The Austrian school and other economic subjectivists criticize mainstream economics for ignoring issues of purpose, intentionality, and meaning (Weber, 1921; Lachmann, 1971; Lavoie, 1991; Kirzner, 1992, 2000). The subjectivity of individual values and preferences prevents Hayek's sensory order from reaching strict determinism, thus preserving free will (Hayek, 1952a, sects. 8.87–8.98). Logical positivists like Otto Neurath "substitute a purely causal treatment for the – as they declare unscientific – teleological treatment" (Mises, 1957, p. 243). Similarly, rigorous behaviorists like B. F. Skinner (1953, 1957) emphasize the mechanical, deterministic aspects of human existence, and avoid addressing intention as too complex or irrelevant. Eschewing teleology is appropriate for the natural sciences, which analytic philosophers sought to emulate – subatomic particles do not have goals or preferences – but represents exactly the wrong approach for the social sciences (Hayek, 1952b). Interestingly, the level of aggregation and frequent recourse to representative agents typical in mainstream economics does much to conceal this fact. A representative agent in an economic model has the autonomy of an electron.

Empirical studies of learning demonstrate that the context of information and prior knowledge influence the way and extent it is learned (Pichert & Anderson, 1977), and generally, the more sophisticated understanding and experience an individual possesses in a given field, the easier that individual finds it to articulate and solve problems in that area (Chi, Glaser, & Farr, 1988). This applies not only to the content of prior knowledge, but also to how it is organized by the individual. Hayek recognizes the subjective models of the sensory order are continually tested against both the objective external reality *and the subjective beliefs of others revealed through social intercourse*. It may seem illogical to claim that the sensory order can be measured against *both* objective reality and the subjective beliefs of

others; however, economists and philosophers have long recognized that commonality of belief systems confers adaptational advantage by facilitating social cooperation. Horwitz (2004) suggests that cooperation in anonymity results from and is facilitated by monetary calculation in advanced economies. Such calculations need not imply identical values, and allow individuals with very different values and preferences to benefit mutually from voluntary exchange. Even when a shared belief turns out to have been objectively mistaken, the fact that it was shared by a group contributed to the predictability of behavior within the group, lowering transaction costs and facilitating exchange and other forms of cooperative action. As Mises (1957, p. 285) writes, "man can succeed only if his actions fit into the trend of evolution." Culture is present only to the extent it is shared. Although economics emphasizes the individual subjective decision, culture suggests a consensus on which among the network of interconnections is most important.

Just as Hayek classifies information as more or less remote from primary sensory experience, formally systematized knowledge is classified according to how it can be applied, as more or less practical. Neves and Anderson (1981) construct a three-stage hierarchy of the domain knowledge relevant to a particular field of endeavor. Their system consists of (1) declarative knowledge (Anderson, 1983, 1993), an understanding of the semantic facts of a domain; (2) procedural knowledge, an understanding of the processes, implying familiarity with relevant causal relationships and an articulate understanding of how to bring about various outcomes; to finally (3) the state of composition, where problems can be solved automatically without thinking consciously about them (Chase & Simon, 1973; Hayes & Flower, 1986). All basic sensory data are declarative, though Neves and Anderson's declarative knowledge also includes conceptual definitions and other abstract knowledge, including automatized, implicit causal understandings. We gain this kind of knowledge both through our own direct experience and through the more social process of formal learning. Procedural knowledge comprises explicit causal relations, and includes our own articulate knowledge of the sensory order. Compositional knowledge is implicit and fully automatized, covering our exercise of the sensory order, being tacit knowledge in the sense of Hayek (1945) and Polanyi (1958, 1967). If an individual could learn the state of composition at which they were aiming when they first set out to amass the lower-level declarative and procedural knowledge, they could construct a sensory order addressing a particular domain, which would not require subsequent revision beyond filling in the blanks as the learner acquires additional declarative and procedural

information, but this is never possible. Education can rely on teachers presenting students with accurate roadmaps to guide them through acquisition of declarative and procedural schemata, to the predetermined and final destination of the compositional state of knowledge. However, except fortuitously, individuals cannot construct these accurate roadmaps for themselves, ignorant as they are of the structure modeling their final destination. Although a teacher can plan a lesson, students cannot map out future experience for themselves with anything resembling certainty.

Construction of the sensory order is a problem-driven scientific inquiry, which normally proceeds subconsciously. We can also apply the sensory order consciously and intentionally (Hayek, 1952a, sects. 6.2–6.13), in which instance we are not merely passive recipients of problems, but often exercise entrepreneurial awareness in defining new ones. Subjective human intentions focus individual approaches to learning and awareness (Newell & Simon, 1972). The “truths” of any scientific discipline, not only economics, are never final but always tentative, contextual, and subject to further refinement, and thus subjective in precisely the same way as the preferences and knowledge of market participants. Scientific knowledge occupies a higher level of generality than the market knowledge of entrepreneurial planners, and nearly always offers deeper understanding than subjective entrepreneurial knowledge and belief, but scientific knowledge can neither be considered nor demonstrated to be ontologically superior. Scientific knowledge is contingent in the sense that it is subject to continual revision. For example in physics, Newtonian mechanics replaced Aristotelian mechanics (Aristotle, 1957), and has been superseded in turn by Einsteinian relativity. Although Newtonian mechanics remains as true as it ever was in its limited context, and being a special case, is nested within Einsteinian physics, we often say that the truth of Einsteinian physics demonstrates the falsity of Newtonian physics. It remains impossible for us to know before the fact whether Einsteinian relativity will ever be superseded – at this point in time, we cannot know that it is not ontologically final, and we can never know that it is.

In Hayek’s view, the sensory order we construct in the mind is a spontaneously evolving order that the organism adapts – for the most part unconsciously – to address its evolving needs. The external environment may change, eventually requiring revision of the sensory order, or the organism may mature through experience to revise the sensory order even in the absence of external change. We cannot manage the burden of processing each new sensory impulse as if it were unrelated to others received simultaneously or in the recent past, and we cannot make sense of existence

without reference to memory of past experience. Conscious thought is inadequate for receipt of all primary sensory impulses; thus, most of the apparatus of perception operates automatically most of the time (Dijksterhuis, Bos, Nordgren, & van Baaren, 2006, p. 1006; Gifford, 2007, p. 270). As the organism maintains this sensory order, the order evolves spontaneously in response to the organism's past experience, choice of classification schemes, and subsequent experience. Our memory is focused by the sensory order, as are our awareness and sensitivity to new impressions.

BRAIN EVOLUTION AS A SPONTANEOUS ORDER

In participating in the initial debate on the theory of evolution, Alfred Russell Wallace (1864) suggested an explanation was necessary for (1) the comparative stability in human physiology from the late Pleistocene epoch (starting approximately 130,000 years ago), during which lower animals had continued to evolve drastically; (2) the emergence of racial varieties among humans without reliance on a discredited theory of successive emergence¹; and (3) the extent to which the human brain drastically altered the prevailing rules of evolution. Wallace argued the reason brain evolution outpaced gross anatomical change, was because once a sufficiently complex brain evolved, brain development offered the most efficient way to gain evolutionary advantage. This was not the case for animals with smaller brains, which continued to diversify in variety in geologically recent times. Human racial characteristics, such as variation in skin color, were the exception, but resulted largely from differences in external stimuli, particularly climate, since few species have ever enjoyed anything approaching the geographic range *Homo sapiens* attained even before historical times. The animal brain confronts us with a physical object that behaves at least partly according to deterministic physical rules, but also seems to display a degree of behavioral indeterminacy, or flexibility – the kind of adaptational advantage that might contribute to heightened entrepreneurial awareness.

Although Wallace's hypothesis is over 100 years old, it not only remains widely accepted, but forms the foundation for modern views of brain evolution. Certain gross features are clear in examining the progression in brain structure from primitive to more advanced animals – as evolution proceeded: (1) brains became structurally more centralized; (2) neurons became more concentrated at one end of the organism (encephalization); (3) the number, size, and distinctness of separate, specialized structures

increased, that is, the brain evolved to display increasing complexity and differentiation; (4) the most recent structures to emerge would be larger and would generally control the more primitive structures from which they emerged (hierarchy); and (5) brains became increasingly able to modify themselves in response to experience, to construct memories and learn new perceptual and motor skills (plasticity).

Our construction of a subjective internal reality, a personal model of external reality and of our own place in it, determines the course of brain development we bequeath our successors. What we observe in the brain is always contextual, dependent both on what we choose to examine and look for, and also colored by the perspective of our experience and assumptions. The brain's physical and chronological evolution proceeds through the animal's behavioral choices. Path dependency imposes a major constraint: "history is an irreversible process (Mises, 1957, p. 234)," and all of us "bear the wounds of evolution (Sir Arthur Keith quoted by Eiseley, 1958, p. 197; 1974, p. 140)." The brain is an organ that grows in terms of connectivity, the more frequently different connections among the various cortices are made and reestablished. It is this growth, which occurs as the living organism learns, which over a longer timeframe has determined the future evolution of the brain and the organism. It is worth noting that the activity that causes individual brains to grow and species brains to evolve is not merely the learning of declaratory knowledge, but thought that results in the construction of the sensory order, as well as results *from* the classificatory schemata, the particular model of external objective reality we have constructed. Connectivity is also enhanced by our evaluation of the sensory order, and our comparison through social intercourse of our own models with those of others.

Kurt Dopfer and Jason Potts emphasize the need for a particularly well-developed brain to make use of tools to serve the ends of the using organism (Dopfer & Potts, 2003, 2008; Dopfer, 2004). Their concept of tools includes not only physical artifacts, but also social institutions, language, rules, behavioral regularities, scientific disciplines and theories, and *the sensory order itself*. These varied tools are supplanted over time as better tools are discovered and used, and survive because they enhance the survival of those who adopt them, and are transmitted through surviving descendants. Potts (2000, 2001a, 2001b, 2007) has developed an explanation of economic behavior based on brain evolution. Various kinds of spontaneous evolution occur simultaneously on different timescales. Individuals might adopt a new product, such as a physical tool or laundry detergent, many times throughout their lifetimes, each time modifying their entrepreneurial plans

to exploit new opportunities. Positive legislation changes rules nearly as often. Spontaneously evolved common and customary law evolve much more slowly (Rizzo, 1985), and overencompassing social organization evolves more slowly still. Biological evolution proceeds with literal geologic slowness, though brain evolution seems to be the most rapid of all forms of biological evolution. The physical structure of the brain cortex “may be conceived of as a historical actualization of a composite of adopted rules (Dopfer & Potts, 2003, p. 4).” Tools and rules are not atomistic – each one confers meaning on and provides context for the others, and the meaning would be different to some degree if one element were different, assuming that were possible. For rules to survive, they must first be adopted by carriers, that is, they must be permitted by the physiological and genetic structure of the organism, they must be tried, and then, the rules must enhance the survival and reproduction of the carriers.

Harvey and Svoboda’s (2007) recent finding of clustered plasticity in nerve tissue is particularly relevant for Hayek’s model, which is founded on the proximity of neurons serving similar sensory receptors (Hayek, 1952a, sects. 3.25–3.34). These researchers found clustered plasticity in the mouse hippocampus. In humans, the hippocampus is thought to be the seat of short-term memory of spatial and declarative learning, where sensory impulses are initially processed, before being “stored” for potential recall in different parts of the brain. It is one of the few parts of the adult brain in which new neurons appear (Eriksson et al., 1998). Long-term memories appear to be like cross-indexed files, with a separate file drawer for each sense, in that long-term visual memory, for example, is stored in a different location from tactile, auditory, olfactory, and gustatory memories that may have accompanied the original experiences. It is likely the hippocampus cross-indexes these memories of disparate sensory impulses to ensure they remain associated when recalled from dispersed locations within the brain.

Examining signal transmission by pyramidal cells in the mouse hippocampus, Harvey and Svoboda found that long-term potentiation at individual dendritic synapses reduces the signal threshold for potentiation at neighboring synapses, thus making them more receptive to a lower threshold of information. This heightened sensitivity was not permanent, but lasted only about 10 min after a neighboring synapse was chemically stimulated. Dendritic spines are filaments ending in synapses that branch out from a nerve cell, and each nerve cell in the brain has an average of 1,000 dendritic spines. Neighboring but not yet activated dendrites may exhibit the kind of heightened awareness like that evoked by a sound in the dark – just as our conscious response to a strange sound is to listen very carefully

and alertly for even weaker sounds. Their conclusion was that such stimulation of dendritic growth and activity could allow for the binding of experientially linked information on the same dendritic branch, implicitly categorizing sensory impulses and our memory of them, by clustering them spatially on particular nerve cells, and around particular locations on a given cell.

Although this phenomenon has not yet been confirmed in humans, or outside the hippocampus, it suggests that use promotes growth and heightened sensitivity, behaviorally taking advantage of available structural variation in nerve tissue. This is particularly important for the sensory order, as the hippocampus is where spatial and declarative memory is initially processed. Our models of external reality reside physically in long-term memory, distributed throughout the frontal lobes and are subject to evaluation and revision. Long-term memories are reinforced but also modified through recall. Furthermore, long-term memories associated with each specific sense organ are located in different parts of the frontal lobes. Locating the information constituting particular memories or categories in proximity would greatly facilitate their revision and periodic removal to subsequent locations. Behavioral advantage accrues to individuals who exploit random physiological variations, which may be transmitted to offsprings if they result from genetic mutation. As with the dendritic spines, the structural modification must occur first, and the behavioral regularity that takes advantage of the modification arises later.

Proust's episode of the madeleine cookie in the *Recherche* (1913–1922[1998]I, p. 60), which triggered a flood of long-forgotten childhood memories, points to and dramatizes the fallability of memory (Lehrer, 2007, p. 79). Although memory is reinforced by recall, the recall of each memory risks, if it does not positively guarantee, that the original memory will be modified, relocated within the brain, and often, falsified. In this way memory becomes more a record of our beliefs and desires than of objective experience of an external reality – it is filtered by the subjective internal reality, our classificatory schemata. Nader, Schafe, and Le Doux (2000) and Duvarci and Nader (2004) found that disrupting memory recall, by preventing secretion of associated proteins, completely erases the original memory. The Harvey and Svoboda findings contribute to confirming Edelman's (1978, 1987) proposal of neural Darwinism. In his formulation, chemical signals between cells, across synapses, strengthen network structure and promote interconnectivity. Si et al. (2003) and Si, Lindquist, and Kandel (2003) found that chemical markers from neural impulses remain on neurons permanently, even after all memory of the original

impulses was forgotten, relocated, modified beyond recognition, or otherwise lost. The more often a network of synapses is activated, the more likely it will be to evoke a similar response, which Edelman argued would allow for relatively rapid establishment of neural networks in response to stimuli.

In contrast to the traditional view that nerve synapses function independently in an atomistic fashion, the Harvey and Svoboda finding suggests the brain stores information in clusters, with related information stored in adjacent cells of the hippocampus, and even in adjacent regions on a given hippocampal pyramidal cell, before transfer for more-or-less permanent storage elsewhere in the frontal lobes. This demonstrates that information storage relies on a classification scheme, and that the path dependency imposed by learned categories results in our desire to experience and preserve apparent logical consistency, resisting paradigm shifts or lesser classificatory revisions that require effort. Both features were suggested by Hayek in *The Sensory Order*. Just like a business enterprise or bureaucratic organization will optimally delay adoption of a new filing system or computer software due to the costs of overfrequent new adoptions, we subconsciously cling to cherished beliefs. Our cost in abandoning prior beliefs and the associated classificatory schemata is the physical energy and mental effort required to relocate that information into new quarters, and incidentally the inevitable loss of some information – equivalent to boxes of files being misplaced during the move – but once we reach a certain threshold of new disconfirming information, we will embrace the new paradigm. The information storage process appears to minimize the possibility of losing information.

On a more sophisticated level, one kind of particularly important classificatory schema within the sensory order is the causal narratives we construct to make sense of the temporal ordering of our experience. Obviously, the extent of an individual's experience enriches our ability to construct these narratives, contributes to their sophistication and explanatory power, and determines the problems we define for causal theories to address. We need these causal theories for our own use, but communication with others suggests a more compelling need. This has important implications for the emergence of cooperative social institutions, particularly religion. Religion is a particularly important institution for explaining behavioral regularities because it generally supports and accompanies emergence of rules of moral conduct. Atran (1990, 2002) argues that religion arose as an evolutionary adaptation, which survived in society because it enhanced our ancestors' survival opportunities. Atran's argument is

two-pronged: first, religion shared by a community allows individuals to rapidly construct “minimally counterintuitive narratives” (Atran & Norenzayan, 2004a, 2004b; Norenzayan & Atran, 2004; Norenzayan et al., 2006) to explain and respond to observed phenomenon, a form of gross, primitive (not to mention communal) causal reasoning; and second, communal religious observances increase the degree of group cohesion, which in turn promotes the development of society and social cooperation. This second hypothesis interfaces intimately with constitutional political economy (Mulligan, 2008). Institutions like firms, legal systems, and governments contribute organization to the social experience, limiting the extent of uncertainty confronting the individual (Loasby, 2007).

Once the sensory order forms a categorization of basic sensory impulses or conceptual abstractions, subsequent experience will necessarily be ordered according to the classificatory apparatus as it exists at any point in time. Most of the developmental sophistication occurring with experience builds uncritically on prior experience and previously learned patterns of connections. Confirmation bias, first identified by Wason (1960, 1966, 1968), refers to our tendency to selectively overemphasize evidence confirming preexisting beliefs and underemphasize evidence that cannot be readily integrated into prior expectations or worldview. Lord et al. (1979) and Evans et al. (1983) also found that prior belief generally dominates logical reasoning. Although the beliefs on which confirmation bias is based may be individually held and subjective, a particularly deep-seated source of bias would likely come from communally shared moral values, as well as Atran’s “minimally counterintuitive narratives.” Furthermore, the evolutionary benefits of minimally counterintuitive narratives help explain why we engage in the empirically observed behavior of confirmation bias.

Some early cognitive adaptations, which clearly had survival value, were a *belief in agency*, *causal reasoning*, and *theory of mind*. Each is embedded in the sensory order and evolves as that order is revised. *Belief in agency* first classifies animate and inanimate objects together, but allows for greater sophistication when it develops to the point where we classify things according to the agencies they exercise; for example, birds and clouds both want to fly across the sky. Belief in agency enables us to detect and avoid predators, but we habitually overinvoke our awareness of agents, both animate and inanimate. When an organism is startled by leaves rustled by the wind, it may waste some energy, which is costly to the organism, but the same timely reaction to leaves rustled by a crouching predator can save the organism’s life and allow for transmission of the adaptation (or at least receptivity to transmission) through the species’ genes. Eventually, we learn

not to overreact to harmless stimuli. Heider and Simmel (1944; Heider, 1958) established that humans instinctively impute agency even to inanimate objects, to the point where it appears to be a fundamental aspect of human behavior, explanation, and language – one artifact of primitive belief in the agency of inanimate objects is the metaphorical imputation of verbs to inanimate objects.

We develop belief in agency during childhood, and later learn to distinguish between the actual agency of living things and our mistaken supposition of agency in nonliving things – a distinction primitive humans were apparently unable to draw. It is especially interesting that this instinctive belief is a successful evolutionary adaptation, though it is often objectively mistaken. The fact that we can recognize some instances of our childhood belief in agency as mistaken, demonstrates how we are capable of revising the sensory order in response to disconfirmation. Young children and primitive humans seem to place a higher reliance on belief in agency than mature humans in advanced cultures. In primitive cultures, belief in agency may contribute to the formation of pantheistic worship of inanimate objects, such as trees and mountains. Such shared beliefs, values, and expectations naturally minimize competition among alternative narratives for a social group, conserving energy in social intercourse and allowing communities to reach consensus on shared goals more rapidly and efficiently. Belief in agency allows us to categorize things we perceive in our environment according to the teleological categories of things that exhibit particular behavioral patterns and things that can be categorized according to different intentions or desires, without yet distinguishing too clearly between the two classes. Sufficient experience allows us to attain higher levels of conceptual development, calling for spontaneous reclassification as we recognize that inanimate objects cannot do or want anything.

Causal reasoning categorizes experience according to narratives we construct, sometimes subconsciously, to explain and make sense out of our experience, that is, to categorize sets of related sensory impulses and the higher-order conceptual abstractions that rely on them. These causal narratives are potentially as subjective and idiosyncratic as our preferences, though they must pass additional tests of internal and external consistency. Failure to pass these tests, which in turn evolve with our experience, occasions further spontaneous reorganization of the classificatory apparatus. More importantly, narratives serve to integrate our experience. Engaging in causal reasoning modifies the sensory order because it is a classificatory activity relating otherwise atomistic sensory impulses

according to varying levels of coherency and logical consistency. Further, from an empirical perspective, causal relationships are not ontologic, and thus the activity cannot be considered deterministic. Degrees of coherence are evaluated subjectively by each individual: “the choices a man makes are determined by the ideas that he adopts” (Mises, 1957, p. 61).

Causal reasoning must be present to support formation of expectations, and thus of entrepreneurial planning. Each person’s expectations are unique and subjective, though social intercourse offers the opportunity to vet, discuss, and evaluate this reasoning and our expectations, at least to the extent they can be articulated. Communicating expectations and theories of causal reasoning must result in some degree of social norming among these otherwise subjective constructs. Feedback received through social norming must provide additional levels of meaning within the sensory order (Hayek, 1952a, sects. 4.53–4.54). However, much of our causal reasoning must be inarticulate in the sense of Hayek (1945) and Polanyi (1958, 1967).

Theory of mind allows us to understand the actions and intentions of others, to predict how others will act in response to a given situation, and to influence and respond to the behavior of others. Very young children have not yet formed a theory of mind, and cannot understand how others can act on mistaken beliefs. As we achieve higher levels of cognitive development, our own minds grow in sophistication along with our appreciation for the mental faculties of others. In applying theory of mind, we are constructing *for others* hypothetical sensory orders of varying degrees of detail and complexity appropriate to the behavior in others we wish to analyze, anticipate, or explain. Such hypothetical constructs imply very little irreversible growth in cortical structure and connectivity, and can easily be revised (indeed, must be revised) as soon as we observe the subject of our speculations act at a variance with our expectations.

In addition to making cooperative behavior easier to facilitate, social interaction allows for development of other evolutionary adaptations like language and abstract reasoning. Once primates began to live in social groups, natural selection began to balance the benefits of cooperative defense, hunting, and child-rearing behaviors against the cost of intragroup social competition. Brain size and social group size evolved in parallel (Dunbar, 1996). Darwin (1872, p. 152; Eiseley, 1958, p. 295) hypothesized that humans evolved from weaker anthropoids who would have gained advantages from socialization and cooperative hunting. More formidable gorilla-like primates were not then thought to be capable of the relatively sophisticated socialization which has since been observed in mountain

gorillas, though it should be noted they live in smaller groups than chimpanzees or humans. Physically weaker animals face greater incentives to live socially, but the social adaptation can result in weaker though better adapted successor species. Particularly large wasps, about whose ferocity and venom we can fortunately only speculate, began to live in proximity for mutual protection against predators. They eventually became true social insects, giving rise to far smaller modern wasps. Some species lost their stingers entirely, and have largely lost their ability of flight. These giant wasps were ancestors to all ants, whose morphological resemblance to wasps remains particularly striking.

The larger brain confers evolutionary advantages on the species, and is necessary to handle the volume, complexity, and sophistication of social interactions, which increase geometrically with the size of the social group. The larger the social group, the greater the species brain weight devoted to the neocortex or neomammalian brain which, among other activities, processes intersocial interaction and communication (Table 1). Macaque monkeys, who interact in social groups of approximately 20, have neocortexes that account for 50% of their brain weight. More advanced, intelligent, and social than macaques, chimpanzees live in groups of approximately 50, and have neocortexes that account for approximately 65% of the brain. In humans the neocortexes account for 80% of brain weight and include the language centers. As primates evolved, enhanced survival and reproductive opportunities accrued to those individuals who were able to construct the most sophisticated and efficacious classificatory schemata in the most efficient manner possible, ensuring growth of the neocortex.

Social interaction relies on and exploits efficient information processing, but also on predominant group behavior. Theory of mind enables much social interaction, which in turn contributes to the experience necessary to apply theory of mind. Primates applying theory of mind to understand each

Table 1. Primate Neocortex Size.

Primate Species	Social Group Size	Neocortex of Total Brain Weight (%)
Macaque (Rhesus) monkeys, <i>Macaca mulatta</i>	20	50%
Chimpanzees, <i>Pan troglodytes</i> and <i>Pan paniscus</i>	50	65%
Humans, <i>Homo sapiens</i>	100–200?	80%

Note: Only mammals possess the neocortex, aka the neomammalian brain. ? Indicate that the upper limit is not well-defined.

Source: Kudo and Dunbar 2001.

other's behavior will likely arrive at similar behavioral rules, though some variation within the group will likely remain. Hauser (2006) concludes that primates are evolutionarily hard-wired to adopt and practice rules of morality. Although the content of rules of social interaction varies across human societies, the survival and reproductive benefits of primate's receptiveness to morality point to an evolutionary source. De Waal (2006) observes a number of moralizing behaviors in primates: empathy, reciprocity, social hierarchy, and peacemaking.

Empathy is displayed by the comforting of the psychologically distressed or physically injured. It is so commonly observed among primates as to constitute the norm. It has been observed in other mammals, but seems fairly exceptional except for primates, and it remains unclear how much this actually occurs, and to what extent human observers anthropomorphize animals (a mistaken overinvocation of theory of mind). In humans it seems intimately related to the simplest possible theory of mind we can assume about other individuals – that their feelings and emotional responses are similar to our own, leading to reciprocity. This is a basic part of the sensory order, which the vast majority of humans form by mid-childhood. Sociopaths lack empathy because they are mentally damaged or incomplete, often lacking even a partial ability to apply theory of mind – frequently accompanied by other cognitive deficits.

Reciprocity, exhibited by nonhuman primates most strikingly in mutual grooming, suggests observance of the golden rule. It does not seem clear that reciprocity presents any implications for the sensory order beyond those already implied by empathy by way of theory of mind, but reciprocity does offer a more sophisticated way to evaluate our own behavior and that of others, than empathy alone. Reciprocity may be thought of as an operationalized empathy, which takes into account the observed behavior of others.

Social hierarchy is observed most strongly in primate societies where the dominant breeding males and females are related to most other members of the social group. Primates' observance of social hierarchy contributes predictability of behavior within the group, and its violation is typically punished. Extreme penalties, such as murder or banishment, can only be imposed on individuals whose behavior displays a dissenting view of the hierarchy. Social hierarchy is clearly observed by social mammals such as horses, wolves, and elephants. An individual's understanding and acceptance of social hierarchy contributes to the sensory order by integrating all,

or at least hopefully most, rewards and punishments received at the hands of other group members. The behavioral regularity is less important for humans as the size of their social group increases, but is very important for social animals living in smaller groups. Its contribution to the sensory order remains indispensable for humans, especially to the extent they interact in hierarchical institutions, such as business firms, military organizations, or the church. Further, we often conceptualize hierarchical institutions as if they existed independently of their component individuals. Though this view is in some sense objectively false, it allows us to process a smaller volume of gross information describing the behavior and characteristics of a collective entity in place of an exhaustive atomistic description of each individual. Rizzello and Turvani (2000, 2002) discuss how innovative behavior emerges and can be successfully diffused throughout an institution, and how norms emerge and evolve. In their view, institutional norms emerge through the subjective process of the social generation of knowledge, contributing structures for facilitating subjective diversity and social learning.

Peacemaking has been observed through reconciliation among male primates who have fought each other, and through females distracting, discouraging, and disarming males to prevent them from inflicting more serious injury when they fight. Peacemaking behavior relies on a highly sophisticated chain of causal reasoning, which also draws on theory of mind, empathy, and reciprocity. If empathy enables us to understand and anticipate how others will feel, and reciprocity enables us to evaluate actions as harmful or beneficial, peacemaking requires us to foresee negative consequences, and act in advance to take preventative action. Collectively, these social behaviors comprise a “stylized morality,” which forms a foundation for human morality and religion (de Waal, 1982, 1996, 1997, 2005). These are cooperative behaviors that serve group interests by minimizing harm to individuals in the group, but each imposes particular information burdens on the sensory order, and thus contributes to developing our consciousness in a particular direction. Hayek views cultural evolution as progressing from a simple and relatively inflexible small-group morality, emphasizing intolerance of others and enforced by ostracism, to a more enlightened, tolerant, and pluralistic open society, where learning from others outside one’s own group is permitted and even celebrated (Gick & Gick, 2001; Gick, 2003).

Behavioral adaptations that underlie morality and religion rely on cognitive adaptations (Table 2). The cognitive adaptations are progressive, in that humans have to develop belief in agency, which can be mistakenly applied to inanimate objects devoid of agency, before we can develop causal

Table 2. Adaptation Matrix.

Behavioral Adaptation	Cognitive Adaptation		
	Belief in agency	Causal reasoning	Theory of mind
Empathy	X		
Reciprocity	X	X	
Social hierarchy	X	X	
Peacemaking	X	X	X

Note: Xs indicate on which cognitive adaptation(s) a behavioral adaptation depend(s). The cognitive adaptations are progressive in that causal reasoning depends on and presupposes belief in agency, and theory of mind depends on and presupposes causal reasoning.

reasoning. A child who believes trees want to live to old age, will empathize with a tree that is cut down, and assume the tree experiences pain and regret. Similarly, theory of mind depends on both causal reasoning and belief in agency, and can be mistakenly projected on insensate objects. Each of the behavioral adaptations relies on some of the cognitive adaptations. Empathy, in and of itself, need only imply belief in agency. Reciprocity and social hierarchy imply applications of causal reasoning, but require an individual to accept the agency of other individuals. Peacemaking behavior almost necessarily requires that individuals attempting it possess some theory of mind about the other actors. This theory of mind must generally be both accurate and fairly sophisticated to support successful peacemaking.

CONCLUSION

Hayek’s construction of the sensory order contributes to integrating a number of varied developments in modern neurobiology and cognitive psychology. The sensory order is itself one of the tools individuals exploit to achieve evolutionary advantage, and with further development and sophistication, contributes to the cortical interconnectivity necessary to attain and understand further advantages. Recent findings in neurobiology show that nerve cells are structured to categorize related sensory data in proximity. This may be the very physical mechanism through which the sensory order works. Humans clearly economize on physical energy and mental effort by attempting to preserve classificatory schemes already established and verified by experience, suggesting at least a temporary justification for confirmation bias. This behavioral principle is also

consistent with Atran's and de Waal's theories of the emergence of religion and morality. A number of early cognitive adaptations seem to have been, and continue to be, widely used to classify experience, including belief in agency, causal reasoning, and theory of mind. Even when these adaptations are objectively misapplied, their application benefits the individual by economizing on scarce energy, cortical capacity, and connectivity.

As our prehuman ancestors developed larger brains, they also gained improved survival and reproductive opportunities through socialization in larger groups. This permitted a greater degree and sophistication of social intercourse. Species socialization also promoted further development of the larger and growing brain, with even more of that evolutionary growth devoted to the neocortex. Once sufficiently socialized, the individual tends to categorize interpersonal rewards and punishments through a stylized morality consisting of empathy, reciprocity, social hierarchy, and peacemaking. Peacemaking in particular relies on a sophisticated causal model, which permits the individual to anticipate negative consequences and plan to overcome them. This behavior could not occur without the sensory order's flexible internal model of external reality.

The sensory order may be considered as constituting the use to which we put the brain structure we inherited from biological evolution. Both confer adaptational advantages, which are further multiplied by the institutional structures we have constructed with them, including behavioral regularities and rules of morality.

NOTE

1. This pathetically Eurocentric theory had two versions (Eiseley, 1958, pp. 300–302). In one, purportedly superior Caucasians evolved last from the earlier, more primitive races. The alternative version had Caucasian man evolving first, but held that the supposedly inferior races degenerated from this beginning. To articulate either version is to apprehend the vast progress in social evolution over the last 150 years. Darwin and Wallace rejected both, but many respected scientists accepted one version or the other.

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HAYEK: COGNITIVE SCIENTIST *AVANT LA LETTRE*

Leslie Marsh

ABSTRACT

This paper conceives of Hayek's overall project as presenting a theory of sociocognition, explication of which has a two-fold purpose: (1) to locate Hayek within the non-Cartesian tradition of cognitive science, and (2) to show how Hayek's philosophical psychology infuses his social theory.

PROLOGUE

It is probably no more justified to claim that thinking man has created his culture than that culture created his reason (Hayek, 1952/1979, p. 155).¹

For Hayek, intelligence is manifest through a reciprocal coalition with the artifactual (social and physical), a causal integration that can take ontogenetic, phylogenetic, individual, collective, cultural, or biological forms. Hayek's abiding insight was to emphasize the cybernetic loop of agent $\leftarrow \rightarrow$ environment $\leftarrow \rightarrow$ agent $\leftarrow \rightarrow$ environment through a perennial and mutual process of modification and conditioning; a reciprocal relation between our conceptual creativity and the environment, to intimate, regulate, and inform concepts and action (Hayek, 1988, p. 9). Mind does not merely respond to a *given* world, mind is *enacted* through a particularized history of environmental coupling: perception is an *act* of

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interpretation and the generation of meaning. For the Hayek agent, to know is to cognize, and to cognize is to be a culturally bounded, rationality-bounded, and environmentally located agent. Knowledge and cognition are thus dual aspects of human sociality.

The notion of the “enactive” mind broadly connotes what I’ve termed the DEEDS wing of cognitive science (Marsh, 2005b, 2006); a loose and internally fluid philosophical and empirical coalition, bound by a non-Cartesian sensibility, and comprising the Dynamical, Embodied, Extended, Distributed, and Situated approaches to knowledge and cognition. Readers should not get too bogged down in the terminology – there is not much stability in the assignments that comprise the acronym DEEDS. “Enactive” coined by Varela, Thompson, and Rosch (1991/2000) and Thompson and Varela (2001), conceives of DEEDS as having overlapping concerns; a family resemblance term. Jaegher and Ezequiel (2007, p. 487) detect five mutually supporting concepts: autonomy, sense-making, embodiment, emergence, and experience. Others prefer the term “situated,” which is taken to be the species: the other assignments, the genera (Robbins & Aydede, 2008). The enactivist stance is a naturalistic nonreductive view of mind as embodied and embedded, giving due emphasis to biological autonomy and lived subjectivity (Froese & Ziemke, 2009). Of particular interest in the current context is its incorporation of the organismic roots of autonomous agency and sense-making into its theoretical framework (e.g., Weber & Varela, 2002; Di Paolo, 2005). It’s high time that the multidisciplinary hub that is cognitive science admit Hayek into the pantheon of non-Cartesian thinkers, taking his place alongside 20th century titans such as Dewey, the later Wittgenstein, Heidegger, Vygotsky, and Merleau-Ponty. To this list, we might add more recent theorists such as Varela, Hutchins, Clark, Wilson and (Gregory) McCulloch, each sources of inspiration for much of the discussion that follows.

By contrast with DEEDS (or situated cognition), orthodox cognitive science has systematically overlooked *not* only the location of thinkers in their geophysical environments, but has also overlooked the interactions *among* thinkers in the ambient social soup. As a DEEDS theorist, Hayek negotiates the extreme polarity of an abstract individualism (or internalism) and an externalism associated with sociological theorizing that posits an inflated social ontology that makes no concessions to the mechanics of the mind and *individualized* learning patterns (Turner, 2007, p. 358). Generically speaking, externalism is the thesis an individual’s environment has some causal determinant on the content of the individual mind. If there were a slogan that I believe captures the Hayekian project it is this: Hayek “*socializes*” the mind and “*cognitivizes*” the social theory.

Writing some 20 years *before* the term “cognitive science” had been coined, Hayek very perceptively understood that a multidisciplinary approach (psychology, physiology, logic, mathematics, physics, philosophy)² to explaining consciousness was called for (Hayek, 1952/1976, vii). In the service of addressing Hayek’s neglect by cognitive science, two couplets of questions should be kept in mind:

- 1(a) Is Hayek’s philosophy of mind anachronistic because he was writing long before the relevant options (i.e., the connectionist vs. the computational model) had been adequately defined? and/or
- 1(b) Have Hayek’s defenders (Weimer & Palermo, 1974, p. 436) been *too* charitable since he does not offer anything precise enough to fit any of the current models?

The second, and more interesting, couplet seeks to assess Hayek’s philosophy of mind in the context of his social philosophy:

- 2(a) Does his connectionist theory of mind *entail* the connectionist model of society?, or
- 2(b) Does Hayek’s connectionist model of society *presuppose* the connectionist theory of mind?

This paper’s primary task is to expand upon 2(a and b), an aspect that others (with the exception of McQuade and Butos) have only hinted at. The reader will be relieved that, in what follows, I desist from presenting “yet another summary of *The Sensory Order*” (Butos & Koppl, 2007, p. 20). Happily, now there are some fine substantive accounts, each emphasizing one or more of the many facets of *The Sensory Order* (Herrmann-Pillath, 1992; Streit, 1993; Tuerck, 1995; Smith, 1997; Birner, 1995, 1996; Boettke & Subrick, 2002; Steele, 2002; Loasby, 2004; Caldwell, 2004b; Novak, 2005; Feser, 2006). An example of a commentator that ostensibly has Hayek’s “cognitive view of society” as a central concern, yet nevertheless does not refer to *The Sensory Order*, is Kerstenetzky (2000). In the service of bringing Hayek to the attention of the DEEDS wing of cognitive science, I show how canonical Hayekian themes such as cognitive closure, decentralization, situatedness, self-organization, and environmental appropriation are derived from his concern about complexity. The Hayekian corpus is an intricate weave of the epistemological, the methodological, and the metaphysical. Though *The Sensory Order* is the focal point to the discussion, to absolve oneself of any consideration of Hayek’s other works, would be to mutilate Hayek, Hayek being subject to the grossest of caricatures over the years by both supporters *and* detractors. Regarding the former category,

Caldwell (2004b, p. 5) rightly points out that given the scope and voluminosity of Hayek's writings and depending upon which part of his work is being trawled, this will account for which Hayek emerges. Though admired by Thatcher and Reagan, it is unlikely that they read much beyond *The Road to Serfdom* and other highly selective readings refracted through others (in Thatcher's case, Keith Joseph; in Reagan's case, Martin Anderson and Paul Craig Roberts). Regarding the latter category, though the term "market" that has come to be synonymous with Hayek, he believed it to be a misnomer: he expresses his discomfort with the term because strictly speaking that is not what he's talking about (Hayek, 1978, p. 183, 1967, p. 164). Michael Oakeshott, arguably Hayek's closest intellectual ally, got Hayek plain wrong. Oakeshott (1962/1991, p. 26) famously took Hayek to task by pointing out that a doctrinal *laissez-faire* attitude is also a species of rationalism, rationalism of course being both Oakeshott's and Hayek's *bête noire* (Marsh, 2010a). Oakeshott's swipe is uncritically taken as a knock-down argument by several commentators (e.g., O'Hear, 1999; Lundström, 1992). Fortunately, Oakeshott's preeminent expositor acknowledges that an ascription of a vulgar atomism to Hayek is wrong (Fuller, 1989, p. 17). Hayek, explicitly and repeatedly, distanced himself from radical libertarianism as early as 1944 (Hayek, 1944/1976, pp. 17, 35, 36, 39, 42, 81; Hayek, 1973, pp. 61–62). Furthermore, for Hayek "Liberalism is not individualistic in the 'everybody for himself' sense" (Hayek, 1976, p. 151).

In the service of presenting a multidimensional Hayek, the binding agent to the discussion is his concern with complexity, Hayek's epistemological leitmotif – it is through this triangulation of mind, society, and complexity that Hayek gets his distinctive philosophical depth.

"Complexity," I contend, is *the* touchstone to Hayek's work. As Caldwell says: "By the 1960s Hayek was seeing complex orders everywhere" (Caldwell, 2000, p. 19; Hayek, 1988, p. 127; Fuster, 2003a, p. 7). "Complexity" is, however, one of those terms that are blithely bandied about: ontological and the epistemological interpretations of complexity tend to be conflated (McIntyre, 1998). Is our understanding a function of the way *that the world is* or a function of our *limitations in understanding* the way the world is? The latter, perspectival, was of course Hayek's concern. McIntyre (1998, p. 31) seems to think that Hayek equivocates between the epistemological and the ontological. I don't think this is the case at all. And this is as true for his social theory as it is for his philosophical psychology. There can be no doubt, the relation between complexity theory and Hayek's theory of spontaneous social order and social evolution is intimate (Vaughn, 1999b, p. 245; Caldwell, 2000, pp. 10, 13; 2004a, 2004b;

Gaus, 2006; also Birner, 1995). It has been suggested that Hayek's work was a precursor of modern complexity theory (Kilpatrick, 2001; Vaughn, 1999a, 1999b; Vaughn & Loren Poulson, 1998). This claim has some plausibility. One subarea of complexity research – multiagent modeling – has taken a great deal of inspiration from Hayek (Baum, 2004) and Vriend (Vriend, 2002; Kochugovindan & Vriend, 1998). Yet others draw upon Hayekian insights to resolve supply and demand issues in a distributed and dynamic web services network (Eymanna et al., 2005). Joita et al. (2007) deploy specialized algorithms to carry out a data mining tasks. Hayek's (1952, # 52) writing here bears a striking resemblance to what is known as Particle Swarm Optimization, a social algorithm that runs on a sociocognitive model of social influence and learning (Kennedy, Eberhart, & Shi, 2001). Indeed, to take the embodied and situated agent seriously as Hayek did "is to invite an *emergentist* perspective on many key phenomena ..." (Clark, 1997, p. 84). The Hayekian leitmotif of complexity turns upon:

1. How (if at all) can we come to characterize what mind actually is? (The first two drafts of *The Sensory Order* were entitled "What Is Mind?" (Kresge's introduction to Hayek, 1994/2008, p. 25)).
2. Is there a problem that can even be formulated?
3. Whatever mind may be, how does it apprehend the natural (and social) world of which it is fully a part (Hayek, 1952/1976, 1.11, 1.2, 8.45)?

These three interlinked concerns are coextensive with the most recent of enactivist concerns: "What are minds, and how do they relate – epistemically and experientially – to the world?" (Torrance, 2006, pp. 358, 360).

Complexity for Hayek offers both *the fabric of possibility and of inherent constraint* – what I term "Hayek's paradox." On the one hand, agents within a *rich* (complex) social tapestry have their conceptual and behavioral possibilities tempered by the partial cognitive and epistemic access to the (complex) manifold that informs the ambient culture. On the other hand, mind is itself constitutionally (and terminally) constrained in fully understanding its own (complex) mechanics – a mind that is significantly constituted by its (complex) social environment. There is the view that many thought experiments that have driven post-War philosophy of mind assume "a naive commitment to the principle that conscious beings must be simple" (Barnett, 2008). The paradox is this: knowledge can become *less incomplete only if it becomes more dispersed* (Loasby, 2004, pp. 101–134). Epistemic and cognitive efficiencies, well beyond the capacity of any one mind, are facilitated through the ubiquity of *sociocultural scaffolding* and dynamic

looping (Hayek, 1967, pp. 34, 42). This is the essence of Hayek's externalism and sets the stage for the discussion that follows.

HAYEK'S *PROTO*-CONNECTIONISM

The large brain, like large government, may not be able to do simple things in a simple way (Hebb, 1958).

There is a nice irony in Dennett's use of this quote: the analogy with big government couldn't be more Hayekian! (Dennett, 1991, p. 209). Complexity, for Hayek, arises from not only what David Chalmers (1995, p. 200) has famously termed the "hard problem" of consciousness, but also from the compounded problem of mind in an ambient dynamic sociality. The "hard" problem concerns the experiential, "felt" aspect to consciousness – in other words, the mind–body problem manifest as cognitive closure, which will be examined in the next section. The corresponding "easy" problems on Chalmers' account are "easy" because they address cognitive abilities and functions amenable to scientific study. (The "easy" problems might well take hundreds of years to be adequately explained). Hayek's task in *The Sensory Order* is, for the most part, concerned with the "easy" problems because as we will see, the hard problem, for Hayek, is forever intractable. *The Sensory Order* (6.4) is concerned with a subset of the easy problems on Chalmers' list (p. 200), namely:

- (1) the ability to discriminate, categorize, and react to environmental stimuli, and
- (2) the integration of information by a cognitive system.

- (1) Discrimination, categorization, and reaction to environmental stimuli

As early as 1920 (Hayek, 1920/1991), Hayek expressed a deep dissatisfaction with the realist myth of the self-differentiating object and the notion of raw datum posited by empiricism. Experience involves both perception and thought, anticipating what Wilfrid Sellars, (1956) would years later call the "Myth of the Given" (and more recently in McDowell, 1994/2000). Experience requires, not just the capacity for sensory awareness stressed by Locke and Hume, but also the capacity to make judgments about what one is aware of: in the current argot, observation is theory-laden (Hayek, 1952/1976, 2.15, 5.19, 6.36–37, 8.4–5, 8.7, 8.10, 8.14, 1952/1979, p. 105; Caldwell, 2004b, p. 277). John Gray (1980, 1984) has stressed Hayek's "skeptical Kantianism," the idea being that though the mind is inherently a

pattern seeker in an undifferentiated metaphysical world (Hayek, 1967, p. 27; Edelman, 1987, pp. 7–8), this structure is itself subject to evolutionary malleability through the idiosyncratic factors of personality, culture, situation, and infinitely fine-grained permutations of other circumstances and considerations (Hayek, 1967, p. 76, 1952/1979, pp. 136–37, 1988, pp. 25, 76; Connin, 1990; Gick & Gick, 2001; Posner, 2007). In contradistinction to Gray, I prefer to call Hayek a “conceptual realist” on the grounds that the mind constructs a reality that impinges upon mind (Wiggins, 2001). Wiggins nicely captures this idea in the slogan that “just as the size and mesh of a net determine, not what fish are in the sea, but which ones we shall catch” (2001, p. 152), so too are the concepts we bring to bear upon experience. This notion has resonance with the quote attributed to Hayek (though no specific work is cited): “Without a theory the facts are silent” (Keegan, 1993, p. 6). My reading of Hayek is that he most definitely sought to maintain the existence of the external world (Caldwell, 2004a, p. 274). By contrast, strong or radical constructivism insists that all facts (artefactual or natural) would cease to exist without the continued presence (and appropriate behavior) of human agents. In other words, there is no independent reality; this generates an unsustainable relativism (Marsh, 2005a; Nooteboom, 2007, p. 138), something Hayek did *not* subscribe to (Hayek, 1952/1979, pp. 156–157, 1967, p. 124, 1978, p. 3). This said, one can see why it has been suggested that Hayek has had his “postmodern moment” (Boettke & Subrick, 2002, p. 58; Connin, 1990; Gick, 2003, p. 162; Caldwell, 2004b, p. 274, note 11). In *The Sensory Order* (8.92), Hayek dismisses the “sociology of scientific knowledge” movement, the heirs apparent to Marx and Manheim. All one need claim is that the cybernetic impact upon the brain has outstripped any adaptive alteration of the genetic code (Wexler, 2006, p. 4). Moreover, hand in hand with his “soft” Kantianism, Hayek rejects the Cartesian idea that underneath the innumerable layers of civilization, there exists a pristine, unvarnished notion of reason (Hayek, 1978, 1967, pp. 82–95). Hayek’s conceptual realism chimes well with the enactive perspective in that enactivism gives due emphasis to the mind’s categorization tendencies modified by individualized experience of environmental coupling (Varela et al., 1991/2000; Varela 1999, p. 13).

It’s trivially true that agents need epistemic access to the world of objects they inhabit and negotiate: intelligent behavior is informationally sensitive (Hayek, 1952/1976, 4.8, 1978, p. 39). Hayek’s discussion could also be seen, as Feser (2006, pp. 289–290) rightly points out, in terms of intentionality, a notion that has been at the center of the philosophy of mind for over a 100 years. Cast in terms of intentionality, Hayek is asking the question: how is it

that mental states have content or representational character? Neural states must in some way be the bearers of semantic and informational content that hopefully (for the most part) has pragmatic truth-value.

Hayek and Hebb's models run on the idea that various cognitive activities are represented by combinations of the firing patterns of individual neurons with memory and learning arising from activity-dependent changes at individual synapses. Changes resulting from a particular pattern of neural activity enhance subsequent instances of that activity pattern (Hayek, 1978, pp. 42–43, 1976, p. 11, 1973, pp. 29–33, 1967, p. 57). For Edelman, these models confront empirical difficulties, namely that such firing is neither necessary nor sufficient for modification (Edelman, 1987, p. 181). Fuster (1995/1999, pp. 89–90, 2003a, 2003b) is of the view that Hayek's knowledge of neural processes was sketchy (Hayek, 1952/1976, 3.1–3.24), *but* that it's incoherent to suggest that it could have been otherwise! Hayek's model does not posit a basic representational element, what in current parlance, is called the "feature detector" or the "submodule" or the "semantic symbol." Hayek denies that this is required: all sensory qualities are strictly relational at all levels. Taken thus, Fuster characterizes Hayek's model as *radically* connectionist in that it does not posit constraints at the neuronal level: it runs the risk of "connective explosion." This in contradistinction to a Cartesian "*binatorial* explosion," the idea that bottle-neck Cartesianism cannot deal with the complexity entailed by the situated agent (Wheeler, 2005, p. 181). In a personal communication, Fuster concedes that he may have exaggerated the risk of connective explosion. He writes:

This only happens in epilepsy. The connectivity away from an active focus of sensation, or memory, decreases rapidly and nonlinearly as a function of two factors: (a) decreasing density of connections, and (b) decreasing synaptic strength. So, even though everything is potentially connected to everything else (a good thing for rehabilitation and alternate retrieval), this does not make the brain subject to fits. Then we are protected by GABA, and negative feedbacks of all kinds." (GABA is the acronym for "γ-aminobutyric acid" neurons).

But Fuster is adamant; whatever *The Sensory Order's* fine-grained failings, it does not invalidate the general characteristics of a connectionist model or as Hayek would put it the "explanation of the principle." Explanation of the principle as used by Hayek is a term of art and was first enunciated in *The Sensory Order* (Hayek, 1952/1976, p. viii, 1.101, 2.18; 8.57–8.92, 1952, #6; and later in 1967, pp. 11–12, 20, 40, 1994, pp. 138, 154). By explanation of the principle Hayek meant that he sought in *The Sensory Order* to flesh out what he'd already started in Hayek (1920) – that is, to provide a scientifically plausible explanation, a *general characterization*

(a schema – Hayek, 1952, p. 6) for a range of phenomena, for the problem of consciousness – whatever the fine-grained physiological detail may turn out to be (Hayek, 1952/1976, viii, 1967, pp. 20–21; Fuster, 2004, 2006). In this sense, Hayek is providing guidance for what seems a promising line of first-order research. This is precisely what Rosenblatt takes to be the value of *The Sensory Order*: a suggestion of what to look for and investigate, rather than as a finished theoretical system in its own right (Rosenblatt, 1958, p. 388). *The Sensory Order* was a work of high-level theory with a limited empirical basis even though Hayek had, 30 years earlier, spent time in the Zurich laboratory of Russian–Swiss neuropathologist Constantin von Monakow, tracing fiber bundles through different parts of the brain (Hayek, 1994/2008, p. 64; Caldwell, 2004b, p. 136)). *The Sensory Order* is necessarily incomplete (Fuster, 1995, p. 89, 2003a, p. 1048, 2006b, p. 126; Başar & Karakaş, 2006). Hence, I have characterized *The Sensory Order* as a work in *philosophical psychology* – a treatise on the *explanation of the principle* as relates to the mind–body problem *and* the problem of mental representation.

If the typical features of connectionism are its relational character, its categorical and hierarchical character, its wide-ranging scope of categorization, and its dynamic interaction between perception and memory, then the model *The Sensory Order* recommends is well and truly connectionist. Fuster makes the bold claim that “Hayek’s model comes closer, in some respects, to being neurophysiologically verifiable than those models developed 50–60 years after his” (Fuster, 1995/1999, p. 89). That Hayek (1920/1991) presents a recognizably, albeit tentative connectionist-like or cortical network theory, undermines the claim that the *earliest* progenitors of connectionism were McCulloch and Pitts (1943). Fuster (1995/1999, p. x, see also 2003a, 2003b, pp. 7, 60) writes that Hayek was “in my opinion the first and unrecognized pioneer of cortical network theory.” Of course, Hayek’s model lacks the fine-grained neurophysiological–mathematical logic sophistication of McCulloch and Pitts – I reiterate: the nature of Hayek’s project was of a different character – it was an *explanation of the principle*.

(2) The integration of information by a cognitive system

The link between mind and epistemology was made as early as 1920 when Hayek served notice that his “linkage” theory of apperception was no less than a naturalistic attempt to explain consciousness with the expectation that epistemology itself would be profoundly impacted. Hayek made good on this promissory 30 years later through his assimilation of Gilbert Ryle’s “knowing how/knowing that” (KH/KT) distinction (Hayek, 1952/1976,

1.57, 2.7, 1.55, 8.42, 1988, pp. 78–79).³ As Hutto (2005) points out the fault line between traditional cognitivist and enactivist paradigms is determined by their respective commitments to understanding cognition as based on KT versus KH.⁴ The KH/KT distinction profoundly informs that Hayek's social theorizing most famously manifest in his critique of the rationalistic tendency inherent in central planning – society is just too complex, has too many variables, local and ephemeral, to offer a predictive science of politics and economics. For Hayek, the greater part of social knowledge (*habitus*, skills, mores, traditions, “forms of life,” and practices – KH) cannot (without remainder) be stated propositionally (Hayek, 1973, p. 11, 1944, p. 14, 1948, p. 155, 1978, p. 8) since it is dispersed across multiple minds in a constantly shifting environment. This dispersion of social knowledge is what Clark calls the “spreading of epistemic credit” (Clark, 1997, p. 69).

Another way of casting the notion of KH/KT complexity is in terms of the “frame problem.” Boettke and Subrick (2002) see a similarity between Searle's (1984, pp. 28–41) Chinese Room thought experiment and Hayek's critique of central planning. Searle's argument, sloganized as “syntax is not sufficient for semantics,” is designed to show that whatever purely formal principles one puts into a computer, they will not be sufficient for real understanding. For Hayek, centralized planning necessarily abstracts from fine-grained worldly experience of local conditions. The common denominator Boettke and Subrick point to is that for both Searle and Hayek interaction with the real world is essential for understanding and intentionality or, as Boettke and Subrick put it, “the argument leads directly to an emphasis on the tacit domain and the contextual nature of knowledge” (2002, p. 56). After almost 30 years of discussion, there is no consensus as to what Searle's thought experiment illustrates. McGinn (1991, p. 211) for one, pointedly says that all that Searle has shown is that semantic properties cannot be had in virtue of the rules of the program. I think that a deeper point emanates from the Heidegger–Dreyfus holism Boettke and Subrick allude to – the infinite richness of everyday experience is a well-honed expertise manifest as a kind of pattern recognition ability, rather than from explicit inferential capacities.

Though the KH/KT distinction is taken to be primarily an epistemological distinction, is as much a claim about the operations of the mind (Hayek, 1952/1976, 2.7). Day-to-day life is one of situated agents perpetually responding to and redefining their environment without having recourse to an explicit range of alternatives. For Hayek abstract rules of which we are *not aware* determine the sensory world presented as conscious experience. In *The Sensory Order*, 6.15 (and 6.22–6.28), Hayek talks of the

“narrowness of the consciousness”; that any one time only a limited range of experience is presented as consciousness. Dennett terms this the “fame in the brain” or “political power” or clout metaphor for consciousness (Dennett, 2005, pp. 136–138, 142), the idea being that a theory of consciousness would need to explain how some relatively few contents become elevated to a position of political power (consciousness). Conscious experience is only a part, or the result, of processes of which we cannot be conscious (Hayek, 1952/1976, 5.55, 5.9, 6.1, 6.3, 6.12). It is only through the super-structure, which assigns to a particular event a determined place in a comprehensive order, that makes it a conscious event (Hayek, 1952/1976, 5.31, 5.55, 1978, pp. 45, 48, 1967, p. 61). For Ryle and Hayek, intelligent performances are not dependent on an antecedent theorizing or knowing. When one does something intelligently one is doing *one* thing and not two things (Hayek, 1978, p. 81, 1973, pp. 18–19, 1952/1976, 4.61). To be intelligent is not merely to satisfy criteria, but to *apply* them; to regulate one’s actions and not merely to be well-regulated, even within the narrow constraints of a game or say an activity such as driving. The careful driver cannot plan for all possible contingencies (Hayek, 1978, p. 7, 1967, pp. 43–44). The driver’s readiness to cope would reveal itself were an emergency to arise, but it is latently there even when nothing critical is happening (Hayek, 1978, pp. 10, 45, 84, 1976, p. 23). Intelligent conduct of serial operations does not entail that the agent is, throughout the progress of the operation, tracking what he or she has completed *and* with what remains to be done. The fine-grained detail and cognitive effort that would be required to undertake even the most banal of sensory-motor activities, let alone activities that are socially complex, would require a level cognitive resources that would (at best) be debilitating (Hayek, 1952, #21–31, 1976, p. 21, 1978, p. 46; Loasby, 2007, p. 1744). Rule-following, evolved through trial and error (Hayek, 1948/1980, p. 11, 1978, p. 81, 1967, p. 130, 1973, p. 43, 1988, pp. 20–21), offers “pre-packed knowledge modules” to reduce the cognitive load. From the enactive perspective, to express KH in terms of KT would be well nigh impossible: KH is the “very essence of *creative* cognition” and necessarily historical (Varela et al, 1991/2000, p. 148).

To constantly “theorize” each and every action, would constitute what Ryle termed the Cartesian myth requiring the positing of a “central theatre,” some central place in the brain where something like an “I” or the “self” attends to and witnesses consciousness (Hayek, 1952/1976, 6.18; Klein, 1999, p. 70). The “I” partakes of both the private and the public realms: the identity of the “I” is, in effect, an emergent property of a complex distributed process mediated by social interactions (Varela, 1999,

p. 62). Positing a substantial private identity, a central authority or “homunculus,” gives rise to “Ryle’s regress”: an observing self must necessarily contain another observing self, and so on *ad infinitum*. The combination of the assumptions that theorizing is the preeminent activity of minds and that it is a private operation, amounts to the postulation of a shadowy additional metaphysical entity – the dogma of the “ghost in the machine,” Ryle’s most famous catch-phrase (Ryle, 1949/1990; Hayek, 1952/1976, 1.91, 8.42). In these terms, the Cartesian abstraction is rendered as nonsense (Hayek, 1952/1979, p. 134). Homuncular explanation is redundant and this seems to be borne out by empirical research done by Fuster (2003b, 2008) on behavioral network dynamics (the perception–action cycle).

One can begin to appreciate the virtues of a connectionist-like model of mind that is posited for the Hayekian situated agent. A connectionist version of representation does not posit mind as a storehouse of representations – Ryle’s the “intellectualist legend” (Hayek, 1952/1976, 5.11). Representation need not be thought of as internal copies or codes, but as an *activity* that individuals perform in harvesting and deploying knowledge by being embodied, environmentally situated and essentially coupled. (This, in contrast, with a symbolic representational system that “encourages” the bifurcation of cognition from its environment and body. It should be noted that the connectionist and the symbolic “paradigms” are not an either/or choice, there are many hybridized versions. Even Vygotsky’s theory is seen as hybridized (Frawley, 1997)). But, as Fuster says: “Connectionism is a useful way of thinking about how, in principle, neural networks could develop and do their job in cognitive function, but it does not solve any specific neural problem” (Fuster, 2003a, 2003b, p. 10). Hayek’s conception is an *exploitative* view of representation that is not simply reactive but *enactive*. It’s a model that that allows us to go beyond our immediate environments to a past through memory, habit and tradition and forward through planning and imagination, neither requiring the direction from paradigmatic cognitive states such as beliefs and desires. This is not to deny the importance of deliberative and reflective thinking, but what needs to be appreciated is the role and relevance of these cognitive modes manifest in Hayek’s critique of rationalism (Hayek, 1952/1979). Hayek’s cognitive agent is *more* in tune or enmeshed with reality: a social and physical reality (Butos & Koppl, 2007, p. 39), consonant with the “conceptual realism” I suggested earlier.

Pinker (2002, p. 292) certainly makes the connection between Hayek’s connectionism and his reliance on the KH/KT distinction. The contrast between rationalistic (propositional) knowledge and non-rationalistic (tacit)

knowledge is a distinction coextensive with what Pinker terms the Utopian Vision and the Tragic Vision; Hayek of course falling within the latter category.⁵ Pinker is of the view that it's no coincidence that this distinction neatly maps onto the distinction between symbolic representation and distributed neural networks.

COGNITIVE CLOSURE

If the human brain were so simple that we could understand it, we would be so simple that we couldn't (Pugh, c. 1938).⁶

A defining feature of Hayek's philosophical psychology is the notion of cognitive closure, an idea that is refracted through his social theory. Cognitive closure falls within the hard problem that was flagged in the last section. The "hard" problem concerns the experiential, "felt" aspect to consciousness – in other words, the mind–body problem.

Hayek admits that the mere conceptualization of the mind–body problem is slippery (Hayek, 1967, p. 22). Hayek's cognitive closure argument goes like this:

1. *Explanation* is delimited by the apparatus of classification⁷ (the mind) (Hayek, 1952/1976, 8.67, 8.81) (premise 1).
2. An apparatus of classification cannot *explain* anything more complex than itself (Hayek, 1952/1976, 5.91, 8.26, 8.69, 8.80) (premise 2).
3. Therefore, the mind cannot fully *explain* itself (Hayek, 1952/1976, 8.91, 8.96, 8.98, 1952).

Hayek takes the view that a unified theory of consciousness (i.e., the hard *and* the easy problems) is forever beyond our grasp (Hayek, 1952/1976, 2.18, 8.88, 8.95). Our minds, constitutionally delimited, can only offer an explanation of consciousness that is condemned to *practical dualism* (Hayek, 1952/1976, 8.87); in effect, Hayek's position falls within what is known as neutral monism (Stubenberg, 2005). Hayek is acutely aware that self-referentiality leads to dead ends, the instrument of explanation simultaneously being the object of explanation cannot get us anywhere (Hayek, 1952/1976, 8.67; cf. Maturana & Varela, 1980, p. 49). Indeed, for Hayek "The whole idea of the mind explaining itself is a logical contradiction" (Hayek, 1952/1976, 8.91, 2.19; Hayek, 1967, pp. 34, 37, 39). This idea has resonance with the quote: "If you work on your mind with your mind, how can you avoid an immense confusion?," a Zen poem known as "Xinxinming" (Hsin Hsin Ming) traditionally attributed to Sengcan

(Seng-Ts'an).⁸ Hayek takes this incompleteness – the constitutional inability of mind to explain itself – to be a generalized case of Gödel's Incompleteness Theorem (Hayek, 1982, p. 292, 1967, p. 62).⁹

Three related points need to be made. First, Hayek is *not* recommending a Cartesian dualism, but simply that despite the underlying physical basis of consciousness, all we really have to work with and through, is a folk psychology that posits two realms – the sensory order and the physical order. In other words, the everyday understanding of “belief and desire” is in contradistinction to a supposedly scientific understanding. Second, Hayek is through and through a naturalist, a position he has consistently held throughout his career (Hayek, 1920; Hayek, 1952/1976, 1:49, 1982). Hayek fully acknowledges that consciousness *is* a natural phenomenon, but determining what the precise relation of consciousness is to the physical world is constitutionally beyond mankind's ken (Hayek, 1952/1976, 1:11). Third, Hayek is *not* a naturalistic agnostic, that is, the view that science *currently* cannot offer an explanation of the mind–body relationship, but in principle it could.

Hayek's cognitive closure position has strong commonalities with Thomas Nagel, Frank Jackson, and Colin McGinn, their conclusions collected under the rubric of “new mysterianism” (Flanagan, 1984). New mysterianism connotes the idea that while naturalism is true, the human mind is terminally constrained in being able to explain itself – whatever we discover about the causal states of consciousness, there will still remain an “explanatory gap.” Hayek has most in common with McGinn. On McGinn's account, we just have to accept what he terms as *cognitive closure*, that is, we need to appreciate that there are limits to our capacity to understand the world (McGinn, 1989, p. 7). McGinn's deflationary tack is that once one accepts the insolubility of the mind–body problem, then this problem dissolves. For McGinn (1989, p. 18), the philosophical view is no less immune to obfuscation than that of folk psychology, except that in the former, there is the assumption that the problem must be scientific. Philosophers of an eliminative materialist stripe depreciate what, in fact, the richness of the human condition (McGinn, 1989, p. 22; Hayek, 1976, p. 127). Even those who are not out of sympathy with the Hayek–McGinn conclusion are concerned that although we might not be able to specify a solution, it is incoherent to suggest that we couldn't understand *what would count* as a solution (Kriegel, 2003, p. 184). And though Kriegel has McGinn in his sights, he would have to let Hayek off the hook if one allowed Hayek's *principle of the explanation* to be extended to all complex explanations. In this sense, Hayek, could without contradiction, say that “we have no clue

about consciousness, but at least we have a clue about why we have no clue” (Kriegel, 2003, p. 188).

Hayek’s discussion of the mind–body problem speaks directly to a topic that has dominated philosophy of mind for the past 35 years – *qualia* (*quale* for singular), a term of art that denotes this subjective “felt” quality to consciousness – the “unexplained residue” (Hayek, 1952/1976, 1.19, 8.85) that physicalism has failed to explain. Indeed, it has been said that, “[t]he problem of consciousness is identical with the problem of qualia ...” (Searle, 1998, p. 28). Qualia-talk went into overdrive in response to an argument presented by Frank Jackson (1982).¹⁰ Jackson’s argument, known as the “knowledge argument,” was conveyed through a thought experiment that I’ve entitled “Monochrome Mary.”

“Monochrome Mary” poses the following question. What, if anything, would be experientially different for Mary on her release into a full color world given that she’d heretofore lived her whole life in a black and white world? Could she anticipate the experience *even though she was in possession of a complete physical description of reality*? Jackson concludes that Mary would still experience something new, in case of the thought experiment, the color red. One line of thought that challenges Jackson’s argument involves an equivocation of “know.” Is Mary’s new-found knowledge propositional or know-how/ability-type knowledge? For Hayek, *qualia* are know-how (Hayek, 1952/1976, 2.7). The Mary puzzle is intended to make for the view that *qualia* cannot be reduced to the level of physics and hence there cannot be a unified theory of consciousness. Dennett (1991) terms these thought experiments as an “intuition pumps,” a pejorative swipe at the a priorism he sees generated by them. Jackson, it should be noted, has since retracted his original conclusion; he is now of the view that the sensory side of psychology is, in principle, deducible from the world’s physical nature.

Jackson’s thought experiment bears a striking resemblance to Hayek’s discussion in *The Sensory Order*, 1.95. Hayek took inspiration from C. D. Broad, the idea that an omnipotent being would still not be able to predict the *qualia* associated with a substance, for example, ammonia (Broad, 1925, p. 71). Here Hayek poses the question: how could one communicate the idea of vision generally and color in particular to the congenitally blind? In *The Sensory Order*, 1.97 and 1.98, Hayek cites physicist Kenneth Mees’ thought experiment as illustrating the distinction between the physical and the phenomenal orders.¹¹ Mees asks us to consider the case of a congenitally and totally deaf person confronted by someone playing a violin. Moreover, he asks us to suppose that this person knows nothing of sound *even* in a theoretical way. Confronted by the actions of the violin player, to the deaf

person the actions will appear irrational. *But*, says Mees, if our deaf person *was* a scientist, he or she would eventually figure out that the movements of the violin bow generated vibrations that could be detected by equipment (the science of acoustics). Now whatever the issues Hayek has with Mees' example, his conclusion is this: "the congenitally blind or deaf can never learn *all* that which the seeing or hearing person owes to the direct experience of the sensory qualities in question, because no description can exhaust all the distinctions which are experienced" (Hayek, 1952/1976, 1.102). The similarity of the conclusion shared by Hayek and Jackson is uncanny.

Hayek's pessimism manifest in his discussion of cognitive closure and *qualia* marks a deep philosophical issue, that is, *the view that science is explanatorily closed* (Hayek, 1952/1976, 1.88, 8.31). Were Hayek an agnostic he would take the view that the ultimate explanations provided by science are in need of supplementation (Hayek, 1952/1976, 1:13; 8:26; 8:31; Hayek, 1952, #35). This is of course not the case. Science has failed miserably at assimilating the irreducible phenomenal aspect of conscious experience (Hayek, 1952/1979, p. 36) and will continue to do so. Humankind is forever condemned to a practical dualism.

HAYEK'S *SOCIAL* CONNECTIONISM

The brain struggling to understand the brain is society trying to explain itself (Blakemore, 1977, p. 185).

This sentiment is echoed in (Hayek, 1952/1976, 1.17): "[M]uch that we believe to know about the external world is, in fact, knowledge about ourselves." Given the intimate tripartite nature of mind, knowledge and society, it is perfectly plausible for one to draw the inference that the connectionist paradigm in Hayek's philosophy of mind has a conceptual analog in the social sphere – hence this section, for the sake of parity, is entitled *social* connectionism. Under consideration first is the general claim of the link between mind and society in Hayek's work. This sets the scene for discussion of Hayek's social externalism.

Mind and Society

That Hayek's view of the mind and his view of the market have strong similarities and continuities has been noticed by many. Anyone who has

read most of the Hayek corpus must surely share this view (Weimer & Hayek, 1982; Boettke, 1990; Boettke & Subrick, 2002; Streit, 1993; Birner, 1995; Tuerck, 1995; Caldwell, 2000, 2004a; Horwitz, 2000; Gick & Gick, 2001; Mistri, 2002; Butos, 2003; Rizzello, 2004; Frantz, 2005; Gaus, 2006; Loasby, 2004; Novak, 2005; Butos & Koppl, 2007; McQuade, 2007; Gifford, 2007; Posner, 2005, 2007; Gick, 2003, 2007, 2010). Indeed, Hayek himself makes the connection: “It was concern with the logical character of social theory which forced me to re-examine systematically my ideas on theoretical psychology” (Hayek, 1952/1976, p. v) and again: “I chose economics, perhaps wrongly; the fascination of physiological psychology never quite left me” (Mahoney & Weimer, 1994; Hayek, 1978, p. 36). At the very least, Hayek’s philosophical psychology was implicit in everything he ever wrote (Hayek 1994, pp. 139, 153). In 1948, he wrote that his 1920 essay was his most important intellectual accomplishment (Novak, 2005; Aimar, 2008, p. 25). Streit (1993, p. 227) goes so far as to say that Hayek, the polymath, “wanted to replace metaphysics with neurophysiology.”

But is it plausible to suggest that Hayek’s work contains a *systematic* answer to any connection? It all depends on how stringent one is in characterizing systemization. One recent commentator (Feser, 2006, pp. 287–288) marks this problem:

That [*The Sensory Order*] foreshadowed connectionism seems at the end of the day a point of merely historical significance; and its status as the “foundation” for Hayek’s economics and politics has, I think, been exaggerated, claims for such a status typically resting on little more than the fact that the book characterizes the mind just as Hayek characterized economics and social systems, namely, as being complex, dynamic and unpredictable in principle. (Hayek would no doubt have characterized the weather in exactly the same terms. Should we therefore regard meteorology as providing a “foundation” for his economics and politics?)¹²

Though Feser’s argument is formally weak, a more charitable reading reveals a fundamental oversight. In addition to mind being complex, dynamic, and unpredictable Feser misses the key idea of mind as an *adaptive classifying structure* necessarily embedded in a social world. It has to be admitted that even if Hayek had a proto-connectionist theory of mind, as I (and others, e.g., Fuster, 1995/1999) claim he definitely did, it does not follow that Hayek *logically* connected the mind with society. To be aware of an implicit connection isn’t the same as drawing out that connection; to signal one’s awareness of the connection in a late work isn’t the same as one’s awareness of it in earlier work (I allude here to the contested provenance of Hayek 1988 – see Caldwell (2000, pp. 17–18)). A case in

point: Edelman (1987) is listed in the bibliography, but is not even hinted at in the body of the discussion. To be sure, Hayek explicitly stops well short of identifying society as some sort of super-brain¹³ (Hayek, 1967, p. 74, 1988, p. 98; Weimer & Hayek, 1982; Birner, 1995; Aimar, 2008, p. 41). *But*, it has to be conceded, Hayek's language does sometimes encourage this vision, notably the functionalism of *The Sensory Order* (Hayek, 1952/1976, 2.27–2.30, 5.75, 1988, p. 42). Functionalism in its most standard form is the idea that mind is only contingently dependent on the brain, the a priori implication being that a mind can be instantiated in any material. Hayek: "So long as the elements, whatever other properties they may possess, are capable of acting upon each other in the manner determining the structure of the machine, their other properties are irrelevant for our understanding of the machine" (Hayek, 1952/1976, 2.28). And again: "[A]lthough ... machines cannot yet be described as brains, with regard to purposiveness they differ from a brain merely in degree and not in kind" (Hayek, 1952/1976, 5.75). Hayek's functionalism is puzzling and is hostage to a raft of objections of which I mention only two. First, it abstracts from the physical details of neurological implementation. This clearly undermines the Hayekian situated agent. Second, it ignores the experiential aspect to consciousness in that functionally identical persons could differ in their experiential states. Hayek's functionalism warrants a detailed discussion, something I'll defer for another time. Interestingly, Hayek's brief functionalist talk anticipates (by 8 years) Putnam's canonical formulation of functionalism (Putnam, 1975, p. 291). Feser (2001) offers a Hayek-inspired discussion of functionalism that accommodates *qualia*. This discussion seems to be an anomaly in Hayek.

The commonalities and disanalogies of mind and society in Hayek is a line of thought that has received its most sustained treatment from McQuade (2007), McQuade & Butos (2006) and Gifford (2009). Alert to Hayek's super-brain admonition, they wish to show that Hayek's theory of mind has analogs in the social domain. McQuade and Butos "do not think that people and neurons are comparable in any other sense than that they can form mutable interaction patterns with each other" (McQuade & Butos, 2006, p. 341; Wang & Sun, 2008).¹⁴ Earlier they claim "their analogs are present, in different physical realizations, in various social systems" (McQuade & Butos, 2006, p. 336). I'm not convinced that McQuade & Butos are not (even unwittingly) smuggling in a form of functionalism: multiple realizability has traditionally been used in support of functionalism.

Putting aside my functionalist qualms, I'll go along with the idea that social systems are *brain-like* in that "the interactions between their

components implement a classifying process on stimuli impinging on the system, and this process can induce real changes in component behavior and interaction that, in turn, engender adaptive reactions of the system as a whole to changes in its environment” (McQuade, 2007, p. 59; McQuade & Butos, 2006, p. 341). Indeed, it seems that Hayek does explicitly countenance this in *The Sensory Order*, 8.45:

What we have tried to do here is to show that the same kind of regularities which we have learnt to discover in the world around us are in principle [*sic*] also capable of building up an order like that constituting our mind.

What’s been emphasized by McQuade and Butos as the common denominator is emergence arising from interminable positive and negative feedback characteristic of adaptive systems.

Extended Cognitive Systems

I want to recast the McQuade–Butos line in different terms. First, as a case for Hayek’s *externalism*; and second, as a case for adaptive systems to be viewed as specific form of externalism – an *enactive* form of externalism. Both jointly displaying the emergent, distributed and adaptive features McQuade and Butos are rightly keen to show can be found in the social domain.

It is hardly contentious that Hayek should be taken as an externalist. Hayek himself could not have put it more plainly (Hayek, 1952/1976, 8.1):

If the account of the determination of mental qualities which we have given is correct, it would mean that the apparatus by means of which we learn about the external world is itself the product of a kind of experience. It is shaped by the conditions prevailing in the environment in which we live ...

Externalism means different things to different people. A somewhat generic characterization is that it is the thesis that an individual’s environment has *some* causal determinant on the content of the individual mind. Externalism provides Hayek with a way of avoiding what McCulloch (2003, pp. 5–9) calls the Demonic Dilemma. That is, do we locate intentionality on the mind side, thereby sealing it off from the world at large, or do we locate intentionality on the world side, thereby failing to explain how there could be any content or subjectivity on the mind side of things. Thus, the discussion of externalism connects with the earlier discussion on cognitive closure and qualia.

As mentioned at the outset of this paper, Hayek's externalism cuts across the various emphases and concerns of the "situated" literature: it speaks to a past and a recent conception. Adam Smith's (1776/1976) idea of the "division of labour" and Hayek's much broader systemic dispersion of knowledge are classic instantiations of the distribution of cognitive labor. Though never referring to Smith or Hayek, Edwin Hutchins' *Cognition in the Wild* (1995) restates important aspects of their insights. All divisions of labor, whether physical or cognitive in nature, simple or complex, involves the distribution of two kinds of cognitive labor – the cognition that is the task and the cognition that governs the coordination of the elements of the task (Hutchins, 1995, p. 176). Hayek's illustration of the distributed nature of cognition in *The Sensory Order* (4.40 and 4.41) bears more than a passing similarity to Hutchins' famous discussion of maritime navigation.

I wish to make a case for the view that society, in Hayek's theory of it, functions as a kind of *extra-neural memory store* – mind and society are relational in a highly qualified way. But first, a disclaimer. Following Adams and Aizawa (2008, pp. 106–132), I want to draw the distinction between an *extended cognitive system* and *extended cognitive process*. The former claims that brain, body, and environment constitute an extended system in the sense that an extended system *augments* cognitive processes. But it is quite a radical step to say that this processing literally *extends* into these artifacts. (The DEEDS literature is as contentious as it is suggestive and I make no claims here to be engaging with it critically. For a sustained critique of DEEDS, see Adams & Aizawa, 2008 and Rupert, 2009.) I don't see any grounds for labeling Hayek as an "extended mind" theorist in this sense. I make the claim that Hayek subscribes to the idea of an extended cognitive system without imputing actual cognitive processing to these structures: "the mark of the cognitive" (Adams & Aizawa, 2008, pp. 76–87; Marsh, 2011a) is for Hayek, very much internalist (Hayek, 1952/1976, 6.8–6.13).

Hayek's externalism rests on the view that extant and spontaneous arising customs, practices and traditions are the sources or *fundamentum* and the *residua* of practical reasoning: to disregard them is to be irrational (Hayek, 1948/1980, p. 24, 1952/1979, p. 163, 1960/1978, pp. 61–62, 1978, p. 18).¹⁵ It's what Hayek terms the "*discipline of reason*" (Hayek, 1978, p. 19, 1973, p. 32, 1988, p. 8). And, whatever a tradition is, by definition, it cannot reside solely within an individual – there is no direct brain-to-brain/mind-to-mind memetic transmission – continuity can only be mediated albeit imperfectly through a web of social artifacts (Turner, 2003, pp. 3, 11). As (Millikan, 1993, p. 170) puts it: "I no more carry my complete cognitive systems around with me as I walk from place to place than I carry the US currency system about

with me when I walk with a dime in my pocket.” For Hayek, knowledge and cognition must be set against a background fabric of cultural possibility: individuals draw their self-understanding from what is conceptually to hand in historically specific societies or civilizations, a *preexisting* complex web of linguistic, technological, social, political and institutional constraints (Hayek, 1973), a “social ecosystem” if you like (Gamble, 2006, p. 130).

Hayek is an “enactive” externalist in the sense that the Hayekian agent *cognitively offloads and harvests knowledge through external sociocultural structures*. This interpretation finds resonance in a recent paper by Gifford (2009). For Hayek, “computational” efficacy is enhanced by allowing information to remain outside the brain, and by thus exploiting environmental and social resources rather than having to encode *everything* relevant internally (Hayek, 1978, pp. 42–43, 1967, pp. 50–53; Bartholo, Cosenza, Doria, & de Lessa, 2009). The mark of advanced cognition depends upon our ability to diffuse propositional (KT) and practical knowledge or wisdom (KH) through external epistemic and cognitive structures offloading the epistemic burden with a reciprocal and cybernetic “enactive” relation between our conceptual creativity and the environment, to intimate, regulate, and inform concepts and action (Hayek, 1973, p. 37; Fuster, 2008; Dupuy, 1994/2000). Organisms, such as ourselves, will appear self-adaptive and purposive and “will in general be ‘active’ in the sense that what at any given moment will determine the character of its operation will be the pre-existing state of its internal processes as much as the external influences acting on it” (Hayek, 1952/1976, 5.65).

The perpetual feedforward and feedback (Hayek, 1952/1976, 2.25, 4.54, 4.9) that is characteristic of the enactive mind undermines the stark polarities of methodological individualism¹⁶ and social holism, which respectively generate the idealized pernicious fictions of the “unencumbered” self and the anthropomorphic society (Hayek, 1988, p. 113; Caldwell, 2004a, pp. 279–287). The Hayekian agent still forms a core part of a specific wide memory system, one in which we serve as a *locus of control* (Hayek, 1952/1976, 5.9, 5.3). Hayek does not see any profit in expunging all vestige of Cartesian internalism: mind and social aggregates are, for Hayek, on an ontological par (Hayek, 1952/1976, 5.3, 5.22, 1979, p. 156). So here’s the rub: Hayek the externalist challenges the view of Hayek the paradigmatic individualist¹⁷ (and a raft of often contradictory assignations such as liberal, *laissez-faire* ideologue, and conservative), interpretations that I contend are skewed – even if one doesn’t factor in *The Sensory Order*.

One way of sharpening up Hayek’s individualist/holist compatibilism is to draw upon the work of Robert Wilson (2004), one of the leading DEEDS

theorists. Wilson formulates what he terms as the “social manifestation thesis.” Wilson seeks, as does Hayek, to mediate radical Cartesian individualism and an implausible Hegelian hypostasized supra-individual social consciousness. For Wilson, much of the group-mind hypothesis can be expressed within an externalist theory of mind: “group consciousness talk” can for the most part be recast as “an aspect of the consciousness of individuals” (Wilson, 2004, p. 290). Wilson’s social manifestation thesis allows that individuals have a disposition to reflect *some psychological states only when they form part of a social group* (Wilson, 2004, p. 299). If for Wilson “The minds that individuals have are already the minds of individuals in groups” (Wilson, 2004, pp. 142, 265, 307), then I can’t see this as being incompatible with the methodological individualist arguing that to ascribe judgments, intentions, and the like to social groups is just a shorthand ascription to the individuals that comprise the relevant groups. The social manifestation thesis views psychological states as “taxonomically and locationally embedded in broader social systems” without having to posit some supra-group consciousness (Wilson, 2004, p. 301; Tuomela, 2007, p. 145). Although individualists and externalists agree that mental states are “in the head” and that they are causally determined, in part, by what lies beyond the head – they disagree about how mental states should be individuated or taxonomized. Hayek, the individualist, through his “soft” Kantianism, posits an intrinsic determination that avoids the notion that two individuals identical in their intrinsic properties, must have the same psychological states (Hayek, 1952/1976, 5.27–28).

An Extended Cognitive System – A Stigmergic Approach

McQuade and Butos’ suggestive emphasis on *adaptive* classifying systems offers grist to the externalist mill. The “coordination paradox” was most famously articulated in Adam Smith’s “invisible hand” metaphor – is essentially *a theory of collaboration via self-interest* – and was endorsed by Hayek (1973, p. 56; Nozick, 1974, pp. 18–22). The star example of an extended cognitive system (or a complex adaptive system) is a “market” since it displays *the* defining feature – emergent behavior.¹⁸ For Hayek, *free* (or open and competitive) markets (Hayek, 1948, pp. 92–106) are in effect “communications systems” (Hayek, 1978, p. 34, 1979, p. 68, 1988, p. 84; Boettke, 1990; McQuade, 2007) that display the following virtues:

- (1) It’s a mechanism for the cooperation among strangers with *differing wants and preferences* in a given environment (Hayek, 1976, pp. 109–111),

- (2) enables activity that has consequences for all its agents, despite the fact that few transactions ever directly take place *in-person* to *in-person* (Hayek, 1988, p. 14),
- (3) it does not rest upon “rational” behavior (Hayek, 1944, p. 64, 1988, pp. 53–54),
- (4) it breeds a certain cast of mind – the entrepreneur (Hayek, 1979, pp. 75–76), and
- (5) it has epistemic (and computational) efficiencies in that knowledge is distributed and dynamic.

Now this is all pedestrian stuff. What I want to introduce here is another way of getting conceptual traction on the coordination problem in a way that emphasizes the “active” externalism of Hayek’s work.¹⁹ The significance of the concept I’m about to expound upon has been made apparent to artificial intelligence (AI)/computational intelligence researchers over the last decade.

Pierre-Paul Grassé, a zoologist, discovered in the coordination and regulation of termite colonies, the *phenomenon of indirect communication mediated by modifications of the environment* – which Grassé termed “stigmergy” (Grassé, 1959). Grassé observed that the coordination and regulation of building activities did not depend on the individual “agents” themselves, but is subject to a cybernetic feedback loop through pheromone traces and environmental modifications made by others (Hayek, 1952/1976, 2.28, 1952, #42; McQuade, 2007, pp. 57, 77; Marsh & Onof, 2008; Marsh, 2011b). In other words, *the environment acts a kind of distributed memory system*. Though the concept of stigmergy has been associated with ant- or swarm-like “agents” with minimal cognitive ability, stigmergy offers a powerful metaphor to be deployed in the human domain. And with his characteristic prescience, Hayek was onto the explanatory possibilities (Hayek, 1967, pp. 69–71; see also Hayek, 1952, #41–43, 51, 1978, p. 53, 1967, p. 73; McQuade, 2007, p. 77). For Clark, “[M]uch of what goes on in the complex world of humans, may thus, somewhat surprisingly, be understood in terms of so-called stigmergic algorithms” (Clark, 1996, p. 279). Stigmergic systems are a ubiquitous feature of human sociality, that include stock markets, economies, traffic patterns, supply logistics and resource allocation, urban sprawl, and cultural memes. Consider this example by Clark. Amazon.com’s “collaborative filtering” technique mimics the stigmergy of slugs and ants, illustrating “patterns of action” (Hayek, 1978, p. 41). Through their activity, humans also lay trails, albeit in this case digital trails, which can be tracked, analyzed, and agglomerated.

It is computationally efficient (and easily scalable) because much of the computation has already been done offline. An item-to-item search generates a “pheromone” trail that gives rise to novel patterns of behavior. The system is computationally efficient since it only searches segments, rather than the complete database. The Amazon system’s great virtue is that suppliers can be finely attuned to consumer behavior. The downside is that there runs the risk of “a kind of dysfunctional communal narrowing of attention” that can be self-fulfilling (Clark, 2003, p. 158). Clark’s point, I think, is that the specter of herd behavior has latent potential and this is somewhat echoed by McQuade (2007, p. 71).

Let us consider the typical features of a stigmergic system (or a complex adaptive system if you like):

1. A context or environment
 - comprised by an indefinite number of local environments and
 - only partially perceivable through an internal dynamics that govern its temporal evolution.
2. Agents
 - There is a multiplicity of agents populating with no one individual or clustering of individuals having global knowledge.
 - Rationality is bounded.
 - Behavior is self-organized.
 - Behavior is stochastic.
 - Behavior is adaptive/dynamical.

Novel features arise from interaction of (1) and (2), features that are neither predictable nor reducible to simpler constituents. I’ve desisted from using the term “emergence” since the concept is highly slippery. For present purposes, it is considered in an agnostic way that may or may not correspond to the philosopher’s conception (a strong variant), typically used to reject the incompleteness of the physicalist picture; or with idea that ostensibly novel phenomena is a function of the current state of knowledge, but can be grounded in underlying simple laws (Chalmers, 2006).

Now, if Hayek was centrally concerned with “communications systems” then he was centrally concerned with the communicative aspect to knowledge. If *social* epistemology²⁰ has the formation, acquisition, mediation, transmission, and dissemination of (for the most part *third-party*) knowledge in complex communities of knowers as its subject matter, then to say that its concern is essentially stigmergic, verges on being tautologous (Hayek, 1944, pp. 33–56). Sociality is stigmergic on the grounds that no one mind has global knowledge – there is no rationalistic master

plan or blueprint; much of the “calculation” is done through social artifacts (the market for one); and last but by no means least, it is stigmergic on the grounds of the iterated looping of behaviors within and through the environment. A stigmergic system is coextensive with any complex adaptive system (Marsh & Onof, 2008; Marsh, 2011b). Take the market as an example, the market would be *stigmergically* superior to all other types of market in that it is the best communications system for all manners of social and other types of knowledge that course through its veins. Of course, for Hayek a free-market best promotes the conditions for moral and political freedoms or autonomy (Hayek, 1973, p. 55).

EPILOGUE

Both freedom and justice are values that can prevail only among men with limited knowledge and would have no meaning in a society of omniscient men (Hayek, 1976, p. 127).²¹

This quote marks an opportune time to take stock: it reminds us of what really was Hayek’s lifelong concern – *liberty*. I don’t share Raz’ confidence in characterizing Hayek’s concern with freedom as exclusively instrumental in nature, rather than intrinsic in nature (Raz, 1988, p. 7). I venture the view that for Hayek the notion of cognitive closure entails the postulation of an open society (Hayek, 1976, p. 127). On my interpretation (or emphasis if you will), there is a somewhat deflationary relation between *cognitive closure* and epistemology: cognition and knowledge are viewed dual aspects of human sociality. This conception has resonance with Alvin Goldman’s “epistemics” – a multidisciplinary understanding of epistemology that gives due emphasis to the psychological processes of the architecture of the mind–brain in belief formation *and* the fact that these beliefs are modulated by a ubiquitous sociality (Goldman, 1986). Again, Hayek’s prescience is uncanny.

Hayek’s notion of cognitive closure, a mark of the human condition, can be ameliorated if the social and artifactual world functions as a kind of distributed extra-neural memory store manifest as dynamic traditions, part of the resources for acting, thinking, or communicating. This cognitive $\leftrightarrow$ epistemological $\leftrightarrow$ liberty tripartite is closely related to a long-standing bone of contention in Hayek centering on the two-fold claim:

- (1) epistemological immodesty is the *sine qua non* of a mixed or socialist economy, and that
- (2) this *inexorably* leads us on *The Road to Serfdom* (Samuelson, 2009).

The manifold ways in which this so-called “inevitability thesis” (Hayek, 1944/1976, Chap. IV) can be interpreted is discussed by Farrant and McPhail (2009). Working from the 1976 edition of *The Road to Serfdom*, Hayek gives out a mixed message. The cover trumpets the book as “A classic warning against the dangers to freedom *inherent* in social planning” (emphasis added). In the foreword, Hayek claims that he has “never accused the socialist parties of deliberately aiming at a totalitarian regime or even to show such inclinations” (Hayek, 1944/1976, pp. xiv, xxi). Hayek is of the view that the source of misinterpreting the inevitability thesis is terminological – that is, socialism at the time he was writing really did mean complete and utter centralization. Thirty years on, socialism in Western Europe pretty much denoted a mixed economy. So what are we to make of Hayek on this issue? (For a fine-grained history and analysis of this issue, see Farrant, 2009.)

Hayek definitely *does* believe that a necessary condition of socialism is a *degree* of centralization, political, and economic, which seriously infringes personal freedom. This looks like a causal claim: socialism cannot operate without this degree of centralization. It’s a quite different (though still causal) claim that a mixed economy either leads to socialism or, for other reasons, itself produces a degree of centralization, political, and economic, which seriously infringes personal freedom. I’d agree that the link between central planning and the kind of socialism Hayek had in mind is logical. One might even see it as definitional. One might think that the diminution of freedom is itself a logical consequence if what is centrally planned, since it is no longer a matter for personal choice. But this line of argument, whether Hayek’s or not, neglects the calculus of freedom. It’s logically perfectly possible for central planning to restrict some freedom, but to create or increase others. Why not? Hayek can’t logically rule it out. It’s a causal matter. In any event, it should be remembered that Hayek’s target was a rationalist *zeitgeist* that infected “socialists of all parties”: this was, after all, the polemical point of the book (note the tongue-in-cheek dedication; p. 35).

Of course it matters whether one is focusing on the Hayek of 1944 or the Hayek of 1967; it is clear that Hayek had refined his views. Consider the later essay “The Theory of Complex Phenomena” (Hayek, 1967, p. 42), where he concludes that:

... we may well have achieved a very elaborate and quite useful theory of some kind of complex phenomena and yet have to admit that we do not know of a single law, in the ordinary sense of the word, which this kind of phenomena obeys ... I rather doubt whether we know of any “laws” which social phenomena obey ... in the field of complex

phenomena the term “law” as well as the concepts of cause and effect are not applicable without such modification as to deprive them of their ordinary meaning.

Hayek rightly admits that the “inevitability” *is* a vague and imprecise expression. So far as I can see, Hayek’s “infelicity” is generated by a lack of philosophical precision – but his critics fare little better on this point. A philosopher would talk about some (specified) kind of necessity. I’d guess Hayek assumes causal necessity, but the covering law(s) would have to contain *ceteris paribus* clauses – which rather undermines the dramatic claim of inevitability. And what is the covering law or set of covering laws? Hayek can, it seems to me, assume causal necessity and does so at various points in his argument. The spontaneous social order emerges causally. *Epistemologically* we can’t predict its features, but it’s not spontaneous in the sense of being *metaphysically* uncaused. Clearly *ceteris paribus* clauses water down a law’s necessity, and in this sense make its operation contingent. And contingency means that the law has a probability of <1 . This is so even if the law “works” with exceptionless regularity: that’s just a contingency. But *ceteris paribus* clauses don’t tell you, without extra assumptions, what the actual probability is between 0 and <1 . If there’s a social law with *ceteris paribus* clauses to support this probabilistic generalization, then we need to know what the clauses are and what in turn their probability is. Central planning leads to the general erosion of freedom unless:

x, y, z where ‘*x, y, z*’ individually or as a disjunctive set have a probability of 0.9 (or whatever).

If, on the other hand, Hayek is offering a social law as an exceptionless generalization, then presumably his whole interlocked social theory will be needed to deliver this law. (By the way, while Marx does talk of the “iron laws of history,” there are other passages where historical transitions are seen as trends of extremely high probability. Epistemologically, of course, Marx never claims chronological precision as to what will happen: he can’t give even the roughest of timelines). The claim might be that there’s a high probability, approaching 1, that central planning will lead to the erosion of freedoms. Not just economic freedom but any freedom that relies on the rule of law, since central planning will need to override the rule of law. What is this probability claim based on? If on enumerative induction, then Hayek cannot make good this claim because his sample base is tiny. Enough said.

It might be said that philosophical fashion has not been kind to Hayek. This is more a reflection of the philosophy of mind (and the philosophy of social science and political philosophy) as practiced under the aegis of a philosophy department than of the more ecumenical philosophy of mind as practiced

under the aegis of a cognitive science department. If as Varela et al. (1991/2000, p. 13) say, “cognitive science stands at the crossroads where the natural sciences and the human sciences meet” then Hayek would be center stage.

Revived interest in certain thinkers or aspects of their work tends to occur because they have been “rehabilitated” (a new interpretive gloss has been put on their work); or because a relatively under-exploited oeuvre provides material to feed academic industriousness. Historically, *The Sensory Order* has simply been overlooked – echoing Hume’s disappointment, Hayek lamented that “it fell dead-born from the press” (Leube, 2003, p. 12). The primary reason is perhaps because disciplinary boundaries as they existed when Hayek was writing, were rigid and myopic (Hayek, 1952/1976, vii; Grenell, 1954, p. 409). This said, *The Sensory Order* did garner some early reviews that, though cursory and lukewarm, were not deeply negative (Chisholm, 1954; Grenell, 1954; Schiller, 1954; Sprott, 1954). Stylistically, *The Sensory Order* is brittle; it has a great deal of repetition and qualification, obscuring Hayek’s usual mellifluous style and the Austrian knack of clarity and crispness of argument. This is partly a function of (a) Hayek trying to articulate concepts that heretofore were only faintly outlined, and (b) Hayek himself may not have been the best judge of what the overriding theme of *The Sensory Order* really was, Hayek acknowledged that even trying to conceptualize the very basics was a fraught enterprise (Hayek, 1952/1976, 1.2). That in Hayek’s view, *The Sensory Order* is centrally concerned with the mind–body problem (Hayek, 1952/1976, 1.10) entails a significant quotient of murkiness (Hayek, 1982, pp. 289–290).

Perhaps the earliest substantive appreciation of *The Sensory Order* came from computer scientist, Frank Rosenblatt whose “perceptron,” an early version of a feed-forward learning algorithm, was influenced by the suggestiveness of Hayek and Hebb (Rosenblatt, 1958). Edwin G. Boring, an experimental psychologist, one time president of the American Psychological Association and founder of Harvard’s Department of Psychology, had some good things to say about *The Sensory Order* (cited in Kresge’s introduction to Hayek, 1994, p. 27). According to Hayek, “... the one man who seemed to have fully understood *The Sensory Order*” was, no less, Schrödinger (Hayek, 1994, p. 139).²² The future A.I. grandee, Marvin Minsky, certainly knew of *The Sensory Order* (Minsky, 1961). Of late, the one person who has consistently promoted Hayek’s achievement has been Joaquín Fuster (1995, 1998, 2003a, 2003b, 2004, 2006a, 2006b, 2008). Fuster (2004) writes that:

It is truly astonishing that its author, in the middle of the ignorance that existed in the first half of the XX century about the anatomical and physiological organization of the cortex, would instinctively coincide with the evidence of the second half of the century.

Hayek's reputation in cognitive neuroscience was significantly boosted by fellow Nobel laureate Gerald Edelman in his influential book of 1987, but was trumpeting Hayek's achievement 5 years earlier (Edelman, 1982). Following in Fuster and Edelman's wake, others in neuroscience are now taking notice (Başar & Karakaş, 2006, pp. 195, 198). Of the better-known philosophers of mind, John Searle, has belatedly acknowledged the value of Hayek generally (Krüger, 1999) and *The Sensory Order* (Searle, 2000).²³ Of course, one should be suspicious of the hagiographical tendencies of the many disciples who claim to be Hayekians, but with such a roster of top-draw names batting for Hayek *and* from beyond the usual constituencies, it seems blatantly disingenuous to suggest that Hayek "dabbled" (Posner, 2005, p. 155) in cognitive science. Hayek supposed dilettantism has its two-fold (and linked) roots in his (a) professional pride and (b) having wide interests. Regarding the former, consider the somewhat neurotic tone here: "After *The Road to Serfdom*, I felt that I had so discredited myself professionally, I didn't want to give offence again. I wanted to be accepted in the scientific community" (Hayek 1994, p. 152); Samuelson (2009, p. 3, note 2) affirms Hayek's neuroses. Regarding the latter, Steele (2008, p. 69) points out that Hayek was as close to being the complete economist as characterized by G. L. S. Shackle:

To be a complete economist, a man need only be a mathematician, a philosopher, a psychologist, an anthropologist, a historian, a geographer, and a student of politics; a master of prose exposition; a man of the world with the experience of practical business and finance, an understanding of the problems of administration, and a good knowledge of four or five languages. All this in addition, of course, to familiarity with the economics literature itself.

Hayek himself wrote: "... exclusive concentration on a specialty has a peculiarly baneful effect: it will not merely prevent us from being attractive company or good citizens but may impair our competence in our proper field ..." (Hayek, 1967, pp. 123, 127). Hayek recalled that at the University of Vienna he was nominally studying law, but was "shifting from subject to subject" (1994, pp. 3, 51). As Streit (1993, p. 256) says:

Reconsidering Hayek's rich scientific harvest, it is probably justified to argue that he could not always penetrate in depth the many fields of research in which he scored striking and often revolutionary results. But in doing so, he opened up new avenues worth further exploration.

No longer can Hayek be dismissed as a mere historical curiosity. Hayek doesn't even warrant a mention even as a historical curiosity in *Talking Nets: An Oral History of Neural Networks* (2000) and the monumental

The MIT Encyclopedia of the Cognitive Sciences (1999). This lacuna has fortunately been redressed by Fuster (2003a). Some Hayekians feel the assignation of Hayek as historical curiosity is perfectly legitimate (Feser, 2006, p. 297); this despite Feser saying earlier (Feser, 2006, p. 288) that “a case could even be made for it [*The Sensory Order*] as the most comprehensive and plausible attempt yet made to carry out the project of ‘naturalizing’ the mind.” Hayek, the cognitive scientist is *the* preeminent theorist of sociocognition. German Chancellor Helmut Schmidt, on the occasion of Hayek’s 80th birthday (1979), sent him a congratulatory telegram (perhaps prematurely) saying: “we are all Hayekians now” (Hoppe, 2004). With the rise of the prevailing technocultural form known as the World Wide Web and its essentially distributed and stigmergic nature²⁴, perhaps only *now* can it truly be said that “we are all Hayekians.”

NOTES

1. Variations of this occur in Hayek (1978, pp. 3–4, 20, 73, 1944/1976, p. 165, 1973, pp. 5, 17, 1952/1979, p. 156, 1988, p. 22). The suggestion “that in order to understand central aspects of cognition we look not to what’s in the brain but what the brain is in” is another more recent variant (Wilson, 2004, p. 212).

2. According to the Cognitive Science Society, current cognitive science comprises artificial intelligence, linguistics, anthropology, psychology, neuroscience, philosophy, and education. These days the term “multidisciplinary” tends to be over-used, but as Rizzello and Schiller point out (Rizzello, 2004, p. 257; Schiller, 1954, p. 534), Hayek was no dilettante. As early as 1919–1920 Hayek spent some time in the Zurich laboratory of Russian-Swiss neuropathologist Constantin von Monakow (Hayek 1994, p. 64; Caldwell, 2004b, p. 136).

3. Even though Ryle (1949/1990) is taken by many to be a behaviorist of sorts (Hayek, 1952/1976, 1.79–1.88), I deal at length with the KH/KT distinction in Marsh (2010b).

4. Another and perhaps more basic way to conceive of the KH/KT distinction is as nonconceptual content/conceptual content – see Bermúdez and Cahen (2008) for an overview. Other ways to make this distinction is as *offline* and *online* styles of intelligence (Wheeler, 2005) and *declarative* and *procedural* memory. A recent paper by Perraton and Tarrant (2007) is skeptical of the extensional and intensional adequacy of KH yet overlooks a whole swath of recent literature: pro, con, and hybrid.

5. Echoed in Reisman (1997).

6. Emerson M. Pugh (1914–1981) cited by his son George Pugh (1977, p. 154).

7. Hayek might well be gratified by Mohan Matthen’s (2005) recent work on the classification process of sensory perception, a treatise that bears a striking resemblance to *The Sensory Order*. Others working on the philosophy of (color)

perception have referred to *The Sensory Order* (Byrne (2009); Byrne & Hilbert (2008)).

8. Thanks to Istvan Berkeley for sharing this.

9. This Gödelean claim and its bearing upon people has canonical status in Hofstadter (1979, pp. 697–698).

10. It would be impracticable to explicate the voluminous debate that has become ritualized. The progenitor of these thought experiments is Nagel (1974). Hayek would find Nagel's "what is it like to be a bat" thought experiment absurd even if he wasn't totally out of sympathy with its conclusion (Hayek, 1944/1976, p. 66, 1952/1979, p. 135).

11. Mees (1882–1960) was an Anglo-American physicist who set up Eastman Kodak's famed R&D laboratory in Rochester, NY. The quote is from Mees' *The Path of Science* (Wiley, New York). It is dated as 1947 in Hayek's footnote (p. 33) and 1946 in the *The Sensory Order* Bibliography.

12. Feser's claim is that if one maintains that the argument (A):

(A) Hayek views the mind as complex, dynamic and unpredictable.

Hayek's view of the mind is the foundation for his views on economics and politics. is valid, then, one accepts the validity of a general argument (B) of the form:

(B) Hayek views the X as complex, dynamic and unpredictable.

Hayek's view of the X is the foundation for his views on economics and politics.

But one can substitute anything for X in (B), which would make the premise true, to obtain a true conclusion. And it is indeed plausible to claim that substituting weather for X, one obtains a true premise. This would then lead to the truth of:

Hayek's view of the weather is the foundation for his views on economics and politics.

The implied *reductio ad absurdum* argument then is that, since the latter claim is patently ridiculous, the original argument (A) is therefore not valid.

The weakness of Feser's argument is that it is not because one claims the validity of the original argument (A), that one accepts the validity of (B). That is because there are lots of hidden premises in (A) that pertain to the importance that an understanding of the mind has for economics or politics (in a general and fairly uncontroversial sense), and are not true for any X (e.g., the weather).

13. Notions of a super-brain or global brain are implied in writings of Heylighen, 2007 and others. One might, in a very suggestive way, say that in much the same way that synapses are strengthened while unused connections weaken and wither away ("neural Darwinism" – Edelman 1987), so too are the social "synapses." Google's PageRank algorithm operates like this. In a recent article, the writer has coined the term "Social Nervous System" (Ross, 2009).

14. Ned Bloch (1978) "Chinese nation" thought experiment was designed as a critique of functionalism. Imagine the population of China (currently 1.3 billion) implementing the functions of neurons in the brain. Each person is assigned some role that reproduces something in the human functional network; in total, the 1.3 billion Chinese are actually reproducing the functional states of a given person. Block's objection is that it is implausible to ascribe that whole Chinese nation should have qualitative experiences while no individual member of the population

experiences any pain. The upshot for Block is that if some part of what being in a mental state corresponds to a qualitative experience, then functional states are not equivalent to mental states.

15. It should be noted that this is not a blanket admonition against social change or social amelioration (Hayek, 1978, p. 19). Posner (2007) takes the view that Hayek offers no indication as to which custom(s) should be rejected. This is precisely Hayek's point – there is no Archimedean point from which to assess the desirability of a given custom. Novelty emerges or custom is made redundant through the dynamicism of social evolution – criticism is necessarily immanent and piecemeal (Hayek, 1976, pp. 24–25, 1978, p. 53, 1967, p. 73) and new rules cannot be arbitrarily laid down (Hayek, 1978, p. 11). As Gick & Gick (2001, p. 153) say “progress and tradition are interlinked.” To obey no tradition is profoundly incoherent (Hayek, 1988, p. 61). Elsewhere Posner (2005) questions how Hayek's rejection of being labeled a conservative squares with his veneration of custom. Posner, as do many, fails to appreciate the inherent fluidity of ideological interpretive categories. Hayek himself (1976, p. 151, 1979, pp. 136–137) marks this problem, at least as it exists in the US context.

16. Hayek's individualism has been subject to ongoing caricature; a typical example being Simon Blackburn's *The Oxford Dictionary of Philosophy* (cited by Steele, 2002). Hayek as “extreme atomist” can be found in Galeotti (1987).

17. Hayek is a compatabilist (Whitman, 2004; McCann, 2002; Gamble, 2006; Butos & Koppl, 2007) – the term not used here as it relates to the free will debate. Of course in social philosophy there is a tension between individualism and traditionalism – the former is corrosive of the latter. As Lewis and Chamlee-Wright (2008) point out, it's not immediately obvious that Austrian atomism is compatible with social embeddedness. But this tension is not to be found in the Hayek – contrary to what Gray (1980) and Smith (1997) claim. Hayek scholars not troubled by a compatabilist reading include McQuade (2007, pp. 67–68), Gick (2010), Nooteboom (2007) and Whitman (2004).

18. Variouslly termed as a spontaneous order, self-generating order, or self-organizing structures (Hayek, 1978, p. 74, 1979, p. xii). In a biological context, these notions have been collectively designated a new term: “autopoiesis” (Maturana & Varela, 1980). Some not obviously familiar with Hayek, have drawn attention to “invisible hand” explanations in social emergence and the simple firing of individual neurons that together accomplish functions unknown to individual neurons (Sun, 2006, pp. 15–16).

19. Jaegher and Ezequiel (2007) sketch out enactive account of social cognition, though for them social cognition is face-to-face encounters.

20. Aside from the occasional article in an analytically orientated epistemology journal (e.g., Sunstein, 2006a), social epistemology has yet to discover Hayek. Elsewhere, Sunstein (2006b, 2008), again in social epistemology mode, examines the phenomenon of the blogosphere through Hayekian eyes.

21. Variations on this theme can be found in (Hayek, 1976, p. 8, 1978, pp. 71, 72, 1979, p. 130).

22. Apparently, Popper was of the view that *The Sensory Order* was, in effect, a causal theory of mind, a stance that for him was untenable (Kresge's introduction Hayek, 1994, pp. 28–29). Hayek responded by saying that he was in full accord with

Popper that point-instant explanations were beyond explanation (assuming that was Popper's intention). If, however, Popper denied that the *general* principle that mental phenomena are subject to physical processes, that's where they would sharply diverge. Popper, it should be noted, went onto to defend a version of dualism whereby consciousness is an emergent property (Popper & Eccles, 1977). Recall, Hayek accepted a *practical* dualism, not a metaphysical dualism: a causal analysis is compatible with a nonmaterialist view of mind.

23. Indeed, it has been pointed out by Runde (2001, p. 20) that Searle and Hayek share a very similar theory of social ontology – see Searle (1995). Someone in mainstream philosophy, who never spoke of Hayek in hushed terms, was Nozick (1974, 1981, 1993, 1997, 2001), though I don't know if he ever read *The Sensory Order*.

24. New forms of stigmergy have been exponentially expanded through the affordances of digital technology (Marsh & Onof, 2008; Sunstein 2006b; Marsh, 2011b).

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HAYEK AND THE EVOLUTIONARY TRADITION AGAINST THE *HOMO OECONOMICUS*

Lorenzo Infantino

ABSTRACT

Purpose – To show the existence of two different lines of thought in economics: the Homo oeconomicus tradition and the evolutionary tradition.

Methodology/approach – Following Hayek, the author adopts the individualistic methodology. This allows to separate the Homo oeconomicus approach, which is a hyper-rationalistic construction concerned with the intentional results of human action, from the evolutionary approach, which is concerned with the unintended consequences of human conduct.

Findings – The Homo oeconomicus tradition incurs the methodological mistake of psychologism, a theory of human nature and a human psychology as they exist prior to society. And yet the nature of individual man itself must be placed within a social context and be explained. As the evolutionary tradition and Hayek suggest, the formation of the Ego and the development of the human mind moves over a range of intersubjective relations.

Research limitations/implications – Homo oeconomicus tries to maximize the result of human conduct. However, the concept of maximization neglects the fact that the exchange occurs as soon as a

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positive-sum game sets in; this is very different from maximization, which does not take into account the 'compensations' that the subject can achieve by means of the other dimensions of human action.

Originality/value of paper – *To speak of 'classical economists', placing evolutionary scholars and strictly utilitarian ones under the same denomination is just as misleading as using the expression 'neoclassical economists' in referring to the evolutionary Menger and utilitarian Jevons and Walras.*

THE SENSORY ORDER IS THE PRODUCT OF AN UNCONSCIOUS PROCESS

The intellectual activity of Friedrich A. Hayek was extraordinarily long. Indeed, almost 60 years elapsed between the publication of his first book, *Geldtheorie und Konjunkturtheorie* (1929), and his last work, *The Fatal Conceit* (1988). During this time, starting from his initial studies in economics, Hayek covered all of the social sciences, never subordinating the methodological aspects and significant issues of theoretical psychology relating to the formation of the Ego and the development of the human mind.

The question therefore arises: Is there a logical thread that runs through Hayek's overall research? The fact is that, whatever areas he explored, Hayek constantly reflected on the conditions that make individual choice possible (or impossible). And the same issues of theoretical psychology addressed in *The Sensory Order* are an integral part of a framework that is always aimed at clarifying the conditions that allow for freedom of choice.

As is common knowledge, *The Sensory Order* represents the fulfilment of a project that Hayek had begun to work on more than 25 years ago (Hayek, 1994, pp. 125–6). His research on the methodology of social sciences then impelled him to complete that project. And, through this, he gave to his extensive theoretical design a psychology different from the associationist one, on the basis of which economics had mostly operated and often continues to operate.

In order to grasp the problem directly, it is useful to begin with some statements that summarize Hayek's thesis. He writes:

The process of experience [...] does not begin with sensations or perceptions, but necessarily precedes them: it operates on physiological events and arranges them into a structure or order which becomes the basis of their 'mental' significance; and the

distinction between the sensory qualities, in terms of which alone the conscious mind can learn about anything in the external world, is the result of such pre-sensory experience. [...] A certain part at least of what we know at any moment about the external world is therefore not learnt by sensory experience, but is rather implicit in the means through which we can obtain such experience; it is determined by the order of the apparatus of classification which has been built up by pre-sensory linkages, [so that] what we experience consciously as qualitative attributes of the external events is determined by relations of which we are not consciously aware but which are implicit in these qualitative distinctions, in the sense that they affect all that we do in response to these experiences (Hayek, 1952, pp. 166–167).

In epistemological terms, it is tantamount to saying that ‘the ideal of science as merely a complete description of phenomena, which is the positivist conclusion derived from the phenomenalist approach, [—] proves to be impossible’ (Ibid., p. 174). In other words, ‘every sensation, even the ‘purest’, must therefore be regarded as an interpretation of an event in the light of the past experience of the individual or the species’ (Ibid., p. 166). Hence the consequence that ‘John Locke’s famous fundamental maxim of empiricism that *nihil est in intellectu quod non antea fuerit in sensu* is [—] not correct if meant to refer to conscious sense experience’ (Ibid., p. 167).¹

This leads to the general conclusion that it is necessary to submit to critical scrutiny all the theories of knowledge (including those in the Kantian tradition) that rest on the notion of a *pre-formed* ‘Ego’, prior to ‘experience or free from (the contamination of) experience’ (Popper, 1977, p. 111). Indeed, such theories all incur a methodological mistake which goes by the name of *psychologism*, understood here as a theory that operates according to ‘the idea of a human nature and a human psychology as they existed prior to society’ (Popper, 1966, p. 93). And yet, as Popper underscored, psychologism

can hardly be seriously discussed, for we have every reason to believe that man or rather his ancestor was social prior to being human (if considering, for example, that language presupposes society). But this implies that social institutions, and with them, typical social regularities or sociological laws, must have existed prior to what some people are pleased to call ‘human nature’, and to human psychology (Ibid.).

As we shall see, Hayek’s opposition to associationist psychology and to behaviourism is an extension of his criticism of the idea that social institutions can be consciously created and shaped at will.² Drawing all the possible consequences from his evolutionary approach, Hayek clarifies that the Ego is formed through the relationship with the Other and by means of a pre-sensory experience, which the individual will never be aware of. What is perfectly in keeping with Freud, whose following words Hayek himself

quotes: ‘mental processes in and of themselves are unconscious and the conscious are merely isolated acts and passages in the total life of the mind’ (Hayek, 1952, p. 25, note 1).

AGAINST ASSOCIATIONIST PSYCHOLOGY

The most widespread notion of political economy is a product of associationist psychology. John Stuart Mill contributed most to shaping such a notion. Even after the blatant success of the evolutionary theory, he remained faithful to the old psychological theory, for he did not renounce its political and cultural premises.³ It is important however to understand that evolutionism shakes the very foundations of associationism. It is no accident that, precisely regarding Mill, Bertrand Russell wrote:

It is rather surprising that Mill was so little influenced by Darwin and the theory of evolution. This is the more curious as he frequently quotes Herbert Spencer. He seems to have accepted the Darwinian theory but without realizing its implications. In the chapter on ‘Classification’ in his *Logic*, he speaks of ‘natural kinds’ in an entirely pre-Darwinian fashion, and even suggests that the recognized species of animals and plants are *infimae species* in the scholastic sense, although Darwin’s book on the *Origin of Species* proves this view to be untenable. It was natural that the first edition of his *Logic*, which appeared in 1843, should take no account of the theory of evolution, but it is odd that no modification were made in later editions.... I do not think that he ever imaginatively conceived of man as one among animals or escaped from the eighteenth-century belief that man is fundamentally rational (Russell, 1956, p. 118).

Mill believed that ‘human beings in society have no properties but those which are derived from, and may resolve into, the laws of the nature of individual man’ (Mill, 1892, p. 531). Therefore, nothing is generated by human interaction and represents a social product. Mill transposed this idea to political economy, saying simply that he only wanted to address one aspect of human conduct: economics ‘does not treat of the whole of man’s nature as modified by the social state, nor of the whole conduct of man in society. It is concerned with him solely as a being who desires to possess wealth [...]. It predicts only such of the phenomena of the social state as they take place in consequence of the pursuit of wealth’ (Mill, 2007, p. 111; Mill, 1892, p. 546). Consequently, political economy is the science that moves from ‘the supposition that man is a being who is determined, by necessity of his nature, to prefer a greater portion of wealth to a smaller in all cases’ (Mill, 2007, p. 112; Mill, 1892, p. 546).

All this is the result of the associationist premises which Mill's analysis proceeds from. In other words, since associationist psychology operates on the basis of a pre-formed Ego, there is nothing that cannot be traced to 'the nature of individual man'. But this does not help us to clarify the terms of the issues. In fact, the 'nature of individual man' itself must be placed within a social context and be explained. It is therefore necessary to focus on some inconsistencies generated by Mill's approach.

- A. *Economics and desire of wealth.* According to Mill, in the individual there is a 'desire of wealth'. And yet, placing such a desire at the origin of the entire range of economic phenomena would mean giving them improper (and consequently fragile) foundations. It is like saying that, if the 'the pursuit of wealth' fails, the economic dimension of action fails. One can therefore understand why in his *Principles of Political Economy* Mill rejected Richard Whately's suggestion to attribute to economic theory the name 'catallactics', in the sense of 'science of exchanges', and went so far as to conceive of *distribution* achieved without exchange (Mill, 1965, vol. 3, book 3, chapter 1).
- B. *Closed system.* By placing Mill's 'desire of wealth' at the basis of economic phenomena, one supports the idea that the economic goal is the ultimate end of some human actions. What leads to the distinction between economic actions and non-economic actions and to the birth of the *Homo oeconomicus* is the subject of a theory that takes the form of a *closed system*: because the actor is disconnected from all social dimensions.⁴
- C. *Maximization, preferences and expectations.* Being driven only by the 'desire of wealth' (discounting all other aspects regarding intersubjective relationships and the *preferences* and *expectations* that such relationships generate), the *Homo oeconomicus* tries to *maximize* the result of his conduct. And yet, if the real objective of each actor consists in maximization, the number of exchanges concluded is reduced significantly and the social bonds of the actors slacken considerably. The concept of maximization neglects the fact that exchange occurs as soon as a positive-sum game sets in (Infantino, 1998, pp. 34–36); this is very different from maximization, which does not take into account the 'compensations' that the subject can achieve by means of the other dimensions of human action.
- D. *Geometry and social sciences.* In his *System of Logic*, Mill polemicized against the representatives of what he called the 'School of geometrical politicians' (Hobbes and Bentham above all) (Mill, 1892, p. 540),

inasmuch as it was unable to see the complexity of social phenomena or to realize of that there is not just one social phenomenon on which ‘innumerable forces do not exercise influence; which does not depend upon a conjunction of causes’ (Ibid., p. 537). Mill also stated that some scholars, ‘from an insufficient consideration of the specific nature of the subject matter, and often because (their own education having stopped short in too early a stage) geometry steads in their minds as a type of all deductive sciences; it is to geometry, rather than to astronomy and natural philosophy, that they unconsciously assimilate the deductive science of society’ (Ibid., p. 536).

Particularly with reference to Bentham, Mill also criticized that ‘general theory’ based ‘upon one comprehensive premise, namely, that men’s actions are always determined by their interests’ because if such an expression is to suggest that ‘men’s actions are always determined by their wishes’, it ‘would not bear out any of the consequences which’ Bentham and other philosophers ‘drew from it’ (Mill, 1892, p. 538).

Despite his clear distancing from the ‘School of geometrical politicians’, in the *Definition of Political Economy* Mill (2007, p. 116) did not hesitate to write that economic theory ‘is built upon hypotheses strictly analogous to those which, under the name of definitions, are the foundation of the other abstract sciences. Geometry presupposes an arbitrary definition of a line, ‘that which has length but not breadth’. Just in the same manner does *Political Economy* presuppose an arbitrary definition of man, as a being who invariably does that by which he may obtain the greatest amount of necessities. ... The conclusions of *Political Economy*, like those of geometry, are only true, as the common phrase is, *in the abstract*’.

As one can see, the problems that Mill’s approach left for the scholars of social sciences are numerous. Obviously, one is not frightened by the fact that he resorted to an ‘arbitrary definition of man’. Every definition has its flaws, although Mill believed that his hypothesis was ‘the nearest to the truth’ (Mill, 2007, p. 112). The point is that his model of an actor is unable to capture something that is particularly significant with regard to the conditions of the real world.

THE SCOTTISH TRADITION

In an attempt to sustain his notion of political economy, Mill tried to find support in the ‘teachers’ of the discipline (Mill, 2007, p. 116). The names

mentioned (and that are of particular interest) are Adam Smith and Dugald Stewart. But the approach of these authors is in blatant contrast with Mill's. Let us dwell especially on Smith's positions.

A. *The forming of the Ego.* In *The Theory of Moral Sentiments*, Smith wrote:

Were it possible that a human creature could grow up to manhood in some solitary place, without any communication with his own species, he could no more think of his own character, of the propriety or demerit of his own sentiments or conduct, of the beauty or deformity of his own mind, than the beauty or deformity of his own face. All these are objects which he cannot easily see, which naturally he does not look at, and with regard to which he is provided with no mirror which can present them to his view. Bring him into society and he is immediately provided with the mirror which he wanted before (Smith, 1976a, p. 110).

This means that in Smith (1976a) the Ego is formed in the midst of other people and through pre-sensory experience.

- B. *Preferences and expectations.* It follows that our preferences and our expectations arise from a range of intersubjective relations. In a condition of isolation, the human mind has no way of developing. It makes no sense to speak of individual preferences if one disregards the individual's social relations. This was amply explained by Smith who also showed, by means of the theory of 'sympathy', that social life takes place within a weave of *expectations* (Infantino, 1998, pp. 9–40).
- C. *Scarcity and dispersal of knowledge.* The individual does not act driven by the 'desire of wealth'. What moves him to action is scarcity (Smith, 1976c, pp. 336–337; Hume, 1930, vol. 2, pp. 199–200). Social action, that is cooperation with others, is generated by man's existential condition, which is one of scarcity, insufficiency. And the dearth we suffer also applies to knowledge, which is highly dispersed in society. That is why Smith wrote:

What is the species of domestick industry which his capital can employ... [E]very individual, it is evident, can, in his local situation, judge better than any statesman or lawgiver can do for him. The statesman, who should attempt to direct private people in what manner they ought to employ their capitals, would not only load himself with a most unnecessary attention, but assume an authority which could safely be trusted, not only to no single person, but to no council or senate whatever, and which would nowhere be so dangerous as in hands of a man who had folly and presumption enough to fancy himself fit to exercise it (Smith, 1976b, vol. 1, p. 456).

Every action therefore has an economic dimension, broadly speaking, involving the procurement of means whereby the individual tries to achieve his ultimate goals. And these are not economic. Only the means are economic.

- D. *Against the maximization of means.* It follows that action cannot be aimed at maximizing means, for this may jeopardize the attainment of one's ultimate goals, that is, the life model everyone tries to fulfil. Smith had dwelled on this issue. And he had stated: 'Pecuniary wages and profit, indeed, are every-where in Europe extremely different according to the different employments of labour and stock. But this difference arises partly from certain circumstances in the employments themselves, which, either really, or the least in the imagination of men, make up for a small pecuniary gain in some, and counter-balance a great one in others' (Ibid., p. 116). And one of the circumstances that, according to Smith, can differentiate wages is honour: 'Honour makes a great part of the reward of all honourable professions. In point of pecuniary gain, all things considered, they are generally under-recompensed, as I shall endeavour to show below. Disgrace had the contrary effect. The trade of a butcher is a brutal and odious business; but it is in most places more profitable than the greater part of common trades. The most detestable of all employments, that of public executioner, is in proportion to the quantity of work done, better paid than any common whatever' (Ibid., p. 117). Therefore, the maximization of pecuniary gains can conflict with the pursuit of ultimate ends.

Mill's positions cannot be assimilated with Smith's. Nor they can be assimilated with those of Dugald Stewart, whose *Elements of the Philosophy of Human Mind* Mill referred to (Mill, 2007, p. 100). It is true Stewart had written that 'it might be possible, by devising a set of arbitrary definitions, to form a science which, although conversant about moral, political, or physical ideas, should yet be as certain as geometry' (Stewart, 1877, p. 361). But this was said with reference exclusively to the consistency that had to exist, also in economic theory, between premises and conclusions. Indeed, it did not escape Stewart that in social sciences there is more of 'a connexion between certain suppositions and certain consequences' (Ibid.), so that 'our reasonings [...] in mathematics are directed to an object essentially different from what we have in view in any other employment of our intellectual faculties' (Ibid.). Adam Smith was aware of this (Smith, 1976a, pp. 313–4). And so was Burke. Together with the latter, Stewart repeated that 'a definition may be very exact, and yet go but a very little way towards informing us of the nature of the thing defined' (Stewart, 1877, p. 364). And he added, still following Burke, that a definition 'ought rather follow than to precede our inquiries, of which it ought to be considered as the result' (Ibid.).

The point is that Mill's attempt to bind with Stewart and Smith has no justification because we are faced with two completely different approaches. Mill's moves within a hyper-rationalistic tradition, while the Scottish moralists gave life to evolutionary theory.⁵ This means that when we speak of 'classical economics' (and, as we shall see, also when we speak of 'neoclassical economics'), we make a serious methodological mistake. In fact, we call by the same name two different ways of dealing with economic and social phenomena. The utilitarian tradition strictly speaking pays attention to single acts. And, as Stewart wrote, 'the frequent appeal to utility as the standard of action tends to introduce an uncertainty with respect to the conduct of other men' (Stewart, 1859, p. 184), since an action to which one attributes less immediate utility is constantly replaced by others to which greater advantage is assigned. And this overturns the system of expectations, undermines the very process of social interaction, which becomes unpredictable.⁶ The utility of actions allows for 'exceptions to the most important rules'; it sanctions too kindly, when the end is somehow attractive, 'the use of doubtful means'; and makes room 'to great a latitude for discretion and policy' (Ibid., p. 184). Political power thus rises to the rank of independent variable and social order is conceived as an intentional fulfilment of that power (Infantino, 1998).

In the approach adopted by Smith and by the Scottish moralists, the rules of morality and justice 'are neither divinely ordained, nor an integral part of original human nature, nor revealed by pure reason. They are an outcome of the practical experience of mankind, and the sole consideration in the slow test of time is the utility each rule can demonstrate towards promoting human welfare' (Bay, 1958, p. 33). The evolutionary tradition privileges the utility of these rules and relies on the *social process*. The order achieved is not the result of a specific will; rather, it is the outcome of a larger cooperation among social actors, which is not planned and cannot be planned by any individual. Just as the individual cannot plan the development of his own mind, he cannot pre-determine the outcome of the social process. There is here a general theory of the unintentional consequences of intentional human actions. And Smith's own 'invisible hand' is merely an application of such a theory. Human action is divided into two parts: what we do to fulfil our projects and what we do to obtain the cooperation of others. Each individual is driven by the need to pursue his goals. And yet, since every actor needs the cooperation of others, he must provide them with the services they request in exchange for it. Each individual is clearly interested in his own ends; in order to achieve them, he must however cooperate with the counterparts. The wellbeing of others is thus fostered, albeit

unintentionally.⁷ Public prosperity is therefore the unplanned result of the actions each individual performs to reach his goals by means of free cooperation.

If what is social is not the intentional product of a pre-formed ego but rather the outcome of a process of interaction, the focus of social sciences must necessarily be the unintentional consequences of intentional human actions.

MENGER AND HAYEK: THE REBIRTH OF METHODOLOGICAL INDIVIDUALISM

The work of Carl Menger, founder of the Austrian school of economics, follows the evolutionary tradition. And yet, although his research focused primarily on the study of the unintentional outcomes of human actions, he made the mistake of including Smith among ‘one-sided’ rationalists. Indeed, he wrote that ‘what characterizes the theories of Adam Smith and his followers is the one-sided rationalistic liberalism, the not infrequently impetuous effort to do away with what exists and not always sufficiently understood with the just as impetuous urge to create something new in the realm of political institutions – often enough without sufficient knowledge and experience’ (Menger, 1996, p. 158). However, since on the same page Menger goes on to praise Edmund Burke for seeing in institutions that have come into being unintentionally as ‘a vast realm of fruitful activity’ (Ibid.), his judgement lacks awareness of the influence wielded by Hume and Smith on Burke’s work (Hayek, 1979, p. 68; Infantino, 2003, p. 173, note 64), an influence also proven by the fact that Burke was often described by Smith as ‘the only person he knew, whose thinking on economic subjects was exactly the same as his own, even without any previous agreement’ (Dunn, 1941, pp. 330–346).

More importantly for the question at hand, what matters is that in Menger the actor, far from being a ‘lightning calculator’, is ‘a bumbling, erring, ill informed creature, plagued with uncertainty, forever hovering between alluring hopes and haunting fears, and congenitally incapable of making finely calibrated decisions in pursuit of satisfactions’ (Jaffé, 1976, p. 521). Since he lives in conditions of scarcity, Menger’s subject is surely an economizing being, but the notion of maximization is foreign to him. ‘Human life is a process’ (Menger, 1994, p. 154) in which man acts with a high degree of ignorance.⁸ This prevents him from foreseeing all the consequences of his own actions. And not only that: to pursue his objectives,

Ego must obtain the collaboration of *Alter*, in other words, to some extent it must place itself at service of *Alter*. The aggregation of actions is an inexhaustible source of unintentional consequences, which often become rules and institutions.

It is therefore not surprising that Menger, after attempting 'every effort to win Walras over to his side' described the relationship between their respective theoretical stands as follows: 'A conformity does not exist between us. There is an analogy of concepts in a few points but not in the decisive questions' (Kauder, 1965, p. 100; see also Antonelli, 1953, pp. 269–287). Therefore, to speak of 'classical economists', placing evolutionary scholars and strictly utilitarian ones under the same denomination is just as misleading as using the expression 'neoclassical economists' in referring to the evolutionary Menger and utilitarian Jevons and Walras.⁹

In the light of all this, one can understand why Hayek's objection to Mill's *Homo oeconomicus* is stronger than the few references to that figure would seem to suggest. In *Economics and Knowledge*, when focusing his criticism on the theory of general economic equilibrium, Hayek states: 'The assumption of a perfect market [...] means nothing less than all members of the community, even if they are not supposed to be omniscient, are at least supposed to know automatically all that is relevant for their decisions. It seems that a skeleton is in our cupboard, the 'economic man', whom we have exorcised with prayer and fasting, has returned through the back door in the form of a quasi-omniscient individual' (Hayek, 1949, pp. 45–46).

Then, in a page of *The Constitution of Liberty*, Hayek adds: 'The *homo oeconomicus* was explicitly introduced, with much else that belongs to the rationalist rather to the evolutionary tradition, only by the younger Mill' (Hayek, 1960, p. 61).

And yet such judgements are but the tip of the *iceberg* of a methodological difference that is present throughout Hayek's work. It is useful to dwell on some theoretical 'materials' Hayek disposed of when he completed his work on the formation of the sensory order.

- A. *The dispersal of knowledge*. It is the idea that, in its most vigorous form, was expressed by Smith on the same page where he speaks of the 'invisible hand'. In *Economics and Knowledge*, considered by Hayek his 'most original contribution [...] to the theory of economics' (Hayek, 1994, p. 79), he writes: 'Clearly there is [...] a problem of the *division of knowledge* which is quite analogous to, and at least as important as, the problem of the division of labour. But, while the latter has been one of the main subjects of investigation ever since the beginning of our science,

the former has been as completely neglected, although it seems to me to be the really central problem of economics as a social science' (Hayek, 1949, p. 50). And we know that starting from such a statement, the dispersal of knowledge became a cornerstone of Hayek's work.

- B. *Social science is not pure logic f choice*. In this regard, Hayek states:

The sociologists used to emphasize that the usual procedure of economic theory involved the assumption of particular ideal types, while the economic theorist pointed out that his reasoning was of such generality that he need not make use of 'ideal types'. The truth seems to be that within the field of the Pure Logic of Choice, in which the economist was largely interested, he was right in his assertion but that, as soon as he wanted to use it for the explanation of a social process, he had to use 'ideal types' of one sort or another (Ibid., p. 47, note 12).

The social process is very different from situations where the pure logic of choice is applied. In social science it is necessary to introduce normative elements, which are the institutional conditions in which the actor operates. And these are the aspects that Hayek increasingly dealt with.

- C. *Against maximization*. Since it was clear to Hayek that action is economic exclusively with reference to means (Hayek, 1944, pp. 91–92), he understood well that the purpose of human action is different in nature, and it cannot consist in the maximization of earnings.¹⁰
- D. *Unintentional consequences and individualistic methodology*. In his most highly methodological work, *The Counter-Revolution of Science*, Hayek states: 'For the social sciences the types of conscious actions are data and all they have to do with regard to these data is to arrange in such orderly fashion that they can effectively be used for their task' because 'the problems which they try to answer arise only insofar as the conscious action of many men produce undesigned results, insofar as regularities are observed which are not the result of anybody's design' (Hayek, 1979, pp. 68–69). In other words: 'It is only insofar as some sort of order arises as a result of individual action but without being designed by any individual that a problem is raised which demands a theoretical explanation' (Ibid., p. 69).

Social sciences must therefore study the unintentional consequences of intentional human actions. In fact, 'if social phenomena showed no order except insofar as they were consciously designed, there would indeed be no room for theoretical sciences of society' (Ibid.). That is why Hayek considered the Keynes's statement that '*in the long term we will all be dead*', not only 'a serious and dangerous intellectual error' but also 'a betrayal of

the main duty of the economist' (Hayek, 1941, p. 409). In other words, Keynes refused to consider unintentional consequences. And, according to Hayek, those words by Keynes have no depth. They cover up the attempt to escape problems. Keynes' failure to consider medium- and long-term outcomes accounts for his hostility towards savings and capital. In other words, the future is denied insofar as the functions of savings and capital as denied. Resorting to a coup de théâtre, he easily represented a scene where, in the long term, we are all dead. And yet, if we step out of stage pretence, the situation is very different because only some will be dead, and the others will have to deal with the worse outcomes of simplistic policies that violate all principles of responsibility.

In addition to what was pointed out in the above paragraphs, Hayek needed one last argument to complete methodologically his critique of the rationalistic *hubris*. His work on the formation of the sensory order is precisely what was missing. His analysis is highly significant because it sheds light on the theoretical elements which Hayek had already put up for discussion. No doubt, if the reflections on the formation of the Ego are placed at the basis of Hayek's work, the latter acquires greater transparency: it clearly shows the logical thread that keeps it together.

We know that Schumpeter defined 'methodological individualism' as the method that departs from the actions of men to explain, by means of unintentional consequence, social processes (Schumpeter, 1908, p. 90). This is the approach that, in a hand-written note, Carl Menger had already named the 'compositive method' (Hayek, 1979, p. 67, note 4). Bearing in mind everything that Hayek says about the sensory order, it is perfectly understandable that at the basis of methodological individualism is the idea that the human mind cannot plan its growth. As Hayek himself puts it:

The conception of man deliberately building his civilization stems from an erroneous intellectualism that regards human reason as something outside nature and possessed of knowledge and reasoning capacity independent of experience. But the growth of human mind is part of the growth of civilization; it is the state of civilization at a given moment that determines the scope and the possibilities of human ends and values. The mind can never foresee it own in advance. Though we must always strive for achievement of our present aims, we must also live room for new experience and future events to decide which of these aims will be achieved (Hayek, 1960, p. 24).¹¹

This shows once again how great is the distance between psychologism and the individualistic methodology.

The conclusions achieved by Robbins (1932), in his *Essay on the Nature and Significance of Economic Science*, are very far from Hayek. As Kirzner

(1992, p. 126) correctly states, ‘despite [...] all its credentials, the subjectivism which Robbins brought from Vienna was only the static aspect of the theory’. And not only that the subsequent development of Austrian economics ‘could hardly have foreseen by an Austrian economist (or any other economist ...) during the 1920s, and was not at all an obvious implication of the articulated perspective shared by Austrian economists of the period’ (Kirzner, 1995, vol. 2, p. X).

THE PRIMACY OF THEORY IN THE CONSTRUCTION OF SCIENCE

One of the ‘tangles’ left by John Stuart Mill is the fact that in his work there ‘coexist’ the permanent fidelity to associationist psychology and the adoption, in social sciences, of the hypothetical-deductive method. In his *Autobiography*, Mill wrote that his *System of Logic* provides ‘what was much wanted, a textbook [...] which derives all knowledge from experience, and all moral and intellectual qualities from the direction given to the associations’ since ‘the notion that truths external to the mind may be known by intuition or consciousness, independently of observation and experience is [...] in these times the great intellectual support of false doctrines and bad institutions’ (Mill, 1971, p. 134). And yet, in the *System of Logic*, Mill did not hesitate to state that the science of society is a ‘deductive science’ (Mill, 1892, p. 536). It is therefore necessary to add that, in the *Definition of Political Economy*, political economy is considered as a science that ‘reasons, and [...] must necessarily reason, from assumptions, not from facts’ (Mill, 2007, p. 116).

What is the point? There is a contradiction in Mill’s texts. But there is an even deeper contradiction at the theoretical level. If indeed associationism is taken as an assumption, one is already (or one must be) inductivist; however, if the hypothetical-deductive method is adopted, it is necessary to abandon associationist psychology.

What Hayek says about the formation of the sensory order helps us to understand that our observations contain elements that we are not aware of. In extreme terms, we can say that, when it has not yet become a mind, the brain can make no observation; and when it has become a mind, its observations are never devoid of assumptions. In such a way the Baconian myth of ‘pure observation’ fails and cannot underpin inductivism.

Adam Smith, who had worked on a highly commendable theory of the formation of the Ego, was sceptical towards statistics, economic statistics in

particular. As is common knowledge, in the *Wealth of Nations* he had stated that he had ‘no great faith in political arithmetick’ (Smith, 1976b, vol. 1, p. 534). His theory of knowledge unequivocally asserts the primacy of theory in the construction of science. In the essay on the *History of Astronomy*, strong emphasis is placed on the role played by imagination. In this regard, Smith wrote: ‘The supposition of a chain of intermediate, though invisible events, which succeed each other [...] and which link together those disjointed appearances, is the only means by which the imagination can fill up this interval, is the only bridge which, if one may say so, can smooth its passage from the one object to the other’ (Smith, 1976d, p. 42). Quoting Tycho-Brache, Smith then stated that Copernicus had ‘moved the Earth from its foundations’ (Ibid., p. 75). And he added that the Copernican endeavour best proves ‘how easily the learned give up the evidence of their senses to preserve the coherence of the ideas of their imagination’: ‘the most violent paradox in all philosophy was adopted by many ingenious astronomers, notwithstanding its inconsistency with every system of physics then known in the world’ (Ibid., p. 77). As for Newton’s work, Smith underlined the role played by imagination and by the hypotheses adopted (Ibid., pp.103–105).

Commenting on the ‘reservations’ expressed by Smith towards ‘political arithmetick’, Dugald Stewart wrote:

I agree with him so far as to think that little, if any, regard is due to a particular phenomenon, when stated as an abjection to a conclusion resting on the general laws which regulate the course of human affairs. Even admitting the phenomenon in question to have been accurately observed, and faithfully described, it is yet possible that we may be imperfectly acquainted with that combination of circumstances whereby the effect is modified; and that, if these circumstances were fully before us, this apparent exception would turn out an additional illustration of the very truth which it was brought to invalidate (Stewart, 1877, p. 523).

Stewart further stated: ‘If these observations be just, instead of appealing to political arithmetic as a check on the conclusion of political economy, it would often be more reasonable to have recourse to political economy as a check on the extravagancies of political arithmetic’ (Ibid.).

Stewart dedicated a chapter of his *Elements of Philosophy of Human Mind* to imagination, which he considered to be ‘the principal source of human improvement’ (Ibid., p. 279). He did not hesitate to sustain that ‘without theory [...] experience is a blind and useless guide’ (Ibid., p. 521). He went on to say that ‘the indiscriminate zeal against hypotheses, so generally avowed at present by the professed followers of Bacon, has been much encouraged by the strong and decided terms in which, on various occasions,

they are reprobated by Newton' (Ibid., p. 498); and yet 'the language of this great man, when he happens to touch upon logical questions, must not always be too literally interpreted. It must be qualified and limited, so as to accord with the exemplifications which he himself has given of his general rules' (Ibid.). Hence the question: 'And did not even the great doctrine of gravitation take its first rise from a fortunate conjecture' (Ibid., p. 245)?

Stewart therefore upholds the primacy of the method of hypotheses or of theory in the construction of science. And this allowed him to understand that Smith's research is precisely a 'theoretical or conjectural history' (Stewart, 1793, p. 293).

As one can see, the theoretical positions of Smith and Stewart, referred to in the above paragraphs, amply corroborate the work conducted by Hayek. Beneath the surface of the most widespread textbook concepts, there is a deep rift that keeps Mill's tradition and evolutionary tradition distant from each other. Even when they seem to converge, they are kept apart by a deep subsoil: *duo, si idem dicunt, non est idem* (if two languages say the same thing, they are not the same thing). With his work, Hayek achieved relevant objectives in the study of man and society. By insisting on the difference between those traditions of research, he was able to reach one of his most fruitful insights, which makes it possible to emerge from the closed system generated by utilitarianism strictly speaking and to understand what is the task and what are the tools of social sciences.¹² And this, as was stated at the beginning of this paper, helps us to identify the conditions which make freedom of choice possible (or impossible).

NOTES

1. Locke's maxim asserts that all we know is a product of our conscious sense experience.

2. Associationism and behaviourism are outcomes of the phenomenalist approach. They are characterized by 'the attitude which [...] we shall call the *objectivism* of the scientific approach to the study of man and society' (Hayek, 1979, p. 78). This attitude has found its most peculiar expression 'in the various attempts to dispense with our subjective knowledge of the working of the human mind, attempts which in various forms have affected almost all branches of social study' (Ibid.). 'They all take it naively for granted that what appears alike to us will also appear alike to other people. Though they have no business to do so, they make constant use of the classification of external stimuli by our sense and our mind as alike or unlike, a classification which we know only from our personal experience of it and which is not based on any objective tests showing that these facts also behave similarly in relation to each other' (Ibid., p. 79).

3. This point will be clarified in the paragraph dedicated to the 'Scottish tradition'. For an in-depth treatment of Mill's political-ideological premises, see Infantino (2008).

4. Durkheim (1895) realized this at once, although he made the mistake of identifying all of political economy with Mill's theory. This caused serious misunderstandings between sociologists and economists. See Infantino (1998, pp. 64–67).

5. Referring to Burke and Savigny, Pollock (1908, p. 42) spoke of 'Darwinians before Darwin'. But the influence of Hume and Smith on Burke is known. This will be discussed in the next section.

6. Ibid. It is worth recalling that Smith was strongly criticized by Bentham for the theory of 'sympathy', which is a tool that places *in fieri* the weave of preferences and expectations (Infantino, 2008, pp. 89–90).

7. As McQuade (2007, p. 62) underscores, 'an individual may deliberately set a particular price, but no individual plans the emergence of the spectrum of market prices that relate different goods and reflect overall appraisals of desirability and scarcity'.

8. That is what Menger taught to Crown Prince Rudolph. See Streissler (1990, pp. 120–121).

9. Mises (1978, pp. 56–57) attributed to Menger the responsibility of not having 'replaced John Stuart Mill's unsatisfactory delimitation of the field of economics with a more satisfactory one'. Actually, Menger's work is, also methodologically, very distant from Mill's. What the founder of the Austrian school did not explain completely is that actions are economic with reference to means and not to ends, which is what Mises then did (1981a, pp. 107–109; 1981b, pp. 156–157).

10. Years later, Hayek (1982, vol. 3, pp. 68–69) would say that the competitive allocation of resources is like 'experimentation in science, first and foremost a discovery procedure [...]. It therefore cannot be said of competition any more than of any other sort of experimentation that it leads to a maximization of any measurable results. It merely leads, under favourable conditions, to use of more skill and knowledge than any other known procedure'.

11. As McQuade (2007, p. 56) uses in a different context, we can say that the human mind and civilization 'are related fraternally rather than filially'.

12. In contrast with the German historical school of economics, which supported a sort of 'romantic empiricism', Menger too upheld the primacy of theory in the construction of science. The representatives of that school denied the possibility of theoretical social sciences and took refuge in a desperate inductivism. Their real objective was to prevent the development of a theoretical science, which could place politics under control, ousting it from the position of an independent variable. This topic is not entirely pertinent here. For a complete treatment of the role played by Menger in that dispute, see Infantino (1998, pp. 100–114). It is also necessary to add that Mises prolonged Menger's critique of the German historical school of economics. He was well aware of the primacy of theory in the construction of science. He went so far as to say in *Socialism* (1981a, p. 28) that 'a theory is already contained in the very linguistic terms involved in every act of thought'. Mises also stated (Ibid., p. 30) that 'a theory that does not appear contradicted by experience is by no means to be regarded as conclusively established'. And yet, over the years, he abandoned this 'fallibilism' and reached a 'geometrical' conception of economics (1998, p. 38), in contrast with his same idea of *Homo agens*, who lives in ignorance and steers his action towards an unknown future (Ibid., p. 105).

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THE SENSORY ORDER AND THE NEUROPHYSIOLOGICAL BASICS OF METHODOLOGICAL INDIVIDUALISM

Francesco Di Iorio

ABSTRACT

Purpose – To show that The Sensory Order is an original effort to support, on a neurophysiologic basis, methodological individualism.

Methodology/approach – Considering that the mind is a complex and self-organized order, Hayek criticizes methodological holism according to which the cause of action has to be sought outside the individual, in macro-laws governing social wholes. He argues that, due to the nature of the mind, the cause of action has to be sought inside the individual.

Findings – The paper stresses that scholars have more or less neglected a very important point in discussions of the Austrian author's psychology. Hayek's psychology supports the idea that the explanation of the action stems from the understanding of its meaning.

Research limitations/implications – The article only discusses some of the epistemological consequences of Hayek's theory of the mind. For instance, it does not analyze in a detailed way the relationship between

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this theory and the idea of distributed knowledge. It left an in-depth examination of this issue for subsequent research.

Originality/value of paper – *Many authors state that Hayek's version of methodological individualism only examines the non-intentional effects of action, neglecting the importance of Verstehen. They argue that the Austrian scholar is not a complete and coherent champion of methodological individualism. The paper shows that this criticism is unfounded.*

INTRODUCTION

The present article analyzes the relationship between the cognitive psychology proposed by Hayek in his book *The Sensory Order*, as well as other works, and his individualistic theory of human action.

There are now a relatively high number of studies on Hayek's theory of the mind. Of course, many of them are very interesting and analyze acutely some of its implications for a number of different fields such as the cognitive sciences, evolutionary epistemology, economics, and political and social theory (see Caldwell, 2007). However, it seems to me that, apart from occasional exceptions (Butos & Koppl, 2006; Caldwell, 2007), scholars have more or less neglected a very important point. The following analysis will try to show that Hayek's cognitive psychology can be considered as an effort – indisputably very original – to support, on a neurophysiologic basis, the two twin ideas claimed by methodological individualism: the indeterminism of action and of the necessity of an interpretative approach (*Verstehen*) to the social sciences.

First and foremost, the epistemological implications of Hayek's theory of mind need to be carefully taken into account due to its undeniable scientific relevance. As highlighted by prestigious neurobiologists and cognitive scientists such as the Nobel laureate Gerald Edelman (1982, 1987), Joaquin Fuster (1995), or Barry Smith (1997), Hayek needs to be regarded, together with Donald Hebb, as the father of one of the nowadays most quoted approaches on the studies of the mind: the “connectionist” paradigm.¹ Such a paradigm is based upon the idea of the self-organization of complex systems which is central to the framework of the overall epistemological and scientific reflection of Hayek (see Dupuy, 1990; Heritier, 1997; Nemo, 1988).

To understand why Hayek's psychology is a defense of methodological individualism one has to take into account that according to it cognitive processes are not conceivable in deterministic terms. For the opponents of

the individualistic approach – the methodological holists – the contrary is true. They consider the action as a mechanical and deterministic effect – an effect produced by factors exogenous to the individual, such as the culture of a given social whole or holistic laws of social change. For Hayek, given that the mind is a complex and self-organized order, it is impossible to know and forecast in a complete and detailed way the formation of thoughts. Consequently, the cause of action cannot be sought outside the individual. Moreover, any kind of deterministic approach has to be ruled out (Hayek, 1952b, pp. 192–193; see also Dreyfus & Dreyfus, 2000; Dupuy, 2000; Petitot, Varela, Pachoud, & Roy, 1999; Varela, 1989). Hayek’s cognitive psychology legitimizes the idea, supported by Weber and other methodological individualists, that the explanation of the action goes through the understanding of its *meaning*. For Hayek, the cause of action has to be sought inside the individual. It is in his evaluations and intentional choices which, due to the complexity of mind, have to be regarded as the “ultimate givens” of social sciences (see Lachmann, 2007).

Hayek’s connectionism can be considered as an attempt to transform the individualistic theory of the indeterminism of action from a mere epistemological principle to what Popper (1979) calls an “objective knowledge,” viz. a knowledge which can be defined as true for either logical or empirical reasons. As we shall see, Hayek argues that the holistic theory of action is not only heuristically unfruitful, but also scientifically inadmissible.

CATEGORIZATION AND INTERPRETATION

Why knowledge is not the mirror of reality and is based on acts of classification

According to Hayek, the mind is an apparatus of interpretation. For him, there is no perfect correspondence between the sensorial world and the external world: “Every sensation ... must ... be regarded as an interpretation of an event in the light of the past experience of the individual or the species” (1952b, p. 166). Perceptions are thus interpretations that depend on memory: both on the biological memory (the way natural selection has shaped the nervous system and the receptive organs of stimuli) and the personal memory (what the individual has learned over his/her life). It follows that sensory qualities, that is, the blue of the sky or the green of the grass, cannot be regarded as objective and ultimate properties of reality, but

as mental constructions. Hayek maintains that the mind reproduces only conjecturally, partially and approximately the external world.

As Butos and McQuade (2005, p. 335) emphasize, Hayek considers the mind an “*adaptive classifying system* – a system which adapts to its environment by forming an internal maintained model of that environment.” The structure of this system “has the potential for maintaining and continuously updating a model of the environment as currently experienced – a model that is capable of anticipating expected changes in the environment” (p. 335; see also McQuade, 2007, p. 57). This system interprets continuously the external reality through “acts of classification” of stimuli (Hayek, 1952b, p. 78). As we will see in detail later on, it is based on a principle of self-organization of complexity. It is bound “to generate knowledge” about its environment which it uses in order to adapt itself (Butos & McQuade, 2005, p. 336).

An adaptive classifying system is able to tie typical sets of stimuli to typical meanings. Hayek expressed this concept in terms of the “Primacy of Abstract” (1978, p. 35): the possibility to acknowledge that a given object is, for instance, a “car” depends on tracing a certain typical set of stimuli back to an abstract class “cars” that he refers to as *pattern recognition* or *rules of perception* (p. 41). It follows that he totally opposes the inductivist and observativist theory of knowledge. The perception of a tangible detail always presupposes interpretative schemes, a sort of theoretical knowledge selecting and interpreting the external reality, linked to past experience:² “In the mind the abstract can exist without the concrete, but not the concrete without the abstract” (p. 37). Also, like Popper,³ Hayek thus maintains that “*all* we know about the world is of the nature of theories and all ‘experience’ can do is to change these theories” (1952b, p. 143).

Hence Hayek holds that abstraction, the tendency to order phenomena in typical classes, is not a purely rational and conscious ability. It is first and foremost a tacit or meta-conscious capability, a property of categories through which the mind operates. As Hayek (1948, p. 44) states: “the richness of the sensory world in which we live ...is not the starting point from which the mind derives abstractions,” but “the product” of meta-conscious abstractions (1978, p. 44).

The classification processes generating perceptions are not procedures of simple classification; they are, instead, modes of “multiple classification” (Hayek, 1952b, p. 50). This implies that, in spite of what is assumed by behaviorism, perceptions never concern a single stimulus, but always groups of stimuli or events. Within the framework of these processes, “at any moment a given event may be treated as a member of more than one class,

each of these classes containing also different other events; and a given event may also on different occasions be assigned to different classes according to the accompanying events with which it occurs” (p. 50). Moreover, the mind also carries out a “a third type of multiple classification: namely one in which successive acts of classification follow upon each other in relays, or on different ‘levels’; in this type the distinct responses which effect the grouping at a first level become in turn subject to a further classification (which also may be multiple in both the former senses)” (p. 51).

The detailed analysis of the functioning of this logic of multiple classification does not constitute part of the scope of the current work. It is sufficient to specify that this particular classifying activity constantly rectifies the interpretations to which it leads according to the continuous flow of new experiences. Moreover, it allows the experience of a large abundance of sensory qualities. This is in accordance with the fact that it allows the overlapping of several abstract schemes of meanings. It is precisely due to the logic of multiple classification, on which the mind is based, that by observing a certain object we can simultaneously gather a large number of particular characteristics. We can realize, for instance, that this item is a house, that it is a yellow house, that it is a house in an art nouveau style, that it is a house with three floors, that it is a house with a garden, and so on. In the child or in the animal this capacity in terms of overlapping of abstract schemes is less developed. The sensory world of the child or the animal is simpler “because of the much thinner net of ordering relations which they possess – because the much smaller number of abstract classes under which they can subsume their impressions makes the qualities which their supposedly elementary sensations possess much less rich” (1978, p. 44).

According to Hayek, the sensory order can be described as a relational order “of our sense experiences in which events are classified according to their sensory properties” (Hayek, 1952b, p. 3). It is a relational order in the sense that it is composed by relations of analogy and difference. It is the way through which sensory qualities (and not the objective facts – the facts of the physical world) differ from each other (in terms of dimension, color, weight, etc.). It is also the whole set of meanings of phenomenal reality.⁴

The difference between the sensory order and the physical or objective world is shown by natural sciences. Natural sciences tend to “revise” and “correct” our sensorial categorizations in order to grasp the features of the physical world. While, for instance, we visually perceive water and ice as different things, they classify them as the same thing by considering their chemical structure. The methodology of natural sciences helps us to understand the relativity and the theoretical nature of our sensorial knowledge.

Even if the methodology of natural science can be considered thus as “objectivist,” Hayek stresses that this does not imply that natural sciences analyze a world that is more real, that the physical world is an “abstract” or theoretical construction, and that even this world is built in a selective way from a particular perspective and is linked to our adaptive needs (1952b, p. 143). It is the outcome of an *alternative* classification compared to the mental one that produces new meanings and operates according to experimental theories (pp. 145–146). For Hayek, it is impossible to conceive neutral and purely objective knowledge.

THE SELF-ORGANIZATION OF THE MIND

Why the sensorial interpretations presuppose a “tacit knowledge” and are linked to connectionist processes

Concerning the most complex sensory phenomena, Hayek maintains that his theory of the mind “leads indeed to conclusions very similar to those of the gestalt school” (1952b, p. 77; see also De Vecchi, 2003). However, while the fathers of the *Gestalt* school hold that the unconscious organization, leading to the recognition of analogous forms also in the case of objects not having any identity in terms of physical structure, concerns elementary sensory qualities that are directly communicated to the mind from basic nervous impulses, which is a kind of basic “qualitative” information, Hayek holds a different point of view. He states that the impulses do not incorporate any sensory quality; in other words, the sensory qualities are not linked to attributes of the impulses. He proposes what today is called a “connectionist” approach.

For the Austrian author the mental interpretations depend solely upon the way impulses are channeled through the neural networks via connections among neurons that the impulses are able to activate. For every single *kind* of perception there is a peculiar *kind* of channelization of the impulses. Both the most and less “elementary” qualities are created via these processes. Impulses do not incorporate any basic “qualitative” information. According to Hayek, the theory of an original pure core of sensation is wrong:

[M]y theory maintains that the sensory (or other mental) qualities are not in some way originally attached to, or an original attribute of, the individual physiological impulses, but that all of these qualities are determined by the system of connexion by which the impulses can be transmitted from neuron to neuron; that it is thus the position of the individual impulse or group of impulses in the whole system of such connexions which gives it its distinctive quality” (Hayek, 1952b, p. 53).

Mental classifications thus depend on the activation of several neuronal chains. They are the product of a system effect. This theory is confirmed by the fact that a precise location of perceptions in the brain does not exist (Hayek, 1952b, p. 148).

According to this connectionist paradigm the way neurons work is not controlled by a central unit, but is simply based upon certain “rules” of activation and on certain rules which govern the global coherence of the system. Such rules define the modalities and the conditions for the activation of the neurons. So whether a neuron becomes part of a chain of connections, carrying nervous impulses, depends solely on these rules.⁵ This means that perceptions *emerge* according to a logic of self-organization. There is no central regulation, nor a central plan (see Varela, 1989).

Through the self-organization of the mind’s neural connections, a “map of the relationship between various kinds of events in the external world” is produced, that “will not only be a very imperfect map, but also a map which is subject to continuous although very gradual change” (Hayek, 1952b, p. 110). It will be partly modified by new experiences. This is due to the fact that perception and memory act according to circular causality logic: perception, created by the memory, affects the latter and changes in the memory loop back on perception (see Fuster, 1995). It follows that a learning process is incessantly running, developing under a *trial and error* fashion and consisting in a substitution of classification modalities, inbred or acquired, with new classification modalities; this substitution is based upon a partial restructuring of the neural connections system (see Besnier, 2005). It is thus relatively easy to understand how an individual, after a time lag, can perceive the same fact or object differently.

By adopting a similar perspective, it is necessary to exclude the existence “of elementary and constant sensations as ultimate constituents of the world” and assume “the inconstancy of sensory quality” (Hayek, 1952b, pp. 173–176). For a connectionist such as Hayek, “[t]o acquire the capacity for the new sensory discrimination is not merely to learn to do better what we have done before; it means new doing altogether. It means not merely to discriminate better or more efficiently between two stimuli or groups of stimuli: it means discriminating between stimuli which before was not discriminated at all” (p. 156). For instance, it makes no sense to state “that, if a chemist learns to distinguish between two smells which nobody has ever distinguished before, he has learnt to distinguish between given qualities: these qualities just did not exist before he learnt to distinguish between them” (p. 156). As “qualities,” they can exist only in our mind; even if their

existence is linked to the features of the external world, they cannot be considered as “given data.”

Mental *maps* are thus partly modified by experience. Given that and taking into account that the biological evolutionary logic makes a perfect correspondence of the individuals’ anatomic structure rather impracticable, a complete identity of human minds is in turn impossible. Human minds will in fact be sufficiently similar to allow the mutual comprehension and interaction among individuals, but “they will not be identical” (Hayek, 1952b, p. 110).

Having analyzed the modalities behind the elaboration of perceptions, Hayek can emphasize the relationship between perception and action. In his judgment, rules of perceptions have to be regarded as closely related to rules of action: the classification of *typical meanings* has to be conceived as functional to an adaptative effort consisting in the implementation of *typical and appropriate answers*. A trivial example is represented by the consequences of a driver’s perception of red traffic lights. Such acknowledgment of a typical meaning allows the driver to develop an adaptive answer that is in turn typical: that is to stop when the traffic lights are red. The abstract rules of perception allow singling out typical problematic situations. They incorporate, in the meanings they create, information about the rules of action that are useful to face them. These action rules can also be defined as “abstract” because they can also be applied to abstract classes of events (Hayek, 1978, pp. 35 ff.; see also Nemo, 1988, pp. 51–53).

Hayek’s analysis leads to a rather important conclusion: know-how is the pillar of consciousness and of the rational thought. There is a tacit dimension of knowledge (see also Polanyi, 1958; Dreyfus & Dreyfus, 2000). Such dimensions will allow us to use certain skills without being able to verbally explain what exactly entails the capability to do them. It embraces not only the meta-conscious competence to apply rules for the elaboration of the sensory world but also the capacity to fully master some *practical* skills (e.g., swimming, riding a bike, and painting). These *practical* skills, not dissimilar from those behind perception, cannot be verbally described (it is impossible, for instance, to illustrate in a manual the guidelines to be followed in order to acquire capabilities to ride a bike without losing balance). They also partly depend on instinct (the biological memory) and partly on learning (the personal memory). Therefore, Hayek’s connectionism lets us comprehend that the mind is not limited solely to conscious logic capabilities. Consciousness is the *tip* of an iceberg: “what we consciously experience is ... the result ... of processes of which we cannot be conscious” (Hayek, 1978, p. 45). The *tacit* dimension of knowledge produces the

necessary assumptions for intentional choice. It defines the basic framework of the process of the conscious decision. Namely, it selects a basic set of meanings and possible alternatives for action which is the foundation of rational reasoning (Hayek, 1952b, pp. 132 ff.).

THE IMPOSSIBILITY OF A FULL EXPLANATION OF COGNITIVE PROCESSES

Why mental processes of interpretation are nondeterministic and why we have to renounce the pretension to know them in a detailed way

Hayek specifies that the theory on the presuppositions of conscience that he proposes represents a mere “explanation of principle” (1952b, p. 182). It only takes into account a general logic of an extremely complex phenomenon. Hayek excludes the possibility for the mind to eventually arrive at a comprehensive self-explanation. This is due to the fact that he denies the possibility for an *explanation of detail* of its running that would permit “to substitute a description in physical terms for a description in terms of mental qualities” (1952b, p. 189; see also Nadeau, 2001). According to the Austrian author, this possibility needs to be discarded for three reasons.

First of all he considers that, given that the mind emerges from the activity of billions of neurons able to interconnect with one another according to a virtually unlimited number of combinations, it belongs to the “complex phenomena” category (1967, p. 55). It is impossible to identify all the interdependent variables contributing to determining such a complex order as the mind is. It is not feasible to master the whole of its physical causes. In such order “the number of variables which would have to be taken into account is greater than those that can be ascertained or effectively manipulated” (Hayek, 1952b, p. 185). Moreover, the chance to only resort to an *explanation of principle* presents severe drawbacks in terms of forecasting. In fact, only for the “simple phenomena,” caused by a limited number of variables, it is possible “to predict particular events” (p. 185). It follows that, “we shall never, by means of the same brain, be able to arrive at a detailed explanation of its working in particular circumstances, or be able to predict what the results of its operations will be” (p. 189).

The second reason is, instead, strictly linked to the adaptive logic of a self-organized order. This kind of order is not characterized by a fully knowable and foreseeable behavior also because, employing a concept widely used by Maturana and Varela (1980), it is endowed with “autonomy.”

As increasingly clarified by connectionist cognitive scientists, the behavior of a self-organized order is neither determined by a program introduced from the outside and followed mechanically (behavior of a self-organized order is not comparable to the one of a machine that has been previously programmed, as a computer), nor by the effects applied by the surrounding environment. Basically an order of this kind does not passively undergo such effects (its behavior has nothing to do, for instance, with the one of the pool balls that is, on the opposite, entirely determined by the external forces acting on it). The cause behind the behavior of a self-organized order is not to be sought outside it (Hayek, 1952b, pp. 122–127). An order like the mind actively employs aleatory novelties that appears in the environment in a continuous and unpredictable manner to constantly self-reprogram itself. Its logic is to safeguard its autonomy from the outer environment and thus its capability to adapt. An order of this kind is not predetermined as it is not possible to foresee the aleatory novelties affecting either its behavior or the way they will impact on the outcomes of its processes of self-organization and upholding of autonomy. These processes are acts of pure creation (see Atlan, 1979, pp. 157 ff.). By virtue of its working modalities, a self-organized order is the “cause of itself” (see Dupuy, 1990).

The third and last reason taken into account by Hayek is of a logical kind. He explicitly speaks of a Goedelian limit. As we have seen, he regards the mind as an apparatus for classification. According to him, no classification can be completely explained. For Hayek, we cannot fully explain our basic interpretative categories because ultimately we rely upon these to create meaning in the first place. In other words, these basic categories are prior to any meaning and so have no place in the order of meanings they create: “There is ... on every level, or in every universe of discourse, a part of our knowledge which, although it is the result of experience, cannot be controlled by experience, because it constitutes the ordering principle of that universe by which we distinguish the different kinds of objects of which it consists and to which our statements refer” (Hayek, 1952b, pp. 169–170; see also Hayek, 1967, pp. 60–63). To get around this problem, we should place ourselves outside our own mind. A classifying apparatus of a higher complexity compared to the one of the human mind would in fact be required – an apparatus that, moreover, would in turn be, for the same logical reasons, unable to exhaustively explain its functioning (Hayek, 1952b, pp. 184–190; see also Nemo, 1988, pp. 60–61). Also due to this motive, we are faced, according to Hayek, with the impossibility to replace an explanation based on mental skills with one characterized by mere physical terms.⁶

FROM THE INDETERMINISM OF COGNITIVE PROCESS TO THE NECESSITY OF *VERSTEHEN*

Why the impossibility to fully explain cognitive process brings into discussion the pretension to explain individual intentions as being a mechanical effect of the context

Hayek calls for *practical dualism*: “in some ultimate sense mental phenomena are ‘nothing but’ physical processes”; “this, however, does not alter the fact that in discussing mental process we will never be able to dispense with the use of mental terms, and that we shall have permanently to be content with a practical dualism” (1952b, p. 191). Being that the mind is an emergent, complex, self-organized system and being that it is impossible to detail its working, it is not feasible to seek the causes of thought in the physical characteristics of the external environment and in the physical processes induced by the former in the nervous system. It stems “that we shall never achieve a complete ‘unification’ of all sciences in the sense that all phenomena of which it treats can be described in physical terms” (p. 191). Given that an *explanation of principle* of the sensory order is the only feasible route, it follows that it is highly impracticable to reduce social sciences to physics.

Considering that mind is based on *autonomy*, as intended by Maturana and Varela, and being impossible to fully comprehend the way it functions, the idea stating that cognitive processes can be conceived in deterministic terms needs also to be ruled out. The prospect according to which action is nothing less than a mechanical product of the context, shared by behaviorism as much as by methodological holism, is not compatible with Hayekian connectionism. By reason of the mind’s complexity and the fact that such order is the “cause of itself,” the “data” for the explanation of the action cannot be external to the individual. This conclusion, Hayek stresses, “is, of course, of the greatest importance for all the disciplines which aim at an understanding and interpretation of human action” (Hayek, 1952b, p. 193). The adoption of the collectivist paradigm and the search for the action’s causes outside the individual is, according to Hayek, the result of a *Hybris*. In other words, it is a kind of scientific and over simplifying conceit, neglecting logical and epistemological limits, that is necessary to come to terms with in accounting for the outcomes of mental operations.

Hayek regards the explanation for the action as necessarily resulting in a reconstruction of the ideas motivating the individuals: he supports that the “data” of social sciences are internal to the actors. For him, the holist

tendency to consider that the “sense” the action has for individuals as being irrelevant, is something illegitimate and misleading: “Unless we can understand what the acting people mean by their actions, any attempt to explain them ... is bound to fail” (Hayek, 1952a, p. 53). The possibility to apply such hermeneutical and subjectivist procedures calls for existence of something common to the researcher and social actors: it needs, therefore, an invariant element in spite of the variability of beliefs and knowledge. Hayek states that such an element does exist by virtue of a genetic predisposition, constituted by the knowledge a priori of the mind’s logical structure and of its perceiving basic categories (1952b, pp. 99–103; see also Butos & Koppl, 2006).

Holism has originally developed in close correlation with naïve realism, one of the mistakes of the positivist vision of science. This helps understanding the reason why such a paradigm has come to consider the mind as a deterministic mechanism instead of an apparatus for interpretation, taking the action’s causes as objective or external rather than internal to individual minds. The naïve empiricism denies the dichotomy between sensory and physical order. Hayek demonstrates, instead, that precisely due to this dichotomy and the nonreducibility of the mental to the physical, it is necessary to rule out the determinism of action and support a “subjectivist method” (see also Butos & Koppl, 2006). By the light of its “practical dualism,” it becomes clear that it is not possible to set aside an explanation of the action in terms of universal a priori and effects of perceptive interpretations: “In the study of human action ... our starting point will always have to be our direct knowledge of the different kinds of mental events, which to us must remain irreducible entities” (Hayek, 1952b, p. 91).

The approach supported by Hayek thus concerns a “*verstehende* psychology.” According to this approach, the powers of human sciences are bounded. The scientist has no possibility but to “use our direct (introspective) knowledge of mental events in order to ‘understand’ ... the results to which mental processes will lead in certain conditions.” However, “such a *verstehende* psychology, which starts from our given knowledge of mental process, will ... never be able to explain,” Hayek writes, “why we must think thus and not otherwise, why we arrive at particular conclusions.” Moreover, “assertion that we can explain our own knowledge involves also the belief that we can at any one moment of time ... act on some knowledge” due to the fact that it is associated to the idea that it is possible to acquire “some additional knowledge about how the former is conditioned and determined” (Hayek, 1952b, p. 192). This is a further reason behind the close association between methodological holism and constructivist rationalism.

According to Hayek, the error of conceiving in deterministic terms cognitive processes is plainly illustrated in Karl Mannheim's approach: "In particular, it would appear that the whole aim of the discipline known under the name of 'sociology of knowledge' which aims at explaining why people as a result of particular material circumstances hold particular views at particular moments, is fundamentally misconceived" (Hayek, 1952b, p. 192). In opposition to Mannheim, Hayek reaffirms that, given the sensory order's characteristics, the causes of actions cannot be traced back to the influences coming from the external context: "To us," Hayek writes, "human decision must always appear as the result of the whole of a human personality – that means the whole of a person's mind – which, as we have seen, we cannot reduce to something else" (pp. 192–193).

Therefore *The Sensory Order* represents the attempt to transform methodological individualism from a mere epistemological principle to an approach centered around a scientific theory of the mind, able to adequately account for the logic behind the creation of sensory perceptions; a theory that legitimizes the indeterminism of human action and matches well with the *interpretative* method of the Weberian tradition.⁷ Both methodological holists and Hayek maintain that consciousness is not the only important aspect of human nature. But, while holists assume the action as being determined at an unconscious level, Hayek denies this possibility. Consider, for instance, Marx's holist theory of false consciousness which is linked to the idea that collective beliefs are unconsciously determined by the economic structure. For Hayek this theory is wrong because of the different way he considers the presuppositions of consciousness. According to him, the *tacit* and *meta-conscious* dimension that is at the basis of knowledge is characterized by processes which are nondeterministic. Their indeterminism rules out the possibility to use Marx's holist methodology and legitimates methodological individualism.

It is worth noting how, by developing a connectionist approach similar to the Hayekian one, cognitive scientists such as Petitot and Varela have recently confirmed the unfeasibility in explaining the action without taking account of the sense that the action carries for the individual. These authors define the connectionist approach as a "phenomenological" approach and question another paradigm of cognitive sciences: the logical-symbolic paradigm. The latter compares the mind to a computer running a program and reducing it to only logical skills. It does not deem the mind as an interpretation apparatus, based on the self-organization logic, and does not take into account the issue relative to the sense's explanation or the role played by tacit capabilities in perceptions and action.⁸ The experimental

research connectionists have implemented over the past few years have reasserted Hayek's intuitions on cognitive limitations connected to the study of the mind and, indirectly, also the epistemological consequences derived by him (e.g., see Dreyfus & Dreyfus, 2000; Dupuy, 2000; Fuster, 1995; Gallagher & Zahavi, 2008; Petitot et al., 1999; Varela, 1989; Winograd & Flores, 1987).

Underpinning the empirical nature of his theory of perception, Hayek points out that it is confirmed by the neurophysiologic knowledge which was available at the time he wrote (1952b, pp. 147 ff.).

SENSORY ORDER AND PRAXEOLOGICAL APPROACH

Why the possibility to understand the others in spite of cultural differences depends on a common a priori knowledge

Even though Hayek is not explicit on this point, it seems to me that the conclusions which he attains in *The Sensory Order* are largely compatible with Mises' praxeological theory of action. As it is well known, Hayek does not fully embrace Mises' apriorism. Unlike Mises, he stresses that market theory is not a priori, but empirical. According to Hayek then, what is a priori is "only the logic of human action" (1994, p. 72).

As we already pointed out, Hayek's cognitive psychology supports an argument against "polylogism," the doctrine that different cultures have different logic (see also Butois & Koppl, 2006). He maintains that biological evolution selected universal categories of human action. This idea is shared by Mises: "The concepts of natural selection and evolution make it possible to develop a hypothesis about the emergence of the logical structure of the human mind and the a priori" (2002, p. 14). However, neither Hayek nor Mises thinks that the empirical research on the evolution of these categories or the attempt to explain them in physical terms can be valid points of departure for social science. Both agree that a theory of biological evolution can only be useful to explain why these categories exist. However, both stress that the fact that we can know them does not depend on experience. These categories are a posteriori for the species, but a priori for individuals. "When we speak of a man", Hayek states, "we necessarily imply the presence of certain familiar categories" (1952a, p. 139). Mind is not "an object which we observe as we observe physical fact." Whenever "we speak of mind, we interpret what we observe in terms of categories in which our own mind operates" (p. 136).

In *The Sensory Order*, Hayek arrives precisely at the conclusion that the ultimate foundations of social science lie in our direct knowledge of the functioning of the human mind. He rules out the possibility of substituting an explanation of action in terms of direct knowledge of the categories of mind for an explanation of action in physical terms (Hayek, 1952b, pp. 165 ff.).

Like Hayek, Mises strongly criticizes monism and reductionism: “There is no explanation,” he states, “in terms of the natural sciences of what causes hosts of people to remain faithful to the religious creed in which they were brought up and others to change their faith, why people join or desert political parties, why there are different schools of philosophy and different opinions concerning a multiplicity of problems” (2002, p. 121). In particular, Mises maintains that contrary to what positivists and scientists state, in order to explain the motivations of actors we cannot avoid the necessity of using a set of categories of action – intentionality, rationality, economic evaluation, and causality – which are tautological or analytical (1998, p. 12). They are implied in the concept of action: “There is no action in which the praxeological categories do not appear fully and perfectly” (pp. 39–40). These categories “are not derived from experience. They are, like those of logic and mathematics, a priori” (p. 32). According to Mises, praxeological categories are fundamental presuppositions of common sense as well as of science: “Without them we should not be able to see in the course of events anything else than kaleidoscopic change and chaotic muddle” (*ibid.*).

A call for the interpretative approach and methodological individualism is also a call for universal categories of action. In order for us to understand others with their different knowledge and culture via methodological individualism, we need a common element between us and them. For Hayek, the use of praxeological and universal categories is legitimate because of the logical and empirical reasons which determine the complexity of mind and the impossibility of explaining action as a pure effect of internal and external physical conditions. Consequently, his theory of mind has to be considered to some extent as a defense of the praxeological point of view.

THE NOMOLOGICAL-DEDUCTIVE EXPLANATION OF ACTION

Why indeterminism of action and causal explanation are not incompatible

Very often theoreticians of methodological individualism criticize the nomological-deductive model of explanation. In other words, they maintain

that the social sciences, as distinct from the natural sciences, are not based on the use of covering laws. They do so in order to defend the indeterminism of human action. Consider, for instance, such authors as Collingwood, Croce, and Dilthey (see Di Nuoscio, 2006, pp. 129 ff.). In my opinion, Hayek, in spite of some ambiguities,⁹ does not follow this line. It seems to me that his position is largely compatible with the nomological-deductive approach as developed by Popper and Hempel.

In *The Sensory Order* he writes that the logic behind knowledge acquisition is universal and that science, like the mind, works through a classification procedure: it explains phenomena by grouping them in typical classes – thus in a nomological fashion. In *The Counter-Revolution of Science*, Hayek explicitly states that trying to explain individual actions implies “subsum[ing] them under rules which connect similar situations with similar actions” (1952a, p. 53).

As clarified by Weber (1975), the *interpretative* method is not in contrast with the nomological-deductive paradigm. Contrary to what is supported by the antinomological individualists, such a paradigm does not inevitably require the use of deterministic laws. It is also compatible with probabilistic laws and hence with the idea that action is nondeterministic. As clearly illustrated by Popper (2000, 2001) and Hempel (1966), the social scientist cannot refrain from using this kind of law in reconstructing the logic of the situation, in other words, the *reasons* of the actor (see also Albert & Antiseri, 2001; Antiseri, 1999; Di Nuoscio, 2004).

Following Mises, Hayek maintains that the only absolute general laws we can apply in the explanation of human action are the praxeological ones. These laws are the basis of understanding and they are compatible with the indeterminism of action because they don’t allow forecasting the specific features of behavior. However, Hayek, as well as Mises, is aware of the fact that these laws are not sufficient to explain action. In my opinion, it is possible to interpret what Hayek calls “our direct knowledge of the different kinds of mental events” (1952b, p. 191) – the knowledge he considers necessary to understand the action – as composed also of nontautological laws, viz., what Popper and Hempel consider as empirical covering laws. Such empirical laws are only probabilistic and we usually use them in an unconscious way because they are, like the praxeological categories, rooted in common sense (see Di Nuoscio, 2004, 2006).

Consider, for instance, this historical explanation: After the Romans won against the Latins, they treated them less cruelly than the others they had vanquished. This was because they knew they would soon have to fight the Etruscans, Samnites, and Gauls, and they would need the help of the Latins

to conquer these peoples (see De Sanctis, 1960, p. 245). This explanation is nomological not only because it is founded on praxeological categories. It is also nomological because it is founded on a trivial psychological law rooted in common sense: "If you want the help of somebody you need to be kind to him." Without this law it would be impossible to find the cause of the Romans' attitude toward the Latins. Categorization and explanation would be impossible. Of course, like all empirical laws which concern human action, this law is only probabilistic because if I want the help of somebody and I am kind with him, it is uncertain that he will give me his help. There is no determinism.

Consider another example: "Carl struck John because John insulted him." This explanation is also implicitly founded on a commonsense-covering law which is only tendential. Sometimes people who are insulted do not strike the person that insults them. However, also in this case, without using this tendential law we would be unable to have any explanation because we could not link the reaction of Carl with what John said. The categorization would be impossible and we could not give a meaning to John's act.

According to Popper (1992), the use of probabilistic laws implies that the empirical control in explanation of action concerns exclusively what he calls the reconstruction of the "situational logic," viz., the reconstruction of the initial conditions of the explanation. Of course, if we know that a law is only tendential, we also know that it is empirically false because there are cases which are contrary to the law. So it makes no sense to try to test it. Consequently, Popper states that the logic of explanation of action is similar to the logic of explanation in applied physics. It is necessary to take care of initial conditions and to use laws, but there is no test of laws.

However, Popper (1992) maintains, like Hempel, that the reconstruction of the initial conditions requires more care in social sciences than in applied physics (see also Nadeau, 1989). This is precisely because of the fact that the majority of the laws used in social science (not necessarily the empirical laws concerning human action) are tendential or probabilistic laws. When it is impossible to use deterministic laws, the phenomenon we want to explain can be compatible with different causes. Consider, for instance, the explanation given by Thucydides about the reelection of Pericles in spite of the negative trend of the war against the Persians. This fact is compatible with different tendential laws. Athenians could have been obliged to reelect him by the use of force or they could have been deceived. Thucydides rules out these hypotheses precisely by analyzing carefully, as a detective, the initial conditions. He shows that Pericles was beloved by the Athenians

and he was able to convince them that it was right to reelect him (see Di Nuoscio, 2004, p. 232).

Popper also agrees with Hempel of the fact that when we use probabilistic laws, the explanation is less certain because there is not necessarily a logical deduction of the *explanandum* (the phenomenon we want to explain) from the *explanans* (the set of initial conditions and laws which are the foundation of our explanation). However, both Popper and Hempel underline that the use of tendential laws is not exclusive to the social sciences and it cannot be considered as the foundation of a different approach than that of the natural sciences. It is only more frequent in the field of social sciences. Clinical medicine, for instance, shows us examples of probabilistic laws: certain treatments against cancer are valid only in $X\%$ of cases. Examples of tendential laws can be taken also from meteorology and other natural sciences (Di Nuoscio, 2004, p. 212; 2006, p. 41).

It seems to me that the nomological-deductive model of Popper and Hempel combines well with the idea of knowledge by classification which Hayek proposes in *The Sensory Order*. Even though Hayek is unclear on this point, his cognitive psychology can be considered in a sense a defense of a nomological conception of methodological individualism.

Moreover, even the idea of “explanation of principle” which Hayek claims for the analysis of functioning of complex orders like mind or market can be easily shown as based on a nomological-deductive approach. It is an explanation based on laws, even though these laws are not quantitative and cannot allow a detailed forecast – like, for instance, the law, “every time the amount of money grows there is inflation” or “every time the mind perceives something it does a classification” (see Hempel, 1966, pp. 359 ff.).

THE NEUROPHYSIOLOGIC FOUNDATION OF THE SUBJECTIVE THEORY OF VALUE

Why “The Sensory Order”’s thesis legitimates Austrian marginalism and is linked to the distributed knowledge paradigm

Hayek’s theory of the sensory order, in addition to legitimizing the *interpretative* method, is relevant for another reason. As Cubeddu (1995) stresses, it establishes the subjectivism of values from a neurophysiologic viewpoint. Hayek’s cognitive psychology claims that value is not an objective feature of things, but a mental construction. Moreover, it excludes the possibility for a complete identity of the ways individuals build it.

According to Hayek, human minds will be sufficiently similar to allow the mutual comprehension and interaction among individuals, but “they will not be identical” (Hayek, 1952b, p. 110). First, a perfect anatomic correspondence of human minds is impossible; second, an identity of personal experiences which influence the mental processes of self-organization must also be ruled out. Consequently, it is necessary to exclude a perfect matching of the individual assumptions of subjective evaluations.

Also for this reason, the relation between the action and the environmental context is more complex compared to what is maintained by those advocating the *objectivistic* and *collectivistic* approach in social sciences. Moreover, the relation between *The Sensory Order* and the Austrian theory of value allows a better understanding of why Hayek maintains that knowledge is dispersed in society. This does not only depend on the fact that there are particular circumstances of time and place which change in a continuous and unforeseeable way. It also depends on the fact that the individual evaluation of these circumstances is based on tacit presuppositions which are partly personal. One who knows a particular circumstance disposes of a kind of information which is not shared by others because his mind can interpret this circumstance in a way which is partly unique (see Di Iorio, 2006; O’Driscoll & Rizzo, 1996; Rizzello, 1999).

In addition, since the mental presuppositions of evaluation are linked to personal experiences, the continuous change of the particular circumstances of time and place in turn partially modifies the individual presuppositions of evaluation. This fact contributes to making the social process nondeterministic and unforeseeable.

THE ANALOGY BETWEEN THE SELF-ORGANIZATION OF THE MIND AND THAT OF THE MARKET

Why it is possible to argue that Hayek uses the same epistemological model in order to analyze those two different phenomena

As we pointed out, according to Hayek’s connectionist paradigm the way neurons work is not controlled by a central unit, but is simply based upon certain “rules” of activation.¹⁰ It follows that perceptions *emerge* according to the logic of self-organization (see Varela, 1989, pp. 60–61).

Due to the above, some authors have correctly underlined that an analogy does exist between Hayek’s theory of the mind and his conception of the

market. Both these phenomena are for him *spontaneous* and *knowledge-generating* orders (see Butos & McQuade, 2002, 2005; McQuade, 2007; Smith, 1997).

It is possible to schematically summarize the analogies which exist between mind and market according to Hayek:

1. Both are adaptive classifying systems which generate knowledge by a sort of categorization. As Butos and McQuade point out, while the mind creates by classification the meaning of reality in order to allow an adaptation of the behavior, the market generates a classification which concerns the goods and their prices (Butos & McQuade, 2005, p. 345). Giving a specific price to every good, it allows the coordination of the economic activities and the cost minimization. Consequently, it works like a cybernetic system, based on feedback regulation, just as the mind does. While in the mind the transmission of information and the categorization are based on chains of neurons, in the market they are based on chains of prices (see Smith, 1997, p. 113).
2. Both are complex systems based on the logic of self-organization. They do not have a single locus of control, but are polycentric. Their functioning depends only on the existence of general rules that bind the behavior of the parts. It does not depend on a central calculation. Hayek himself underlines this similarity. Commenting on the nature of market society, he states “the brain of an organism which acts as the directing centre for that organism is itself in turn a polycentric order, that is, that its actions are determined by the relation and mutual adjustment to each other of the elements of which it consists” (Hayek, 1967, p. 73).

This fact has important epistemological consequences. As Varela underlines, self-organization is a necessary attribute of complexity (Varela, 1989, pp. 61–77). The performance of complex self-organized orders would be inconceivable as the outcome of central direction. Both the idea to consider the mind as a computer and the idea that it is possible to centralize the direction of economic activities are fallacious for systemic and cognitive reasons. No computer or central planner can use the distributed and tacit knowledge which mind and market can use (see Petitot, 2002).

3. As self-organized orders, both adapt themselves to continuing and unpredictable changes and both have consequently a dynamical structure. Inside these systems, order always coexists with a certain degree of disorder because they re-adapt themselves continually. In this way, they can preserve their functioning (see Dupuy, 1990). In the case of mind the

re-adaptation that results from new experiences concerns the systems of connections, while in the case of market the re-adaptation concerns the structure of production and division of labor. As Atlan (1979) points out, the time of this kind of system is indeterministic. Their future is not inscribed in their past. It depends on the way in which future changes will influence their capacity of self-organization – it is unforeseeable. This phenomenon has been well analyzed concerning the market by O'Driscoll and Rizzo (1996) who criticize the theory of general equilibrium because of the fact that it is based on a *Newtonian* time, underlining that for understanding the market it is necessary to use a “dynamic conception of time” as intended by Bergson (O'Driscoll & Rizzo, 1996, p. 51).

4. Both are emergent phenomena. Mind emerges from the brain and especially from the activity of neurons in conformity to the rules that regulate their functioning. Market emerges unintentionally from the intentional actions of individuals who follow specific ethical principles and civil laws. As we will analyze further, a fundamental characteristic of the emergence lies in the fact that it creates systems that are endowed with attributes irreducible to the attributes of their basic components. Mind cannot be reduced to the brain. Similarly market cannot be reduced to the intentions of individuals (see Nadeau, 2003).
5. Both are evolutionary orders, the outcome of a long process of trial and error (Smith, 1997). These orders developed in an indeterministic way by adaptation to unforeseeable facts or conditions. This fact is one of the sources of Hayek's criticism of the constructivist mentality. “*Man*” – he states in the epilogue of *Law, Legislation and Liberty* – “*is not and never will be the master of his fate: his very reason always progresses by leading him into the unknown and unforeseen where he learns new things*” (Hayek, 1979, p. 176).

SENSORY ORDER AND SOCIAL CHANGE

Why Hayek's theory of cultural evolution is not incompatible with methodological individualism

Considering action as determined by social factors, the holistic paradigm sees the source of the social change outside of individuals and residing in laws of historical evolution that govern the social collectives as wholes and enforce their control on actors. Methodological individualism considers the holistic theory of social change as untenable because of the existence of two

fundamental sources of indeterminism that avoid the possibility of establishing laws of historical evolution. It stresses, on the one hand, the indeterminism of human action and, on the other hand, the emergence of unintentional, unforeseeable consequences (see Boudon & Bourricaud, 1990; Antiseri & Pellicani, 1995).

Vanberg (1994) and other scholars (Boehm, 1989; Gray, 1986; Hodgson, 1991, 1994) have criticized Hayek's theory of cultural evolution, centered on the idea of group selection, underlining its inconsistency with the pillars of methodological individualism. They maintain that Hayek, by heavily relying on the role of group selection and on action governed by acquired rules, has ended up in deriving a theory of cultural determinism which considers the group not as the effect of individual behavior, but as its cause. Vanberg states that Hayek uses a "*functionalist* type of argument, according to which its contribution to the 'maintenance' of a social system explains the existence of a social pattern or institution, a type of argument has had some popularity in sociology and social anthropology" (Vanberg, 1994, p. 84). This collective-functionalism "explicitly rejects the idea that the relevant social process can be explained ... in terms of *individual actions*" (p. 85). According to Vanberg, "[a]pparently and strangely enough, Hayek appeals to such a *collectivist-functionalist* notion when he stresses that, in cultural evolution, a process of *group selection* is of 'greatest importance'" (p. 85). His "appeal to the collectivist-functionalist notion of a process of cultural evolution that operates at the group level, as such, stands in contrast" Vanberg concludes, "to the explicit methodological individualism that otherwise informs his work" (p. 85).

This kind of criticism does not take into account the fact that Hayek, as already illustrated, considers the mind in a rather different way than the holistic sociologists. He neither regards it as a machine that mechanically implements a program previously acquired via the process of socialization, nor as an order which is passively affected by external factors.

According to Hayek's cognitive psychology, the fact that an individual complies with cultural rules which are shared inside a given social group cannot be explained in a holistic and deterministic way by assuming the group as being the cause of the action. For Hayek, the fact that an individual follows, for instance, ethical rules is always linked to tacit processes of evaluation and interpretation of the advisability of his action. Since the mind is a complex self-organized order, those processes are nondeterministic. This implies that in explaining why an individual follows certain moral customs we have to assume that he accepts them and that he will keep using them as long as he believes that they are an appropriate way

to act or that they can be useful for solving problems. In other words, we have to assume that this individual follows certain ethical rules because they make sense to him and that this fact is the cause of his action (see also Boudon, 1994, 1995, 2001; Antiseri & Pellicani, 1995).

As we mentioned, Hayek also develops another argument against cultural determinism. He considers the mind, being based on the logic of self-organization, as endowed with the capacity to modify, at least partly, the effects of prior learning and thus also the effects of the socialization process. Consequently, he rules out the possibility that the mind is “programmed” by culture as certain Durkheimian sociologists suppose. Its connectionist approach regards the mind as having the capacity to develop, according to new experiences, new perceptive, and adaptive rules in line with the safeguarding of its autonomy (see also Boudon & Bourricaud, 1990, pp. 527–534).

Moreover, not only the acceptance, but also the application of a cultural rule appears in the light of Hayek’s psychology as something more complicated than holists suppose. Also this fact is not characterized by a mechanical nature. For Hayek, applying a certain cultural rule in specific situations relies on interpretive processes of this situation. Moreover, according to him, those interpretative processes can concern the same meaning of cultural rules and the way they can be applied because this meaning can sometimes be ambiguous and contradictory (Hayek, 1973, pp. 55 ff.).

Of course, this does not mean that for Hayek there is no influence of social factors. In particular, for methodological individualists like him, social conditioning also exists and has to be taken into account (a trivial example of social conditioning is given by the fact that a professor entering the class dressed as an ancient roman would make his students laugh at him). But from Hayek’s point of view, the study of social conditioning has to consist in the analysis of the way in which this conditioning is *interpreted* by the actors. In order to understand how it influences them, it is necessary to understand how they perceive it (see also Boudon & Bourricaud, 1990, pp. 412 ff.). According to his psychology, there is no way to assume social conditioning simplistically and mechanically as being the source of action.

Because of the way Hayek conceives the mind, for him the group cannot be the cause of individual behavior, but only its effect. But then, as Di Nuoscio (2000) emphasizes, according to Hayek, groups do not exist ontologically. Due to Hayek’s nominalism and his cognitive psychology, “group selection” must be considered as a selection of systems of rules that are taken up by individuals because of an evaluation. The conceptual part of

this evaluation – the part which is built on the tacit skills of the mind – can be explained in terms of what Boudon calls a *subjective* or *bounded rationality* (Boudon, 1992; Di Iorio, 2009). Following Hayek's epistemology, one has thus to rule out cultural determinism and conclude that the holists overstate the importance of the social context. In other words, one needs to assume that action cannot be explained without understanding the *motivations* of individuals.

It is also necessary to stress that it is wrong to assume that Hayek is not a methodological individualist because he is not a social reductionist, that is, the idea that it is necessary to reduce all societal attributes to individual attributes. The tendency to associate methodological individualism and social reductionism is based on a misunderstanding (see Boudon & Bourricaud, 1990; Di Nuoscio, 2000, 2006). In fact, despite some occasional lexical ambiguities, none of the main exponents of methodological individualism call for social reductionism. In the same way, Hayek, as scholars as Boudon, Merton, Popper, Spencer, or Weber, correctly maintains that it is simply impossible to deny the necessity to use the reference to societal attributes.

On one side, it is evident that a methodological individualist is also obliged to consider the groups as systems, viz., as non-aleatory combinations of elements. He has to suppose them as something more than the simple addition of many individualities (see Boudon & Bourricaud, 1990; Di Nuoscio, 2000, 2006). As Hayek explains, "the existence of those relations which are essential for the existence of the whole cannot be accounted for wholly by the interaction of the parts" (1967, p. 71). On the other hand, social reductionism is incompatible with the concept of the emergence of unintentional consequences. Consequently, it is incompatible with a central concept in the individualistic paradigm. By definition, unintended and aggregate effects cannot be reduced to actors' intentions: they are precisely emergent properties.

Social reductionism makes methodological individualism epistemologically useless and scientifically indefensible. How could we analyze, for instance, the functioning of price signals and the changes they involve in the structure of production without implicitly using nonreductionist concepts as those of a "system," a "self-organized order," and an "emergent property"?

The difference between holism and individualism is not given by the reductionist nature of the latter. It is given rather by the tendency of holism to hypostatize the groups following the naïve empiricist error to mislead the theoretical nature of the social phenomena as well as of the other phenomena (see Di Nuoscio, 2000, 2006). In other words, this difference is

given by the tendency of holism to consider the groups as social data, that is, as concrete entities not explainable theoretically in terms of intentionality and emergence, but only as “given wholes.” As we pointed out, holism supposes those “given wholes” as being governed by macro-laws which determine the actions of the individuals, who are intended to be nothing else but emanations of the group.

A last point needs to be considered. As a foundation of his theory of action, Hayek’s connectionism rules out not only the idea that his theory of social group is linked with cultural determinism, but it also rules out the chance of identifying laws governing social changes. First, it is incompatible with the historicist point of view of authors like Comte, Hegel, and Marx which assumes the existence of laws governing the evolution of mind and, as a consequence of this fact, of society (Hayek, 1952a, pp. 111 ff.). If the mind is a self-organized order, the existence of these laws as well as the possibility of predicting the future evolution of society by forecasting the evolution of mind are inconceivable. Second, Hayek’s connectionism is incompatible even with the most recent holistic theories of social change. These theories refrain from predicting the future evolution of the whole society. They are more modest and only establish laws governing the change of specific parts of it (see Boudon, 1991). These laws affirm a deterministic relation between a certain type of variation in the environmental conditions and the coming of a distinct type of social structure. Consider, for instance, Parsons’ theory on the tendency toward the “nuclearization” of families in industrial societies. Hayek’s approach helps us in understanding why this kind of theory does not work and why it is possible to find opposite historical cases which falsify them. Considering in particular Parson’s law, what indicated that it “is a *non sequitur* is simply the fact that in *certain* societies, as Japan, industrialization has occurred with, rather than against, the extended family and has tended to strengthen it, at least over a long period” (Boudon, 1991, p. 22).

Finally, Hayek’s theory of the mind purports that social change cannot be forecast on the basis of deterministic laws. It argues rather for the idea that it must be analyzed *ex post* via methodological individualism, that is, by the reconstruction of the reasons of the individuals and of the aggregation effects of their actions.

CONCLUSION

What is usually regarded as Hayek’s major epistemological contribution to methodological individualism is the way he links the “invisible hand

explanation” to the ideas of distributed knowledge, self-organization, and cultural evolution. Many scholars share the conviction that Hayek did not really contribute to the development of the individualistic theory of action. Also, an author like Boudon (2005), who does not deny by principle the compatibility between the idea of group selection and that of methodological individualism, expressed dissatisfaction about the way Hayek deals with micro-sociological issues. According to Boudon, Hayek’s version of methodological individualism only “insists on the non-intentional effects of individual action” neglecting the importance of the *Verstehen* (Boudon, 2005, pp. 74–99).

As we have had the opportunity to see, the claims that Hayek did not contribute to the individualistic theory of action and that he is not a complete and coherent champion of methodological individualism have to be rejected. Those mistakes are linked to the fact that the implications of his cognitive psychology have been quite neglected. It is true that Hayek gives, on the whole, more importance to the analysis of the macro-social phenomena than to the study of intentionality. It is also true that he never tried to develop a detailed sociological theory of collective beliefs. However, by using his connectionist theory of the sensory order, he strongly argued the necessity to assume the indeterminism of action and to use the interpretative method in social sciences. In this way, he offered an interesting and highly original thesis against the holistic epistemology of action.

NOTES

1. It has to be stressed that Hayek had worked out the core of the innovative theses in *The Sensory Order* in a short essay of 1920 entitled “Contributions to a Theory of How Consciousness Develops.” In *The Sensory Order* the theories contained in his student paper will be taken up again and enriched, also in line with the contributions from cybernetics and the systems’ theory.

2. There is some obvious analogy between Hayek and Kant. However, an important difference lies in the fact that Hayek’s categories are, unlike Kant’s, fallible and evolutionary (see Agonito, 1975, pp. 62–71; Butois & Koppl, 2006, pp. 30–31).

3. Regarding analogies between Popper’s criticism toward *observationism* and the connectionist approach, refer to Besnier (2005, pp. 75–79). As stressed by Herrmann-Pillath (1992), Hayek upholds an *evolutionary epistemology*.

4. Concerning the idea establishing that the sensory order is a sort of relational order, Hayek has been influenced by Mach (see 2001).

5. Hayek does not illustrate in depth the activation modalities of neurons. He goes as far as to agree with Hebb (1949), who was the first to analyze in detail this issue (see Hayek, 1952b, note 1, p. 64 and note 1, p. 114).

6. Thomas J. McQuade argues an interesting criticism of this logical limitation pointed out by Hayek. “The weakness in this line of argument,” he writes, “is the unstated assumption that the only classificatory system involved here is the human brain. But the classifications of science are not produced by a single human brain; they are the result of the interactions between many human brains in a system whose overall complexity exceeds that of a single brain – a system which, therefore, does not violate the condition set by Hayek’s opening proposition. The classification produced in the mind of an individual scientist, even though it includes feedback influences from his observations of the classification produced by science, is not the same thing as the emergent classification we call ‘scientific knowledge’ and should not be conflated with it” (McQuade, 2007, p. 69).

7. This thesis is criticized by Smith (1997) who states that in Hayek’s psychology there is no “room for planning, for self-control and for the deliberate self-shaping of the conscious subject.” As Butos and Koppl (2006, pp. 33–34) stress, Smith’s position is untenable. Hayek “shows us,” they write, “that any reduction of the mental to the physical can be made only ‘in principle’. We cannot describe thought and action without using words such as ‘plan’ and ‘purpose’ Hayek’s methodological dualism vindicates our use of the language of planning and purpose, which Smith curiously imagines to be inconsistent with Hayek’s theory of mind.”

8. Because of the “phenomenological nature” of connectionism, it may be possible to consider Hayek as a little bit more close to Alfred Schütz than what would appear at first sight. As it is well known, Schutz tried to base his methodological individualism on Husserl’s thought and to develop a phenomenological method for the social sciences (see Schütz, 1967).

9. Hayek states that the concept of law is useless for social science (Hayek, 1967, pp. 3–42). However, this does not allow me to consider Hayek an antinomological individualist because his criticisms are against the idea of determinist laws of social phenomena. It seems to me that his approach is not incompatible with the idea of tentential laws claimed by Weber, Popper, and Hempel.

10. Hayek does not illustrate in depth the activation modalities of neurons. He goes so far as to agree with Hebb (1949), who was the first to analyze this issue in detail (see Hayek, 1952b, note 1, p. 64 and note 1, p. 114). For more details, please see Barthelemy, De Glas, Descles, and Petitot (1996); Petitot et al. (1999) and Petitot (2002).

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THE SENSORY ORDER, THE ECONOMIC IMAGINATION AND THE TACIT DIMENSION

Peter E. Earl

ABSTRACT

Purpose – This paper examines Hayek's view of the mind to see if it provides a useful and unifying foundation for understanding both deliberative choices that involve conscious information processing (the 'economic imagination') and choices that are not determined by conscious processes such as those involving 'gut feelings' or knowledge that the chooser is unable to articulate (the 'tacit dimension').

Methodology/approach – The paper analyses Hayek's view of the mind from the standpoint of evolutionary economics and biology. Because of the significance of pattern detection in Hayek's analysis, the paper examines parallels with key ideas in personal construct psychology and artificial intelligence. As well as exploring the evolutionary advantages of behavior based on programmed responses to the detection of particular patterns, it also explores possible evolutionary and neural origins of dysfunctional heuristics and biases.

Findings – Hayek's theory of the mind provides useful foundations for analyzing choice in a evolving, pluralistic and context-based manner rather than seeing all choices as made in much the same way on the basis

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of 'given preferences' that obey the axioms of rational choice theory. His theory complements work in psychological economics based on Kelly's personal construct psychology, cognitive dissonance theory and Maslow's hierarchy of needs. The analysis leads to questions being raised about the conventional faith in the notion of a diminishing marginal rate of substitution.

Originality/value of paper – *The paper shows how very different ways of choosing can be understood in terms of Hayek's analysis of the mind.*

INTRODUCTION

This paper explores the possibility of using an evolutionary perspective on Hayek's (1952) book *The Sensory Order* as a basis for understanding how people try to cope with the challenges of a complex, changing world. These challenges can be addressed in a variety of ways, and different ways of choosing may be appropriate depending on the challenges that different contexts present. On each occasion for action, decision makers thus face a multilevel problem, the first level of which is to make a 'meta choice', the choice of a means of choosing. Most economists ignore both the context aspect and the 'meta choice' issue and use a one-level, 'one-size-fits-all' approach to choice. Such an approach simplifies analysis and avoids a potential problem of infinite regress: how do people choose how to choose? This question might seem safe to ignore on the basis that people in practice do not normally end up paralysed by the need to choose: their minds work so as to ensure the decision process is truncated even if this means that a lot of effort goes into making decisions come right rather than making the right decisions in the first place. However, perhaps if we actually consider how the mind might be expected to have evolved to work at this 'meta' level of choice to remove the infinite regress problem, we may be able better to understand how people choose at the level at which economists habitually focus.

This paper uses the theory of the mind set out in *The Sensory Order* in such a role. Put very simply, the underlying argument is as follows: (i) Hayek provides us with a theory of how the mind perceives events by classifying them in relation to previous perceptions, (ii) occasions for choice are events in their own right that our minds likewise classify, and (iii) among the ways in which the mind classifies these occasions for choice is with respect to the kinds of decision-making processes that they require. The paper uses this line of thinking to explore how *The Sensory Order* can provide underpinnings for

two broad decision-making processes: conscious deliberation ('the economic imagination', as per the subjectivist/behavioral perspective offered in Earl, 1983) and for decisions for which the chooser would be unable to offer a verbal account that went beyond 'intuition' or 'gut feeling' ('the tacit dimension', as per Polanyi, 1967).

The rest of the paper is divided into the following sections. "A pluralistic perspective on choice" explores further the plurality of strategies for choosing and highlights their contrasting evolutionary properties. "Connections, characteristics and constructs" focuses on how consumers learn what products have to offer and examines the relationship between Hayek's view of the workings of the mind and the analysis of choice in terms of the characteristics possessed by rival products. "Cognition and instinctive behavior" explores how Hayek's thinking can help us understand instinctive behavior and choice in unfamiliar situations, while "Consciousness versus the tacit dimension" focuses on the development of both conscious decision-making capabilities and tacit knowledge. "The fallible decision maker" considers perspectives from *The Sensory Order* and from evolutionary psychology as a basis for understanding the origins of heuristics and biases that can compromise the quality of decisions. "Brands and criteria-based decision rules" considers the complexity entailed in distinguishing between different brands of competing products and addresses the question of what the mind's apparent ability to cope with this implies for its ability to make complex computational trade-offs when choosing. The paper ends with "Concluding comments".

A PLURALISTIC PERSPECTIVE ON CHOICE

Survival in a complex and changing world requires the ability either (i) to *respond* sufficiently rapidly via forms of behavior that deal well enough with the challenges currently at hand or (ii) to *predict* the challenges that may be faced and put into place strategies that are effective for preventing them from eventuating or for dealing with them if they eventuate. Both kinds of behaviors can be seen 'as if' they result from people doing cost/benefit estimations about a variety of possible courses of action, from which it becomes clear which one is the best. This simplifies matters analytically but is potentially misleading if, in some contexts, the actual process of choosing does not involve deliberating about alternatives and instead just entails classifying the situation at hand and instituting some kind of programmed response without assessing opportunity costs.

Programmed responses to challenges may not be the choices that a globally rational decision maker would reach but, as Alchian (1950) pointed out, survival does not require the best possible response, merely one that is good enough as a means of dealing with the threat that is at hand. An appropriate programmed response could thus be perfectly adequate for survival purposes. Nondeliberative, programmed behavior can also be the means an organism uses to construct a more secure environment in the face of uncertainty. For example, an animal may end up storing food in several places because that is what it does instinctively, rather than by sizing up the pros and cons of being able to concentrate on defending a single store (with some probability of losing it outright) versus the risks of having multiple stores and losing one or more of them while others go undetected by predators. Human decision makers may make rather similar decisions about their retirement investment strategies, using simple rules rather than in the light of careful analysis of financial data and economists' forecasts. For example, even a monetary economist as distinguished as Charles Goodhart (2008, p. 7) admits to using a very simple rule, namely, 'Martin Wolfe of [*The Financial Times*] is always right'.

Though rule-based responses to the detection of particular stimuli may go against the mainstream economists' focus on opportunity cost/benefit calculations they may in some contexts increase the chooser's survival potential. As Winter (1964) pointed out, if the environment keeps changing, the longer it takes to work out the best way of responding to an initial change, the less suited that response may be to the environment that the organism faces by the time the response is implemented. Furthermore, deliberation itself consumes time and other resources that could have been used for doing other things. In some cases, delaying action is an efficient strategy (e.g., when a standards battle is going on and there is the risk of an expensive loss if one selects the technology that fails to dominate). However, in other cases delay may constrict the organism's future range of options, either because it produces major negative consequences (e.g., death due to dithering instead of immediately stepping out of the way of a vehicle that one has just noticed heading rapidly in one's direction) or because first-/early-mover advantages are foregone.

Once we recognize the distinctions between reactive and anticipatory/strategic behavior, and between programmed and deliberative behavior, along with the possible efficiencies in some contexts of reactive or nondeliberative choices, there appears to be a case for analysing human action in a pluralistic manner. To force-fit all choices into a single framework of optimization among alternatives subject to various constraints may be misleading if programmed forms of action result in decision makers failing to

respond to market incentives in ways consistent with the predictions of the traditional framework. Instead, we need a framework for understanding the distribution of decision-making methods among different kinds of contexts. Let us now see what Hayek's *The Sensory Order* has much to offer toward the construction of the latter kind of framework.

CONNECTIONS, CHARACTERISTICS AND CONSTRUCTS

Hayek's focus in *The Sensory Order* is not on decision making but on how the mind perceives events, either external phenomena or imagined as sets of qualities. As such, it has the potential to provide cognitive underpinnings for Lancaster's (1966) characteristics-based view of choice. The latter was offered with an emphasis on its ability to provide a way for economists to analyse how consumers cope with the introduction of new commodities. Lancaster did not see consumers as having any need to form new preferences in such contexts so long as their preferences were over characteristics rather than goods, and so long as new goods were nothing more than new combinations of characteristics that were already incorporated into consumer preference orderings. New products might have unprecedented features but the services they perform may be entirely familiar ones, delivered to standards not previously available. However, Lancaster did not attempt to provide a theory of how consumers come to locate new products in characteristics space: he proceeds as if their qualities are objectively given.

Austrian economists, by contrast, would see consumers as subjectively constructing in their imaginations personal perceptions of what products have to offer. Different people may thus reach different conclusions. Consider Fig. 1, which shows an advertisement for Raleigh bicycles that was used in Africa in the early 1950s. Some might construe it (as the designers of the advertisement probably intended) as implying that an all-steel Raleigh bicycle would be much more dependable than other brands because it could be pedaled at speed over rough ground without falling apart – fast enough, indeed, to be a means of escaping from a lion. Others might simply infer that those who own a Raleigh bicycle are likely to be chased by lions. Yet others might draw a more sophisticated inference leading to the latter conclusion, namely, that a Raleigh bicycle enables its user to go so fast that lions would mistake them for a gazelle, and so on.¹

Hayek's view of the mind has at its core the idea that events in the world (including possible events that we imagine in our heads) are not perceived in



Fig. 1. The Ambiguous Consequences of Owning a Raleigh Bicycle in Africa.

a manner determined by incoming sets of sensory stimuli. Rather, what we see depends on the patterns that we can find in the flow of sensory inputs that overlap with patterns stored from past experience. Events are classified into particular categories because the sensory inputs associated with them fire up patterns of neural connections similar to those that have previously been fired up by other events. For example, we can make sense of an 'airbag' as an automotive safety feature if we have mental concepts relating to notions such as 'a bag', 'pneumatic devices' and 'what happens when a car stops suddenly'. The uniqueness of a particular event resides in the unique combination of neural patterns that its unique combination of sensory ingredients triggers compared with previously experienced events. It is its particular *combination* of qualities, not the possession of unique kinds of qualities, that makes an event stand out as something unique compared with other events and against background 'noise' in the chooser's perceptual field, for if its qualities bear no resemblance at all to anything we have previously experienced, our minds will be unable to make any sense of it at all (Hayek, 1952, p. 142).

Hayek's argument about our blindness to qualities we have not previously experienced may be illustrated with the example of how two different people may assess a particular flow of sound emanating from a radio. Neither may have heard this particular flow of sound before. One of them, who has

previously listened to a lot of modern popular music, has developed mental pigeonholes for a wide variety of genres and hears it as a piece of music that is a fusion of heavy-metal and hip-hop. This may be precisely how its performers, their producer and their record company saw it. However, the other person is not someone who listens to modern popular music and instead listens to classical music, jazz and songs from musicals. The sound fails to trigger any of the latter person's past musical memory patterns, but does trigger some of the same neural patterns as are triggered by news reports from a war zone or dangerous urban environments. This latter person is probably going to call it, at best, 'a load of noise'.

Though Hayek's analysis of how the mind comes to see events as patterns of qualities has an immediate resonance in relation to a characteristics-based view of choice, his connectionist viewpoint can also be used for understanding how people get their ideas about causal relationships. To achieve this, our minds need to be able to sense patterns made up of multiple patterns: a cause is one event in the external environment and an effect is another. If our minds are to be able to sense which events are linked, and how, they may need to be programmed to identify and then connect pairs of patterns that tend to occur together, or where one patterns repeatedly tends to precede another. Either way, to recognize one event as the cause and the other as the effect requires seeing which of them fits the patterns the person has previously used to classify a 'cause' and an 'effect'. When multiple possible causes are identified the person may be expected to apply further classificatory rules to decide which patterns are plausible and which are not. Hayek sees these patterns of connections as a viscous mass: our understanding of the world around us is something that we continually modify, both in terms of causal relations between events and the nature of the events themselves (Hayek, 1952, p. 175).

Hayek's view of human perceptual processes has much in common with the personal construct psychology of George Kelly (1955, 1963) whose ideas were first used in economics a quarter of a century ago by Earl (1983, 1986a) and Loasby (1983). Kelly sees people as if they are like scientists: they theorize about events in their lives by constructing models of what they believe they are looking at in terms of their existing templates for categorizing events. These templates comprise bipolar construct axes (e.g., music vs. noise, safe vs. dangerous and so on) and Kelly suggests that they are organized into complex hierarchical systems in which what people can see in a particular situation is shaped by the rules of their construct systems as well as by the set of construct axes they have at their disposal. If two possible ways of construing an event clash, people will resolve the cognitive dissonance by seeing the event in a manner consistent with the construct

they rank the highest (hence, e.g., the difficulties that some have accepting the loss of a loved one around whose continued presence they had built many of their expectations). This view is very similar to the thinking of Lakatos (1970) in the philosophy of science: he suggests that some of a scientist's ideas will be 'core' to their research programme and not open to challenge, with the burden of adjustment to problematic data being placed on 'auxiliary hypotheses' in their research program's 'protective belt'.

Where Hayek goes beyond Kelly, however, is by offering an underpinning neurological analysis of how events come to be categorized via the patterns of neuronal connections formed by the patterns detected in their received sensory inputs. This enables a more physiological view of learning than that offered by personal construct psychology. Both approaches see people as learning by revising the connections they make or creating new ones but Hayek's analysis allows for underlying neural connections to be strengthened by the repeated firing of particular combinations of sensory inputs. By extension, it also allows for memory decay due to established connections not being fired up to make sense of new inputs because these particular connections have not lately been useful for finding patterns. (There is a clear parallel with social networks here: if we don't keep using parts of our social networks because they seem to have become less useful to us, they tend to be harder to reactivate at a later date as means to bring about particular new connections.) As a consequence of the latter, the corresponding construct axes or template configurations of constructs cease to be available.

At a conscious level, there is a case for initially being tentative about the connections one is forming: it may be unwise to jump to conclusions about what things comprise on the basis of a single set of sensory inputs being compared with patterns from one's memory (e.g., a single review of a product, unless one has, over repeated samples, come to see the particular reviewer as very reliable), or about patterns of cause and effect. From an evolutionary standpoint, then, there is a good statistical reason for our brains to have evolved, so that they only gradually firm up underlying neural connections when forming perceptions, just as they do when developing the connections that give us fine motor skills. There may also be evolutionary advantages to the brain resisting new connections that require existing ones to be unravelled: to flip between different ways of looking at the work, and, hence, to different ways of behaving, may preclude in-depth learning about anything and is likely to cause major problems for social coordination, because those with whom we interact will have trouble predicting our behavior. In some situations, however, evolutionary advantages come from

the brain being able to make very rapid connections. Seemingly instinctive responses to threats or opportunities are a sign that this capacity exists.

COGNITION AND INSTINCTIVE BEHAVIOR

Hayek's argument that new events are only capable of being perceived if they have similarities to what we have seen before needs careful examination in the case of infants: how can they make any sense of events around them when everything is new to them, and how can they progressively develop more adult-like ways of discriminating between different events? With no personal experience to call upon initially, the infant's perception must begin on the basis of experience that, as Hayek (1952, p. 168) puts it, has been acquired previously 'by the species' – in other words, programmed sets of neural connections that are hard-wired from birth to use as templates. These templates need not be particularly sophisticated and their application need not require any human sense of 'self' or any ability to envisage consciously the implications of discovering a particular class of object in the sensory field; all that evolution requires is that organisms have reliable method of matching objects to actions such that chances of survival are increased in the environments they are prone to encounter.

Studies of nonhuman behavior show the evolutionary importance of basic pattern-detection skills with matched behavioral responses. A famous case from Tinbergen's (1951) work on animal behavior is the need for young geese to run for cover if a hawk is overhead: if they do not respond in this way the moment they see it, the hawk may swoop on them. To survive, they do not need to have an abstract concept of 'a predator', or related concepts such as 'violent death'; they just need a combined stimulus detection/response rule that makes them run for cover when a hawk is overhead, but does not divert them from foraging if a threat is not present. (A perceptual system that caused them to run for cover when threats were not present would hinder their chances of survival by interrupting their feeding – just as in humans excess anxiety and paranoia can greatly interfere with everyday life.)

Seen from the ground and without good resolution, the silhouettes of a hawk and a goose are similar, even though a hawk has a long tail and short neck and a goose has a long neck and a short tail. A bold crucifix shape can represent either in silhouette form. To assign such a silhouette to one category or the other a further cue is required, the direction of movement: an efficient rule in this case is the one that results in the gosling running for cover in the presence of such a silhouette if the shorter part of the central

axis heads it, with the long part trailing it, but not if the movement is in the opposite direction. Possession of such a simple classification/response rule puts the gosling into a far better position for surviving than it would be in if it had a complex rule requiring it to deliberate, assemble a list of possible courses of action and choose the best one. There is an obvious parallel here between the survival value of fast and frugal rule-based action in the animal kingdom (cf. Bekoff, 2005, p. 36) and Winter's (1964) arguments about the potentially superior survival capabilities of a firm that uses decision rules rather than marginalist methods to cope with changes in its competitive environment.

An analogy in relation to artificial intelligence is also instructive here, even though, as Steele (2002, p. 130) notes, artificial intelligence generally works with serial processing of inputs and particular storage locations for particular patterns whereas Hayek's view of the brain is very much that of parallel processing and distributed storage. Broadly speaking, modern sensing technologies work in much the same way as we are supposing the mind of a gosling works to produce behavior without any conscious, abstract understanding of the situation at hand. Hayek (1952, pp. 47–49) himself clearly sensed this parallel even though the technology of his time was much less sophisticated. In trying to clarify his vision of how cognitive processes work, he drew parallels with mechanical sorting machines such as those then being used to sort census cards in which punched holes represented the data. Such machines were the forerunners of modern digital computers. Had he been writing today, Hayek could have noted parallels with visual recognition devices, such as scanners that are programmed to do optical character recognition from text, registration-plate recognition devices used by police to search for vehicles of interest among the rest of the traffic on a particular stretch of road, or electronic quality checking machinery. Incoming light inputs are pixilated and the machine checks to see whether certain patterns of pixels are present. In some cases, this works hierarchically with search for different kinds of patterns being done very rapidly in sequence (e.g., first find and lock on to registration plates on passing vehicles, then check them for each possible letter and digit). What the machine 'sees' thus depends on what it is programmed to look for. Other modern devices imitate other senses. For example, a smoke detector can serve to some degree as a nose substitute, while the pickup and signal processor of a guitar synthesizer can rapidly count the frequencies at which the guitar's strings are vibrating, in effect working as an ear's cochlea.

Having detected the presence of particular objects, such devices can in turn initiate some form of action if this has likewise been programmed: a

registration-plate recognition device may issue fines to motorists who have failed to pay a congestion charge, a smoke detector may emit a high-pitched note if smoke is present, and a guitar synthesizer may instruct digital samples of particular instruments playing particular notes be sent to an amplifier. At no point do these sensing devices need to have any conceptual understanding of what they do, any more than the gosling needs to think 'hawk' or 'predator' before running for cover.

In all of these cases, exactly as in Hayek's theory, the devices recognize only what they are programmed to recognize. For example, suppose hybrid petrol-/electric-powered vehicles are not liable for a congestion charge within an inner city zone. The registration-plate recognition system will only know not to send a fine to the owner of a hybrid if it has been programmed to recognize the distinctive shapes of such vehicles (as would be possible with a Toyota Prius) or, more likely, particularly since hybrid versions of some conventional vehicles are available, if it has been programmed to ignore the registration plates of hybrid-powered vehicles. These devices remain oblivious to all other patterns, not just in other sensory modes (e.g., ignoring sounds or smells and only looking for visual patterns), but also within their particular sensory mode (e.g., if programmed to recognize rectangular objects, they will ignore pixel patterns that consist of curves).

The selective vision of such devices makes it easier to see why Hayek objected to what he called 'mosaic psychology' (1952, pp. 76, 153). A modern pixilated image is akin to a mosaic, but for it to be seen as an image of anything ordered rather than mere 'noise' requires that combinations of pixels of similar colors within it must be found to match up with some preexisting pattern. Hayek sees the mind as working in a holistic manner as a complex system, whereas the mosaic approach is an aggregative, reductionist one.

Adults will rarely be in situations in which they cannot make any conscious classification at all of the events they are facing. Even so, they will experience something close to that of the infant human or gosling if they find themselves in a country where both the language and culture follow different rules from their home territory and they are unable rapidly to infer what the alternative rules may be. In such situations, one's survival chances may be increased by behaving in a manner that involves copying the local population – 'When in Rome, do as the Romans do' – with refinements insofar as one can discern people who seem to be the local equivalent of oneself.

New kinds of consumption activities or a newfound ability to choose in unfamiliar market segments resulting from significant changes in purchasing power may present rather similar 'new territory' challenges even on one's

home turf. Products may have a language of their own in some respects and even some of the brands may be unfamiliar and difficult to judge relative to those that are known from other products and market segments. Here, too, following the herd is a simple means of survival, reducing the risk of being asked to provide the sort of account that a deviant choice might provoke. The leading brands may have got established in such a market segment on the basis of rules that are no longer currently appropriate. However, if new members of the population of potential buyers also suffer from a lack of familiar landmarks for making comparisons, other brands may have trouble dislodging them because the latter are not favored by herd-based rules (cf. Choi, 1993).

Those who lack decision rules for a new situation may be unable to experiment due to time pressures, barriers to obtaining access to the product for trial (e.g., by renting it if it is an indivisible durable) or because the initial trial involves a crucial choice in Shackle's (1972) sense (e.g., because there is a risk of acute social embarrassment). For them, the alternative to following the herd is to enter the 'market for preferences' (Earl & Potts, 2004) and access the insights that others have to offer about how well different products match with different consumer lifestyles. This is not without its challenges: in working out which authorities should be trusted one may still have to fall back on simple social rules.

CONSCIOUSNESS VERSUS THE TACIT DIMENSION

The acquisition of language is clearly one route to developing a more complex sensory order involving the emergence of the ability to engage in abstract, conscious thought. Language provides elements for making new connections and importing useful connection sets from others with a greater pool of experience. Whether a language is a spoken one or a symbolic one, such as the language of mathematics, it provides a means for thinking logically about causal chains and the implications of particular propositions or actions. From the standpoint of Hayek's theory, we can see that the human brain may develop a set of patterns that make up a working language via repeated exposure to particular words being spoken by other people in combination with other replicated sensory inputs. But the ability to recognize particular words does not necessarily imply any abstract understanding of what they mean in themselves: a machine can be programmed to recognize words and act upon what it hears, just as some students are able to memorize

lecture note handouts verbatim and download them in exams without showing any sign they have understood the arguments in what they have memorized.

Understanding and consciousness of what one has understood are emergent phenomena in the sense that the brain is making new kinds of connections. The mind does not merely attach, say, 'cornflakes' to one kind of physical event (so that each time cornflakes are encountered the sound of the word is triggered in one's head as part of the pattern the brain assembles) and 'rice bubbles' to another kind of physical event. It also connects 'food' to particular physical events and begins to see sets of sub-categories: cornflakes and rice bubbles are thus in the category of breakfast foods, and breakfast is an event that happens soon after waking and so on. A young child may acquire and use words without being fully aware of their meaning: breakfast food may include both 'cereal' and 'fruit', but 'muesli' might be seen by some children as 'a cereal' and by others as 'a breakfast food made up of cereal, nuts and fruit', but none of them may have any notion of what 'cereals' are in the sense of what they have in common as plants. Regardless of how far humans develop the linguistic side of their perceptual processes, however, it is clear that the ability to make linguistic connections of a complex kind is something that human brains are unique among living organisms in being programmed to be able to do.

The development of language skills and conscious connection-forming abilities are vital for having the kind of 'economic imagination' that can engage in deliberative problem-solving activities and provide accounts of how a decision has been reached that refer to the characteristics of rival options in cost-benefit terms. However, this should not distract us from the potential for a wider range of experience to permit a greater capacity to take decisions in ways that cannot be readily put into words.

What we are talking of here is the growth in what Polanyi (1962, 1967) called 'the tacit dimension' and the basing of choices on 'tacit knowledge'. For example, most people would be hard pressed to say exactly how they can instantly recognize a person they know as matching their prior template for that person rather than having a set of features that comprise someone new to them. Another familiar illustration of the tacit dimension is how children normally find it something of a struggle to learn how to ride a bicycle without stabilizer wheels or to swim, despite their parents' and friends' best attempts to show them how. But eventually they get 'the knack' for performing these tasks: their brains establish workable sets of connections to coordinate sensory inputs with the movements made by their bodies (e.g., to adjust their weight distribution from side to side when

cycling, or breathing at the right time when swimming). In the process of acquiring such skills, their brains are, in effect, developing complex algorithms in which one set of connections triggers another. What children are able to do, without conscious thought to avoid losing control of their bicycles when they are surprised by a bump in the road or have to pull up sharply, is very similar to what electronic stability control and antilock braking systems are programmed to achieve in modern cars.

In situations of great complexity where there are many things to weigh up and considerable uncertainty, or where there is a lack of knowledge of the kind that can be processed consciously, decisions will require an instinct for what to do. It seems that some kind of basic instinct to drive action is a vital aspect of choice even when decision makers are very knowledgeable: Damasio (1994) studied the decision making of patients with brain injuries and showed that particular kinds of brain injury leave people able to articulate differences between alternatives in great detail, but completely unable to make choices between them. But 'gut feelings' may also be based upon the brain's unconscious processing of tacit knowledge, where it likens current options to past patterns in ways that cannot be put into words.

If we think of people as having evolved, for survival purposes, a tendency to keep asking themselves, tacitly if not consciously, 'is there any need for me to change what I am doing?' then from Hayek's perspective we would also see them as continuously classifying decision occasions (including, in effect, 'no change of action is required', vs. 'I need to do something as my environment has changed'). Different kinds of choice strategies will have been attached to different kinds of events: for example, an imminent threat of a particular kind may have a simple fight-or-flight response, whereas other situations may be labeled, in terms of fit with previous ones, as, say, 'the sort of decision where I ask a friend or check Amazon.com for a recommendation', 'the sort of situation where I carefully search for options, weigh up the pros and cons and don't rush to a verdict' and so on.

With our minds continuously scanning to see if action is required there may be periodic shifts between what we are attending to and our modes of choice. In some situations, rather like a modern computer that can have several programs running at once, we may even be simultaneously choosing both consciously and tacitly in different domains. For example, when driving a familiar route we may at one moment be thinking about what to cook for dinner when we arrive home, without consciously reflecting on our use of the vehicle's controls or the route we are taking, but at the next moment we may abruptly cease thinking about the choice of meal and without hesitation step on the brakes because our minds have detected that

the car in front is braking, or we may have noticed it is starting to rain and have to choose whether to turn the screen wipers on yet and in what mode.

From Hayek's standpoint, we would expect that people who develop expertise in particular areas do so by forming connections regarding much bigger samples of pertinent events in their area than nonexperts do. When faced with additions to their sample, they have a much bigger set of templates to call upon for making judgments about the fresh sets of sensory inputs. This is in line with the work of Herbert Simon and coworkers (Chase & Simon, 1973; de Groot, 1965) on the skills of chess players. As Simon (1976, p. 145) comments,

[W]hat appears to distinguish expert from novice is not only that the former has a great quantity and variety of information, but that his perceptual experience enables him to detect familiar patterns in the situations that confront him, and by recognizing these patterns, to retrieve speedily a considerable amount of relevant information from long-term memory. It is his perceptual experience that permits him to play, and usually win, many simultaneous games against weaker opponents, taking only a few seconds for each move.

In other words, a chess master does not possess a superhuman ability to work out the tree of possible consequences for each of many possible rival moves; rather, the chess master can call upon many more experiences of patterns on a chess board and a bigger repertoire of strategies that have proved successful in similar situations. Simon (*ibid.*) argues that there is no reason to believe that the brain of the experienced business professional works any differently from the experienced chess player, so the executive can 'react "intuitively", without much awareness of his cognitive processes, to business situations as they arise'.

THE FALLIBLE DECISION MAKER

The ability of humans to use intuition to cope with complex problems and ignorance does not guarantee that the decisions based on gut feelings will be of high quality. Evolutionary psychologists such as Barkow, Cosmides and Tooby (1992) argue that unconscious processing mechanisms in use today evolved early in human existence to deal with choice environments very different from those that people now face. If so, we might expect that in some modern contexts people will be prone to making errors. For example, we may be programmed to be more impulsive in some cases than the pressure of the situation actually requires: stone-age people facing predators who could also be food sources needed to be able to make 'fight-or-flight'

decisions very rapidly, whereas modern consumers could often readily defer reaching decisions about which new consumer durable to buy and yet may be prone to leap impulsively because of the stress of dealing with pushy sales staff. Within modern behavioral economics we find US scholar, such as Myers (2002), emphasizing the shortcomings of intuitive decisions in much the same way that in the work of Kahneman, Slovic, and Tversky (1982) heuristics and biases are seen as leading to departures from mainstream economic views of rationality. By contrast, European scholars such as Gigerenzer (2007) marvel at 'the intelligence of the unconscious' and emphasize scope for training people how to avoid major errors.

Back in 1952, of course, Hayek himself had nothing to say about how the sensory order may operate to produce the kinds of heuristics and biases that have since been identified in empirical work. Clearly, some of these findings – such as the tendency to treat high probabilities as certainties and to ignore very low probabilities altogether – imply that the process of making neural connections has evolved a tendency to simplify in this context. The editing of probabilities looks irrational from the standpoint of expected utility theory. So, too, does the kind of focusing on the most attention-arresting pair of gains and losses that a strategy might offer that is central to Shackle's (1979) nonprobabilistic 'expected surprise' analysis of choice under uncertainty. However, it seems plausible that evolutionary processes have *selected* this aspect of bounded rationality for its survival-enhancing capabilities. Clearly, it is cognitively possible for people to avoid these probability-editing biases and assess risks in the manner of expected utility theory rather than operating in a manner consistent with Shackle's potential surprise theory or Kahneman and Tversky's (1979) prospect theory: they just need to be trained to use their brains' potential rather better when dealing with hazardous situations. (Help is now readily available, from, e.g., Gigerenzer, 2002.) From the standpoint of evolutionary psychology, and again from the standpoint of Winter (1964), simplified ways of thinking about risk might be highly functional. Even though they might not be perfectly reliable (unlikely events may actually happen occasionally with dire consequences for those who ignore them as possibilities), they may nonetheless increase species survival chances overall by enabling faster responses to threats.

Further clues as to what may be happening come if we map Hinkle's (1965) work in personal construct psychology on to Hayek's view of how the brain works. Hinkle attempted to make sense of resistance to change on the basis that changing some constructs in a person's construct system may necessitate changing many others that are subordinate to the changed constructs. He called these changes 'implications' and developed a technique

known as ‘implication grid analysis’ to map them. Hinkle found that resistance to change in a construct is a function of the number of subordinate implications it carries. This can be used as a basis for understanding elasticity of consumer demand (see Earl, 1986b). For example, most consumers would see few negative implications associated with switching between brands of tinned tomatoes to save a few cents, whereas to *fail* to switch might be at odds with their self-constructs (e.g., as rational, thrifty shoppers) and potentially carry many negative implications. However, we might expect that the demand for a lone brand of *organic* tinned tomatoes to be much more inelastic due to its purchasers being people who see organic products in relation to core areas of their lives, such as their health or their views of themselves as eco-friendly individuals.

From Hayek’s standpoint, major changes in mental constructs and their organization would entail major changes in underlying patterns of neural connections. Hinkle’s research findings seem to imply that human brains are programmed to preserve connections that have been firmly established and to reject cognitions that would unravel connections previously firmed up as means for coping with life’s challenges.

This perspective is very much in line with the conservative nature of the findings of the heuristics and biases research, such as the endowment effect, sunk-cost bias and status-quo bias. If faced with a cognitive dilemma – that is, a situation of cognitive dissonance involving inconsistent prospective perceptions – the human brain appears to have evolved an ability to come up with a cognitively harmonious perception that limits the amount of neural reordering as far as possible. This may not entail pragmatically compromising until one is able to find ‘a happy medium’ as one might expect from traditional trade-off notions from mainstream choice theory but, rather, *reframing the situation so that no trade-off appears to be being made* (cf. Steinbruner, 1974, chapter 4). Thus, for example, suppose we have ‘set our heart’ on obtaining a particular product that we see as having a great capacity to enhance our social standing and buttress how we see ourselves. If we then find that it is more expensive than expected this will not result in us unraveling these expectations and their underlying neural connections. Rather, we will attempt to expand our budgets by trying to obtain credit even if a credit-based purchase is a source of cognitive dissonance because we have also previously formed connections between, say, our self-images and the idea that debt is something to be avoided. Dissonance will be removed by severing such connections and creating ones consistent with going ahead and borrowing if this is the easier form of neural reconfiguration than one involving abandoning the plan (cf. Maital, 1982, pp. 142–145; Earl, 1992). For example,

the person might suddenly start being willing to perceive that being in debt with a new car is actually more financially responsible than continuing with an existing vehicle after raising their assessment of the likelihood of big maintenance bills on the latter, something that would not have happened had it not been necessary to find the extra funding to buy the new vehicle.

Taken together, these two perspectives on the origins of heuristics and biases in the mind's operations – that is, that of evolutionary psychology and the Hayekian one of dominance of well-established, strongly coupled sets of neural connections over nascent and loosely coupled ones – do not bode well for timely responses of consumers to challenges presented by ecological issues. Potentially catastrophic events whose prevention requires major lifestyle adjustments (and hence major cognitive adjustments) seem prone initially to be simply ignored if they are presented as having low probabilities. Subsequent attempts to demonstrate that such events are actually on the way are likely to be argued away until enough evidence mounts to produce a cognitive tipping point in which the continuation of the existing lifestyle becomes seen as a bigger threat to the person's cognitive system than a change to a new 'sustainable' lifestyle.

BRANDS AND CRITERIA-BASED DECISION RULES

Although consumers have in-built tendencies toward making decisions at odds with mainstream economists' notions of rational choice, these shortcomings should not distract us from their abilities to develop very refined powers of discrimination that enable them instantly to recognize people that they know and classify products as distinct combinations of characteristics. The height of the latter capacity is perhaps epitomized by the 'anoraks' who can immediately identify particular models of vehicle – for example, a 'Toyota Camry Ateva 2005 facelift', rather than simply a 'Toyota Camry from the generation prior to the current one' on the basis of whether or not it has body-colored door handles and alloy wheels (which rule out the base Altise model), a nonchromed grille (which rules out the top-end Grande model), no body kit (which rules out the Sportiveo variant) and the color configuration of the rear lights (silvered tops and bottoms rather than silvered middles rule out the prefacelift 2002–2005 model).

An important thing to notice here, however, is that the knowledge used to make such distinctions may largely be codified and assembled as checklists. Such checklists are actually quite simple combinations compared with the

combinations of curves and lines that less car-obsessed consumers need to be able to use to distinguish one car model from another. In the latter case, their knowledge may be more of a tacit kind, and where products are in many respects similar, puzzles may be resolved by looking at the manufacturers' badges.

For a particular manufacturer's products (in general) and its individual models to be instantly identifiable to consumers, they must appear to be composed of *particular* combinations of visual cues, in exactly the same way that machines programmed to separate passing items on a production line into different categories need to be able to determine which *particular* combinations of lines and curves they comprise. (The mathematical underpinnings of object recognition systems are challenging, especially if it is necessary to be able to recognize a particular form from different angles. For some examples, see Hann, 2001; Hann & Hickman, 2002.) People and machines classify events into categories on the basis of invariant aspects that comprise some kind of signature at a general or more particular level. Car manufacturers can thus create distinctive looks for their products by combining codified features, such as their signature front grills with much more tacit styling combinations. (For example, how the wheel arches are flared, and the relationship between the wheels and wheel arches may be chosen in an attempt to give their products a particular stance consistent with the image they are trying to create.) If they make a particular set of design cues a constant, they may be able to ensure instant recognition even though they allow other design cues to change. For example, during the evolution of the Volkswagen Golf from the 1974 Mk1 to the Mk6 35 years later, a Golf's external appearance remained distinctively 'Golfish' despite moving from an angular design to a much more aerodynamic shape based more on curves. The one obvious invariant that no other car in its class seemed consistently to have was its very thick, rather parallelogram-shaped rear pillar, but in other respects it is hard to put into words exactly what its signature is.

Now, if consumers can distinguish between products in these complex ways, a question that follows is whether or not, in choosing between them, their minds will go through a complex n -dimensional trade-off process akin to that envisaged in Lancaster's characteristics-based theory of choice. Though this seems perfectly natural to most economists, the economic imagination could work very differently: for example, the mind could go through a choice process that is itself a template-based activity that initially classifies products as suitable or unsuitable in terms of whether they meet a set of checklist requirement and then applies tie-breaking or priority ranking template if the need arises (see Earl, 1983, 1986a).

However choices *are* made – and the forms of decision criteria/preferences may vary between different contexts – the clear message from Hayek’s work is that while people may all be born programmed to make certain basic choices in the same way, they grow to differ in what they do because they develop different systems of neural connections as their pools of experience increase and diverge. Thus, if it is meaningful to speak of a consumer as having a ‘preference system,’ it is one that the consumer has constructed and which may be very much a work in progress that consists of a set of rules for dealing with particular kinds of situations. As time passes, some rules may drive out others due to relative frequency of their actual use affecting the relative strengths of the neural connections between types of situation and each of the rules that have been associated with them. To put it another way: if we start to perceive that a particular rule is prone to prove unworkable and force us to fall back on an alternative from our repertoire of rules for a particular kind of situation, we are more likely to use one of the latter in the first instance on future occasions of that kind.

There seems to be no necessary evolutionary basis for assuming that the classification of product or characteristic bundles as better than, worse than, or as good as other bundles will be done in a manner that implies a diminishing willingness to make marginal substitutions. This contention may be better appreciated if we consider the key role of attention as part of the process of decision making. The brain needs to be programmed to allocate its finite attentive capacity to focus on a limited set of events at any one time while also scanning for potential patterns with greater attention-arresting properties. Rival possibilities represent competing claims on the brain’s attentive capacity, so to have a hope of being selected they must stand out against the others. From the standpoint of evolutionary psychology, we would expect that what will stand out is not ‘outstanding overall value’, but a stellar performance on a dimension that is important for immediate survival or longer-term reproductive success. The obvious means for firms to try to ensure this is to have (or claim to have) something outstanding to offer and advertise this, while trying to divert attention from the compromises that have been incurred to make it stand out in a particular area. This would make it hard for a product that is a solid all-rounder, but in no way *outstanding* to get serious consideration unless the brain is programmed to look for product configurations that have no glaring shortcomings – or unless consumers have developed a strategy for doing this and ignoring advertising ‘hype’.

Clearly, many people do end up buying products that are unremarkable but functional, and they manage to filter out the advertising messages that aim to promote a focus on a single characteristic as if there were no such

thing as diminishing marginal substitution. (It is tempting to assert that this seems particularly evident among the middle-aged and senior consumers who have had more time to develop skepticism and become aware of the potential downsides of products that have an 'exciting' allure in other ways; the younger generation, meanwhile, look upon their preferred products as 'boring'.) However, cognitive strategies that involve not being taken in by the loudest or brightest stimuli need not involve making trade-offs: as already indicated, they could instead involve being open to considering any products that met particular sets of criteria. Such a way of thinking is cognitively far simpler than assessing marginal substitutions across many dimensions.

From an evolutionary standpoint, the bigger the threat to our survival, the more attention-arresting something should be. Maslow's (1954) famous suggestion that people have a 'hierarchy of needs' goes against the grain of mainstream notions of substitution. However, it can be seen as perfectly reasonable in these terms, as there is an evolutionary reason for a hierarchy of needs being hard-wired at the species level: without water we die sooner than we die without food, so if we have neither, a set of neural connections pertaining to thirst will grab hold of our attention and we focus first on finding enough water. In Maslow's analysis, it is only when all of our basic needs are covered that we are in a position to make trade-offs and set about engaging in 'self-actualization', fashioning our lifestyles by choosing between those bundles of goods/activities that can be obtained without jeopardizing our basic needs. Before this stage, bundles of goods that are more efficient at getting the basic needs met will be the ones that command attention, and this is where the consumer's experience will be significant for classifying possibilities.

It is possible that a trade-off way of ranking possibilities could evolve as part of the process of dealing with a Maslowian hierarchy. From experience, the consumer might form templates that classify some mixes of products as more efficient than others for meeting particular basic needs. Such mixes might be constructed on the basis of rules that conflict with the trade-off notion by having satiation thresholds for some characteristics (e.g., 'too showy/obvious' regarding some strategies aimed at meeting a need for social membership), but others might be consistent with the notion of continuously diminishing marginal rates of substitution in characteristics space. Thus, for winning a member of the opposite sex, a male may judge that it is, say, unwise to have terrible clothes and grooming but a great car, or vice versa, rather than a 'happy medium'. If attention is focused on the most important currently unmet basic need, the focus would be on assessing how

product characteristics of various kinds could help toward that particular need: with a car as a potential means toward winning a partner, sleek styling might be seen as carrying good implications, but not if achieved at the cost of a potentially embarrassing risk of unreliability. 'Implications' here can be viewed in the Hinkle (1965) sense, as discussed earlier, as a kind of common unit of measurement for different characteristics, to make an additive approach to choice possible. We could thus have a hybrid of programmed and learned attention-focusing preferences that mixes lexicographic and trade-off ideas.

We should not, however, jump to the conclusion that, subject to constraints of an evolutionary kind, as per the Maslow hierarchy of needs, consumers will necessarily evolve decision-making systems that approximate to conventional trade-off notions. Trade-offs become problematic for the consumer to compute in any conscious sense where there is a large range of products between which to choose and the products perform in very different ways across significant numbers of characteristics. Consumers who try to reason their way to a choice in such a situation will need a means of ranking products that permits simplification, such as a checklist of requirements to generate a shortlist, with trade-offs being performed, if at all, only in relation to the short-listed products. Otherwise, it will be necessary to choose on the basis of simpler decision rules, such as those involving familiar brands, copying others, or on the basis of intuition. In the latter case, the consumer's brain might unconsciously work in a manner approximating to trade-off notions as per the 'implications' approach to avoid undoing established neural pathways. If the way that cognitive dissonance gets removed involves denying at the conscious level that a trade-off has really had to be made, introspection may provide few clues about the underlying process, so debates about what actually happens might require an experimental approach to reveal underlying 'preferences'.

CONCLUDING COMMENTS

Hayek's theory of the mind provides a unifying foundation for analysing choice in a evolving, pluralistic and context-based manner rather than seeing all choices as made in much the same way on the basis of 'given preferences' that obey the axioms of rational choice theory. If read from an evolutionary standpoint, *The Sensory Order* should encourage economists to recognize the role of intuitive thinking rather than conscious processing as a potentially efficient basis for coping in contexts where

there are competitive reasons for taking decisions quickly or where complexity and uncertainty make it problematic to rank rival possible courses of action. Its focus on finding familiar patterns in the mid of incoming sensory inputs should also encourage reflection about the possible role of template-based decision making in contexts where people are trying to solve problems in a conscious manner and have built up complex perceptions of the differences between a large number of rival possible solutions.

Since there are limits to what we can know about each individual's perceptions and their underlying patterns of neural connection, Hayek's analysis leaves room for Austrian economists to continue to emphasize limits to predicting very precisely the choices that people make. Even so, his view that these patterns are developed from interpretations of past experiences on the basis of even earlier experiences opens up potential for predicting behavior in a broad enough sense to limit coordination problems. What we need to be able to do is to know, as the expression goes, 'where they are coming from'. This line of thinking is consistent with that suggested by Heiner (1983), who argues that prediction of behavior is only possible in a world of singular events because people cannot work out the optimal way to deal with the singularity of an event, but instead try to see what pattern it fits into and then apply a behavior strategy developed for that form of pattern.

Finally, and perhaps controversially, it might also be argued that Hayek's analysis should make subjectivist economists more open to the writings of modern behaviorist consumer researchers, such as Foxall (1990, 1997). Hayek (1952, p. 44) took issue with the strict behaviorist position that it is not necessary to understand mental processes in order to account for behavior. To do so was to dodge the question of how stimuli came to be interpreted in one way rather than another. Behaviorists' tendencies to focus on the frequencies of individual stimuli also entail neglecting the significance of patterns of stimuli. This goes against what Hayek had picked up from the Gestalt psychologists for whom, say, it is the relationship between notes – that is, the tune – that shapes a person's reaction to them, not each note as an individual stimulus (see De Vecchi, 2003; Caldwell, 2004). However, his emphasis on the importance of past experience for how current events are perceived, and on the time it takes for neural connections to firm up or decay, is not at odds with what behaviorists believe about learning and behavior. If we can discover which past experiences have been seen as rewarding and which have been viewed by consumers as punishing, then this, too, may help us anticipate their behavior.

NOTE

1. A full color version of this advertising poster can be found in electronic format at <http://flickr.com/photos/dreamtargets/1084477112/> ('Vintage Raleigh Poster – Lion', by Dreamtargets). The Indian version of it featured a tiger instead of a lion and, according to Professor Peter Payne, in a talk on the history of advertising that he gave at the University of Stirling in the early 1980s, many Indian consumers actually saw it as implying they would get chased by tigers if they rode a Raleigh.

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HAYEK ON PRICES AND KNOWLEDGE: SUPPLEMENTING “THE USE OF KNOWLEDGE IN SOCIETY” WITH *THE SENSORY ORDER*

Samuli Leppälä

ABSTRACT

Purpose – To show that The Sensory Order is useful for understanding Hayek’s position on the informational role of prices.

Methodology/approach – Hayek’s psychological theory argues that every sensation is interpreted in the light of past experience. This idea is applied to Hayek’s view on the price system by arguing that, similarly, every price is interpreted in the light of local knowledge. The usefulness of this approach is tested by addressing some common mainstream interpretations.

Findings – Prices perform their informational role in interaction with local knowledge. The standard view, in which prices convey the same information to everyone, ignores the fundamental importance of local knowledge and varying interpretations.

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Research limitations/implications – *The paper only discusses some of the central insights given in Hayek's theory of the mind. Furthermore, implications for connected issues, such as entrepreneurship and market process theories in general, are left for subsequent research.*

Originality/value of paper – *While the connections between Hayek's thought in different fields and the importance of interpretation has been suggested by others, this paper contributes to the Austrian price theory and suggests the relevance of The Sensory Order to economists by making this connection more pronounced.*

INTRODUCTION

Among mainstream economists, Hayek is probably best known for his ideas concerning the manner in which market prices convey dispersed knowledge as outlined in his 1945 AER article “The Use of Knowledge in Society.” However, his 1952 book *The Sensory Order*, is a very different story. Even though cognition and psychology are now central issues in the field of economics, Hayek's idea of the mind as an adaptive classificatory system has not found much use in economic theory (Butos & Koppl, 2007). Leaving things open to interpretation, Hayek himself has said that *The Sensory Order* is extremely important for understanding his later work (Caldwell, 2004, p. 7).¹

Although Hayek worked with a very broad set of questions, there are several unifying features in his work (Rizzello, 1997, p. 13). Butos (2003) and Butos and McQuade (2002, 2005), among others, have suggested that Hayek's theory of the mind can be applied to his theories of market and social order, as well as to his view on social science methodology. Here, an attempt is made to study one such relation concerning his theory of the mind as a classificatory system in combination with the theory of the price system as a communicator of dispersed knowledge. More specifically, we will apply the main insights found in *The Sensory Order* to deliver a more comprehensive theory of how market prices overcome the so-called knowledge problem to make effective use of the knowledge that is dispersed throughout a society. Combining a theory concerning the informational role of prices with one of learning would seem to be a match worthy of studying.

The next section will briefly review Hayek's (1937, 1945) most important works on the price system. While Hayek outlined mainly his “price theory” in these two papers, he has returned briefly to these questions in his later

work. Mainstream economists, however, have mostly concerned themselves with these papers, and therefore we will concentrate on them as well as widen their perspective with *The Sensory Order*.

The third section will summarize the main points of *The Sensory Order* that are of interest to our current analysis. This book, however, mainly states how sense data is interpreted by the mind, and revolves around the issues of cognition and learning. When considering economic decision-making, choices are based on beliefs of a wide variety and not only on sensations. Furthermore, while belief is a psychological concept, we are also interested in the quality of beliefs, that is, their truthfulness, and how individuals are able to improve that quality by justification, which are indispensable components of the concept of knowledge. Therefore, we need to broaden Hayek's psychological theory to form a proper epistemological theory. It will be argued that Hayek's theory fits well together with the coherence theory of justification, making it more adaptable to rational decision-making.

The fourth section will demonstrate how we can supplement Hayek's price theory with *The Sensory Order*, and in the following fifth section, we will show how this merger may help us to scrutinize some common Hayek interpretations. Although Hayek's price theory has reached mainstream audiences, many Austrians complain that Hayek's ideas have been misunderstood or not fully comprehended. The perspective provided by *The Sensory Order* will be applied in an effort to compose a unified framework for studying the informational role of prices. The last section of this paper will summarize our discussion.

THE KNOWLEDGE PAPERS

Hayek presented his ideas concerning the informational role of the price system in his 1945 *American Economic Review* article "The Use of Knowledge in Society." This article has been cited numerous times by mainstream and heterodox economists alike, thereby establishing Hayek's authority in the matter. As a result, when it comes to price theory, it is mostly this article we refer to.

Rounding out the pair of articles comprising the "knowledge papers," is one of Hayek's earlier works, "Economics and Knowledge," published in *Economica* in 1937. While it has also been acknowledged by many, the previously mentioned paper has received far more attention, especially among mainstream economists. Here, Hayek spends more time discussing

the problem, particularly the nature and assumptions related to knowledge. The price system, as a potential solution, is mentioned only briefly, and in this respect the paper serves as a prelude to "The Use of Knowledge in Society." Hayek has returned to the informational role of the price system in passing, but since these papers established Hayek's authority on the matter, it is here where our interests lie.²

Hayek (1945) begins with "the economic problem of society," which was first identified in Hayek (1937). That is, how is a rational economic order possible? In such an order, resources should be optimized according to the existing needs and production possibilities, while the knowledge of these needs and possibilities is in an imperfect form and dispersed throughout the society. As such, the problem becomes that of "the utilization of knowledge not given to anyone in its totality" (Hayek, 1945, p. 520).

To make use of dispersed knowledge, Hayek (1945, p. 521) identifies two options: (1) all the relevant knowledge should be at the disposal of a single central authority or (2) additional knowledge should be conveyed to individuals enabling them to coordinate their plans among themselves. The first option considers the possibility of central planning, while the second option, which is the focus of the present study, considers how the market economy is able to achieve coordination through the price system.

To further clarify the nature of the problem, Hayek describes the properties of dispersed knowledge. The knowledge in need of utilization is not (only) scientific knowledge, that is, knowledge of general rules of the world or technical knowledge about the available production techniques and possibilities (Hayek, 1945, pp. 521–522). Dispersed knowledge is above all "knowledge of particular circumstances of time and place." An appropriate further qualification for such knowledge could be "market knowledge,"³ which is knowledge of one's wants and immediate surroundings, including knowledge of others' beliefs, intentions, and actions. Therefore, each individual knows something that the others do not. Technical knowledge, however, can become market knowledge when it is included in one's decision-making framework. But technical knowledge alone is not sufficient for decision-making, since one has to also evaluate, based on other beliefs, which options are worthwhile pursuing.

More contemporary interpretations of Hayek's theory (e.g., Lavoie, 1995) stress the tacit nature of knowledge. While Hayek appreciated Gilbert Ryle's distinction between "knowing that" and "knowing how," Hayek (1937, p. 50) also stressed that not only skills or "know-how" constitute the knowledge problem. While the debate whether Ryle is a behaviorist is ongoing (Hacker, 2006; Soames, 2006), the current interest among Austrians

to combine Hayek's and Ryle's insights should be approached with some caution. Although Hayek (1952, p. 19, 39) said that our *knowledge of the mental order* is largely that of "know-how," he also clearly resisted the behaviorist notion "that psychology can entirely dispense with any knowledge of the subjectively experienced mental qualities, and that it ought to confine itself to the study of bodily responses to physical stimuli" (Hayek, 1952, p. 26). However, different skills do affect the relative scarcities of goods and services, and are thus a *part* of the knowledge problem.

Another issue is that the knowledge of particular circumstances of time and place do not always correspond to objective facts. Thus, they are personal beliefs that individuals have, and could well be false.⁴ Hayek in particular warned against assuming simple correspondence between subjective beliefs and objective facts (Hayek, 1937, pp. 39–40; Hayek, 1945, p. 530), which is what mainstream economics does by assuming that all beliefs are necessarily true (Faulkner & Runde, 2004, p. 424) and that this is known by the agents themselves. Aggregation of dispersed knowledge is impossible since knowledge consists of opposite expectations and valuations, and is thus often contradictory (Hayek, 1945, p. 519; Reisman, 1997, p. 34). The veracity of many beliefs is revealed only later. The argument proceeds by suggesting that it is not only laborious to aggregate knowledge, but indeed impossible (Hayek, 1945, p. 524; see also Parsons, 1997). To make the aggregation even more challenging, the knowledge of local circumstances is constantly changing, which is the reason why we have economic problems in the first place (Hayek, 1945, pp. 523–524).

In the market economy, the aggregation is unnecessary since everyone already knows their immediate surroundings. But individuals need to know more to be able to take into account the local knowledge of others. This is done by the price system (Hayek, 1945, p. 525). Changes in local circumstances are reflected in prices, which allow others to take into account the changed prices and respond by adjusting their decision; a process which is illustrated by Hayek's tin example (*ibid.*, p. 526). Prices do not convey all knowledge (particularly the origin of the change), something that would indeed be impossible, but rather convey the relevant knowledge indicating that a change has taken place. If brought about deliberately, the designer of such a system would need to know everything that is known by every member of the society. With the price system, however, particular uses of any kind of resource, for example, need only to be known by one individual in order to be reflected in the price (Hayek, 1937, p. 52). We will return to the question of how prices execute their informational task after we have reviewed Hayek's *The Sensory Order*.

FROM THE SENSORY ORDER TO AN EPISTEMOLOGICAL THEORY

Hayek presents his theory of the mind in *The Sensory Order*, the book which has its origins in a paper that Hayek wrote in 1920 as a psychology student (see Caldwell, 2004). In the preface of the book (Hayek, 1952, p. v), he explicitly denies that his competence in the field of economics has anything to do with this theory. Eventually, however, Hayek would acknowledge the connection between his works in the two fields (Hayek, 1994, p. 153).

The basic idea, which we will use in this study, is that the mind is a relational order. Every sensation is classified according to its likeness with past experiences. There are no pure sensations, since every new sensation is understood only within the framework of past experiences. Any quality can thus be described only in terms of its relation to other qualities (Hayek, 1952, p. 37). The system is therefore not a static one, as a new sensation may bring forward the need for reclassification if it does not easily fit into the already existing classification. Therefore, the same event can be both an object and a classifying act (Hayek, 1952, p. 65).

Between the physical and mental order there exists no simple one-to-one correspondence (Hayek, 1952, p. 3). Our minds have their own order and physical objects have theirs. Physical objects also have different effects on one another while to us the same objects appear to be either alike or different (the former phenomenon being explained by physics while the latter by psychology). We are only partially aware of these connections, since the mind cannot explain itself in detail, as the explaining device would require a higher degree of complexity than the target of the explanation itself.

Hayek (1952, p. 28) accuses the traditional schools of psychology, including behaviorism, of treating “the problem of mind as if it were a problem of the responses of the individual to an independently or objectively given phenomenal world.” As a result, the physical world does not unilaterally determine responses since it is the existence of a phenomenal world in between stimulus and response, which needs to be explained. Latsis (1972) accuses economists of similar “situational determinism.”

While there is a sort of path-dependency in Hayek’s explanation, the responses are not path-determined (see also Rizzello, 2004). The sensation is interpreted according to the previous experience, resulting in a continuously adapting sensory order. Thus, neither a past experience nor the physical stimulus alone determines the response, but rather the complex interaction between the two.

There is a problem with this theory that Smith (1997, p. 22) has observed. Hayek's theory seems to leave no room for conscious, deliberative thinking.⁵ It is thus not directly applicable to the economic decision-making process as it does not account for individual autonomy in the reasoning and decision-making processes.⁶ *The Sensory Order* concerns itself largely with beliefs based on sense data, whereas in economic decision-making various types of different and possibly inferential, beliefs are used. Sensory beliefs are essentially given to the individual as one has little power over what one senses; only control of the meaning of such experiences. We would, however, like to think that individuals have some freedom in choosing, inferring and modifying nonsensory beliefs. In addition, while many beliefs are acquired through cognition, the actual decision-making is based on reasoning. Therefore, the quality of beliefs should be addressed with a theory of truth and improving the quality through reasoning with a theory of justification. Therefore, we need to bridge the gap between cognition to (economic) reasoning and from (sensory) beliefs to knowledge.

To make Hayek's theory of mind more applicable to market behavior, it needs to be expanded to form a more complete theory of knowledge (and not merely of cognition). Knowledge, or expectations for that matter (cf. Buto & Koppl, 1993, p. 315), is not a mere consequence of sensory inputs. It will be argued here that the coherence theory of justification, as discussed below, is compatible with *The Sensory Order*.⁷ By combining standards of knowledge with belief acquisition processes, we are complementing traditional with naturalized epistemology (see Hands, 2001; Kim, 1988; Rooney, 1998 for discussions about naturalized epistemology).

In traditional epistemology, knowledge is defined as justified true belief.⁸ This means that in order to know (a proposition) that p , the individual needs to believe that p is true. Furthermore, this belief needs to be true, as false beliefs do not count as knowledge. For not having a true belief by mere accident, the individual should be justified for holding the belief. Thus a central feature of rational decision-making is justification, the concept of which we will address next.

Bicchieri (1993) introduces a useful framework for understanding the link between (rational) choices and beliefs, which is the difference between *practical rationality* and *epistemic rationality*. Practical rationality entails that an agent chooses optimally, in light of his desires and beliefs. If S desires q and believes that p is the best way to arrive at q , he is practically rational by choosing p . Practical rationality therefore says nothing about the content of beliefs, only that the agent acts accordingly. Epistemic rationality, to the contrary, is concerned with the content of those beliefs.

It refers to beliefs by saying that *rational beliefs* are such that they are internally consistent, and are formed appropriately in light of available evidence (Bicchieri, 1993, p. 2).

Bicchieri (1993, p. 13) discusses whether the double rationality requirement is necessary for explaining and predicting individual behavior, since one could do without epistemic rationality by assuming perfect knowledge. Her conclusion is that perfect knowledge is neither the most common nor the most interesting case. For Hayek (1937, p. 45; 1945, p. 530) assuming perfect knowledge is to avoid the most central question of economics: how can less than omniscient individuals achieve coordination?

Central to epistemic rationality is justification. The general problem that any theory of justification needs to address is how to avoid infinite regress or circular reasoning if a belief is justified by another belief. The two main theories of justification that try to overcome this apparent dilemma are coherentism and foundationalism.

Foundationalism divides beliefs into two categories: basic (or foundational) beliefs and nonbasic (or nonfoundational) beliefs. Basic beliefs are such that they do not require justification, as they are self-justified. Usually with basic beliefs one means our own immediate experience or our sensory states (Dancy, 1985, p. 53). Nonbasic beliefs, however, receive their justification from basic beliefs.

Coherentism adopts another strategy. Rather than pursuing linear justification, it takes on a holistic approach, which considers that a belief is justified if it is a member of a coherent set. Coherence is a matter of how well the beliefs in an individual's belief set dovetail, fit, agree, or hang together (BonJour, 1985, p. 93). Hayek's theory of mind seems to have several key elements in common with the coherence theory of justification. First, Hayek states that not even sensory data are given in the sense that foundationalism assumes. These inputs are also interpreted in the light of the past experience, that is, the previous sensory inputs. Experience therefore, precedes every sensation or perception (Hayek, 1952, p. 166). Second, classification is a matter of coherence. Hayek (1952, p. 146) claims that all the higher and conscious mental processes follow the same principle of classification. Here the classification becomes more complex and is more conveniently expressed in terms of coherence. New inputs are interpreted in the light of the past experience as a whole and are integrated into the set in this fashion. Sometimes, however, to retain the coherence, the classification needs to be modified (Hayek, 1952, pp. 145, 168).

Rational beliefs, as presented above, need to be coherent and revised appropriately in light of new evidence. Hence, justification becomes an

integral part of decision-making processes. While these beliefs can also be false, their veracity has at least practical importance. Therefore, a brief consideration of the truth condition is appropriate.

Since we already proposed the coherence criterion of justification, it would be convenient to adopt the coherence theory of truth as well. Justification and truth would be defined in the same terms, and a true belief would thus be one that is a member of a coherent set. That would, however, force us to accept that all rational, hence justified, beliefs are true. While there is, and should be, a logical connection between truth and justification, they should not be thought of as being the same.

A better candidate for the truth criterion would be that of correspondence. While Hayek does not mention the correspondence theory of truth per se, he (1937, p. 145; 1952) repeatedly mentions the correspondence between subjective beliefs and objective facts. Indeed, fundamental to Hayek's theory of the mind is the interaction between the internal environment of the human brain and the external world (Loasby, 2004, p. 105). According to the correspondence theory, a belief is true if it corresponds to an external fact that makes it true. In other words, the state of affairs described by the belief exists in the actual, objective world. This allows us to explain why justified beliefs sometimes bring disappointments.

The connection between the coherence theory of justification and the correspondence theory of truth may not be that apparent. We can assume, however, that if a belief set remains coherent for a long period of time, it is due to its correspondence with external facts (BonJour, 1985). Likewise, if it does not correspond to external facts, it is unlikely to remain coherent. "[O]ur knowledge of the phenomenal world raises problems which can be answered only by altering the picture which our senses give us of that world, i.e. by altering our classification of the elements which it consists" (Hayek, 1952, p. 173).

INFORMATIONAL ROLE OF PRICES

In Austrian theory prices are quantities of money which individuals choose as expressions of their plans and which therefore transmit information about these plans. In real life, information is transmitted through various other institutions as well, but the price system is obviously the most important of such institutions. Vihanto (1989, p. 92)

Next, we turn our attention to the informational role of prices and attempt to utilize *The Sensory Order*, as reviewed in the discussion above, to

form a more comprehensive theory. Later, we will test this interpretation by attempting to analyze some selected responses to Hayek's work.

To understand the informational role of prices, we should first discuss what prices are. Economists usually define prices as realized exchange ratios for specific goods where the ratio is an amount of money. Therefore, it is easy to see that the existence of the price system is dependent on many other market institutions, most importantly money, property rights, and the freedom of contract. Understood as realized exchange ratios, mutually agreed by the exchanging parties, prices are as Mises (1949, p. 336) said, facts of the past (see also Ebeling, 1990, p. 189). For some economists, the term "price" applies only to cases where we have one equilibrium price, while in reality this does not happen even in the most basic of markets (see Kirman & Vignes, 1991 for references). Whether Hayek assumed equilibrium prices will be discussed later.

Market prices reflect that in the past the buyer and the seller were both willing to execute the trade at the negotiated price, with both parties considering the deal to be agreeable for individual reasons. Market prices themselves only reflect that both the buyer and the seller found the terms of trade agreeable, not their reasons for it. Market prices are a result of the interaction between scarcity and desirability, and report changes in either of these two factors (McQuade, 2007, p. 75).

What we now understand as the core thesis of *The Sensory Order* is that "[e]very sensation, even the 'purest', must therefore be regarded as an interpretation of an event in the light of past experience of the individual or the species" (Hayek, 1952, p. 166). For convenience, we rewrite it as:

(1a) every sensation is interpreted in the light of past experience.

While this cannot cover all the insights provided by *The Sensory Order*, it should be more transparent to give a concrete perspective of what *The Sensory Order* can deliver to price theory. The core thesis of Hayek's knowledge papers (1937, 1945) was that the price system helps to convey and utilize knowledge of particular circumstances of time and place. We label that knowledge as "local knowledge" to mean that it consists of personal, rational beliefs.⁹ It refers to a wide variety of beliefs while at the same time stressing the diversity found in a society arising from the differences in individual circumstances. It does not imply that these beliefs only vary between different locations, but also between individuals in the "same" location. Price, on the other hand, can be understood as a special kind of sensation and local knowledge as a product of past experience, which also includes various inferred beliefs. Modifying (1a) by substituting

“sensation” with “price,” and “past experience” with “local knowledge,” we get that:

(1b) every price is interpreted in the light of local knowledge.

This assertion implies that no price is a pure “sensation” in the sense that it would inform all individuals similarly. The local knowledge base possessed by the individual provides a meaning for the price. Furthermore, it is this local knowledge that is used to justify beliefs and the actions that result from them. Thus, what the price implies varies between individuals, who may have different local knowledge. There is no unique meaning attached to the price itself and no unique response to a change in the price. Strictly speaking, it is not only the price that requires interpretation, but the conditions and intentionalities of men as reflected in prices (Ebeling, 1990, p. 187).

It is fairly obvious that a price without context is only a numerical value, which does not reflect anything (Lavoie, 1985, p. 80; Ebeling, 1990, p. 187). Prices by themselves are nothing but information that is the product of interacting individuals, requiring an individual who will incorporate it into a framework to become knowledge (Holcombe, 2003; Ebeling, 1995, p. 143). While prices help to compare different options, such comparison is impossible if nothing is known about the options themselves. Receiving information that the price of x was p does not say anything more than that if nothing is known about x . Furthermore, since all prices are relative prices, we need to know how x is related to the other options that we have at our disposal. As such, prices do not convey any particular knowledge from an individual to another per se (Velthuis, 2004), since what is known by i is not automatically known by j through the price system. Knowledge is thus not copied from one agent to another as economists’ models usually presume (Lavoie, 1990, p. 361), but rather when interpreted from the standpoint of local knowledge, prices reveal more precise knowledge of the differences between time and place, even though this knowledge varies among individuals. Therefore, in contrast to the signaling theory, prices reflect many different meanings simultaneously (Velthuis, 2004, p. 382).

Another insight provided by *The Sensory Order* concerns the adaptive nature of the order. This sentiment can be summed up as:

(2a) a sensation may bring forward a need of reclassification.

The sensory order is not only used to interpret incoming sensations, but is also subject to change. This happens when the new sensation does not easily

fit into the existing classification. We can transfer this idea to the manner in which prices inform individuals:

(2b) a price may bring forward a need of belief revision.

This proposition attempts to stress the fact that prices (like any other type of information) do not lead to the accumulation of beliefs only. When interpreted based on the individual's belief set, prices may reveal that the set is no longer coherent. Some beliefs may need to be discarded or new ones adopted in this process, but exactly what happens depends on the local knowledge at hand. Since local knowledge varies between individuals, so do the results of belief revision. Usually some beliefs have less epistemic value than others, and by making minimal changes the individual can attempt to restore the coherence.

Kirzner (1985) typically stresses the entrepreneurial function of market prices. This means that prices provide incentives for entrepreneurs to discover new profit opportunities. It will be argued here that this feature of the informational role of prices results in response to certain circumstances. What is outlined here is an overall picture of the informational role of the price system, including entrepreneurial alertness as elaborated by Kirzner.

When an individual becomes informed of a new price, he compares that to his belief set. If the price is consistent with those other beliefs, he will do nothing other than adapt his behavior to the changed circumstances. He has, however, become informed of the changes and his reaction will restore, in part, the overall coordination of the economy. If, the price seems to be inconsistent with what the individual believes, he may come to recognize this as a signal of unexploited profit opportunities. Thus, when compared to an individual belief set, prices can spontaneously reveal profit opportunities without requiring a deliberate search. This is not the only option, however. Discovering such inconsistencies can entice the individual to reconsider other beliefs that he has about the situation. Prices can reveal gaps in individual knowledge, for example, if beliefs about marginal costs or utilities seem to be inconsistent with the observed prices. Furthermore, this may lead individuals to search for additional knowledge in an effort to fill these gaps. While reconsidering their beliefs to retain coherency, individuals may also generate completely original knowledge that others with necessarily dissimilar local knowledge have not.

As Hayek (1937, p. 49) observed, the division of labor is connected to the division of knowledge. At the initial stage individuals only know of their own preferences and skills, resulting in a division of labor that is bound to be disappointing. But gradually, when parties begin to negotiate through the

exchange of goods and services, the situation improves as others are able to observe the prices that the goods and services were exchanged at. The prices alone tell little of the local circumstances of the individuals involved in the transaction, but by comparing the prices to their own local knowledge, everyone gradually becomes aware of where their comparative advantage in the group lies. Hence the primary role of the price system may not be conveying tacit knowledge (what cannot be told) but implicit knowledge (what is not told) that is implied by the local knowledge.

Finally, a comprehensive theory of the informational role of prices requires that it presents prices “as signals influenced by choice” and “as signals influencing choice” (Lavoie, 1985). Thus, prices are both inputs and outputs of market action. If not inputs, prices would not inform anyone. If not outputs, no knowledge would get into the prices. This assessment supports the market process view of the price system.

HAYEK INTERPRETATIONS

As many Austrians (e.g., Kirzner, 1992, Ch. 8; Thomsen, 1992) have noted, Hayek’s 1945 article has raised many interpretations that are less than faithful to Hayek’s original meaning. This is partly because of Hayek himself, as his views on the market system gradually began to distance themselves from those of mainstream economists. The following is an analysis of these interpretations from the point of view of Hayek’s own writings and our contention that every price is interpreted in light of the local knowledge at hand. Thomsen (1992) has dealt with a variety of interpretations by explaining that there are several different informational roles that the price system performs. Here we will attempt to unify the informational role of prices with the above-mentioned contentions.

Mainstream economists (e.g., Grossman, 1981) usually credit Hayek with saying that:

(3) prices aggregate all information.

For example, Dasgupta (1980, p. 115) says that Hayek argued that prices make all private information public. For Hayek, all that a buyer or seller needs to know is summarized by a single number (McAfee & McMillan, 1987, pp. 699–700, p. 721). These assertions appear to suggest Hayek advocates the efficient market hypothesis, according to which prices always fully reflect all available information (Fama, 1970).¹⁰ Hayek, however, did

not argue that prices convey all information, but rather the relevant information that individuals need in addition to their local knowledge.

Hayek (1937, pp. 48–49) spends some time pondering what “relevant knowledge” means, but later (Hayek, 1945, pp. 525–526) concludes that it is the relative use and production of particular things outside these local circumstances, not the causes of them, which one needs to be aware of. This addition to local knowledge is sufficient for the coordination of market activities. Grossman, Stiglitz, and others, however, mistakenly confuse the situation in which prices convey relevant knowledge to one where the individual infers all information from market prices (Thomsen, 1992, p. 40).

Hayek (1937, pp. 49–50) also clearly states that too much informational value should not be placed on the price system alone, which is worth quoting in full:

It has become customary among economists to stress only the need of knowledge of prices, apparently because—as a consequence of the confusions between objective and subjective data—the complete knowledge of the objective facts was taken for granted. In recent times even the knowledge of current prices has been taken so much for granted that the only connection in which the question of knowledge has been regarded as problematic has been the anticipation of future prices. But, as I have already indicated at the beginning, price expectations and even the knowledge of current prices are only a very small section of the problem of knowledge as I see it.

- (4) if (3) is true, there is no need to search or to produce additional information.

This is the conclusion that Dasgupta (1980, p. 115) draws from premise (3) above. While it applies to the efficient market hypothesis as well as to rational expectation models (Radner, 1979, p. 656), it is less effective in comparison to Hayek’s price theory (as presented here). First of all, as noted earlier, prices have no meaning and hence no informational role without local knowledge. Second, one does not need to search for this knowledge, which one already possesses. This knowledge is not technical knowledge, but rather market knowledge, which is not produced through R&D activities as Dasgupta (1980, p. 116) argues. Rational expectations theorists have struggled with the problem of why one’s own information would matter if prices automatically reveal all information possessed by everyone (Hellwig, 1980). This is due to their assumption that prices can yield all the information required without local knowledge playing any role in the first place. Prices as such are considered to be (using our terminology) pure and all-revealing “sensations.” It is not that prices are assumed to give

individuals relevant knowledge, but rather that prices alone are able to give them all the knowledge they require.

In his review of the development of economics of information, Stiglitz (2000) credits Hayek for understanding how prices convey information about scarcity. However, he points out the interdependence between prices and quality (see also Stiglitz, 1987). Stiglitz claims that Hayek did not take notice of this fact, which to some degree undermines his contribution. When in some situations prices are not able to fully reflect the quality of the product, they do not allocate the goods efficiently, thus failing to deal with scarcity. The conclusion is that (3) is not true since:

(5) prices are only about scarcity and fail to include information on quality.

While it is true that Hayek did not consider this situation in the knowledge papers,¹¹ it is less problematic than Stiglitz assumes it to be. The asymmetric information literature begins with a premise that is very descriptive in many situations: producers or sellers know the quality of their products better than their customers. After that, however, they include a second, much less realistic premise: buyers know the overall distribution of quality, but not the quality of any given product. No explanation is offered as to how the individuals have acquired the knowledge of the parameters of the equilibrium probability distribution (Frydman, 1982, p. 653) or the parameters of the economy's structure, which are the answers to the questions that economists have debated for more than 200 years (Bellante & Garrison, 1988, pp. 213–214).

Hayek's concern was that while individuals are familiar with their local circumstances, they should also take into account circumstances elsewhere. That is, they are more likely to know the quality or other attributes of the products that they are actually exchanging locally than of those that exist elsewhere. Of course, when they do not know any differently, the price can be used to form beliefs about the quality of the product. Nonetheless, when individuals' tastes differ, information on how others value a product is not as sufficient as when individuals have different perceptions about the quality. As such, there is no single parameter of quality that everyone can adhere to.

If no buyer has the necessary local knowledge that can be conveyed through the price system, there is no knowledge that prices could convey. This should not be seen as a failure of the price system since prices are not established in such situations, as is the case for higher quality automobiles in Akerlof's (1970) lemon markets. Such cases are of course quite valid on their own grounds, but they confirm rather than disqualify Hayek's price theory.

Prices have no value if there is no local knowledge to be used to interpret them. Announced offers have no informative value for the buyers if there is no local knowledge that can be used to evaluate them. In such a case, the market price will not be established. If buyers do not know and are not ever able to learn the quality of a product, it is clear that such nonexistent knowledge cannot be conveyed through the price system. If, in addition, one assumes that there is no entrepreneurship or institutions such as reputation or warranties, sellers are neither able to use nor convey their local knowledge. The core message of Akerlof's findings should then be how important these factors really are.

A common complaint by Austrians is that mainstream economists miss Hayek's point when they do not take into consideration that he was talking about disequilibrium prices. While the issue of equilibrium versus disequilibrium prices has not been addressed by Stiglitz, Grossman, or others, the lack of acknowledgement strongly indicates that they assume that:

(6) Hayek implied equilibrium prices.

Admittedly, this assumption is easily made. Throughout his career, Hayek gradually distanced himself from Walrasian equilibrium economics and moved toward a market process view of the economy (Desai, 1994). By the 1930s and 1940s, the differences between Austrian and neoclassical economics only really began to reveal themselves.

Hayek's much cited tin market example (1945, p. 526) is very much a description of an equilibrium price change. In this example, an exogenous shock causes the price of tin to rise and all the individuals within the economy to change their consumption and production patterns according to the new equilibrium price.¹² Following the example, Hayek makes a reference to the law of one price. While he considers that transportation costs create price differences, he does not make any reference to actual arbitrage opportunities or how they could be eliminated.

However, Hayek also says that these adjustments are less perfect than equilibrium analysis imply, and that such standards should not be used when analyzing the efficiency of the market system (see also, Demsetz, 1969). Moreover, it should be remembered that Hayek's knowledge papers are in part critical contentions toward equilibrium analysis.

Disequilibrium states are important for the informational role of the price system,¹³ but not because of arbitrage opportunities alone. As discussed earlier, because a given price is interpreted by individuals based on their local knowledge, no single pure interpretation emerges that prices would convey. When two individuals have negotiated on a price that is based on

their local knowledge, others interpret this based on what they know. If what they know is coherent with the observed price, and they in a sense “agree with it,” only then can they adjust their consumption and production patterns to it. If no individual believes to know better than the status quo, then the current aggregate knowledge is already incorporated in the price. If on the other hand, someone believes to know better and “disagrees with the price,” they will attempt to benefit from this superior knowledge.

Noticing an arbitrage opportunity is an example of a case where an individual’s belief about a situation is inconsistent with what is implied by the price. However, noticing such an inconsistency does not always imply that the individual believes that there are arbitrage opportunities present. It can be that the inconsistency arises from false beliefs and observed prices can induce the individual to revise some of his beliefs to retain coherence.

In this respect, Hayek’s insistence on the subjective nature of knowledge is also important. In his tin example, the rising (or conversely falling) market price for tin might not be due to actual economic changes but rather mistaken beliefs about them. In addition to false beliefs, the seller could also be taking advantage of private knowledge, and thus deliberately sending false signals (McAfee & McMillan, 1987, p. 733). But if someone believed this to be the case, they would surely try to exploit such a discrepancy. Since individuals are able to remember past prices, they generally do not accept any price change as given (Cortés & Rizzello, 2007, p. 102). The price system ensures that it is sufficient if only one individual has the relevant knowledge. Through the price system, this local knowledge is taken into account, reducing the need for everyone to know the same thing. Thus, in the course of a market process, both errors of overoptimism and errors of overpessimism would be corrected (Kirzner, 1963; Leeson, Coyne, & Boettke, 2006).

This can be compared with Grossman and Stiglitz’s (1976) claims that prices are only important when knowledge is costly, and that costly knowledge will produce noisy prices, thus hampering the efficiency of the price system. Prices could be noisy or plainly false, but in such a case someone would observe this and the prices would be gradually corrected. Grossman (1976) asserts that in a noisy price system individuals would also want to know the reason for the price change. Surely, if a producer can offer a lower price than the competitors, this would entice others to find out how this has been achieved. Higher prices though would not suffice if other producers do not already see the need for raising prices. Individuals are not completely ignorant, but have and need to have their local knowledge to interpret prices.

Grossman and Stiglitz's (1976, p. 252) conclusion that a central planner with all the information could improve on what noisy prices do is a clear misunderstanding of Hayek's statement. How is the central planner supposed to have all the information? Prices set by one party (a monopoly or a central planner) only convey, at best, the trade-offs seen by that party. This will lead to distortions in the economy, as depicted by the Austrian business cycle theory, that are not easily corrected, whereas the market provides an opportunity to take into account all beliefs considering all local circumstances.

CONCLUSION

The impetus of this study was to examine whether *The Sensory Order* can be useful to form a more comprehensive theory of the informational role of prices. The key insight that Hayek's theory of the mind provides was taken to be the claim that every sensation is interpreted in the light of past experience. Thus, there are no pure sensations, as every sensation is classified according to its similarity with previous sense data. The classification is not immutable, however, as new sensations might signal the need for reclassification.

Economic decision-making also involves other beliefs beyond our immediate sensory experience. Thus, there is a need to extend the theory to include nonsensory beliefs. It was proposed that a convenient way to do this would be to consider rational beliefs that are justified by the coherence theory. Coherence, as a complex form of classification, explains how individuals interpret and judge information based on their belief set. To retain the coherence of a set, an individual is constantly modifying it in the light of new evidence.

Hayek's contention was that the market economy is able to overcome the knowledge problem because prices convey to individuals the relevant knowledge that they require in addition to their local knowledge. By applying the assertion of *The Sensory Order* to the informational role of prices, we concluded that every price is interpreted in the light of local knowledge. Therefore, prices need local knowledge since only this can give them meaning. Since local knowledge varies between individuals, so does the manner in which individuals interpret prices. This explains how prices can convey different knowledge to different individuals and how acting on their beliefs they can convey this knowledge through the price system.

Many interpret Hayek as saying that prices convey all the information. This is to overlook the importance of local knowledge, for which prices are

not a substitution. It is as a result of local knowledge that prices convey any knowledge at all. Since local knowledge is something that the individual already possesses, there is no need to search for it or produce it through R&D. Observed prices, however, can reveal gaps in local knowledge, requiring an individual to search for additional information. An individual can sometimes lack the knowledge of the quality of a good and may use prices as a source of knowledge. Thus, as individuals become more knowledgeable, someone might discover the quality, and by acting on these beliefs convey this information to the price.

Rather than an equilibrium theory, Hayek's price theory supports a market process view of the price system. This is because individuals can only evaluate the price in the framework of their belief set. Through prices, individuals are able to take into account factors outside their realm of expertise. Sometimes prices require individuals to revise their beliefs, or to exploit observed discrepancies in the market. Equilibrium prices only refer to a theoretical situation where there are no inconsistencies and deviations between individuals' belief sets. However, it still requires an explanation of the process that brings it about. In every case, what knowledge prices reveal and how individuals act upon them ultimately depends on their local knowledge.

NOTES

1. Hayek considered both, that is, his knowledge papers together with his other articles collected in Hayek (1949) as well as *The Sensory Order*, as his most original contributions (Hayek, 1994, p. 79, p. 138).

2. Hayek later mentions the informational role of the price system in, at least, Hayek (1988, Chapters 5 and 6). The earliest indication of his ideas on the matter, before "Economics and Knowledge," can be found in his 1928 paper on intertemporal price equilibrium (Desai, 1994, p. 27). While out of the scope of this study, especially Hayek (1978) is relevant for his later theorizing of the market process.

3. Also Lachmann (1976, p. 55) uses the same term, if a bit more narrowly. This term also separates it from "economic knowledge," which has stabilized its meaning as knowledge possessed by economists. Hayek (1937, p. 33), though, observes that the two, market and economic knowledge, are not unconnected.

4. "It is important to remember that the so-called 'data', from which we set out in this sort of analysis, are (apart from his tastes) all facts given to the person in question, the things as they are known to (or believed by) him to exist, and not in any sense objective facts" Hayek (1937, p. 36).

5. Hayek (1952, p. 132) himself is willing to apply his theory to both conscious and unconscious events, but the room for conscious reasoning seems to be minimal, if not nonexistent. See, however, Butois and Koppl (2007, pp. 32–34).

6. There are also further complications with the mind-market analogy (Smith, 1997, p. 24).

7. Butos and Koppl (1993, 1999), who use Hayek's theory of mind to generate a theory of expectations, also see coherence as a property of Hayekian expectations.

8. While the traditional definition was challenged by Gettier (1963), no universally agreed definition has arisen to meet the challenge. While Gettier showed that justified true belief is not always sufficient for knowledge, there is no good reason not to retain them as necessary properties, however.

9. Some knowledge can, of course, be common to everyone, but that does not need to be conveyed by definition.

10. It is unclear how Grossman and Stiglitz (1980, p. 404) identify Hayek as compatible with efficient market theorists.

11. Elsewhere, Hayek (1949, pp. 92–106), however, discusses the economists' fiction of assuming homogenous commodities. The necessary differences between commodities and services are far from defects of competition, but should be understood as dynamic benefits of competition.

12. Thomsen (1992, pp. 59–60) reviews the example from a market process angle.

13. As Fisher (1983, p. 7) has noted, equilibrium economists should not avoid disequilibrium analysis, because "[i]f 'equilibrium' is to have any substantive meaning, one must be willing to countenance the possibility of encountering disequilibrium states." In addition, utterly important for equilibrium analysis is not to assume such factors constant, which are supposed to be the equilibrating forces (prices, beliefs, etc.).

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THE SENSORY ORDER AND ORGANIZATIONAL LEARNING

Steven Horwitz

ABSTRACT

Purpose – To show how Hayek’s work on cognitive theory can inform our understanding of the operation of the firm, more specifically the ability of firms to engage in organizational learning.

Methodology/approach – After reviewing Hayek’s cognitive theory, the paper provides a brief overview of modern organization theory, focusing on the post-Coasean literature and the “resource-based view” of the firm specifically. I then offer several analogies between the two theories, arguing that the “map” and “model” of Hayek’s work on cognition can be analogized to the balance sheet and the current budget/business plan of the firm.

Findings – I find that the “map” and “model” of Hayek’s work on cognition can be analogized to the balance sheet and the current budget/business plan of the firm. The feedback between the model’s encounters with the external world and the map’s structuring of the classificatory process of the mind parallels the way in which firms’ success or failure feedback to the evaluation of their assets. The paper also discusses the way in which the competitiveness of the individual’s and the firm’s environment might matter for the speed and effectiveness of learning processes.

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Research limitations/implications – *I conclude with some suggestions for further research, focusing on questions of political economy and how a more politicized marketplace might undermine organizational learning when seen through the lens of Hayek's work. Specifically, the paper argues for comparative explorations of different organizational structures in similar institutional environments and similar structures in different institutional environments to see which sort of factors matter for organization learning. One implication of the paper is that less competitive environments should demonstrate weaker and slower organizational learning.*

Originality/value of paper – *Hayek's work on cognitive theory has never been applied to organizational learning in any sustained way before. In doing so, this paper draws out some original research questions and political economy implications.*

"Many of the individual and unique features of a particular corporation which make for its success are of the same character as the similar features of an individual person; they exist largely as an intangible tradition of an approach to problems, based on a tradition which is handed on but ever changing, and which, though it may secure superiority for long periods, may be challenged at any time by a new and even more effective corporate personality" (Hayek, 1967 [1959], p. 288).

Over the last decade or so, scholarly examination of Hayek's work has increasingly focused on the role played by *The Sensory Order* (1952) in his overall theoretical framework. One key strand of this work is that Hayek's vision of the operation of the mind was that it was a spontaneous order, specifically that it could be understood as a rule-structured, ends-independent, complex, emergent order that arose from the more simple biological connections of the human brain. Because Hayek's first work on the mind was a student paper in the early 1920s, only to be revisited in the late 1940s in the writing of *The Sensory Order*, some have argued that his vision of the mind as spontaneous order was perhaps in some ways always present as a guiding idea for his work in economics and then his later development of spontaneous order theory more specifically.

There is another way to see the place of *The Sensory Order* in Hayek's work. Rather than mind being central, the concept of spontaneous order was always there, but frequently unarticulated until his work in the 1950s. Instead, Hayek proceeded by discovering in various applications analogous forms of argument and examples of spontaneous order, all of which interested him even if he could not precisely identify what they all had in

common. In fact, in an essay of his in which he describes his own approach to his work, he says of people like himself: “many of their particular ideas in different fields may spring from some more general conception of which they are themselves not aware, but which, like the similarity of their approach to the separate issues, they may much later discover with much surprise” (Hayek, 1978 [1975], p. 54). It would not be surprising to find in Hayek a number of parallel or analogous approaches to objects of inquiry, especially before he could develop a common framework for understanding them (e.g., Horwitz, 2008, on Hayek’s theories of mind and capital).

Because Hayek’s theory of mind is essentially a theory of learning, it can be analogized and applied across a wide range of phenomena. As the head quote of this paper indicates, Hayek glimpsed some ways in which the firm could be understood as a process of learning. In such a firm, ways of doing things and structures would come to take the form of a knowledge-containing tradition of approaches to problems that could define the firm much as similar processes define individual people. Written in the late 1950s, this notion of the firm as an organization that learns through repeated practices and processes was ahead of its time. A great deal of work on the theory of the firm in the 50 years since, and especially in the last few decades, has focused on the way in which firms learn and embed routines into their organizational operations. With the even faster pace of economic change that characterizes our own time, the ability to learn and adjust quickly to new circumstances, or to apply existing structures and processes to new areas, has become a defining feature of successful firms.

In this paper, I want to explore more deeply the ways in which Hayek’s theory of mind and its corresponding theory of learning might relate to more recent work on organizational learning and the theory of the firm. The latter literature is far too vast for me to give it the treatment it deserves, but I will try to draw from some canonical sources, as well as recent extensions of those ideas, that give an accurate flavor of the work in question. My goal is mostly to be suggestive. Organizational theorists have not, to my knowledge, taken a serious look at Hayek’s argument in *The Sensory Order*, despite the fact that they are aware of Hayek’s more narrow economic contributions. My hope is that pointing out some of the relevant parallels will inspire them to do so. A more conscious application of Hayek’s work on mind and learning might provide a stronger foundation for many of the observations in the organizational learning literature as well as opening up new lines of inquiry.

The paper opens with a discussion of Hayek’s vision of mind and its implicit learning theory. That is followed by an excursion into the recent literature on organizational learning, which leads to several sections

exploring the analogy between minds and firms. The final substantive section takes a brief look at the effects of the institutional environment on the ability of organizations to learn.

THE IMPLICIT LEARNING THEORY OF *THE SENSORY ORDER*

Detailed summaries of the core ideas of Hayek's theory of mind as presented in *The Sensory Order* can be found in a number of places (e.g., McQuade & Butos, 2005; Steele, 2002; Horwitz, 2000; and Butos & Koppl, 1993), as well as contributions to this volume. For my purposes here, I will provide a very brief summary of the general approach of the book and then focus on the aspects of Hayek's theory that are relevant to the question of how individuals, and by analogy organizations, learn.

Hayek's self-imposed task is to explain how the mind creates a sensory order, that is, our subjective experience of the world, which is different from what we know comprises the physical order of the world based on our understanding of science. The world is made up of a jumble of atoms, molecules, waves, and particles, yet our mind somehow presents those to us as a largely orderly and organized set of perceptions. Hayek argues that the mind is primarily a process of classification, where incoming sensory data get sorted based on the organism's history. The mind "pre-classifies" incoming stimuli based on a built-up set of neural connections that are the result of both our biological inheritance and our unique individual interactions with the external world. These connections reflect classifications and associated behavior that have "worked" in forwarding the individual's ability to navigate the external world. In addition, this classification is contextual and multiple. How the mind classifies any particular input will depend on the other inputs that arrive with it, implying that there is no simple one-to-one relationship between any given stimulus and the sensory order the mind constructs. Rather, the sensory order is the emergent outcome of the particular pattern of stimuli being classified by a specific set of neural connections that, while having much in common with that of other individuals, is nonetheless unique to the individual in question.

As a way of understanding this process, Hayek uses the metaphors of "map" and "model." These two concepts are particularly important for understanding the theory of learning that is implied by Hayek's theory of mind. The "map" refers to the established neural connections that the mind has created over the course of the individual's lifetime. That is, it is the entire

“network of connections” among the various external stimuli the organism is processing and the “structure of external events which it can be said to reproduce” (Hayek, 1952, p. 109). The map is the set of neural and physical phenomena in the brain that are the product of the learned associations that the individual’s mind has built up over time. Important for our purposes is Hayek’s insistence that this is a “very imperfect map, but also a map which is subject to continuous although very gradual change” (p. 110). The map then is a sort of storehouse of historically constructed relationships that is the result of the interplay between the physical structure of the brain and the particular stimuli that comprise the individual’s life experience. The map is a reflection of the individual’s long-term learning about the nature of the external world.

The model draws from the map to represent the “particular environment in which the organism finds itself at the moment and which will enable it to take account of that environment in all of its movements” (Hayek, 1952, pp. 114–115). The map reflects the individual’s past learning, while the model is the application of that past learning to the present situation. Like any model, it is a less-than-complete representation of the current environment, but it attempts to capture, in a skeletal form, the basic outline of the situation so that it can guide behavior appropriately. The model can only be constructed from the elements of the map, hence our understanding of our current situation at any point in time is necessarily constrained by the physiology of the brain and our past experiences as an individual. Unlike the semi-permanent neural connections of the map, the model is far more dynamic as it will change as the environment changes and the organism draws upon other aspects of the map. It is a snapshot of the current neural “roads” that are being “traveled” as opposed to a depiction of the whole network of routes available. One might analogize it further to the difference between a chart depicting the patterns of traffic flow on a major city’s streets over the course of some long period of time and a graphic depiction of the actual flow of traffic through the mapped streets at the current moment, or a radio or GPS traffic report.¹

This analogy points out another important difference between the map and model. As we have noted, the map is a product of the organism’s historical learning and experience. The model, however, is what guides the individual’s behavior in the moment. It is “forward looking.” Of course, the ability of the model to help the actor “look forward” is dependent on its relationship with the map, which reflects what the actor has learned in the past. However, the map itself provides no direct guidance for action. Knowing the history of the traffic flow cannot by itself provide sufficient guidance if you wish to navigate the streets right now precisely because it is

the product of experience only. The model of the current flow is what you need in order to navigate successfully, and you need a depiction of the specific areas in which you plan to travel. Of course, the model of the current flow can only draw from the same set of streets that comprise the historical map, but it is also true that the information from the model feeds back to the map. In the case of our traffic analogy, the current state of traffic flow will be added to the data of the historical map and perhaps, if enough change accumulates over time, change the frequencies of that long-run map in significant ways.

The model is best understood as a set of expectations about the future that guide the organism's immediate behavior. It is "predictive" in an "if-then" sense, as all models aspire to be. Hayek (1952, p. 121) explains this point in further depth:

The representation of the existing situation in fact cannot be separated from, and has no significance apart from, the representation of the consequences to which it is likely to lead. Even on a pre-conscious level the organism must live as much in a world of expectation as in a world of "fact," and most responses to a given stimulus are probably determined only via fairly complex processes of "trying out" on the model the effects to be expected from alternative courses of action. The reaction to a stimulus thus frequently implies an anticipation of the consequences to be expected from it.

The process by which the organism "tries out" various possibilities on the model often happens, of course, at the pre-conscious level. But even on the conscious level, humans might choose to imagine alternative courses of action based upon their current understanding of the world. Such imaginings are also about the consequences to be expected from the various alternatives. In either the pre-conscious or conscious case, the results of the actions taken comprise a new set of stimuli to be classified by the mind.

What we are describing here is the organism's learning process. As evolutionary psychologists have argued, the human brain as a physical organ already has some degree of "preset" neural connections. Some portion of the Hayekian "map" is therefore part of our biological heritage. However, the rest of it is the product of the learning that is the product of the individual's interaction with the external world. As the model is validated or invalidated by successful or unsuccessful navigation of that external world, the results are part of the historical experience that is reflected in the map. Hayek notes that the map is "semi-permanent," meaning that those neural connections can change as the individual learns and grows. If Hayek's views are correct, and modern neuroscience more or less suggests he was on the right track (e.g., Fuster, 1999; Edelman, 1987), then human learning "redraws our maps" and thereby affects the model through which we act in the world. Hayek's work

here refers to the sort of experiential learning involved with success and failure of action in the world.² This also ties nicely with evolutionary approaches to the mind, which argue that our “mental systems” were forged in the long evolutionary process that produced *Homo sapiens*.

We can also tie Hayek’s cognitive theory and its associated theory of learning to a very general understanding of what we mean by a “personality.” If the sensory order is largely a set of rules by which the brain in the form of the mind mediates between external stimuli as inputs and human behavior as outputs, it seems fair to say that humans are essentially “rule-following animals.” This is true even if, as with the mind, we cannot articulate all of the rules that we follow.³ Who we are, and what our personality is, is in some sense nothing more than the collection of rules by which we classify the world and act within it. This does not imply a kind of crude biological reductionism, as we can learn and we can, at least on the margins, change some of those rules, both intentionally and through experience. It is, however, a recognition that humans are rule-followers and that often those rules are ones we are not aware of as they operate on us. Self-reflection and self-criticism are thus the essence of our humanness as we try to understand and change the rules we do not like, to the extent we can. This differentiates us from other species for whom learning can take place, but only through experience, rather than by self-reflection as well. The set of often tacit mental rules that comprise our identity are what Hayek refers to in the head quote as the “intangible tradition of an approach to problems” that characterize both individuals and corporations.

THE FIRM AS A COLLECTION OF ROUTINES REFLECTING ORGANIZATIONAL LEARNING

As noted in the introduction, over the last few decades economists have developed a variety of new ways of looking at the firm. For much of the 20th century, the firm was conceptualized as a production function that translated inputs into outputs subject to constrained maximization of profits. This way of thinking treated the inner workings of the firm as a “black box” and took little notice of how that view of the firm did not correspond with the real-world competitive environment in which firms actually operated. The “firm-as-production-function” approach certainly allowed for mathematical tractability, particularly in the context of the ongoing growth of general equilibrium theory. However, the static environment necessary for the mathematics created enormous challenges for applying the theory to a world

of disequilibrium, uncertainty, and changing knowledge, and, by implication, how firms and other organizations might learn.

Moving away from the black box view required that firms be understood as social institutions that solve specific kinds of organizational problems. For early contributions, such as Coase (1937), this meant that the institution of the firm could be seen as a lower-cost option for coordination than using the market. Bringing the direction of resources under the conscious control of the firm and its managers could reduce the information/transactions costs of using those resources to satisfy human wants. Coase's work, along with renewed interest in Schumpeter's (1961) work on the entrepreneur, prompted a reexamination of the inner workings of the firm. Work in this direction really begins with the contributions of Penrose (1959) and Richardson (1972). Penrose argued that the firm was best understood as a "bundle of resources," and she used this perspective to look at how firms changed and grew over time, with a particular emphasis on the inner organizational processes that forwarded such growth. As Mathews (2006, p. 74) points out, Penrose's approach enabled her to distinguish between the "entrepreneurial" role in gathering the distinct resources needed to pursue the firm's objectives and the "managerial" role in ensuring the most efficient use of those resources possible in service of those objectives. The so-called resource-based view (RBV) of the firm has emerged from this work.

The current RBV literature tends to focus on the ways in which a firm's distinct resources provide it with rents because those resources are specific to the firm in ways that prevent them from having the equivalent value to other producers. The goal for management is to amass a set of resources that fit together in ways that are highly effective in producing the desired output, but whose value is tied up with that particular firm's production process in ways that make their opportunity cost significantly lower than the value of their marginal product to that firm. Firms acquire long-run competitive advantages this way when opportunities for imitation by competitors have ceased, that is, when the market reaches equilibrium. As Foss and Ishikawa (2007, p. 750) describe it, the RBV is "a competitive equilibrium model with factor market imperfections." Implicit in the RBV approach is that actors have full relevant information about consumer demands and the value of their own resources. The challenge they face is the maximization under constraints problem that is at the core of modern microeconomics.

However, despite the insights that the RBV approach generates, its equilibrium orientation limits some of the questions it can ask about how organizations learn in the face of novelty, which is a key element of real world, as opposed to "perfect," competition. Infusing the RBV approach

with some disequilibrium foundations from the Austrian tradition can help to overcome these limits and enable us to explore issues of learning more richly. Of particular importance to doing so is the Austrian theory of capital, particularly in the hands of Lachmann (1978), building off of earlier work by Hayek (1941). Lachmann saw the firm as a set of complementary capital goods, where capital is understood as anything that contributes to human plans. In the Austrian view, capital goods are heterogeneous or have “multiple specificity.” Capital goods and human capital are neither specific to only one plan, nor so general as to serve equally well in all plans. This fits nicely with the RBV emphasis on the uniqueness of resources. However, the key difference is that the Austrians bring in the entrepreneur as the organizer of those capital combinations and see his task as attempting to peer through the fog of uncertainty that surrounds the production process (Kirzner, 1973). Rather than assembling capital goods of known value to make an output of known value, the entrepreneur is the one responsible for deciding *which* capital goods will be combined in complementary ways to produce *which* output (Lewin, 1999).

Entrepreneurs operate in an environment full of errors waiting to be corrected and opportunities for profit waiting to be exploited. They bring together the physical and human resources that they believe will “fit together” in a coherent plan that will produce the desired output. From this perspective, the firm exists as a way of “pulling together” a variety of entrepreneurial discoveries by all the parties to the contracts that define it (Sautet, 2000, p. 76). It is the entrepreneur who imagines how a group of resources might, when combined, bring into existence a set of capabilities not currently in place, and how such capabilities might enable the firm to exploit a market profit opportunity hitherto unexploited. Sautet (2000, p. 75) argues that the firm as a set of entrepreneurially envisioned contracts comes into existence in part:

to secure the capabilities of the inputs that will come into existence when they are used in common. The only way the entrepreneur-promoter can exploit his/her discovered opportunity is by the implementation of a firm.

Setting the RBV approach in a disequilibrium context allows us to see that the value of a firm’s resources can never be known with certainty and that it is the task of the entrepreneur to imagine ways of using undervalued resources to exploit noticed profit opportunities in the market.

The focus on entrepreneurship moves the firm beyond just being a set of resources, but that is not quite far enough. In the eyes of more recent work in strategic management, the firm can be seen as a set of *resources* in the way Penrose describes, a group of *activities* that the firm engages in, and a

collection of *routines* that link the resources to the activities (Mathews, 2006, p. 75). The concept of “routines” comes from the work of Nelson and Winter (1982), who explicitly tie the idea to disequilibrium approaches such as Schumpeter’s (1961). Routines are the various rules, procedures, behavioral patterns, and the like that define how a firm operates. As the name suggests, they grow from repetition, and can become increasingly effective as the firm evolves. Effective routines allow the firm to engage in increasingly complex tasks by, essentially, increasing their capabilities. Rather than just seeing the firm’s resources as a static collection of things, the emphasis on routines provides a “dynamic capabilities” perspective on the firm’s behavior (Teece, Pisano, & Shuen, 1997). The firm is not just a collection of resources, but an institution that can learn to improve its own operation so as to enhance the productivity of those resources.

In this alternative to the production function view, firms are seen as institutions that are created to solve knowledge problems and are engaged in an ongoing process of learning. They face an uncertain future where they must entrepreneurially anticipate what their customers want (their “activities”), imagine the physical and human capital necessary to meet those wants (their “resources”), and then develop linkages between their resources and activities through the evolution of routines. Those routines become the crystallization of the firm’s learning processes. The knowledge they both represent and create can be the learning of knowledge that is “out there” but unknown to the firm, or it can be genuinely creative and innovative discovery. Routines are knowledge formed into rules, procedures, and practices that have passed the market test.

THE ANALOGIES OF MINDS AND FIRMS

Minds and Firms as Learning Processes

As is probably clear from the preceding section, there are a number of fruitful analogies between Hayek’s theory of mind in *The Sensory Order* and the understanding of organizations and how they learn that is associated with the RBV view of the firm. Before exploring those analogies, it is worth pointing out some important differences between the two cases. The most important of those differences is the fairly obvious point that the learning that takes place in firms is all ultimately processed through the minds of human actors. Certainly the change that comes from the top down in the form of new organizational forms or changes of mission must be the conscious product of

entrepreneurs and managers. A good deal of learning comes “from below” as well, in the form of discoveries made by employees (what is often termed “intrapreneurship”). Though such discoveries are themselves often serendipitous, the choice to repeat them and the choices by others to imitate them such that they become part of the firm’s routines are the production of intentional human action. The final routines that emerge will likely be things that match no one person’s intentions exactly, but are nonetheless a product of various human actions (if not human designs).

The rules that comprise the mind are, by contrast, not always consciously chosen by the actor. As we saw in the discussion of *The Sensory Order*, some number of the rules that guide our behavior are the “hard-wired” products of evolution, while others come into being through our interaction with the world through learning processes that occur “automatically.” Neither of these is under the direct conscious control of the actor associated with the brain in question. Again, this is not to say that we can do nothing to change how we understand and act in the world. To the contrary – as we saw earlier, we do have the ability to change the rules by which we understand the world. However, that ability is limited, and certainly more limited than the analogous ability to change the way in which firms operate.

As Hayek has emphasized in other contexts, the very nature of spontaneous orders is such that their complexity prevents anyone from having a complete understanding of the details by which they operate. We are limited to offering “explanations of the principles” (Hayek, 1967 [1955]) by which spontaneous orders operate. The entire reason that certain institutions have the characteristics of spontaneous orders is that they could not achieve their degree of complexity without making use of such processes. The most obvious example of this point is evolution by natural selection. Rather than seeing the complexity of the products of nature as evidence for some sort of “design,” an understanding of spontaneous order and evolutionary theories more generally points to the fact that higher degrees of complexity require processes that can sort and filter quickly and effectively, and can thereby process more knowledge through shorthand than any supposed designer could.

Firms, by contrast, allow for more forms of conscious human intervention into their design and operation. In Hayek’s earliest work on the differences between spontaneous and constructed orders, the firm was one of his first examples of the latter, perhaps unsurprisingly given Coase’s dichotomization of the “firm” and the “market.” However, as our discussion of recent work in the theory of the firm illustrated, firms are now thought to have more elements of complexity and spontaneous order in them than the earlier dichotomies of Coase and Hayek would suggest. They are better understood

as a hybrid perhaps best described as being the product of conscious human design at the higher level of organization and mission, but more like a spontaneous order in the interstices of the detailed production processes that comprise them. For many of the same reasons that Hayek argued that spontaneous ordering processes are necessary for societies of a particular degree of complexity (or for a complex natural phenomenon such as the human mind), so would we expect that firms of a certain size would have to take advantage of local knowledge, rule-following, and individual initiative, rather than centrally directing every aspect of the firm's actions. Very large firms, such as Wal-Mart, do precisely this by creating a very powerful corporate culture that inculcates its values into every employee, which in turn allows senior management to trust that such employees can and will exercise individual initiative in ways that forward the firm's goals, both day-to-day and during times of crisis (Horwitz, 2009).

When such hybrid firms learn through their spontaneously ordered elements, the results of that learning can be consciously applied to the way they are structured and organized in a way that is much more difficult when we speak of individuals. The argument in this paper's head quote that a firm's "corporate personality" is always subject to challenge and capable of being increasingly "effective" should be read in this light. The analogies we draw between minds and firms are powerful, but like all analogies they do have their limits. In the next two sections, I explore two of those analogies in some detail.

Rules and Routines

The most obvious of the analogies between minds and firms and their learning processes is the parallel between the rules by which the sensory order operates and the routines that guide the operation of the firm. A firm's routines link its resources and activities, that is, they provide the rules by which its inputs/resources are converted into output-producing activities. To understand why some firms were more successful than others, it was soon found necessary to move beyond just examining their objective resources (including human capital) and the activities they decided to engage in. Attention began to be paid to the less visible ways in which those were connected through "approaches to problems," as Hayek put it, that effectively translated the resources into successful activities. The opening of the "black box" of the firm brought the role of routines into greater focus as an explanation of how firms do what they do.

The analogies between routines and the rules of classification by which brains and minds operate should be fairly clear. Mental rules link the sensory stimuli that come into the brain to the actions that the organism takes based on them. How the mind classifies those stimuli determines how the organism acts upon the external world. Like firms, people have “activities” or goals that they wish to achieve, whether conscious or automatic (e.g., breathing or other nonconscious activity), and they have “resources” that go into the process of achieving those goals in the form of their perceptions of the world around them. The mind’s rules of classification are, in essence, routines that guide the organism’s behavior by turning stimuli into action on the external world. The classification process is a matter of “faced with this set of sensory inputs, what is the best model of the current world and what does that imply about how to react, given the goal at hand?” Of course the process is not intentional in the sense that question suggests, but that does capture the mental classification process with some accuracy.

Firms process inputs to outputs in similar fashion. The particular constellation of inputs the firm has at any one point and the specific goal it has in mind will trigger a particular routine for turning those inputs into outputs. Such routines can also be understood as classification processes, as they are learned responses to how particular combinations of inputs, including human labor, should best be turned into effective (i.e., profitable) outputs. How a specific input will get used will depend on the context it appears in: what are the other inputs with it and what is the output the firm wishes to produce? Thus, just as with the sensory order there is no one-to-one correspondence between a stimulus and a response because the appropriate response depends upon the context in which the stimulus arrives, within the firm there is not necessarily a one-to-one relationship between any specific input and any specific output.

Although the opening quote from Hayek points out the tacit nature of both the rules that guide individuals and the routines that guide firms, neither is exempt from the gaze of critical assessment. Individuals frequently strive to improve themselves through introspective processes that attempt to figure out (sometimes with the help of a professional) why they have developed the rules that they have and how they might change them. Firms can and do engage in a similar process. One can think of business process documentation as a way of trying to articulate a piece of a firm’s operation and subject it to organizational introspection to see if it can be done better. Sometimes improvements in routines happen serendipitously as noted above, but they can also be the result of management creating an environment in which routines are subject to collective examination with

the goal of improving them. What a firm cannot do is to throw everything out the window and start completely from scratch. It must always start from a set of practices that are part of the “intangible tradition” that Hayek noted. Organizational routines, like mental rules, are not beyond the reach of intentional human action.

Maps and Models; Balance Sheets and Budgets

Another way to think about the parallels between firms and the mind is to analogize the map and model to the balance sheet and budget/business plan, respectively. As we saw earlier, the map, for Hayek, represents the semi-permanent neural connections that the brain builds up through experience. The map is, to a significant degree, a product of the individual’s history, in combination, of course, with the physical structure of the brain and its evolved capacities. Although it is “semi”-permanent, it does change as the individual’s experiences accumulate in ways that provide evidence that the established ways of doing things are no longer successful.

The firm’s balance sheet functions in a similar way. It would not be accurate to describe a balance sheet as “semi-permanent;” however, the balance sheet is a periodic snapshot of the deployment of resources within the firm. In that sense, it is not something the firm totals up minute to minute, but rather it is a tool to be looked at over somewhat longer runs of time to examine the firm’s effectiveness and potential. The value of the firm’s current assets, liabilities, and net worth represent both the inherited situation at the start of the period under examination as well as the results of the actions taken during that period. A year-end balance sheet will reflect both the value of what the firm started with and the value it added or subtracted over the year. Like the sensory order’s map, it represents the “built-up” set of relationships among the firm’s assets and liabilities. Hayek (1952, p. 115) describes the map as representing “the kind of world in which the organism has existed in the past...it provides by itself no information about the particular environment in which the organism is placed at the moment.” As firms attempt to evaluate their profitability using the balance sheet, they are describing the world in which the firm has existed in the past. By itself, it tells us nothing about how to move forward, although, like the map does for the model, it serves as the backdrop for constructing the more forward-looking budgets and business plans.

The budget for the next period, and the business plan that accompanies it, is the analogy of the model in Hayek’s theory of cognition. More so than the

balance sheet, the budget and business plan are constructed to be forward looking, reflecting the firm's expectations of how the next period will unfold. Much as the model represents those parts of the map that are "in play" in the current environment, so does the budget/business plan represent the uses to which the items in the balance sheet will be put in the relevant period and the environment in which it will unfold. Like the individual, the firm does live in world of expectations, where its behavior is guided by budgets and plans that reflect historical learning that has been translated into a set of implicit predictions about how inputs will be transformed into outputs. What Hayek (1952, p. 121) says about the use of the model in the mind has clear parallels to the way in which firms behave: "most responses to a given stimulus are probably determined only via fairly complex processes of 'trying out' on the model the effects to be expected from alternative courses of action." The mind, as Hayek also notes, performs these processes "pre-consciously," but within the firm they are the ongoing conscious decisions of managers and employees. Managers and entrepreneurs now frequently speak of "budgeting models" and will often create a variety of models to "try out" before deciding how to proceed.

For example, if the price of an input changes (a new stimulus), the firm's managers are likely to be guided in determining what, if any, action to take by assessing the effects of that price change on the current budget and business plan. They are likely to do this by exploring possible substitutes for the more costly input and seeing whether the substitutes will work in complementary ways with the other inputs. In other words, the firm will "try out on the model the effects to be expected from alternative courses of action." The results of this process of "trying out" will inform the firm's final decision. Hayek (1952, p. 121) says of the mind, "The reaction to a stimulus thus frequently implies an anticipation of the consequences to be expected from it." This seems to be how firms react to new "stimuli" as well.

Even the map and the model do not get us all the way to the most basic level of the sensory order, which is the physical process of exciting neurons. The map and model remain metaphors for the underlying physical processes. Can we find an analogy to the excitement of neural connections in the operation of the firm? Perhaps the closest we can get is a sort of tracking of the firm's cashflow on a daily basis. Where are the funds coming in and where are they going out? Which products are customers buying and which inputs need to be replenished by the firm? What prices are being paid for all of these? The cashflow is the most immediate indicator of which places within the firm's budget and business plan are being activated at any given moment,

or over the very short run of a day, for example. To a large degree, what “trying things out” via the model leads the firm to do is to adjust its own spending or to perhaps alter the prices and quantities of its outputs, both of which change the cashflow, just as the action an organism takes requires that certain neural connections be activated.

Finally, the map and model relationship can also be seen by the ways in which the actions that emerge from trying things out through the model eventually feed back to the map. Like the semi-permanence of the map, the snapshot reflected in the balance sheet is open to change as the firm acts through time. As cashflows change and actions guided by the budget and business plan are taken, the value of the assets and liabilities captured by the balance sheet change as well. Although within the firm the snapshot might only be taken quarterly or semi-annually or annually, the movement captured by that still photograph is ongoing. The moments at which they are taken represent the firm’s past at those moments, where that past is informed by the outcomes of the choices firms make on an ongoing basis. At the moments reflected in the balance sheet, the firm uses that past to inform its future, just as in the dialectical relationship between the map and model. At the highest level, this is how the firm learns. It watches the success of its expectations as embedded in a model, which in turn informs the value of its capital, which in further turn informs the next round of action.

If we bring our earlier discussion of routines to bear on these issues, we get a more complete picture of how Hayek’s cognitive theory might serve as a template for organizational learning. The dialectic learning process that is represented in the relationship between the balance sheet (map) and the budget (model) is what informs the development of the routines (rules) that guide moment to moment behavior in the firm (individual). Good organizational routines are both the cause and effect of successful behavior by the firm. Patterns and rules that connect resources and activities well, that is, staying within budget and carrying out the current business plan, will generate positive results on the balance sheet and thus be imitated and replicated. If the current budget/plan is not successful, and is so reflected in the balance sheet, then existing routines will need to be critically assessed (as may, perhaps over a longer run, the firm’s resources and activities). Humans who continue to find their attempts to navigate the world to be unsuccessful will eventually have to realize that the rules that guide their behavior are in error. Firms have the same relationship to their routines, assuming that the institutional environment is one in which the incentive structure rewards such learning.

ENVIRONMENTAL EFFECTS ON ORGANIZATIONAL LEARNING

All of these considerations about organizational learning need to be placed in one other explanatory context. For firms, the competitiveness of their economic environment will be a key factor in how quickly and effectively they learn, where “effectively” refers to doing a better job at linking their resources and activities. If we assume that the firm’s general goal is to profit by providing the goods and services consumers want at a price they will pay, learning how to do that, and do it well, will be more likely when the firm faces significant competition. The idea that firms will perform better under competition than monopoly is hardly new in economics. However, linking that claim to a view of firms as organizational learners in a Hayekian manner provides some additional insights.

For learning to be quick and effective, the learner must receive quick and effective feedback from the external environment *and must also bear fully the costs and benefits that the feedback provides*. In Hayek’s cognitive theory, it is assumed that when the individual (or any organism) acts successfully in the world that the very same individual receives the benefits of that success (or at least a substantial portion of them). Hayek’s argument also assumes that when action is unsuccessful then the actor bears the costs of that failure. Without that assumption, there is no way to link the accuracy of the expectations implicit in the model with longer-term changes in the map, which in turn help to assure the future accuracy of the model. When we prevent individuals from garnering the benefits of successful action of any sort, or when we cushion them from bearing the losses of unsuccessful ones, we break these connections and prevent them from developing rules that provide accurate guidance for navigating the external world.⁴ These same features can be seen in the behavior of firms.

Firms that operate in highly competitive environments are more likely to develop effective routines and respond well to the unexpected. In such environments, economic survival is tightly linked to finding effective routines that bridge resources and activities. Firms whose routines cannot do so, or cannot do so well, will quickly be replaced by firms whose routines can. Success in a highly competitive market is highly suggestive of a high degree of organizational learning generated by the constant contact with an external world in which the success of the firm’s actions is judged frequently and powerfully. Learning requires that both individuals and firms have the incentive to learn and improve and feedback processes that provide

them with the knowledge necessary to know *what* to do in order to learn and improve.

Firms that have monopoly protections and other organizations, such as government agencies, that do not operate in competitive environments will learn much more slowly, if at all. Moreover, operating in different environments with different sets of incentives and different forms of feedback can alter the very structure of such organizations. For example, a firm that now benefits from some sort of monopoly protection will likely add to its activities various attempts to lobby to keep or expand that protection. In turn, this will likely mean acquiring new resources that are more specific to securing political rents. With these new resources and activities, and operating at least part of the time in an environment that rewards them, the firm is likely to develop new routines and shift others, all in the direction of political rents rather than meeting the wants of customers. This diversion of resources is well-noted in the rent-seeking literature, but it would be a useful project to try to understand it in terms of organizational learning. The same can be said of government agencies themselves, which operate in a world of votes, budgets, and power where the feedback of that environment encourages learned routines that link resources and activities that are not concerned with genuinely meeting the needs of the citizenry.

A comparison of the performance of Wal-Mart and other private sector retailers with that of the Federal Emergency Management Agency (FEMA) during Hurricane Katrina illustrates some of these issues (Horwitz, 2009). Wal-Mart and others were able to get resources into New Orleans and the rest of the damaged areas along the Gulf Coast days, if not weeks, before FEMA. Given the competitive environment in which Wal-Mart operates, it has well-honed routines for managing its supply chain and getting supplies to stores, during both crises and normal competitive times. Faced with the somewhat novel circumstances of Katrina, it was fairly easy for Wal-Mart to apply those routines to a slightly different situation and to get the job done well. FEMA, by contrast, operates in a totally different environment that both undermined its ability to learn in general, thanks to a high degree of political uncertainty, and diverted any learning that did take place to satisfying the goals associated with bureaucracy. When the drastic and severe conditions of Katrina were upon them, they were unable to respond effectively without the sorts of routines for moving resources that the private sector firms had developed. The emphasis on environment for the effectiveness of organizational learning should create skepticism that “better leadership” is the automatic solution for poorly performing organizations.

Better leadership is of little help where the incentives and knowledge needed for learning in ways that get the job done are absent.

SUGGESTIONS FOR FURTHER RESEARCH AND CONCLUSION

If the Hayekian cognitive perspective on organizational learning that I have sketched above is worth pursuing, the next task will be to apply it empirically. Perhaps the most obvious research agenda would be to explore in detail how firms react to change and what sorts of organizational structures and cultures seem to enable better adaptation. Such studies could focus on how new routines are formed and/or how old routines get adapted to new circumstances, with particular attention paid to the speed at which such learning takes place. Examining how firms react in crisis situations can also reveal important elements of the effectiveness of its organizational learning, as the prior discussion of Wal-Mart's performance during Hurricane Katrina suggests. Detailed analyses of these sorts of episodes could benefit from the Hayekian perspective presented herein. A third set of internal-to-the-firm questions would revolve around the relationship between organizational learning and organizational structure. Do "flatter" organizations develop "better" rules and routines? Do organizations that give more scope for individual initiative learn better? If so, what features of such organizations make such initiative possible? We know that for humans, exploration and risk are central to learning. Does the same hold for organizations?

Another set of research questions would be comparative across types of organizations. This essay touched on some of the reasons that private firms might learn more effectively than public agencies, but there is much more to be explored there. In addition, the question of where nonprofits and organizations such as universities fit into a Hayekian view of organizational learning is pregnant with possibilities.⁵

A final set of research questions would be ones involved with larger political economy issues. In particular, it would be valuable to understand how various institutional arrangements affect the incentives facing organizations and thereby divert or distort the sort of learning that takes place. The brief discussion of rent seeking earlier provides a hint at where such analyses might go: do highly politicized markets lead firms to develop routines that are geared more toward *politically* effective activities and uses of their resources, which mostly benefit the firm itself, rather than the *economically* effective ones that would benefit consumers as a whole as well?

This question is especially pressing with the various bailout and stimulus packages now in place that are dramatically politicizing the environment within which private sector firms operate. The analysis above suggests that these packages may lead to a great deal of microeconomic discoordination, even if such inefficiencies do not show up in conventional macroeconomic variables. A similar analysis might be made of the effects of environmental concerns, both legislative and cultural. How do the powerful cultural forces that reward “green” firms affect the ways in which firms learn and the kinds of routines they develop? One might think of this as an organization theory analog of child development theory in psychology: under what conditions do firms and children both learn and thrive?

Although all of these questions are at a high level of generality, I believe they point toward an agenda for understanding organizations that makes use of Hayek’s cognitive theory. That this would be a next step in Hayekian research should not be surprising as the long-standing emphasis in Hayekian political economy on the way in which competition and markets generate both the incentives and knowledge necessary for the effective allocation of resources in society as a whole can be understood as a generalization of Hayek’s own cognitive theory. When Hayek (1978 [1968]) referred to competition as a “discovery procedure” and argued that freedom is justified not by our rationality but by our ignorance, he was seeing the market, or the “catallaxy,” as an example of human learning. Much of his work in the 1930s and 1940s, from the core argument of “Economics and Knowledge” (1948 [1937]) that markets are a learning process, to his work on the possibility of socialist calculation (1948 [1940]) that emphasized the ways in which markets generated knowledge that planners could not access, was the beginning of a vision that connected individual human learning and cognition to the ways in which social processes, such as the market, were also epistemic ecosystems. Included in all of these ideas is the importance of tradition and tacit knowledge as the sedimentary results of those learning processes.

Hayekians have had much to say about the cognitive processes within the individual and the way in which social systems involve learning. They have had less to say about those organizations that sit “between” the individual and large-scale social processes, such as the firm or the family. Hayekian approaches to the theory of the firm have become more numerous in recent years, but few have approached that topic from the perspective associated with Hayek’s cognitive theory and *The Sensory Order* in particular. The Hayek quote that opens this paper provides some sense that Hayek glimpsed some of these connections, and I hope that I have outlined some of the ways in which such an investigation might proceed. It is not new to view firms as

organizations that learn, nor is it new to explore the social-scientific implications of Hayek's cognitive theory. By bringing them together, however, something new and valuable might well emerge.

NOTES

1. I thank Roger Koppl for discussion of this analogy.
2. Whether Hayek's views extend in the same way to formal education is an interesting question to pursue. Garnett (2009) explores the idea of "Hayekian pedagogy," but is more interested in the role that spontaneous order and decentralized knowledge play as opposed to Hayek's cognitive theory.
3. Here is where Hayek's cognitive theory matches so well with his broader social theory. Much of his work from *The Constitution of Liberty* forward focused on the way in which humans are rule-following animals in the social order, even though many of the social rules we follow are ones we cannot articulate. The idea that rules lead to an order more complex than we can understand applies both to our own minds and to the social world. The relationship between these two orders is explored in Horwitz (2000).
4. This argument also is suggestive of a way to think about child development from the perspective of Hayekian cognitive theory. The tricky part is finding the right line between ensuring that children feel the consequences of their own mistakes but not so severely as to harm them or discourage them from rational risk taking. For Hayek, the connection between freedom and learning was a lifelong theme, most clearly articulated in *The Constitution of Liberty* (1960).
5. I thank the referee for raising the question of how nonprofits fit the theory.

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INSTINCTS AND INSTITUTIONS: THE RISE OF THE MARKET

Jean-Paul Carvalho and Mark Koyama

ABSTRACT

Purpose – How did cooperation emerge in large-scale, fluid societies? Standard theories based on direct and indirect reciprocity among self-regarding agents cannot explain the high level of impersonal exchange observed in developed market economies.

Approach and findings – Drawing upon recent research from across the behavioral sciences, we attribute the emergence of cooperation in early trade to an evolved characteristic of human psychology that makes revenge sweet: people are willing to pay a price to punish those who betray their trust. Once cooperative expectations became fixed, institutions such as the law merchant and ethnic trading networks, as well as certain “bourgeois virtues,” helped sustain and extend trade during the medieval period.

*Contribution of the paper – Our argument continues the tradition begun by F.A. Hayek in *The Sensory Order* (1952), by providing an integrated explanation for the rise of the market based upon the coevolution of human psychology, culture, and institutions. In our conclusion, we revisit Hayek’s (Hayek, 1976, 1978, 1988) analysis of the conflict between our instincts and the institutions that have created the market order.*

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There is a transformation which is antecedent to Marx's Rise of Capitalism, and which ... looks like being even more fundamental. This is the Rise of the Market, the Rise of the Exchange Economy (John Hicks (1969, p. 7)).

INTRODUCTION

The Rise of the Market and the Problem of Exchange

The rise of the market was a gradual process: "It takes us back to a much earlier stage of history, at least for its beginnings; so far back indeed that on those beginnings (or first beginnings), we have little direct information" (Hicks, 1969, p. 7).¹ From an evolutionary perspective, however, this process, the emergence of impersonal trade, large-scale cooperation between strangers, and a world-wide division of labor, is a recent phenomenon. But it is a topic we understand only dimly.² Market institutions were "stumbled upon" inadvertently; they were not the product of design or even of conscious choice.³ It was only after the event that we realized what had occurred, or as Hayek put it, the developments that "contributed greatly to the growth of an extended order, were little understood at the time, or indeed for centuries afterwards, even by the greatest scientists and philosophers" (Hayek, 1988, p. 38). In this paper, we provide an integrated explanation for the rise of the market based upon the coevolution of human psychology, culture, and institutions. In this way, we continue a tradition that Hayek initiated in *The Sensory Order* (Caldwell, 2004).⁴

The rise of the market is closely connected with what Avner Greif has called the *fundamental problem of exchange*. Exchange is universal, rooted in the human desire to "truck and barter," it has always existed. Impersonal exchange, however, is quite different. It is anonymous, traders do not personally know each other, nor is there a promise of repeated engagements. Thus, there exists little threat of punishment for cheating, in terms of ostracism or noncooperation in the future. As a result, in the absence of the requisite institutional framework, impersonal exchange is beset by problems of commitment and coordination. A coordination problem arises because it is costly to match buyers and sellers. When transaction costs are high, markets are thin and the rewards to investing in market exchange are small. Similarly, there is a commitment problem because exchange is almost always sequential "namely, some time elapses between the *quid* and the *quo*" (Greif, 2002, p. 169).

Therefore, impersonal exchange resembles a *trust game*, or a one-sided prisoner's dilemma game, in which the player who moves first must decide

whether or not to trust the second player before giving the second player an opportunity to cooperate or defect. The game is typically modeled as follows: player 1 is given $\$x$. She can transfer a proportion $p \in [0, 1]$ of her endowment to player 2. In the process of transfer, the amount is multiplied, say by a factor 1.5, so that player 2 receives $1.5p\$x$. In other words, there are potential gains to trade. Player 2 can then choose to transfer a proportion $q \in [0, 1]$ of the amount she receives back to player 1. Every efficient strategy profile involves player 1 transferring all of her endowment to player 2, that is, $p = 1$. However, in a one-shot setting, a purely self-regarding player 2 would return nothing to player 1 ($q = 0$). So player 1 would not trust at all, setting the most socially inefficient level of “trade” $p = 0$. Similarly, in a real one-shot setting, what reason would a trader have to trust a stranger?

For precisely these reasons, before the rise of the market, trade among strangers was difficult and rare. Trade still occurred, but most of it was small scale, personal exchange between individuals who knew each other. The exceptions to this prove the rule: in many early societies small and easily transportable luxury goods could also be exchanged between strangers and nonmarket rituals and customs arose to formalize and sanctify these exchanges. Commerce was concentrated in the hands of particular ethnic minorities who specialized as middlemen. But these customs could not generalize; they were not scalable because under such arrangements only a small subset of goods had value-to-weight ratios high enough to be worth trading.⁵ The markets that did exist were flea markets characterized by high transactions costs. In these types of markets, the costs associated with inspecting quality, deterring theft, and avoiding trades with strangers greatly reduced the volume of trade and additionally distorted the types of goods that were traded (Fafchamps & Minten, 2001). A distinct set of institutions were required for the emergence of generalized impersonal exchange and these institutions are those associated with the market economy. Market societies are governed by innumerable instances of impersonal exchange yet cheating or defection is extremely rare. What institutions make cooperation on such a large scale possible? And how did they emerge?

An Outline of the Paper

This is the fundamental problem with which this paper is concerned. Hayek (1952) initiated a research program in which mind and society are produced by the perennial *coevolution* of cognition, culture, and institutions. We argue that the rise of the market and the transition to impersonal exchange is the

outcome of this same coevolutionary process. Relatively recent research in evolutionary psychology, experimental economics, and economic history broadly supports Hayek's theory, while filling in the details of his framework (see Rubin & Gick, 2004). To explain how large-scale cooperation is sustained, we develop a synthesis of the theoretical literature on cooperation in repeated games, historical evidence on medieval trade, and the revival in cultural arguments among economic historians of the growth of impersonal exchange in Europe.⁶ However, we point to a deficiency in the literature: we still do not understand how early trade could have *emerged* where institutions were poor and the level of trust among strangers was low. In this paper, we attribute the *emergence* of cooperation in early trade to an evolved *psychology* that supports *strong reciprocity* (Gintis, 2000; Bowles & Gintis, 2004), under which agents sacrifice resources to reward a kind act and punish those who betray their trust. Specifically, we draw upon the experimental work of de Quervain et al. (2004) to argue that instinctive preferences for punishing cheaters enabled merchants to credibly commit to cooperative trade in the absence of formal contracts or state protection. Feuding in medieval Germany is one example of how strong reciprocity can lead to the emergence of trade where institutions are poor and levels of trust are low.

We proceed as follows. The first section draws on Hayek's (1952) theory of the mind to distinguish between instincts shaped by evolution, cultural beliefs or morals, and the economic institutions they support. In the next section, this framework is used to examine the emergence of cooperation, first between relatives and within small groups, and then in larger, more fluid societies. We argue that direct and indirect reciprocity on their own are unlikely to sustain cooperation in large, fluid societies. In such societies, evolved human instincts and formal institutions for enforcing contracts play a crucial role in supporting trust and cooperation. The next section reviews the historical and experimental evidence on the psychological and institutional mechanisms supporting impersonal exchange. In the final section, we return to Hayek's argument that institutional change has been so rapid that our instincts have not caught up. We conclude by considering some of the economic and political implications of this insight.

THE FRAMEWORK

We begin by outlining a framework for analyzing the mechanisms that help solve the fundamental problem of exchange, and thereby contribute to the rise of the market.

Hayek's Tripartite Theory

Beginning with *The Sensory Order*, Hayek (1952, 1960, 1973, 1978, 1979, 1988) develops a framework for understanding the rise of the market, by distinguishing between the evolution of instincts for cooperation, the cultural transmission of morals and the emergence of institutions supporting cooperation. These three distinct (but interdependent) mechanisms are as follows:

1. Evolution and Development of Mind
2. Cultural Transmission and Evolution
3. Evolution of Informal and Formal Institutions.

These processes operate at different levels (and time scales), from the infra-individual level of neural architecture, to the level of kin and cultural groups, to entire societies and collections of interlinked societies.

The Evolution and Development of Mind

Human psychology was largely shaped during the Pleistocene period 1.8 million to 11,500 years ago (Cosmides & Tooby, 1992).⁷ A central question in this paper is: how are individuals who evolved primarily in small, kin-based, hunter-gatherer groups able to sustain large-scale, technologically advanced societies that support anonymous trade? What *instincts* or *institutions* make this possible? We present evidence in the next section that evolved human instincts are “scalable” in the sense that they entail neurophysiological and cognitive adaptations for supporting cooperation among strangers in large-scale societies. In particular, we argue that the taste for exercising violence against those who violate our trust was imperative in the emergence of trade among strangers.

According to Hayek (1952), the mind can be interpreted as an evolved organ for classifying and filtering sensory data. Particular neural networks map this raw data onto mental states.⁸ This mapping is itself partly inherited and partly shaped by experience, particularly during development. The strength of synaptic connections evolves in response to new sensory stimuli. New sensory data are interpreted based on existing mental categories. So cognitive processes are always path dependent; the historical context always casts a shadow on cognition and decision making. As such, the social and institutional forces shaping experience create the possibility of variation in mental models and rules of behavior across cultural groups. The mind is shaped by its environment.

Cultural Transmission and Evolution

The “mind is embedded in a traditional impersonal structure of learnt rules.” Human beings are dependent on learnt rules because our “capacity to order experience is an acquired replica of cultural patterns that every individual mind finds given” (Hayek, 1979, p. 157), and “abstract concepts are a means to cope with the complexity of the concrete that our mind is not capable of fully mastering” (Hayek, 1973, p. 29). This means that mind itself is partly a product of cultural evolution: “What we call mind is not something that the individual is born with, as he is born with his brain, or something the brain produces, but something that his genetic equipment helps him to acquire, as he grows up” (Hayek, 1988, p. 22).⁹

The categorization function of the human mind plays a major role in human cultural evolution by enabling the generalization of abstract rules learned in a particular context, through vertical, oblique, or horizontal transmission (Cavalli-Sforza & Feldman, 1981), to a range of new and unfamiliar contexts. Cultural evolution at this level operates through a process of social learning that is conditioned by preexisting instincts (e.g., Boyd & Richerson, 2005). Cultural beliefs and values that support cooperation can therefore emerge over a comparatively rapid time frame. Hayek argued that it is this transmission of abstract rules that enabled a major departure from our evolved instincts, causing the transition from small-scale, collective, kin-based societies to complex, technologically advanced, large-scale societies:

This exchange society and the guidance of the co-ordination of a far-ranging division of labour by variable market prices was made possible by the spreading of certain gradually evolved moral beliefs which, after they had spread, most men in the Western world learned to accept (Hayek, 1979, p. 164).

The Evolution of Formal and Informal Institutions

Cooperation can emerge within societies as a self-enforcing social norm. Due to the multiplicity of equilibria in social interactions, the same cultural transmission mechanisms operating within societies or groups can lead to the emergence of *different* social norms and formal institutions in different groups, and can thus support diverse forms of cooperation. Historically, trade is ubiquitous across very different societies and markets of some form have been supported by a wide variety of different institutions. There is no a priori reason to suppose that the institutions chosen will be optimal. In other words, many different institutional arrangements are capable of sustaining *some* limited level of impersonal exchange, but the institutional path a given society stumbles upon will not necessarily be the one that

maximizes trade. However, Hayek argued that selective pressure at the level of human groups acted so as to choose specific institutional arrangements and norms over others (see Andreozzi, 2005). Over the course of centuries moral traditions or social norms that emerge within groups are subject to a group-selection process.¹⁰ Competition between different societies means that social norms and formal institutions that result in the greatest populations or the highest standards of living are more likely to persist.

SOCIAL COOPERATION: ITS BASIS IN EVOLVED PSYCHOLOGY

In this section, we evaluate how cooperation is achieved, first between relatives and small groups, then in larger societies.

We can explain cooperation among genetic relatives. The theory of *kin selection* (Haldane, 1955; Hamilton, 1963) is the primary explanation in evolutionary biology for why individuals take actions that are costly in terms of their own reproductive fitness but benefit the group as a whole. If an individually costly action sufficiently enhances the reproductive potential of the individual's close genetic relatives, then the individual's genes can proliferate through his/her relatives' reproductive activity, despite the decline in the individual's reproductive fitness. Therefore, kin selection at the level of the gene might explain trust and cooperation within hunter-gatherer societies, which were composed of close genetic relatives. However, kin selection cannot explain the widely observed phenomena of trust and cooperation among strangers with weak genetic-relatedness, nor can it explain the emergence of an open society.

When the same set of agents interacts repeatedly, cooperation among nonrelatives can be sustained through *direct reciprocity*. This principle can be used to explain cooperation in small groups. Under an extreme version of this norm, players reciprocate trustworthy behavior by trusting their partner again in their next interaction, while punishing cheating partners by refraining from trade with them in all subsequent interactions.¹¹ Therefore, a player who is trusted in an exchange must weigh up the cost of perpetual autarky with the once-off benefit from defection. The folk theorem shows that if the same set of players interact frequently enough and are sufficiently patient, then perpetual trust and cooperation can be supported as a subgame-perfect equilibrium of the repeated game (e.g., Friedman, 1971; Rubinstein, 1979; Fudenberg Maskin, 1986).¹² But, while direct reciprocity can explain trust

and cooperation in long-run relationships formed in small communities, it hardly applies to large-scale societies in which a large proportion of exchange occurs between individuals who have not met before and will probably never meet again (e.g., Boyd & Richerson, 1988).¹³ For example, a borrower not intending to repay a loan could relocate to avoid personal enforcement of sanctions by the lender or simply borrow from someone else in the future.

Cooperation in Close-Knit Societies

To explain trust and cooperation in large populations composed of close-knit groups, we turn to *indirect reciprocity* (see Nowak & Sigmund, 2005). The key idea here is that an individual's actions affect their *reputation* for trustworthiness. Since any two individuals rarely, if at all, interact more than once, punishment of a defector must be carried out by agents who were not directly cheated, but who can observe their current partner's reputation. Therefore, the shift from direct to indirect reciprocity coincides with a shift from personal to community enforcement (see Greif, 1992). As such, the issue is not whether exchange is conducted in long-run relationships with fixed partners, but the extent to which information about reputation flows freely though the population.

If agents are only able to observe whether there was a defection by any player during the period, and all agents condition a trigger strategy on this information, a single erroneous defection can lead to the permanent breakdown of cooperation, and the punishment of many innocent cooperators. Clearly, a different institution for sharing information is required to enable individuals to target punishment. Kandori (1992) shows that *local information processing*, in which individuals carry a label that summarizes their reputation and is observed by their trading partners, can facilitate trust and cooperation, even in the presence of occasional errors.

Greif (1989, 1993, 1994) illuminates how ethnic trading networks sustain trust and cooperation by facilitating the flow of reputational information in the middle ages. Today credit rating agencies, clubs that monitor members' actions, social status markers, and gossip networks can fulfill this information sharing function. However, among a fluid population in which individuals are sufficiently unlikely to observe the reputation of other players, indirect reciprocity will not sustain large-scale trust and cooperation (Nowak & Sigmund, 2005).

Large-scale trade between strangers becomes possible once expectations of social cooperation have evolved. And this trade, once it gets going creates

a positive feedback process that is self-supporting.¹⁴ But the existence of this positive feedback mechanism does not explain how trade gets started in the first place. Moreover, these institutions presuppose an existing system of impersonal exchange. Though their development can explain improvements in the efficiency of this framework, it cannot explain the emergence of this framework itself. It does not explain how traders converge on expectations of mutual cooperation.

A further problem with indirect reciprocity is that, in the absence of elaborate information sharing, punishment of defectors with noncooperation is *costly* in terms of an individual's reputation. This creates a new social dilemma: who punishes? Trust and cooperation cannot be sustained among *self-regarding* agents by indirect reciprocity, unless "justified" defection (i.e., punishing a partner for prior defection) can be distinguished from "unjustified" defection (Nowak & Sigmund, 2005; Ohtsuki & Iwasa, 2006). For this, individuals need to have information not only on their partner's prior actions, but also on the actions of their partner's prior partners, the actions of the partners of their partner's prior partners, and so forth. This level of information sharing is unrealistic except in very close-knit networks.^{15,16} Therefore, direct and indirect reciprocity alone cannot explain the emergence of impersonal trade in large-scale societies. In the next section, we consider some of the historical institutions that emerged during the medieval revival of trade and made possible the rise of market institutions and the beginning of modern economic growth.

HOW DID MARKET-SUPPORTING INSTITUTIONS EMERGE?

There is still no generally accepted theory of how trust and cooperation are sustained in *fluid*, large-scale societies. We address this problem in the following way: first, we examine the historical *emergence* of the institutions that made the market order possible; second, drawing upon recent work in experimental economics, we argue that early trade was enforced by the socially sanctioned threat of bilateral violence. This argument suggests that the explanation for large-scale cooperation has its basis in human psychology.

We have established that the kind of personal-reputation-based mechanisms capable of supporting trade among small groups or tightly knit communities could not call forth an extended order of trade among

strangers. “Yet somehow, however slowly, however marked by setbacks, orderly cooperation was extended, and common concrete ends were replaced by general, end-independent abstract rules of conduct” (Hayek, 1988, p. 31). There were a number of elements to this transition and we can only provide a cursory treatment here.

First there was the beginning of long-distance trade that, though it predates written history, chiefly concerned luxury goods with a high value-to-weight ratio, and thus could not itself form a basis for the later expansion of trade in basic commodities. Even this trade was always extremely vulnerable to the depredations of raiders, pirates, nomads, or states. Nevertheless many of the institutional forms or organizational innovations such as the bill of exchange or the joint-stock company initially developed for use in long-distance trade later became the basis for a broader based and more general expansion of markets.¹⁷ Second, this transition was mediated by a variety of institutions that developed either in antiquity or in the period of the medieval commercial revolution. These institutions were not necessarily designed for the particular purposes they served, nor were they necessarily efficient (see for instance Ogilvie, 2007).¹⁸

Some of these institutions created conditions under which the information required to sustain cooperation could be shared among strangers while others drew upon instincts for reciprocity. For instance, Milgrom, North, and Weingast (1990) detail how during the Champaign fairs of the 12th and 13th centuries an institution known as the law merchant provided information on each merchant’s trading history for a fee, thus sustaining a cooperative equilibrium even among a large and fluid population of otherwise anonymous merchants. But the geographical extent of the medieval law merchants was limited, and its historical significance has been questioned (Volckart & Mangels, 1999; Boerner & Ritschl, 2002). In comparison to the work on the law merchant and on merchant guilds, less attention has been paid to cases in which market-supporting institutions seem to have emerged by exploiting the underlying human instincts that can support trade and exchange on a large scale.

Here we can consider one example: late medieval Germany, where, in the absence of either a single overarching legal authority or anything analogous to a law merchant, impersonal trade appears to have been sustained by the informal institution of *feuding* (Volckart, 2004). Feuds were a form of regulated violence that enabled merchants to commit to fulfilling their contractual obligations. Merchants who were cheated could punish the perpetrators through physical violence or by disrupting their trade. There are two important points here. First, this form of punishment usually

imposes a significant cost upon the punisher, the cheated merchant. Knowing this, the trustee merchant in a trade would not face a credible threat of punishment by a cold, self-regarding agent. However, if a merchant derives sufficient satisfaction from punishing those who cheat him, then the threat of retaliation becomes credible, and trust and cooperation can be supported even in a once-off, anonymous interaction (see for example Rabin, 1993). Therefore, feuds can be a “cheap” way of supporting some level of trade if they draw upon instinctive preferences for punishing defectors.¹⁹ The second point is that feuding appears to have been a socially sanctioned response for an agent who is cheated. So not only must a cheated merchant experience emotions of anger and hate to cause him to feud, but the broader society must deem his retaliatory actions (and perhaps emotions) to be justified. We can now consider exactly what these instinctive preferences for punishing cheaters are and what role they have to play in supporting large-scale impersonal exchange in fluid societies.

Instincts for Trust and Cooperation in Large-Scale, Fluid Societies

Fehr and Gächter (1998) survey the extensive experimental evidence that individuals tend to reciprocate like-behavior for like: punishing defectors and rewarding cooperation even when such actions are costly. Human behavior is shaped by concern for the *intentions* of others. Positive reciprocity involves rewarding perceived good intentions, while negative reciprocity requires perceived bad intentions to be punished.

There is strong experimental evidence for the widespread existence of negative reciprocity and its role in maintaining cooperation. The game played by subjects has the same structure as the (one-shot) trust game. Recall that the standard game-theoretic prediction, with self-regarding agents, is distrust and noncooperation. In contrast, experiments indicate that around half of subjects trust their partners, and three quarters of those trusted do not violate this trust (e.g., Berg, Dickhaut, & McCabe, 1995; Smith, 1998).²⁰ Even higher levels of cooperation are supported when a third move is added to the game in which the first player can incur a cost to reward or punish the second player. A significant number of subjects incur costs to reward cooperators and punish defectors. The experimental evidence suggests that negative reciprocity reflected in informal institutions that support impersonal exchange, such as feuding in medieval Germany, is deeply rooted in human psychology.²¹

These results are consistent with the presence of strong reciprocity, which is a predisposition to *sacrifice* resources in order to (i) reward a kind act and (ii) punish perpetrators of unkind acts (Gintis, 2000; Bowles & Gintis, 2004). Fehr, Fischbacher, and Gächter (2002) claim that “the fraction of subjects showing strong positive reciprocity is rarely below 40 and sometimes 60 percent whereas the fraction of selfish subjects is also often between 40 and 60 percent.” In addition, Gürer, Irlenbusch, and Rockenbach (2006) show that players who can choose between interacting in a regime with punishment and a regime without punishment, begin switching to the punishment regime as (costly) punishment boosts cooperation, while cooperation declines in the regime without punishment. By the end of the sequence of exchanges, all subjects have migrated to the regime with punishment and, once there, strongly cooperate. This demonstrates that institutions that leverage the influence of strong reciprocators can evolve via group selection.

Why do human beings engage in costly cooperation and punishment? What are the proximate neurophysiological mechanisms that lead to behavior consistent with strong reciprocity? Studies have shown that the striatum, an area in the midbrain, is a key part of neural circuitry that processes information on rewards, while the dorsal striatum processes rewards resulting from decisions. Using neuroimaging technology, de Quervain et al. (2004) demonstrate that subjects with higher activation in the dorsal striatum engage in more costly punishment. Subjects also tend to experience lower activation when they are only able to register disapproval using a symbolic token, rather than punish by deducting a monetary amount from partners who cheat. These observations suggest that human beings who are cheated find punishment satisfying. This explains the negative reciprocity component of strong reciprocity. Rilling, Sanfey, Aronson, Nystrom, and Cohen (2004) show that mutual cooperation with a human partner generates higher striatum activations than either mutual cooperation with a computer partner or earning a similar monetary reward in a trivial decision task. These observations suggest that human beings find mutually beneficial social exchange rewarding for nonpecuniary reasons. This explains the positive reciprocity component of strong reciprocity.

Recent studies have discovered a further neurobiological mechanism for trust. The neuroactive hormone oxytocin has been linked to prosocial behavior in nonhuman animals. Oxytocin receptors are located in brain regions associated with behaviors such as pair bonding, maternal care, sexual behavior, and the ability to form normal social attachments. Therefore, oxytocin enables animals to overcome their natural aversion to

proximity. In experiments conducted by Zak, Kurzban, and Matzner (2005), subjects who receive a monetary transfer that reflects an intention of trust exhibit higher oxytocin levels than subjects who receive an unintentional monetary transfer of the same amount. Subjects who have higher levels of oxytocin also exhibit more trustworthy behavior. Evidence presented by Kosfeld, Heinrichs, Zak, Fischbacher, and Fehr (2005) suggest that higher oxytocin levels cause higher levels of trust, but not trustworthiness. This suggests that the hormone oxytocin facilitates trust among strangers by making social exchange with nonrelatives feel safe and familiar, resembling interactions with kin or close acquaintances.²²

Positive reciprocity can also be supported by the desire for esteem. Adam Smith in *The Theory of Moral Sentiments* described how commercial society was founded on each individual's concern for how he was regarded by others (Smith, 1759, 2002). Numerous experimental studies have suggested that human beings care about the welfare of others and about what others think of them (see Fehr & Schmidt, 1999; Sobel, 2005).²³ The desire for esteem can help to resolve the problem of trust and cooperation in the type of sequential exchange we have been considering. If the second player is sufficiently concerned with how the first player regards him, he will cooperate. Anticipating this, the first player will trust the second and trade will take place. If the second player does cheat, then emotions of anger and revenge may drive the first player to sacrifice resources to punish him, without any expectation of future benefit. In our example, the cheated merchant may go to great lengths to track down a cheating agent in order to ruin his reputation or otherwise impose a cost upon him. Anticipating this, even a self-regarding agent may prefer to cooperate.

Accordingly, Bowles and Gintis (2004) demonstrate that the presence of strong reciprocators sustains cooperation in large, fluid populations. Their simulations also indicate that under assumptions approximating likely human environments over the 100,000 years prior to the domestication of animals and plants, strong reciprocators could invade a population comprised of unconditional cooperators who cooperate but do not punish, and self-regarding agents who neither cooperate nor engage in costly punishment. Substantial frequencies of all three behavioral types can be sustained in a population. The experimental evidence, therefore, suggests that a significant proportion of human beings have evolved instincts that generate behavior consistent with strong reciprocity.

Human instincts that evolved in small kin-based communities of hunter-gatherers play a critical role in supporting trust and cooperation among nonrelatives in fluid, large-scale societies. Our instincts have in this sense

made the market order possible. But this relationship is a contingent one. Both the historical and the experimental evidence suggest that a multiplicity of different institutional arrangements is compatible with our evolutionary heritage. The question is how or why have we stumbled upon institutions that have made capitalism possible? There is only one piece missing from the jigsaw and this is the evolution of culture and social norms.

The Evolution of Culture and Social Norms

Social norms and culture matter. In the early 1990s, economists supervising the transition to market-based economies in Eastern Europe and the former Soviet Union appeared surprised when the formal market institutions and legal system they had transplanted in these countries *ex novo* failed to function as they do in the west. They failed to appreciate that markets and the economic and legal institutions of market-based economies are embedded in, and supported by a wider network of social norms and moral beliefs (Boettke, 2001). Furthermore, moral beliefs and social norms are themselves emergent phenomena, in turn, conditioned upon preexisting institutions and levels of market activity.

Hayek offered a theory of institutional or cultural selection according to which certain beliefs are favored over others via tradition, myth, and religion.²⁴ This theory suggests why it was that beliefs favorable to market-based exchange could survive and spread over the very long run.²⁵ In this section, we argue that the growth of impersonal exchange in Europe, in the course of the Middle Ages, initiated a gradual process in which social norms themselves changed, and where circumstances were favorable, recognizably bourgeois values emerged. Commerce came too esteemed. And this, in turn, created an environment in which the volume of trade could increase.

Hostility to commerce in antiquity and in the middle ages was almost universal.²⁶ This hostility manifested itself in the form of prohibitions on nobles involving themselves in trade, in sumptuary laws that restricted what kinds of clothes members of each class could wear, in usury laws that hindered capital markets, in guild laws that restricted the mobility of labor, and in widespread monopoly privileges. Aristocratic blood, ability in war, and religious devotion were admired, trade and market exchange denigrated. As Max Weber put it, the capitalist spirit of a Benjamin Franklin “would both in ancient times and in the Middle Ages have been proscribed as the lowest sort of avarice and as an attitude entirely lacking in self-respect” (Weber, 1930, p. 21).

The direct consequence of this was that from the perspective of a modern historian: “Most of the wealth produced under the old order was simply squandered” (Doyle, 1992, p. 33). It was spent on conspicuous consumption, on servants, expensive clothes, and ostentatious entertaining rather than reinvested. The nobility were distinguished by their “obsession with maintaining existing status” and their “aggressive pursuit of wives of superior status” (Hurwich, 1998, p. 178). Aristocratic status involved great expense, particularly as Adam Smith argued, through the maintenance of a large number of retainers (Smith, 1776, 1976, Book III, chap. IV).

The moral values that upheld impersonal exchange were slow to emerge because they differed dramatically from those that had been favored during most of mankind’s evolutionary history. These new moral values involved “withholding from the known needy neighbors what they might require in order to serve the unknown needs of thousands of unknown others” (Hayek, 1979, p. 165). These moral norms comprised a learnt web of “non-instinctive rules of conduct that enabled mankind to expand into an extended order” (Hayek, 1988, p. 12). These rules survived and slowly spread precisely because they made possible larger and more successful societies.

This was the essentially Mandevillian insight that what had previously been considered private vices were in fact public benefits. This notion spreads slowly through the acquisition of new “bourgeois values.” Sociologist Benjamin Nelson (1969, 1949) argued that commerce and religion eroded traditional beliefs over the course of the Middle Ages. The clannish distinction between “brother” and outsider or “other” had been eroded by Christianity, so that:

The ground was thus cleared for the establishment of a new sort of “brotherhood,” universal rather than tribal, competitive rather than cooperative, which we have here been lead to call “Universal Otherhood,” a distinctive society, wherein-if we may anticipate-all men are “brothers” in being equally “others” (Nelson, 1969, 1949, p. xxiv).

This new form of society emerged first and in a partially developed form in the commercial republics of Italy and in the Low Countries, in Venice and then later in Genoa, Pisa, Florence, Bruges, and Ghent (Pirenne, 1925). From there, itinerant merchants traveled all across Europe stimulating trade and commerce (Hunt & Murray, 1999). Italian traders and bankers, in particular, were active across Europe: they “founded a kind of hegemony over European commerce and finance” (Bergier, 1979, p. 107) and the “Italian business-man made his influence felt from London to Peking” (Tawney, 1955, p. 291). With the rise of a merchant class commercial mores

spread slowly and unevenly through Europe with the growth of cities and towns (McCloskey, 2006). Where the volume of trade was large enough, itinerant merchants settled down and became sedentary, forming merchant colonies in the cities located along important trade routes (Spufford, 2002). These cities made possible the existence of a flourishing merchant class, for whom trade and active participation in the market was a way of life.

Unlike those peasants, laborers, or craftsmen whose engagement with the market was incidental or insulated by guild law, merchants, in general, and those involved in finance in particular, had to embrace the new values of commercial society. These bourgeois values, which privileged patience, prudence, and probity, in turn helped to make market institutions self-enforcing. This improvement was gradual, and as Smith observed it radiated through society slowly. Successful merchants who exited commerce in order to become country gentlemen raised agricultural productivity as they applied the habits they learned in commerce to farming (Smith, 1776, 1976, Book III, chap. IV).

But commerce, and the cities and city states that were vital to commerce, were always vulnerable to predation from larger princely states.²⁷ From the 10th to 17th century, market institutions, bourgeois values, and the freedom and prosperity they brought were only possible so long as “political anarchy” divided Europe (Baechler, 1975; Jones, 1988). Only once the ethics of the mercantile city state had been transferred to a nation state were the institutions supporting market exchange truly and permanently entrenched. Only then was the rise of the market complete and irreversible.²⁸ This occurred most successfully in the Netherlands and in England in the 17th and 18th centuries, paving the way for an industrial revolution. The transition was not, however, truly completed, nor can it be, because we retain many of the moral beliefs and values that regulated life in small-scale societies. In the final section of the paper, we explore why money, finance and capitalism “remain unremittingly suspect to moralists” (Hayek, 1988, p. 102).

THE CONFLICT BETWEEN INSTINCTS AND INSTITUTIONS

The rise of the Great Society is far too recent an event to have given man time to shed the results of a development of hundreds of thousands of years (Hayek, 1976, p. 146).

We can see that *The Sensory Order* plays a vital role in Hayek’s overall research project because it offers an explanation of how experiences are categorized (Hayek, 1952). If the filtering/ordering process the theory

outlines remains relevant, it is because it bears the important implication that the meaning we impart to our social experiences depends on their cognitive context. Hayek argued that when it comes to politics or economics, the first model in our heads that we reach for is essentially wrong. The immediate context we draw upon is that of the small-scale hunter-gatherer society: "We have not shed our heritage from the face-to-face troop, nor have these instincts either 'adjusted' fully to our relatively new extended order or been rendered harmless by it" (Hayek, 1988, p. 17). This has a number of important implications.

The sharing of meat is universal in hunter-gather societies because it is an efficient way to reduce the risks that face each individual hunter. But a communal sharing ethic cannot prevail on a large scale in an extended order because its imposition would cause its collapse. "If we were to apply the unmodified, uncurbed, rules of the micro-cosmos (i.e., of the small band or troop, or of, say, our families) to the macro-cosmos (our wider civilization), as our instincts and sentimental yearning often make us wish to do, we would destroy it." (Hayek, 1988, p. 18). There is thus a mismatch between our evolved inclinations and the institutions that are responsible for modern society. And though, these two models are of course capable of coexisting, a psychological dissonance too often results.²⁹

In this paper, we have presented the argument that the emergence of large-scale cooperation cannot be explained on the basis of direct or indirect reciprocity alone. If this is the case then institutions like the law merchant or the merchant guild can only have played a limited role in explaining the rise of impersonal exchange.

We have further argued that trade was often supported by institutions that relied on norms of strong reciprocity. The moral rules that we have inherited and hardwired into minds thus made possible the emergence of larger and more expansive networks of cooperation.

The subsequent rise of market, however, led to the spread of new moral attitudes, of bourgeois values. The growth of these values made market institutions self-enforcing and led to the demise of many of the rules and restrictions that had curtailed trade through the medieval period in particular. Usury laws, for instance, were first undermined by the repeated attempts of merchants to evade the strict letter of the law before interest below 10 percent became legal in England, in the second half of the 16th century.³⁰ Furthermore, since the spread of commercial values lowered the cost of enforcing contracts it meant inefficient institutions like feuds were replaced by formal systems of enforcement. In other words, the spread of bourgeois values made Adam Smith's Great Society possible.

But these values remain problematic because they are opposed to our inherited moral instincts.

We have argued that human sociability is scalable, but it is not perfectly scalable. Cooperation between strangers is possible, but relations among members of a large and dispersed and anonymous society characterized by a complex division of labor are *qualitatively* different to the kinds of relationships that comprise a small-scale society. This is necessarily true, as Hayek realized, because of the divided and dispersed nature of knowledge occasioned by the division of labor. The information required to achieve coordination between agents is never “given” to a single mind that could work out the implications, and can never be so given (Hayek, 1945, p. 519).

Individuals typically underestimate the benefits of the market order.³¹ One reason for this is that as Hayek noted, the market order is complex and intangible “based on purely abstract relations which we can only mentally reconstruct” (Hayek, 1973, p. 38). This is one of the marvels of trade. We do not see or know all of those who benefit from the exchanges we make, nor do we see or understand how all the goods we consume are produced; the visible link between inputs and outputs is obscured.

This, however, is precisely what is alienating and discomfoting; markets seem chaotic, unordered, inequitable, and even random. Their arbitrary nature offends and demands management or correction. Market-based societies are open-ended, vast yet disparate networks utterly unlike anything our Pleistocene ancestors would have known. Hayek located the atavistic longing Rousseau, Marx, Marcuse articulated in precisely this incongruity. It is often impossible to keep track of all the different agents involved in even a simple market transaction, to count who is benefiting and who is losing out. All we see is the overall pattern, how the system seems to reward winners and losers.

Typically, this will not match our evaluations of desert. It will seem unfair. Therefore, it must be made fair, that is, they must be made to fit a pattern compatible with our moral intuitions. But this corseting is inherently corrosive of the properties that make markets valuable, their ability to convey knowledge, provide incentives, and coordinate human action. Hayek believed that the market order was inherently fragile for this reason. To the extent that it is a complex phenomenon, that is, a system characterized by nonlinear relationships between a large number of variables that are capable of generating patterns that were not *ex ante* predictable, then attempts to manage or govern it that do not take this into account will fail. And attempts to impose preconceived patterns onto its outcomes will prove destructive. A market order cannot be reconciled with “solidarity” and is

thus constantly endangered by “the predilection for the concrete” (Hayek, 1976, p. 49).

This argument, if it is accepted has important consequences for political philosophy since according to Hayek: “Though our sense of justice will generally provide the starting point, what it tells us about the particular case is not an infallible or ultimate test. It may be and can be proved to be wrong” (Hayek, 1976, p. 41).

NOTES

1. Markets predate written history. We do not subscribe to the views of Karl Polanyi on early trade (Polanyi, 1944, 1957). The “Rise of the Market” refers to the slow transition from societies in which trade between strangers was a rare occurrence to societies in which daily life would be inconceivable without our dependence on trade with individuals we do not know personally.

2. “We know surprisingly little about the institutional foundations of exchange in past societies” (Greif, 2000, p. 252).

3. As Hayek put it “We have never designed our economic system. We were not intelligent for that” (Hayek, 1978, p. 164).

4. Butos and Koppl (2007) make the case that *The Sensory Order* occupies a crucial role in establishing the cognitive basis for Hayek’s critique of scientism, the theory of markets as discovery mechanisms, and his perspective on institutions. Here, we focus on Hayek’s explanation of institutional development.

5. In their archeological study of the ancient Maya, for instance, Tourtellot and Sabloff (1972), found that most artefacts were exchanged only within a community. Only prestige items were traded across different communities.

6. See Landes (1998), Lal (1998), Jones (2006), Greif (2005, 2006), and Clark (2007) for recent attempts to refine Max Weber’s famous claim that a gradual change in values precipitated the emergence of capitalist institutions in Europe (Weber, 1930).

7. It was likely also shaped by evolution during the preceding Pliocene period (5.3–1.8 million years ago) (Foley, 1996).

8. As Butos and Koppl (2007, p. 23) put it, the brain is a structured organ and it is the structure of the neural connections within the mind that creates the classification system.

9. By culture, Hayek refers to the collection of heuristics, customs and traditions, personal experiences, and socially transmitted information an individual uses to make a decision in a particular context. Cultural evolution can proceed either by changing individual preferences or instincts, or by shaping and selecting the institutional environment.

10. In his later work, it was precisely Hayek’s emphasis on group selection that met the most hostile reception. The reason for this is that, as is well known, group-selection mechanisms are subject to severe free-rider problems. Such incentives to free ride meant that “group advantage, as such, simply cannot explain why the

individuals within the group will actually exhibit such group-beneficial behavioural regularities" (Vanberg, 1986, p. 86). Recent work, however, has reassessed this view (see Sober, & Wilson, 1998; Zywicki, 2000; Andreozzi, 2005; Gaus, 2006).

11. There is evidence that human beings have specialized cheater-detection cognitive programs for this purpose (Cosmides & Tooby, 1992; Stone, Cosmides, Tooby, Kroll, & Knight, 2002).

12. Still, many other equilibria remain, including perpetual mistrust and defection.

13. In his classic example of men draining a meadow, David Hume (1739) articulates how cooperation becomes geometrically more difficult as the size of the group increases.

14. For example, Henrich et al. (2004) present evidence from experimental games conducted in 15 small-scale societies spread over five continents, that people in more developed market economies exhibit more cooperative behavior.

15. See Young (1998, p. 101) for a formal definition of a close-knit social network.

16. Accordingly, Putnam (2007) presents striking evidence that trust (even of one's own race) and cooperation are lower in ethnically diverse neighborhoods in the United States.

17. For the history of bills of exchange, see de Roover (1946, 1967). The history of sea insurance is particularly interesting as many of the institutional innovations of antiquity were rediscovered in the later middle ages having laid dormant for hundreds of years (Hoover, 1926).

18. As the work of Acemoglu (2003, 2006) has indicated, geographical accident, military or political power or some initial distribution of resources can have a decisive effect on how particular institutions in particular regions developed.

19. However, it is a relatively inefficient way of supporting trade, because it imposes significant external costs upon third parties, via the disruption of trade.

20. Hundreds of subsequent experiments generated similar results, in countries with different demographic characteristics and with stakes up to 2–3 months' income (Fehr & Fischbacher, 2003).

21. In their experiments, Kurzban and Houser (2005) show that subjects divide into three groups: cooperators, defectors, and reciprocators. This is evidence of a stable polymorphic equilibrium of types.

22. In their experiments, 45 percent of human subjects who were administered oxytocin through a nasal spray trusted their partners maximally, compared to 21 percent for the placebo group. There was no significant difference in trustworthy behavior between the two groups.

23. Brennan and Pettit (2000) assess the role esteem plays in underlying market exchange.

24. Thus, in *The Fatal Conceit*, Hayek emphasized that "the only religions that have survived are those which support property and the family" (Hayek, 1988, p. 137).

25. The evolution of social and cultural values is faster than biological evolution, but it is still gradual in a historical context. As Boyd and Richerson describe it: "the wheels of cultural evolution roll on the time scale of millennia, even though, when we look closely at any one society over short periods of time, change is often readily perceptible" (Richerson & Boyd, 2008, p. 109).

26. See Finley, 1999, 1973) for evidence of the widespread contempt for trade in ancient Athens and Rome.

27. Notable examples of this are the Norman conquest of Sicily and Southern Italy in the 11th century under the d'Hautevilles, the invasion of Italy by Charles VIII of France in 1494, and the subjugation of the German city states during the Thirty Years War.

28. There is consequentially an element of historical contingency in this story as Hayek put it: mankind chanced, inadvertently onto the path of development that led to the market order.

29. "[D]espite the advantages attending our limited ability to live simultaneously within *two* orders of rules, and to distinguish between them, it is anything but easy to do either. Indeed our instincts often threaten to topple the whole edifice" (Hayek, 1988, p. 18).

30. On the impact of usury laws in general, see de Roover (1967), Goff (1979, 1988), and Koyama (2008). For an analysis of their demise in England, see Jones (1989).

31. Caplan (2007) provides statistical evidence that relative to economists, and controlling for income and education, noneconomists are more pessimistic about the state of the economy in general, and systematically underestimate the gains associated with voluntary exchange in the marketplace.

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AN INSTITUTIONAL SOLUTION FOR A COGNITIVE PROBLEM: HAYEK'S SENSORY ORDER AS FOUNDATION FOR HAYEK'S INSTITUTIONAL ORDER

Nikolai G. Wenzel

ABSTRACT

Purpose – To show that Hayek's cognitive theory sheds light on Hayek's institutional theory.

Methodology/Approach – Although F. A. Hayek contributed richly to many fields of economics – from capital theory to monetary theory, and from institutions to spontaneous order – one theme is omnipresent in his work: the knowledge problem. This paper examines Hayek's work in psychology, The Sensory Order, and argues that there exist strong parallels between Hayek's cognitive and institutional theories.

Findings – Hayek's institutional (or social) theory makes a lot more sense when understood as a necessary consequence of his cognitive theory. Furthermore, Hayek's cognitive theory allows for rational individuals making choices that are socially embedded.

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Research limitations/implications – *Three Hayekian themes are explored: (1) institutional implications of limited knowledge; (2) learning and knowledge generation; and (3) mental models. The paper then uses a challenge within Hayek – the tension between microlevel methodological individualism and macrolevel institutional evolution – as a starting point toward resolving the lingering individual-culture methodological puzzle in contemporary economics. These are mere starting points for further research.*

Originality/value of paper – *Horwitz (2000) writes that “Hayek’s thought will have come to fruition when the social sciences abandon rationalist and constructivist explanations of social phenomena in favor of ones that recognize the roles of tacit and contextual knowledge, institutional evolution, and spontaneous order. Such an approach would dramatically improve our understanding of the human mind.” This paper offers a step in that direction.*

Hayek is at all times an epistemologist.

– Weimer (1982)

All institutions of freedom are adaptations to [the] fundamental fact of ignorance.

– Hayek (1960, p. 30)

INTRODUCTION

F. A. Hayek’s thinking and writing span over multiple disciplines. From capital theory to monetary theory, and from tacit knowledge to institutional emergence, the list is long and rich. Within Hayek’s multiplicity of contributions, a theme emerges – the problem of limited knowledge, and, more specifically, how economic activity and social coordination are even possible given the dispersion of knowledge among the hundreds of millions of decision-makers who interact on a daily basis.

While Hayek is recognized as a theoretical economist and as a social theorist, he is less well known as a theoretician of the mind. His foray into theoretical psychology, *The Sensory Order: An Inquiry into the Foundations of Theoretical Psychology* (Hayek, 1952), is certainly not his most known work; it receives short shrift within the corpus of Hayek studies, and explicit studies linking the Hayekian sensory and institutional orders are few and far between. Miller (1979) thus laments that *The Sensory Order*

is “largely neglected by Hayek’s interpreters” and Krecké and Krecké (2007, p. 6) explain that “for a long time, economists have been skeptical about the relevance of Hayek’s [cognitive] work for their discipline” – although, encouragingly, “*The Sensory Order* is nowadays rediscovered by more and more economists.” Garrouste (1999) notes that *The Sensory Order* “is much more known and quoted by psychologists and neurobiologists than by economists.” Rizzello (1999, note 7, p. 24) summarizes the neglect: *The Sensory Order* “has been little considered by most scholars. A few authors even consider it as separate and different from the rest of Hayek.” The reasons for this neglect are a subject for another paper.¹

Horwitz (2000) explains that “much of the voluminous literature on the social theory of F. A. Hayek has focused on his economics or his politics, or the relationship between the two. A far smaller proportion has explored the relationship between his work in the Austrian tradition in economics, his defense of political liberalism and his work in theoretical psychology and philosophy.” For such explorations (in addition, naturally, to Horwitz (2000)) see, for example, Miller (1979), Weimer (1982), Buto and Koppl (1993), Böhm (1994), Rizzello (1999), Buto and McQuade (2002), Steele (2002 and 2004), or McQuade and Buto (2005).

The present paper agrees with these explorations that there exist strong links between Hayek’s cognitive and institutional theories – and sets forth to present systematically ideas that largely exist within the literature, if diffused among multiple papers (how appropriate, and ironic!). The main parallels are three.

First, the sensory order is an imperfect representation of the physical order, and there are limits to what the human mind can know, as knowledge is acquired from experience. For both these reasons, knowledge is limited – a point emphasized in Hayek’s social theory, and the institutional consequences of the knowledge problem, that is, reliance on “local knowledge of time and place” (Hayek, 1948) over centralized planning.

Second is the importance of learning and generation of knowledge. The sensory order involves a constant learning process, as the mind updates its understanding of the environment. Learning at the cognitive level finds its place in Hayek’s institutional theory: institutions survive, disappear, or evolve, depending on how useful they are at generating and communicating knowledge. In fact, the very case for liberty made by Hayek (1960) rests on the superior potential of free societies for overcoming the knowledge problem.

Third are the mental models and epistemological communities. Much of our understanding of the world comes through the representations we make

of it. Hayek explains this in his cognitive theory, and then extends the reasoning to his institutional theory. In addition to filtering the world through individual mental models, the human mind also learns from its environment, for example, through language and norms. Thus, knowledge has a social dimension, and much of what we know we cannot articulate, as it is embedded in institutions, norms, and customs.

This paper demonstrates the parallels between Hayek's cognitive and social theories, and then takes the reasoning one step further: not only do there exist strong parallels between the two theories, but both revolve around the same foundation, that is, the knowledge problem. Thus, Hayek's institutional order can be seen to flow from the sensory order, and Hayek's very defense of liberty rests on a cognitive – rather than utilitarian or deontological – foundation.

The first section "Hayek's Sensory Order" outlines and summarizes Hayek's theory of the mind. The second section "The Cognitive Foundations of Hayek's Social Order" explores the three themes (limited knowledge, learning, and mental models) that are prevalent in both Hayekian theories, and which help explain the Hayekian social order, as understood from a cognitive perspective. Just as the second section examines themes explicit in Hayek's work, the section "Beyond the Text: Post-Hayekian Applications" moves into the realm of speculation, using the parallels between the two orders to shed light on an ongoing methodological challenge in economics, the individual–group analytical relationship. The final section concludes.

HAYEK'S SENSORY ORDER

A Bit of Historical Context

The Sensory Order, although written in 1952, had been gestating in Hayek's mind for the better part of 30 years. In a textbook case of Kirznerian (or Hayekian?) serendipity, Hayek's cognitive theory emerged by accident. Caldwell (2004, p. 136) explains how "the winter in Vienna was terrible [in 1919–1920]: many in the city were near starvation, and the weather was so blisteringly cold that fuel shortages forced the university to close. Hayek's family arranged [a trip to Zurich] both to get him out of that dismal environment and to help him recuperate from the lingering effects of his [World War I] bout with malaria." It was during this winter in Switzerland that Hayek "worked for a few weeks in the laboratory of [a] brain

anatomist..., tracing fiber bundles of the brain.” Although he received encouragement on the ensuing paper, “Contributions to a Theory of How Consciousness Develops” (Hayek, [1920]1991), Hayek was distracted by other responsibilities upon his return to Vienna. In “September 1920, Hayek put the paper aside in order to prepare for some law exams, and it lay in a desk drawer for a quarter of a century. Hayek returned to it in 1945, and it would ultimately form the basis of... *The Sensory Order*” (Caldwell, 2004, p. 137). In addition, the major psychology professors at the University of Vienna had died during the world war, or shortly thereafter, so Hayek gave up his psychological aspirations in favor of the law (Caldwell, 2004, p. 138).

The Sensory Order

I will offer a brief summary of what Caldwell (2004, p. 261) calls “a technical scientific text whose didactic prose is, for those uninitiated in the jargon of psychology, at times impenetrable”; Hayek himself “once reported that his ‘colleagues in the social sciences’ generally found his book to be ‘uninteresting or indigestible’” (Hayek, 1979, p. 199, n. 26, as quoted in Caldwell, 2004, p. 261, n. 2).² As even *habitués* and all neophytes to Hayek’s writings will profess, *The Sensory Order* is not Hayek’s most accessible work. In the words of Boring’s (1953) review, if “Hayek wrote to gain disciples, he had better have written more persuasively and less precisely, for one can get bored by the deadly logical progression of this book long before he can find a cogent phrase to which he dares register vigorous dissent.” And yet, for all the legendary obscurity surrounding the book, the central point of Hayek’s sensory order is quite straightforward.³ For concise, tidy summaries of Hayek’s cognitive theory, see Butos and Koppl (1993), Horwitz (2000), or McQuade and Butos (2005).

Hayek starts his inquiry by noting that we live in a world with two orders⁴: the physical order of the world around us, and the sensory order (Hayek, 1952, 1.7 and 1.9),⁵ which is our perception of the physical order. Hayek notes that the physical order is objective, while the sensory order is subjective (1.10). He sets out to explain the relationship between the two: “the task of psychology is to explain the correspondence between a physical event and its mental picture” (1.19; see also 1.3).

Hayek notes that we erroneously tend to confuse the physical and the sensory order, although the latter is merely a mental representation of the former. “It is the existence of an order of sensory qualities and not a

reproduction of qualities existing outside the perceiving mind which is the basic problem raised by all mental events,” he writes (Hayek, 1952, 1.18). “Psychology must concern itself,” he continues, “with those aspects of what we naïvely regard as the external world which find no place in the account of that world which the physical sciences give us.” He gives a general statement of the problem at hand: “Psychology must start from stimuli defined in physical terms and proceed to show why and how the senses classify similar physical stimuli sometimes as alike and sometimes as different, why different physical stimuli will sometimes appear as similar and sometimes as different” (Hayek, 1952, 1.21).

As a rudimentary answer to the problem, Hayek defines the mind as “a particular order of a set of events taking place in some organism and in some manner related to, but not identical with, the physical order of events in the environment” (Hayek, 1952, 1.49); in other words, the mind is a classificatory apparatus of external stimuli – or, in the phrasing of Steele (2002), “mind is that subjective mental order of events that prevails in that part of the physical universe that is self.” This mental order lies within an organism, which is a part of the physical order; it offers an imperfect mapping, but nonetheless allows the organism to function in the physical order (Hayek, 1952, 1.49). Glossing over the technical, neurological details,⁶ Hayek emphasizes the importance of the classification of neural stimuli – not the individual neural impulses, but their place within a general, mental order of classification, that is, which neural firing leads to which other, and so on, based on an overall classificatory order.

The core, then, of the sensory order is one of linkages, of classifications of physiological events – based on the event itself, to be sure, but also its place within a classificatory order borne of past experience (Hayek, 1952, 2.7). The sensory order is thus variable (1.57) and dynamic (2.21), as it builds on past knowledge (2.24). Hayek explains that “there will be implicit in all sensory experience certain relationships determined by earlier linkages (*i.e.* by the influence of the external world on the organism) which have never been the object of sensory experience in the ordinary meaning of the term; and that the order of sensory qualities will be subject to continuous modification by new linkages between impulses” (Hayek, 1952, 5.14). New impulses (*i.e.*, new experience) are thus both classified by, and have an influence on, the classificatory mechanism: “even after relatively simple systems of connexions [*sic*], effecting some measure of classification, have been formed, this system will be constantly modified by new linkages. But as the existing system of connexions becomes more and more complex and more firmly embedded, any new linkage will be less likely to alter its general

character” (Hayek, 1952, 5.13). In other words, there are diminishing marginal returns to learning.

Hayek is very clear that the sensory order is a “very imperfect” and “even a definitely erroneous reproduction of the physical order” (Hayek, 1952, 5.20–5.24). Indeed, as the organism goes through the world and learns about it, the mental order evolves into a gradual approximation – but just an approximation – of the physical order (5.17–5.19). Within this approximation, Hayek distinguishes between a semipermanent “map” of the physical order, and a fleeting “model” of the immediate environment within which the organism is operating. The “map” is thus the more stable aspect of the mental order, while the “model” is its more evolving facet.

The Map and Model

Hayek describes a mental “map” of the physical order (Hayek, 1952, 5.25). The map is imperfect, and subject to continuous (if gradual) change, as it is merely a representation of the physical order; not only does the map represent “merely some” of the physical order, but some mental relationships simply differ from those which exist, objectively, in the physical world (5.26). The map is a semipermanent structure of connections, an apparatus for classification that exists independent of particular impulses at a given moment (5.42). It represents the organism’s cumulative knowledge about the “kind of world in which the organism has existed” and the stimuli received in the past (5.42). The map, a semistatic apparatus of classification (5.42 and 2.21), provides the framework for evaluating and classifying new impulses, that is, new situations (5.43). The map can be updated, but in a diminishing marginal manner as the organism gains more experience. As Hayek explains, the “structure itself... is liable to change as a result of the impulses proceeding in it, but relatively to the constantly changing pattern of impulses, it can be regarded as semi-permanent” (5.43). In sum, the map provides “a theory of how the world works rather than a picture of it” (5.89). Finally, Hayek emphasizes that the map is subjective; the maps of two individuals can be similar, but never identical, as this would, of course, presuppose identical history and anatomical structures between the two individuals (5.27; see also 1.69). Ever the alpine mountaineer, Hayek explains that “two persons discussing the same walk, with different maps of the region before them, will in general encounter no difficulty in understanding each other, although particular points on their route may have different significance for them” (5.28).

The “model” of the physical world operates, on a more immediate level, within the map (Hayek, 1952, 5.44). This model, a “continuous representation of the environment” (5.49), represents the present-time environment within which the organism is operating. It allows for a continuous self-updating from new information (impulses) received, and, in a much slower process, updates to the underlying map (5.49). Thus, the model represents both the actual state of the environment and expected changes (5.50–5.62), in a constantly adaptive, dynamic process through which the organism can constantly check and correct expectations. In fact, Hayek emphasizes that the organism lives as much in a world of expectations as a world of fact. “Most responses to a stimulus are probably determined only via fairly complex processes of ‘trying out’ on the model the effects to be expected from alternative courses of action. The reaction to a stimulus thus frequently implies an anticipation of the consequences to be expected from it” (Hayek, 1952, 5.61). Furthermore, the model implies that “behavior will be guided by representations of the consequences to be expected from different kinds of behavior” (5.68); the model predicts the effects of different courses of action and selects among the effects of alternative courses the one that is apt to be desirable (5.68–5.70; see also 6.26). As Hayek (1982, p. 291) later wrote, “stimuli and responses thus become merely the input and output of an ongoing process in which the state of the organism constantly changes from one set of dispositions to interpret and respond to what is acting upon it, to another such set of dispositions.”

Summary: Main Themes

Six themes bear repeating as a synopsis of this somewhat technical summary.

1. *The overall theory.* The sensory order is a representation of the physical order; it is a classification that takes place via a network of impulse connections (Hayek, 1952, 7.2).
2. *Subjectivity.* Hayek reminds us that “perception is always an interpretation” of the physical world (6.36), and thus, *ipso facto*, subjective (generally, see Hayek, 1952, 6.33–6.38).
3. *Incomplete and imperfect representation of the physical world.* The model of the physical world is incomplete, and gives a distorted reproduction thereof; “the classification of ... events by our senses will often prove false, that is, give rise to expectations which will not be borne out by events” (Hayek, 1952, 6.44; generally see 6.39–6.43). Indeed, some classifications “might indeed prove later not a help but an obstacle to orientation and appropriate behavior” (Hayek, 1952, 7.32).

4. *Learning and updating.* Hayek reminds us that “to acquire the capacity for new sensory discrimination is not merely to learn to do better what we have done before; it means doing something altogether new ... It means discriminating between stimuli which before were not discriminated at all” (Hayek, 1952, 7.32). Since knowledge is contingent on the classification process, a change in classification implies a change in a portion of the knowledge embodied in definitions (Hayek, 1952, 8.18–8.19). We thus see an increasing tautology of knowledge (Hayek, 1952, 8.12–8.27) in a confirmatory bias of sorts. In addition, as expectations are dashed or confirmed, the process adapts, in a “Darwinism of ideas” of sorts.
5. *Social embeddedness of knowledge.* The organism’s past experience, the environment within which it has operated, will affect the structure of classification. The “apparatus by means of which we learn about the external world is itself the product of a kind of experience” (Hayek, 1952, 8.1; see also 5.1–5.16). As the sensory order is shaped by the environment, “we interpret any new event in the environment in the light of ... experience” (Hayek, 1952, 8.1). Language increases the capacity for classification, as the individual, through the acquisition of language, can use the classification and experience of humanity (or at least his social group), in addition to his own (Hayek, 1952, 6.10).
6. *The map and model.* Whereas the map is a semipermanent structure of understanding (which is ultimately dynamic but updated only very slowly), the model is an immediate, fleeting representation of the immediate environment.

To these, I would add a seventh theme, as important as it is implicit in Hayek’s writing. As an adaptive, classifying system, the sensory order is a spontaneous order. It is not guided by an outside force, nor is it consciously guided by the mind: the sensory order “can be recognized as present in actions which are not directed by consciousness or by a human mind” (Hayek, 1952, 1.68). Böhm (1994, p. 163) likewise explains that knowledge and order are generated by the process itself, rather than externally. Generally, see Butos and Koppl (1993) on the mind as spontaneous order.

THE COGNITIVE FOUNDATIONS OF HAYEK’S SOCIAL ORDER

This theory of the sensory order clearly flows from the mind of a thinker who is deeply concerned with the interactions of human actors with their

environment, and among each other. It comes as no surprise, then, that Hayek's cognitive theory helps explain his institutional theory. Indeed, Hayek himself explains that *The Sensory Order* was intentionally designed as a complement to his social theorizing: "In the end, it was concern with the logical character of social theory which forced me to re-examine systematically my ideas on theoretical psychology" (Hayek, 1952, p. v).

The claim made here, however, is stronger than a mere demonstration of similarities. To be sure, there are parallels between Hayek's cognitive and institutional theories. But the story does not end there. Rather, the two are complementary. Hayek's psychological theory fits comfortably within his overall social theory, as it ends up forming the very foundation thereof. In turn, Hayek's institutional theory – on the importance of generating and transmitting limited knowledge – can be seen as a solution to the cognitive problem he identifies. The knowledge problem flows from the sensory order and the mind's limited capacity. And Hayek's institutional theory flows from the knowledge problem.

On the cognitive side, I rely, naturally, on *The Sensory Order*. On the social side of the comparison, my analysis centers, unsurprisingly, on *The Constitution of Liberty*, as Hayek's central statement of his constitutional theory.⁷ But I also draw more generally on Hayek's institutional writings and essays on the knowledge problem, videlicet:

1. Writings on knowledge from the 1930s and 1940s, written before *The Sensory Order* (at least in its formal incarnation) and collected in *Individualism and Economic Order* (Hayek, 1948).
2. Writings on knowledge published around the same time as *The Sensory Order*, and compiled in *The Counter-Revolution of Science: Studies on the Abuse of Reason* (Hayek, 1979[1952]).
3. Writings on knowledge and institutions from the 1960s and 1970s, compiled in *Studies in Philosophy, Politics and Economics* (Hayek, 1967) and *New Studies in Philosophy, Politics, Economics and the History of Ideas* (Hayek, 1978).
4. Miscellaneous writings by Hayek, in which he himself comments on the link between his cognitive and social theories, such as *The Sensory Order after 25 Years* (Hayek, 1982) and *Hayek on Hayek* (Hayek, 1994).

Based on the themes that emerge from these writings, I draw parallels among three major themes present in both Hayek's sensory and institutional orders, and show the cognitive foundations of Hayek's social order.

The Knowledge Problem

The limits of knowledge form the most basic theme of both Hayekian orders. Hayek is quite clear that the sensory does not offer a perfect or accurate representation of the physical order; in addition, there are limits to what the human mind does (and can) know, as knowledge is acquired from experience. For both these reasons, knowledge is limited. It is not surprising that Hayek should thus emphasize the limits of knowledge in his social theory: knowledge is limited and diffuse among the economy's actors.⁸

The capacity for human reason is limited, as Hayek so painstakingly emphasizes in his studies on the abuse of reason (Hayek, 1979[1952]). Hayekian institutional theory stems from Hayekian cognitive theory. The fundamental accomplishment of civilization is overcoming the knowledge problem (Hayek, 1960, p. 22); civilization allows us to profit from knowledge we do not individually possess (Hayek, 1960, p. 24), and to make use of evolved "rules known by none and understood by all" (Hayek, 1967, pp. 44–46).⁹

Because knowledge is diffuse and limited, we must limit coercion and planning. The case for liberty is based on human ignorance (Hayek, 1960, p. 29): no individual has the knowledge, and therefore the capacity, to determine the ends of others (Hayek, 1960, p. 88). Government (or, writ large, central planning and coercion) should thus limit itself to general principles, rather than specific planning based on the incomplete knowledge of central planners (Hayek, 1948, p. 18; Hayek, 1960, pp. 205–209). Furthermore, because of the limits of human reason, there is only so much institutional design possible; in fact, most institutions emerge as "the result of human action, but not of human design" (Hayek, 1967, p. 96). No individual has – or can have – knowledge of the overall order of rules (Hayek, 1967, pp. 68–69); rules are thus best left to emergence, as design is impossible (Hayek, 1979[1952], pp. 149–154; Hayek, 1960, p. 70; Hayek, 1967, pp. 96–101). Because of limited knowledge and limited reason, the best we can do is create the conditions for growth, rather than trying to plan the details of economic activity (Hayek, 1979[1952], p. 160). The market process, rather than central planning by officials with limited knowledge, allows for a process of economic coordination (Hayek, 1979[1952], pp. 165–182).

If left free, humans will achieve more than individual reason could design or foresee (Hayek, 1948, p. 10), and institutions that facilitate and allow use of local knowledge of time and place will help overcome the knowledge problem (Hayek, 1948, p. 14). Conversely, much harm has come from the hubristic (and false) notion that any individual, or group of individuals,

possesses sufficient knowledge to create the rules and institutions under which we live (Hayek, 1967, p. 107). Whereas a wise approach to institutions will simply seek to limit harm (Hayek, 1948, p. 11), rationalist constructivism will inevitably decay into tyranny, because no central planner has enough knowledge to advance the goals of others, but will inevitably end up imposing his own goals (Hayek, 1948, pp. 1–32; Hayek, 1960, pp. 54–70).

In sum, Hayek himself emphasizes the links between the sensory and the institutional order (Hayek, 1994, pp. 30–35), based on the problem of limited knowledge and thus limited planning ability. There is thus an explicit epistemological basis to social organization (Hayek, 1979[1952], p. 161); “all institutions of freedom are adaptations to this fundamental fact of ignorance” (Hayek, 1960, p. 30). Weimer (1982) opines that the central problem in Hayek is one of knowledge: dispersed knowledge, and coordination in spite of such dispersion. Likewise, Horwitz (2000) argues that “Hayek’s argument was ultimately one about the comparative epistemological properties of alternative sets of social and economic institutions.”

Learning

Hayek states clearly in *The Sensory Order* that the mind is not instantly formed and instantly developed. Rather, the sensory order is dynamic, and involves a constant feedback loop, a process of trial and error, as the mind updates its understanding of the environment. In Hayekian language, the mind’s “model” of the environment is constantly updated, whereas the semipermanent “map” of the world is updated slowly (Hayek, 1952, 5.49) – but both evolve, as the brain receives new stimuli, and engages in reclassification to correct mistakes and update its understanding of the world based on new information. This learning process at the cognitive level parallels Hayek’s explanation of institutional evolution: institutions survive, disappear, or evolve, depending on their utility, as societies “learn” based on new information.¹⁰ To be sure, Hayek explicitly offers the disclaimer that “society is not a brain” (Hayek, 1967, p. 74); but the parallels are there. For example, Hayek (1960, p. 23) explains that humans are not initially endowed with a mind for civilization, but gradually learn.

In fact, the very case for liberty made by Hayek (1960) rests on the superior potential of free societies for trial and error, and learning from mistakes in an environment of limited and diffuse knowledge. Civilization itself is the result of a long evolution of learning (Hayek, 1978, p. 38). Current knowledge includes past human adaptation to the environment

(Hayek, 1960, p. 26). In light of our limited knowledge, efficient institutions are those that facilitate transmission of knowledge (i.e., learning), both over time and among contemporaries (Hayek, 1960, p. 27). The advance and preservation of civilization is dependent on the maximum opportunity for accidental learning (Hayek, 1960, p. 30). Freedom is a learning process, under which free institutions allow for individuals to learn from their mistakes (Hayek, 1948, pp. 21–22; Hayek, 1960, pp. 31, 37, and 76). In fact, institutional evolution through self-correction – learning on a larger scale – can take place only within a framework of freedom (Hayek, 1960, p. 67). Within that framework, institutional growth can occur via a process of trial and error (Hayek, 1978, p. 260), in an institutional Darwinism of sorts, which would not be possible if institutions were frozen by the rigidity of *dirigisme* (Hayek, 1978, p. 43; see also Hayek, 1948, pp. 8–9 and Hayek, 1967, pp. 70 and 100–101).

This learning process extends beyond the institutional level to markets, where information is constantly updated via such instruments as the price mechanism, allowing for coordination of plans among actors with incomplete knowledge. The individual, who possesses incomplete knowledge, is guided by the knowledge of others, via institutions, markets, the price mechanism, etc. (Hayek, 1960, p. 28). McQuade and Butos (2005) argue convincingly that social orders share much in common with the sensory order, as both involve complex emergent phenomena of classifying, adapting, and learning.

The limits on knowledge thus imply a second institutional role, beyond facilitating the communication of knowledge and learning by individuals: the *generation* of knowledge (in a sense, a sort of “social-level” learning). Knowledge in society is dispersed – hence the importance of communication via institutions. But knowledge in society is also limited – hence the importance of generating knowledge through institutions. Perhaps the single, greatest, most obvious example is the price mechanism as knowledge generated by the market order. For the same reasons that the institutions of a free society are superior to central planning for *communicating* institutions, they are also superior at *generating* knowledge. For details, see Butos and McQuade (2002), as well as Rizzello (1999, p. 27).

Mental Models, Epistemological Communities, and Subjectivity

In many ways, mental models and expectations seem like they ought to be treated within learning. Indeed, much of learning involves confirmed or

dashed expectations – does the world conform to my understanding thereof? Does experience fit my expectations? Is my heuristic model useful in the face of reality? If so, I strengthen my expectations. If not, I revise my expectations, thus learning and revising my inaccurate representation of the world. But this angle is already covered in the previous paragraph. Instead, I focus in this section on the importance of mental models, a mental filter, for our understanding of the world, and the ensuing subjectivity of human knowledge.

The sensory order helps us make sense of the physical world by providing a representation, if incomplete and only partially accurate, thereof. The “map” is a semistatic apparatus of classification (Hayek, 1952, 5.42). Whereas the “model” undergoes constant updating as the organism interacts with its environment, the “map” provides a framework for evaluating impulses (Hayek, 1952, 5.43).¹¹ Or, in other words, the Hayekian map provides a mental model – a way of understanding the world and making sense of it, that is changeable, yes, but more likely to sort events to suit the framework (in a confirmatory bias of sorts) than to be updated (Hayek, 1967, p. 75). Thus does Hayek (1982, p. 290) emphasize the central importance of individual “dispositions” to interpret stimuli, and the changes in disposition as the organism evolves (see also Butos & Koppl (1993) on the effect of accumulated knowledge on classification). Moving from the sensory to the social order, this helps explain Hayek’s notion of “the facts of the social sciences,” which he defines as the opinions and beliefs that people hold (Hayek, 1979[1952], pp. 47 and 64 and the eponymous essay in Hayek, 1948, pp. 57–66). Individuals build up an understanding of the world (the sensory order) based on their “views and concepts” of that world (Hayek, 1979[1952], p. 40). In sum, the objective nature of the world is irrelevant for explaining behavior (Hayek 1979[1952], p. 51), because the sensory order – along with the human action that stems from it – is subjective. While the “real world” of the physical order is objective, our understanding thereof, our interpretation thereof, our knowledge, are all subjective – with ramifications, naturally, for the social sciences (see especially Hayek, 1979[1952]).

Finally, we are reminded of the social dimension of knowledge and learning. The human mind learns from its own experience, and from its own updating – but also from its social environment. Likewise, knowledge – “the facts of the social sciences” – has a social dimension, as humans learn not only explicitly, but also implicitly, from language, customs, institutions, etc.; much of what we know we cannot articulate, as it is embedded in institutions, norms, and customs (Hayek, 1960, pp. 68–78 and 101–102; Hayek, 1967). The human mind is a product of civilization, shaped not only

by experience but also by custom (Hayek, 1960, p. 24). Hayek's sensory order thus helps explain cultural similarities and differences: the sensory order can be similar, but never identical among individuals (Hayek, 1952, 1.69 and 5.27). Indeed, similar "maps" will lead to *similar* descriptions of the world among individuals with similar backgrounds, but no two individuals will have exactly identical backgrounds and will thus never have exactly identical minds (Hayek, 1952, 5.28). This theory matches our observation of cultural tendencies by country and other groups; while differences will vary by individual, members within a given philosophical or language community will tend to view the world similarly, and often in contrast to, members of different communities. Böhm (1994) explains that knowledge is "embedded in the social practices of a particular community"; see also Steele (2002) and Butos and Koppl (2007) on the social embeddedness of individual reason.

BEYOND THE TEXT: POST-HAYEKIAN APPLICATIONS

The Hayekian *opus* already exhibits volumes about the links – obvious and implicit – between Hayek's sensory order and his social order. Moving beyond Hayek into "post-Hayekian" thinking, insights from the two orders can shed light on a lingering challenge within contemporary economics: rational choice by fully informed, utility-maximizing *Homo economicus* versus the growing sense that "culture matters" (see the eponymous book by Harrison and Huntington (2000), as well as Pejovich (2003) or North (1990)).

I am profoundly dissatisfied with the neoclassical model of perfectly rational, perfectly informed agents maximizing utility with zero transaction costs. To be sure, it is helpful as a pedagogical exercise, but loses traction quickly in a comparative cultural context. Indeed, under these assumptions, culture is irrelevant, and individuals respond rationally and uniformly to incentives. But that is hardly the case. Instead, I agree enthusiastically with a growing literature on the importance of cultural context. But I am also dissatisfied with sociology's squishy methodology, which goes to the other extreme in ignoring incentives and methodological individualism, that is, the premise that individuals – and only individuals – act and choose. Both culture and incentives matter in determining behavior; the question is how. Evans (2007) laments that:

The study of culture nestles uneasily within economic literature, on account of its inherent immeasurability. Despite earlier calls (Boulding, 1974) it has only begun to be taken

seriously by economists fairly recently, along with the necessary methodological change from quantitative techniques (such as econometrics) toward qualitative methods (such as ethnography (Chamlee-Wright, 1997; Tilly, 2003) and analytic narratives (Bates, Greif, Levi, Rosenthal, & Weingast, 1998; Rodrik, 2003). Some studies attempt to use various proxies for culture to establish its causal influence on economic outcomes (Guiso, Sapienza, & Zingales, 2006) but aggregate data can't hope to understand the imbedded, subtle and local knowledge that constitutes society. Cultural issues are complex and will yield emergent phenomena, and must therefore be studied as a process, rather than an essence.

Therefore the incorporation of culture into economic theory will not be straightforward, because it is not merely an extra variable to be added to an existing framework. On the contrary, "culture is not another factor to be considered in addition to rational incentives, it is the underlying meaning of the specific content of any rational choice" (Lavoie & Chamlee-Wright, 2000, p. 42).

In more general methodological terms, this debate is not new. Instead, it parallels an internal Hayekian tension, between methodological individualism and institutional evolution at the broader, group level. Space considerations prevent me from going into this debate. Fortunately, the problem has already been explored, for example, by the half dozen contributions to the symposium on group selection and methodological individualism in Koppl (2004). The problem is also nicely summarized by Rizzello's (1999) discussion and explanation that the sensory order is "influenced by genes as well as by the experience of the individual and his species" (p. 25) or that an individual's cognition is "personal and unique" but "also the result of the interactive process which takes place with [his] cultural and social context" (p. 106). If cognition is thus both individual and environmental, it should come as no surprise that choice and decisions are too, as captured so nicely by "institutional individualism" – an approach that acknowledges the role of the institutional environment, while maintaining methodological individualism. In other words, environment isn't irrelevant (as is the case for radical methodological individualism); simultaneously, action is not predetermined by the environment (as it is in, say Marxism, much of sociology, or other aggregate-based theories of human action).¹²

Economic Choice: Culture and Economics

As a first attempt, perhaps we can map culture as price-elasticity of demand in neoclassical price theory: a change in price, like a change in incentive or an institutional change, will have an effect. But the extent of that effect will depend on elasticity or culture. Alternatively, culture may be seen as the utility curve interacting with the budget constraint of institutions. Not to

get lost in analogies – both are important in predicting and influencing outcomes. It is clear that both culture (the informal) and institutions (the formal) are important and mutually influencing. Perhaps a resolution to these questions – and a step toward bridging the insights and methodologies of sociology and economics – will come from application of a Hayekian cognitive framework (Garrouste, 1999). Indeed, the economics–sociology divide can be grossly characterized as follows: to the methodological individualist economist, individuals are rational utility-maximizers, whereas to the sociologist, thinking and values are determined by cultural context. Once we incorporate Hayek’s cognitive theory, we have insights from both: yes, the sensory order is largely determined by environment (language, institutions, customs, etc.), and yes, reason is limited. But no two individuals are exactly the same (Hayek, 1952, 1.69, 5.27, 5.28), and each individual mind has its unique sensory order. We thus have socially embedded rational individuals, who make choices and respond to incentives, but according to their contextually defined mental models – and perhaps a resolution to a methodological battle.

Building on the cultural theory of Thompson, Ellis, and Wildavsky (1990), Schwarz and Thomspson (1990), and Wildavsky (1998), Evans (2007) describes four cultural types – *individualist*, *egalitarian*, *hierarchist*, and *fatalist*:

The individualist is a near-relation to *homo economicus* and believes that choice, action and consequences are all borne solely by the decision-maker ... Resources are defined beyond the planet’s natural stock, extending to human ingenuity and technical progress. He requires internalised incentive mechanisms, and believes that return follows effort ... There’s a constant general equilibrium and hence *laissez-faire* is the optimum attitude. Essentially an optimist, but the individualist will only calculate his personal returns to assess whether an action has been a success ...

In stark contrast to the individualist’s benign vision, the egalitarian is a pessimist foreseeing impending disaster ... [He] defines resources solely as “natural resources” which are finite (and therefore depleting), requiring urgent action to manage the needs of society. [His] inherent belief in equality, together with the free-rider problem, means that everyone has to abide by the same policy, and this requires enforcement. *Homo egalitarian* can only achieve [his] desired action by eliciting a change in the behavior of others, and will therefore engage in persuasion (educational campaigning) as well as coercion (mandatory policies) ...

The hierarchist is not in the business of altering behavior – needs are given and the simple job is to allocate resources fairly. *Homo hierarchist* is a public official who will manage goods and capital because [his] vision falls in between the individualist and the egalitarian: nature is forgiving within limits. The hierarchist will set those limits using regulation and management. [He] will acquire information and authoritative advice in

order to make social activity accountable and recorded, and expect experts to deliver correct advice ...

If life is essentially random, and our actions have little consequence, there is scant point in managing needs or resources. *Homo fatalist* is poor, and lacks the resources to effectively alter his situation. He is also disenchanted and falls outside collective behavior. He will therefore react to situations rather than create them, and merely try to cope.

The effects on behavior and social choice follow from cultural type, as each type will demand different institutions, different political goods, different levels of state intervention, etc. – and react differently to the same “objective,” so-called, incentives like price changes. Again, each Wildavsky–Thompson type can be summed up broadly as a mental model, a collection of expectations and a vision of the world – or, in Hayekian language, a “map” of the world, that can be updated (or corrected) by new information, but that will all too likely lead to confirmatory bias, as new data is viewed through a subjective filter. An application of Hayek’s cognitive theory to the understanding of the four types will likely strengthen the (already impressive) explanatory power of the Wildavsky–Thompson framework, as we gain greater insight into the motivations and mental models that drive actors on the stage of political economy.

In language more poetic than the prose employed by most economists, we might paraphrase Hayekian insights on the tension between individual and contextual choice with the following passage from Elias (1978[1939], p. 262; see also, generally Elias 2001[1939]):

What is meant by the concept of the figuration can be conveniently explained by reference to social dances. They are, in fact, the simplest example that could be chosen. One should think of a mazurka, a minuet, a polonaise, a tango, or rock ‘n’ roll. The image of the mobile figurations of independent people on a dance floor perhaps makes it easier to imagine states, cities, families and also capitalist, communist and feudal systems as figurations. By using this concept we can eliminate the antithesis, resting finally on different values and ideals, immanent today in the use of the words “individual” and “society.” One can certainly speak of a dance in general, but no one will imagine a dance as a structure outside the individual or as a mere abstraction. The same dance figurations can certainly be danced by different people; but without a plurality of reciprocally oriented and dependent individuals, there is no dance. Like every other social figuration, a dance figuration is relatively independent of the specific individuals forming it here and now, but not of individuals as such. It would be absurd to say that dances are mental constructions abstracted from observations of individuals considered separately. The same applies to all other figurations. Just as the small dance configurations change – becoming now slower, now quicker – so too, gradually or more suddenly, do the large figurations which we call societies.

An Example: Constitutional Culture

I have long been dissatisfied by contractarian explanations of constitutions (see, e.g., Buchanan (1975), for all its brilliance and explanatory power), and prefer instead a coordination approach, such as Hardin (1999), which fits more neatly within the emergent Hayekian tradition (Voigt, 1997). Specifically, I emphatically reject the notion that good constitutional parchment is sufficient for successful constitutionalism (Wenzel, 2007a, 2008a, forthcoming a, forthcoming b). The most glaring example is that of Argentina, which adopted, almost verbatim, the US Constitution in 1853. The constitution stuck for about half a century of prosperity, but was then rejected, as Argentina fell into a string of military coups, populism, and ravaging of the economy by both military and civilian governments. Same parchment, different results – the explanation must lie in the underlying culture, which rejected the constitutional parchment. In a series of papers (Wenzel, 2007a, 2007b, 2008d, forthcoming a, forthcoming b) and working papers (Wenzel, 2008a, 2008b, 2008c), I develop a theory of constitutional culture and constitutional stickiness. Constitutional culture encompasses the norms, attitudes, and beliefs, explicit and implicit, that individuals (and preponderant groups of individuals within a country) hold about the nature, scope, and function of constitutional constraints. Constitutional culture encompasses such things as willingness to be bound, willingness to share power (even with a disliked opposition), sense of entrenchment of the constitution, willingness to sacrifice constitutional procedure for a desired result, philosophical vision of political goods to be delivered by a constitution, the balance between majoritarianism and constitutional constraint, etc. Constitutional culture is a complex emergent phenomenon, as it arises from the interactions of individuals, through learning, feedback, socialization, social interactions, etc.

The second step in my proposed framework is simple: in order to stick, (formal) constitutional parchment must match the underlying (informal) constitutional culture; if the two are slightly mismatched, compromise will ensue in the form of constitutional change (as was the case, for example, in France's Fifth Republic, or Russia's second post-Soviet court of constitutional review). If they are radically mismatched, the country is apt to reject constitutionalism entirely (as happened in the Philippines after World War II, and in Argentina at the dawn of the twentieth century, to name but two examples).

Constitutional culture can be treated as a mental model, a series of expectations and understandings about the constitutional order, how it is

and how it ought to be. I thus see potential fruitful applications of the “map” and “model” approach from Hayek’s sensory order, to understand how individuals and (cautiously) groups of individuals approach constitutionalism – rather than a one-size-fits-all approach where all respond uniformly to constitutional incentives, willingness to be bound, and constitutional culture are tied up in the individual’s Hayekian map of the world.

CONCLUSION

The links between Hayek’s theory of mind and his theory of social–economic coordination (i.e., between his sensory order and institutional order) are omnipresent. Although *The Sensory Order* has been a traditional second fiddle to Hayek’s institutional work, many writers emphasize the links between the two orders. Weimer (1982) sees “several recurrent themes that provide continuity in Hayek’s social, political and economic thought, on one hand, and his psychology, on the other”; “Hayek is at all times an epistemologist, especially when doing technical economics, and even in his historical and popular writings.” Miller (1979), likewise, reminds us that “in a number of his writings ..., Hayek discusses the close connection that has existed since the seventeenth century between theories of politics and theories of knowledge.” Steele (2002) explains that “it is unsurprising that the exposition and conclusion of *The Sensory Order* are implicit in much of Hayek’s other work.” In the same spirit, Kresge (in Hayek, 1994) sees an “obvious” link between Hayek’s cognitive and social–institutional theories, and Krecké and Krecké (2007, p. 9) conclude that “even if Hayek has not explicitly linked his cognitive approach developed in *The Sensory Order* with his later work in the fields of political, legal and economic theory (in terms of “spontaneous orders”), the parallels between [the two theories] can hardly be overlooked.” Horwitz (2000) argues that *The Sensory Order* “is crucial to understanding both [Hayek’s] economics and his politics.” Rizzello (1999, p. 2, repeated on pp. 11 and 24) puts the point even more plainly: “In my opinion, the breakthrough in Hayek’s thought is that he shed light on the link between the micro-dimension (the mental process of acquisition of knowledge) and the macro-dimension ... (the development of a spontaneous order)”; in fact, Rizzello (1999, p. 26) sees *The Sensory Order* as the very “physiological bas[i]s of methodological subjectivism” and Posner (2007) thinks that the development of a rationale for the price system (based on institutions and knowledge) is Hayek’s “most important contribution to economics, and it comes directly from his cognitive theory.”

And it hardly seems coincidental that the only work by himself that Hayek references in *The Sensory Order* is “Scientism and the Study of Society” (Hayek, 1979[1952]), an essay on the abuse of reason.

Beyond the insights teased out and developed by Hayek himself, there are many more lessons to be drawn from this prolific thinker’s mind. Posner (2007, p. 266), for example, sees implications of Hayek’s cognitive theory for organizational economics, Austrian economics, law and economics, and behavioral economics. More broadly, Horwitz (2000) concludes that

Hayek’s thought will have come to fruition when the social sciences abandon rationalist and constructivist explanations of social phenomena in favor of ones that recognize the roles of tacit and contextual knowledge, institutional evolution, and spontaneous order. Such an approach would dramatically improve our understanding of the human mind. Hayek’s *The Sensory Order* provides just such a beginning.

A deeper understanding of Hayek’s sensory order, and especially the cognitive foundations of his social theory, helps shed light on Hayek’s institutional theory, and, more broadly speaking “the constitution of liberty.” But the applications of *The Sensory Order* have still just begun, and the promise and potential of Hayek’s relatively obscure essay in psychology are myriad for our analysis of the human mind, human institutions, and the general arrangements of human affairs.

NOTES

1. Posner (2007, p. 264) suggests the following explanation: Hayek’s cognitive theory makes “an impressive theoretical case, now amply confirmed by experience, against central planning whether confined to the economy or extended cross the entire range of public issues in a society. But with central planning ... now so thoroughly discredited, the practical significance of Hayek’s cognitive theory and of the political-economic theory that he derived from it has diminished.” One might add that, with the ongoing rise of what Hayek dubbed “pseudo-competition” (aka “market socialism”), we would do well to return to Hayek’s cognitive insights.

2. Bertalanffy (1968) has been recommended as an antacid for the interested reader. I thank an anonymous reviewer for pointing this out.

3. In fact, with a hint of Monday-morning hubris, I suggest that Hayek could have confined his work to sections I, II, V, and VIII, relegating sections III, IV, VI, and VII (or about 40% of the book) to a technical appendix or long footnotes. And, while I am second-guessing the great Hayek from the height of my scholarly youth, in for a penny, in for a pound, I wish Hayek had expanded section VIII, on the philosophical consequences of his cognitive theory. In a sense, this paper provides a modest contribution in that direction; see also Horwitz (2000).

4. Technically, there are three orders: the physical, the psychological, and the neurological. For the purpose of this paper, I conflate the latter two (psychological and neurological) as the distinction is not necessary here. The interested reader is invited to review, specifically, chapters III and IV of *The Sensory Order*.

5. Out of simplicity, I use Hayek's section headers, rather than pages numbers, throughout, for Hayek (1952).

6. See footnote 4.

7. Hayek's social theory spans more than half a century – essentially from *Economics and Knowledge* (Hayek, 1948) in 1936 – and arguably earlier, in the calculation debate – all the way to the *Fatal Conceit* (Hayek, 1988). I am thus casting an intentionally broad net when I refer interchangeably to Hayek's social theory, his institutional theory, his constitutional theory, or his theory of spontaneous order.

8. What Hayek labels “local knowledge of time and place” in his seminal essay on “The Use of Knowledge in Society” (in Hayek, 1948).

9. The interested reader (who is up for the more technical challenge) is directed to Cantor's Theorem; see Hayek (1967, pp. 60–62, especially note 49).

10. I use this sleight of hand with great trepidation. As a good methodological individualist – and student of Hayek's studies on the abuse of reason (Hayek, 1979[1952]) – I am well aware that groups, including societies, do not learn or make choices.

11. Hayek uses different language to express the same idea when he talks of fixed strategies and variable tactics (Hayek, 1967, p. 56, note 39).

12. I thank Anthony Evans for his clear phrasing of institutional individualism – both in conversation and on his blog (see http://thefilter.blogs.com/thefilter/2006/12/institutional_i.html). See Agassi (1975), as well as Boettke (1996), Toboso (2001), and Evans (2008).

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THE ROLE OF DISPOSITIONS IN HAYEK'S COGNITIVE THEORY

Evelyn Gick

ABSTRACT

Purpose – The goal of this chapter is to contribute toward an understanding of Hayek's book "The Sensory Order." It focuses on his concept of dispositions which influence perception as well as action. The dichotomy between the phenomenal and the scientific world which Hayek stresses throughout his works is based on the existence of such dispositions.

Approach – Hayek's cognitive writings such as The Sensory Order, "Rules, Perception and Intelligibility" (1963), "The Theory of Complex Phenomena" (1964), and "The Primacy of the Abstract" (1969) are the main resources to explain his cognitive theory.

Findings – Hayek's concept of dispositions facilitates re-interpretation of the term "dispersed knowledge." Also, Hayek's Theory of Cultural evolution is seen as an evolution of dispositions, which stands in line with his view on the concurrent development of mind and culture.

Import of findings – The chapter offers a cognitive interpretation of Hayek's theory of government, depicting governmental action and rule-setting as spontaneous as well as deliberate processes.

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1. INTRODUCTION

Among all of the works of F. A. von Hayek, economists seem to have the most difficulty in understanding his cognitive theory as described in *The Sensory Order*. An unusual contribution, this book has so far received little attention and even less appreciation. Since its publication in 1952, Hayek has added a series of subsequent articles, such as *Rules, Perception and Intelligibility* ([1963] 1967), *The Theory of Complex Phenomena* ([1964] 1967), and *The Primacy of the Abstract* ([1969] 1978), where he elaborated in more detail on some specific issues raised in the preceding book. Despite this, Hayek's cognitive writings were typically ignored by economists. Recently, and years after his death, a group of scholars (Rizzello, 1999; Butos & Koppl, 2007; Birner, 1999, 2002; Horwitz, 2000) have focused anew on Hayek's cognitive theory opening a fruitful discussion that now interprets Hayek's later work in the context of his early cognitive contributions.

Although hardly mainstream, the attempt to combine economics with psychology as Camerer (2008) emphasizes is not new. The roots of neuroeconomics can be traced back to Pareto. The neglect of psychological influences of neoclassical theory may well be due to the "pessimism of its time," accounting for treating the human mind as a black box (Camerer, 2008, p. 357). Today, fMRIs and related tools bring neuroscience and economics together and thus offer a new foundation for behavioral economics; this in spite of continuing skepticism about the usefulness of some of its results (Gul & Pesendorfer, 2005).

Hayek's *The Sensory Order* is a contribution to cognitive psychology, a branch of psychology that studies the fact that humans do not directly perceive impulses from the outside world, nor that they directly respond to them. Instead, human perception, thoughts, and actions are studied as the result of internal transformations and computations, originating from processes that are repeatedly carried out in the brain (Gazzaniga, Ivry, & Mangun, 2002). Hayek's central argument in *The Sensory Order* is that human perception and human action is a purely subjective phenomenon. The mind does not represent physical or external events but is an interpretation of these events that we call reality. In this chapter, I argue that Hayek's work on social evolution cannot be fully understood without relating it to his cognitive theory.¹ This, at first, may seem counterintuitive since we commonly think of cognitive processes taking place at an individual level, whereas the cultural point of view usually encompasses groups and societies as units of selection, rendering Hayek's social theory

apparently inconsistent with the concept of methodological individualism (Vanberg, 1986). Societies are not only subject to group selection but have developed through a process in which individuals choose the rules that form the social order. Rule-following behavior – conscious or not – is not restricted to its social aspect. By emphasizing dispositions as rules, both at the individual level (mind) and at the social level (group, society), Hayek's central argument on the significance of rule-following behavior permits a new interpretation of his social theory.

The discussion is organized as follows. Section 2 gives an overview of Hayek's cognitive theory. Section 3 focuses on the role of dispositions, and Section 4 explains social evolution through the alteration of dispositions. Section 5 offers an application; Section 6 concludes.

2. HAYEK'S COGNITIVE THEORY IN A NUTSHELL

The aspect of the central nervous system that interests us most is its capacity to channel information. The channeling or classification of information occurs through “the transmission from neuron to neuron within the central nervous system” (Hayek, 1952, p. 52). This classification apparatus is of a dual nature. It has the quality of a “map” as well as of a “model.” The map shows all possible neuronal connections, it reflects the *static* aspect² of the nervous system, which operates on impulses but exists “independently of the particular impulses proceeding in it at a particular time” (Hayek, 1952, p. 115). Static, used in this context, means semi-permanent: external effects may have an imprint on the central nervous system and may eventually change it. This option to extend the map through new connections has evolved during human history and still continues: new connections are formed during an individual's lifetime.³

The *model* describes the “particular environment in which the organism finds itself at the moment,” (Hayek, 1952, p. 114) which means it resembles the current neuronal connections that a stimulus evokes. Essentially, in Hayek's view, neuronal connections will only take place when a stimulus can be brought into relation to another *similar* stimulus that has occurred in the past. In this way, stimuli and combinations of them are identified as belonging to a certain *class of stimuli*.⁴ If no such similarity can be established, no new stimulus can be perceived (Hayek, 1952, p. 64). This view corresponds with the findings of recent cognitive theories as shown by LeDoux (2002, p. 52): an activity of a receiving neuron is triggered when a stimulus, the “excitatory input” arrives *in conjunction with another stimulus*.

A single stimulus is filtered out by inhibitory interneurons that prevent projection neurons to fire and therefore connect to the next neuron at the synapses. Inhibitory interneurons, writes Ledoux (2002, p. 49) “are involved in information processing within a given level of a hierarchical circuit. One of their main jobs is to regulate the flow of synaptic traffic.”

Even if a single stimulus would arrive repeatedly at a receiving neuron, it would not trigger this neuron to fire. The map is, therefore, a reproduction of the relations of stimuli that occurred in the past, whereby “the past” refers to experience both of the species and of the single individual under consideration.⁵ It mirrors a subjective explanation of the external world and shows the discrepancy between the physical and the *phenomenal world*. Hayek explains the importance of the map as follows (Hayek, 1952, p. 131):

What the apparatus of classification provides is more a sort of inventory of the kind of things of which the world is built up, a theory of how the world works rather than a picture of it.

Hayek however locates different maps at different levels of the neuronal order. Lower-level maps will “serve as a guidance of merely a limited range of responses, and at the same time act as filters or pre-selectors for the impulses sent on to the higher centers, for which, in turn, the maps of the lower levels constitute a part of the environment” (Hayek, 1952, p. 111). That is, a very quick and predictable action refers to a category that operates on a “lower level” such as the message “fire is dangerous.”⁶ An action that follows a stimulus that is more complex such as “the teacher does not look at me. Should I cheat or should I refrain from cheating because cheating is bad per se?” will be the result of a multiple categorization. In fact, as Hayek describes it, it involves multi-level categorization (Hayek, 1952, p. 92):

While it is on the whole more likely that responses via the lowest centres will be innate for the individual, that is, acquired by the race in the course of evolution, while the responses effected by the higher centres will be largely based on individual experience, this cannot be regarded as a universal rule. Probably some inherited responses are effected on fairly high levels, while some learned responses may, after sufficient repetition, become almost completely automatic and be effected at low levels.

The process of classification has one important feature: the original stimulus can never be perceived by the individual in its pure form, but only in a categorized, classified, or abstract way. We never perceive all the objective properties of a stimulus but “only certain ‘aspects’, relations to other kinds of objects which we assign to all elements of the classes in which we place the perceived objects” (Hayek, 1952, p. 143).

Stimuli brought into connection with past experience lead to what we call perception. Hayek (1952, p. 143) speaks of “abstract perception.” Abstract perception is never accurate since it is rooted in past perception, which put stimuli into a certain *class*. Stimuli do not give us an exhaustive list of qualities of the objects, but at the phenomenological level we perceive them as stimuli belonging to a class of stimuli we know from the past. In modern cognitive science (Gazzaniga et al., 2002, p. 205), this phenomenon is called object constancy. People are able to recognize a bicycle when seen from above or able to discern the essential features of objects that are otherwise strange: pink elephants, for example, or striped apples. Older stimuli act therefore as “molds” for new incoming stimuli.

The following section will show that individuals may put a stimulus into different relations or classes at different times: the models can be changed constantly, in adaptation to current environments.

3. THE CORE ELEMENT OF COGNITIVE PROCESSES: THE DISPOSITIONS

Stimuli do not evoke predictable reactions. When responding to a stimulus, different people act differently, and some do not react at all. Moreover, the same stimulus may be answered by the same individual differently at different times. The explanation for this phenomenon lies in human “dispositions.” A disposition describes an inclination of an organism to react in a certain way when exposed to stimuli (Hayek, [1969] 1978, p. 40). The organism is not only prepared to react to a perceived stimulus in a certain way, its perception also prepares it to expect to receive future stimuli. That is, it expects to perceive stimuli that relate to stimuli perceived earlier, as well as it expects itself and other individuals to react in a certain way (Hayek, 1952, p. 98).

In an interview (Weimer & Palermo, 1982, p. 290), “*The Sensory Order* after 25 years,” Hayek refers in particular to the role of dispositions:

It seems to me now that I could have greatly simplified my exposition in the book if I had throughout used the term *disposition*. Perhaps I refrained from doing so because I feared then that it would be understood as referring primarily to dispositions to act or to move, whereas of course what I had in mind were as much dispositions to interpret further stimuli and dispositions to change dispositions and also various long chains where dispositions succeed other dispositions, with actions coming in at a very late stage only as potential events that might have been produced if certain other stimuli had occurred.

Here Hayek clarifies that dispositions are not limited to rules of action, where action is understood as the last step in the cognitive process, but also refer to rules of perception. The most comprehensive treatment of dispositions is given in Hayek's ([1963] 1967) article "*Rules, Perception and Intelligibility*" which deals in-depth with dispositions as rules of perception.

Rules of perception refer to the rules that other people follow; the human mind thus *perceives the rules of action of other people* (Hayek, [1963] 1967, p. 45). Without this ability, the interaction between mind and society would remain unidirectional, going from mind to society, without any feedback from society to mind.

The relation of rules of perception and rules of action is shown in Hayek's "*Notes on the Evolution of Systems of Rules of Conduct*" ([1963] 1967, p. 45). Movement patterns are used to exemplify a disposition to act:

From these instances where action is guided by rules (movement patterns, ordering principles, etc.) which the acting person need not explicitly know (be able to specify, discursively to describe, or 'verbalize'), and where the nervous system appears to act as what may be called a '*movement pattern effector*', we must now turn to the corresponding and no less interesting instances where the organism is able to recognize actions conforming to such rules or patterns without being consciously aware of the elements of these patterns, and therefore must be presumed to possess also a kind of '*movement pattern detector*'.

In this example, the neuronal system is seen as an organism that is both able to perceive patterns and to act according to patterns. The interesting question is then *which* pattern will we perceive and *which* action pattern will we follow as the neuronal system offers a huge variety of possible classifications of stimuli, and therefore, of patterns. Hayek differentiates between dispositions that an individual uses first and dispositions that he uses in addition to them. A classified stimulus will not evoke a specific response but a tendency toward a class of responses or a response "possessing certain properties" ([1969] 1978, p. 40). It is not until this first or primary disposition gives a rough direction that other dispositions will overlap this first very general and abstract disposition. Overlapping dispositions are "adaptations to typical features of the environment" ([1969] 1978, p. 41).⁷ Abstract rules of action operate "thus as moulds into which the various effects upon it [the organism] of the external world are fitted" ([1969] 1978, p. 41).

The sensory or phenomenal order is based on rules that have the following characteristics: they can be innate (genetic material), they can be acquired by personal experience, and they can be based upon the experience of the human species adopted during several stages of cultural evolution.

It is this triplet of characteristics which ensures that individuals behave similarly but still remain “personalities” and are therefore by no means “rule-following animals.” The way an individual classifies a stimulus cannot be predicted; the subjective aspect of individual experience makes any prediction impossible. The same holds for individual responses to stimuli. As Hayek states in “*Notes on the Evolution of Systems of Rules of Conduct*” ([1967] 1967, p. 68) “The concrete individual action will always be the joint effect of internal impulses, such as hunger, the particular external events acting upon the individual (including the actions of other members of the group), and the rules applicable to the situation thus determined.” Put differently, an observer cannot know what the internal impulses of another individual are; nor can he state how strong they are, nor can the observer know the impact of an external event since each individual has his own personal experience and has formed his own neuronal pathways.

However, it is not completely impossible to predict a single, concrete action because of the phylogenetic quality of dispositions.⁸ During human evolution, our species has acquired some hard-wired dispositions, such as helping one’s kin or helping the old or the sick (Rubin, 2002).

Besides these rules that are common to all individuals, there are dispositions that can be found only in particular societies, and that are difficult to understand for members of other societies, for example, the rites of initiation for African women. These dispositions can be thought of traditional rules and values embodied in this society during millennia.

Both the hard-wired dispositions and those dispositions referring to cultural uniqueness constitute the individual neuronal map. They are the primary dispositions or abstract rules which the individual will refer to in the first place before taking any action. These rules, passed on through generations, have no specific purpose but build the very frame of a society and make general predictions of behavior (pattern prediction) possible. Traditional rules or primary dispositions embody knowledge that individuals are not typically aware of – knowledge about how to do things in a way best adapted to the environment. Traditional rules store the knowledge and the experience of the past, or, as Birner (2002) stresses, the “wisdom of ages.” They incorporate the knowledge that exists in “formulas, symbols, and rules whose meaning we do not understand and through the use of which we avail ourselves of the assistance of knowledge which individually we do not possess” (Hayek, [1945] 1948, p. 88). These relatively stable parts of the neuronal map are complemented by personal experience of the individual. Perception is a process rooted in experience and is therefore “colored” by that past experience (Hayek, 1952, p. 98). The information

received together with a stimulus is of a personal nature and since it was categorized, it cannot be called “objective.”⁹ Actions based upon this information will trigger a feedback that eventually increases the knowledge of the individual: his dispositions – primary as well as overlapping – will be reinforced or altered by adapting them to the past event. An important consequence of this process is that knowledge is personal and therefore subjective and dispersed.

In this light “dispersed knowledge” leads to a new interpretation: Hayek stresses that it is the part of the map which is shaped by individual experience that is the reason for dispersed knowledge. More precisely, Hayek emphasizes that “the concrete knowledge which guides the action of any group of people never exists as a consistent and coherent body. It only exists in the dispersed, incomplete, and inconsistent form in which it appears in many individual minds” ([1952a], 1979, pp. 49, 50). We rely on this limited, subjective knowledge when interacting with other people *as well* as we rely on knowledge embodied in primary dispositions. This latter knowledge is taken into consideration when we activate so-called mirror-neurons (Gazzaniga et al., 2002, p. 469) using our own classificatory system to explain the behavior of others. We expect people to respond in a certain way to certain stimuli not only because we explain their behavior using our own classifications,¹⁰ but also because we have learned that certain dispositions are prevalent in the environment we know. The probability of seeing expected behavior in a *known* social environment is higher than in an unfamiliar one. Stated differently: it is easier (and less prone to mistake) to perceive the rules of action of other individuals in a familiar environment than to perceive the rules of action of individuals with different traditions. Of course, the individually shaped dispositions add uncertainty to the presumed security of a known environment. Subjective and dispersed knowledge connects to the knowledge embedded in traditional rules or primary dispositions. In the next section, we will see how rules can evolve and how individual and societal knowledge can increase and thereby spur social evolution.

4. SOCIAL EVOLUTION AS EVOLUTION OF DISPOSITIONS

Hayek’s emphasis on group selection, understood as the selection process in cultural evolution, has so far generated a large literature (Vanberg, 1986; Zywicki, 2000, 2004; Whitman, 2004) that aims at bridging the gap between methodological individualism and group selection.

I argue that Hayek has delivered a framework that permits us to re-evaluate group selection from a new perspective. Already Langlois (2004, p. 263) reminds us that Hayek, although embracing Wynne-Edwards' (1962) version of group selection, does not deal with biological groups but with "systems of rules of conduct." Hayek ([1967] 1967, p. 66) states this fact in Footnote 1 of his article "*Notes on the Evolution of Systems of Rules of Conduct*": "... [W]e shall occasionally use the pair of concepts 'order and its elements' and 'groups and its individuals' interchangeably." Methodological individualism can be reconciled with group selection once the units of selection are no longer understood as groups of individuals, but as *orders* or *institutions*. Hayek clearly had in mind to use bundles of "abstract rules" or "institutions" as units of selection instead of groups as a whole when arguing:

"The results of cultural evolution are structures such as language, law, morals, – all that what we usually call institutions in the broader sense. It is about the evolutionary process of the institutions..."¹¹ Hayek separates biological from cultural evolution in his final work *The Fatal Conceit*, stressing that "[c]ultural evolution operates largely through group selection; whether group selection also operates in biological evolution remains in the long run an open question" and "[w]hat is not transmitted by genes is not a biological phenomenon" (1988, p. 25).

Mind and social order are in constant interaction since classification is embedded in the social structure.¹² The mind is both product as well as producer of cultural evolution. Dispositions link the mental to the social order. The social order is built on rules that the human mind will perceive and to which it will react. On the other hand, individuals are endowed with "dispositions to change dispositions," permitting them to develop new rules of action eventually altering the social order.

Cultural evolution, seen as the change of the social order, takes place in two steps. First, an individual realizes an inconsistency between his use of dispositions (rules of action) and the expected results from his actions. As Vromen (1995, p. 164) correctly observes, expectations can be frustrated due to unusual changes in the "natural" environment as well as alterations in the "social" environment "because of unforeseen changes in the behaviour of others." Inconsistencies between the expected and the actual results of his actions can be solved either consciously or unconsciously; small unintended violations of rules and practices will allow the inconsistencies disappear. Individuals who change their behavior change their dispositions as well; they alter and add knowledge that is stored in their dispositions. Since the neuronal map of the individual offers more options to react to stimuli, a learning process in form of trial and error will result in the selection of a different disposition, or in the development of a new one,

therefore leading to new neuronal connections and to an alteration of the neuronal map.¹³

The generation of a new disposition is “always a discovery of something which already guides its [the mind’s] operation.”¹⁴ To state it differently: even a new disposition needs to be brought into relation with stimuli that occurred in the past and needs to be compared with an established pattern of action. The new rule of action that an individual uses will be observed by other members of his group. If this deviation from the common rules (the main dispositions) of action proves to be successful, he will be imitated by others. Vromen distinguishes furthermore between “within-group imitation” and “between-group imitation” (Vromen, 1995, pp. 172, 173). The more individuals follow the new rule of action the more likely the new rule will become part of the social order. It is possible that an individual is not allowed to deviate since breaking a widely respected rule is usually subject to punishment by other members of the society. This tendency to conservatism, of course, contributes to the relative stability of the social order. The “innovative” individual needs to evaluate the costs of breaking the rules in question. He may give up on following a new rule or, in order to escape punishment, he may migrate to a group or to a society (between-group migration) that either already practices the new rule or offers more tolerance toward deviators.¹⁵

A new disposition or rule of action first established by one member of the group and later followed by a sufficiently large number of other group members changes the social order. Note that this first step in cultural evolution has brought about a new rule, embodied now in the remaining system of rules. This single rule therefore had an impact on the social order. The next step focuses on this altered social order. Hume’s “slow test of time” will tell which social order is better adapted to the environment. In the end, those groups or societies using a set of rules adequately adapted to their environment will survive and thrive.

Thus, cultural evolution is the result of an active process of exchange between the phenomenological order and the social order whereby an individual’s perception of rules and his action according to rules is the key element. We may therefore conclude with Hayek (1979, p. 156) that: “[m]ind and culture developed concurrently and not successively.”

5. THE ROLE OF GOVERNMENT IN THE PROCESS OF SOCIAL EVOLUTION

So far, we have stressed the importance of individual action with respect to the development of rules of behavior. We have described social evolution as

a result of individual action, followed by both within-group selection and group selection. This section adds an analysis of rules that can be deliberately set by government, together with an explanation of the role of dispositions in establishing a legal framework.

Hayek's emphasis on restricting governmental action because of the knowledge problem can be better understood if we bridge his theory of an ideal government with his cognitive theory.

Following the Scottish moral philosophy, Hayek defines government as a "government of law and not of men or of will." That is, the government acts like any other member in the society, in accordance with the legal framework that has largely emerged spontaneously.¹⁶ Such a legal framework provides the "rules of the game" that are valid for every member of the society and therefore for government, too. It consists of the rules of just conduct, those general principles that – as Hayek ([1944] 1994, p. 80) writes – enable members of society "to foresee with fair certainty how the authority will use its coercive powers in given circumstances and to plan one's individual affairs on the basis of this knowledge." The similarity between the "general legal rules" and the rules of perception and action is no coincidence as they are both part of the system of rules that form the social order but are of explicit nature. These rules determine the conditions under which the available resources may be used but they do not tell the individual for what ends they should be used. Examples of such rules can be found in private and criminal law. In addition, there are rules of public law as well as rules that are not legally set but nevertheless obeyed such as moral and customary rules. All these rules form the framework for individual and governmental actions and form the backbone of the social order.¹⁷

The position of government is a particular one: government plays the role of a "primus inter pares," or "first among equals," with the duty to protect the legal framework so that a spontaneous order may develop further. Judges, for example, have the duty of "finding" the law by looking at the consistency of rules, legal as well as traditional. There are parallels between the development of new dispositions and the "finding" of a new legal rule. This is no coincidence: legal rules are rules of behavior that are part of a larger set of social rules. As much as new dispositions need to be developed on the basis of already existing dispositions, a new legal rule can only be discovered if it fits the rest of the rules. Accordingly, judges use "immanent criticism"¹⁸ the same way as private individuals do. If none of the existing general rules of behavior fulfills the needs of a situation, the judge may alter an existing rule or develop a new rule capable of overcoming inconsistencies (Hayek, 1973, pp. 118, 119). The judge is not seen as "the creator of a new order but a servant endeavouring to maintain and improve the functioning

of the existing order” (Hayek, 1973, p. 119). This process is not dissimilar to the development of a new rule of action in Hayek’s cognitive theory. A new legal rule, now part of a given framework, will become part of the individuals’ set of dispositions. While both individuals and judges spur the process of social evolution, the main difference between them lies in the coercive power that government has in implementing a new rule. Hayek emphasizes that “coercion is justified only in order to provide such a framework within which all can use their abilities and knowledge for their own ends so long as they do not interfere with the equally protected individual domains of others” (Hayek, 1979, p. 139). In order to achieve this goal he elaborates a model constitution that secures a division of powers between two different and mutually independent representative bodies (Hayek, 1979, p. 109). The Legislative Assembly lays down the rules of just conduct as explained above. The Governmental Assembly, finally, cannot “issue any orders to private citizens which did not follow directly and necessarily from the rules laid down by the former [Legislative Assembly]” (Hayek, 1979, p. 119). The social order is thereby a result of rules that are followed by individuals and, when stressing the importance of the role of government in affecting this order. Hayek refers to rules that are deliberately altered, improved, or set anew. The resulting order is of a spontaneous character, since government introduces only rules that have no explicit end, the rules of law and of just conduct. These deliberately set rules will fit into the existing order, with judges finding new rules in order “[t]o maintain and improve the functioning of an existing order” (Hayek, 1979, p. 119). We may argue that these deliberately set rules will become part of an individual’s main dispositions. However, which actions the individual will choose depends on the circumstances it finds itself in. Rules may well be set deliberately; the resulting social order is subject both to within-group selection and group selection.

6. CONCLUSION

The chapter illuminates Hayek’s work on social evolution and the role of government from a cognitive standpoint. The tool to do that is the term “disposition.” As Hayek stressed late in his life (1982), the term “disposition” is helpful to clarify his cognitive writings. In taking this hint the chapter has set out to connect the phenomenological order and the social order with the help of dispositions.

In doing so, I have stressed the validity of Hayek's cognitive theory laid down in *The Sensory Order* as well as in his later cognitive writings in the light of new contributions in cognitive psychology. Although Hayek could not rely on the many tools that are available to current neuroscience, his cognitive theory shows some remarkable results. This chapter thus adds a new explanation to why knowledge is dispersed, and how knowledge can emerge in a process that connects individual actions to the social order. Individuals are thought here of being actively involved in both the sensory and the social order, as they alter the society in which they live, given their restricted cognitive framework. This argument, as the last section has pointed out, is at the core of Hayek's rejection of planned economies.

The main outcome of connecting Hayek's cognitive theory with his social theory is that because of the sensory order, categorization is a successful tool to make individual behavior dependent of individual perception. As illuminated in this chapter, individuals perceive the patterns of rules that underlie the society in which they live. This is where "path-dependency" (Rizzello, 1999) has its place. Since government is composed of individuals who are subject to the same cognitive restrictions as everyone else in society, Hayek call for politics without discretionary measures, implying a plea for a strict division of powers. Hayek's normative view is in favor of a society governed by general rules, an arrangement conducive to the transfer of knowledge from society to every single individual. In that sense, legal rules and other set rules (explicit knowledge) set by government complement traditional rules (implicit knowledge).

NOTES

1. Hayek stresses the importance of *The Sensory Order* in his famous epilogue (1979, fn. 26, p. 199): "My colleagues in the social sciences generally find my study on *The Sensory Order: An inquiry into the Foundations of Theoretical Psychology* (London and Chicago, 1952) uninteresting or indigestible. But the work on it has helped me greatly to clear my mind on much that is very relevant to social theory. My conception of evolution, of a spontaneous order and of the methods and limits of our endeavours to explain complex phenomena have been formed largely in the course of the work on that book."

2. Hayek (1952, p. 115): "The semi-permanent map, which is formed by connexions capable of transmitting impulses from neuron to neuron, is merely an apparatus of classification...It is the apparatus of classification in what we have called its static aspect...". Also: The map resembles Kant's static categories but differs from them because of the semi-permanent nature. Horwitz (2000, p. 25) gives a nice explanation on how Hayek reconciles Kant and Hume in his writings.

3. See also Rizzello (1999, p. 106): "At any given point in an individual's life, a great part of his/her cerebral circuits is *personal and unique, since it reflects the history and events of that particular organism*, which is also the result of the interactive process which takes place with his/her cultural and social context."

In an interview given 25 years after the publication of *The Sensory Order* Hayek used the term "linkages" to explain how the map is formed. Linkages are "purely physiological connections between impulses." (Weimer & Palermo, 1982, p. 240, and Hayek, 1952, p. 104). These linkages add significance to, or classify, stimuli in the following way. Whenever new stimuli arrive, they may build new connections to other stimuli that were perceived in the past. New linkages modify the existing neuronal structure, resulting in a different mental structure.

4. Hayek (1952, p. 64): "Each individual impulse or group of impulses will on its occurrence evoke other impulses which correspond to the other stimuli which in the past have usually accompanied its occurrence."

5. LeDoux (p. 3): "Genes and different experiences (nature and nurture) shape the synaptic organization of the brain. The particular patterns of synaptic connections in an individual's brain, and the information encoded by these connections, are the keys to who the person is."

6. Posner (in Gazzaniga et al., 2002, p. 98) showed in his seminal experiment in 1986 at the University of Oregon that higher-level categorization takes a longer time than lower-level categorization.

7. Hayek ([1969] 1978) gives a clarifying example: "The particular movement of, say, a lion jumping on the neck of his prey, will be one of a range of movements in the determination of which account will be taken not only of direction, distance and speed of movement of the prey, but also of the state of the ground (whether smooth or rough, hard or soft), whether it is covered or open territory, the state of fitness of the lion's various limbs-all being present as dispositions together with its disposition to jump." The disposition to jump is in this case the abstract rule of action or primary disposition which is "superimposed" by the other overlapping disposition that the lion takes into account before jumping.

8. See Hayek (1952, p. 42): "It will be contended that in the course of its phylogenetic and ontogenetic development the organism learns to build up a system of differentiations between stimuli in which each stimulus is given a definite place in an order, a place which represents the significance which the occurrence of that stimulus in different combinations with other stimuli has for the organism." See also Gazzaniga et al. (2002, p. 547): "Choosing how to act does not simply require discriminating between incoming stimuli. When choosing how to act, we must integrate incoming stimuli with our values, current goals, emotional state, and social situation....The orbitofrontal cortex seems to be especially important for processing, evaluating, and filtering social and emotional information. The result is that damage to this region impairs the ability to make decisions that require feedback from social or emotional cues."

9. This refers to the argument of Hayek that social science is not a natural science. (Hayek [1952a] 1979).

10. Hayek ([1942] 1948, p. 63): "In discussing what we regard as other people's conscious actions, we invariably interpret their action on the analogy of our own mind: that is, that we group their actions, and the objects of their actions, into classes or categories which we know solely from the knowledge of our mind."

11. Translated [EG] from “Die ueberschaetzte Vernunft” (Hayek ([1982]1996)) in *Die Annassung von Wissen* (1996).

12. Howard (1994, p. 225): “Social structures can neither emerge nor be sustained without human cognition: human cognition can be practiced only through situated social activity.”

13. Hayek (1952, p. 95): “In the first instance, the sensory representation of the environment, and of the possible goal to be achieved in his environment, will evoke a movement pattern generally aimed at the achievement of the goal. But at first the pattern of movement initiated will not be fully successful. The current sensory reports about what is happening will be checked against expectations, and the difference between the two will act as a further stimulus indicating the required corrections. The result of every step in the course of actions will, as it were, be evaluated against the expected results, and any difference will serve as an indicator of the corrections required.”

14. Hayek ([1969] 1978, p. 46) and Hayek ([1969] 1978, p. 44): “...the richness of the sensory world in which we live...is...the product of a great range of abstractions which the mind must possess in order to be capable of experiencing that richness of the particular.”

15. The possibility of deviation depends on the degree of reputation as Hayek (1979, p. 204; Fn.48) says: “Though present morals evolved by selection, this evolution was not made possible by a license to experiment but on the contrary by strict restraints which made changes of the whole system impossible and granted tolerance to the breaker of accepted rules, who may have turned out a pioneer, only when he did so at his own risk and had had earned such license by his strict observation of most rules which alone could gain him the esteem which legitimized experimentation in a particular direction.” See also Vromen (1995).

16. Members are subject to the same restrictions of knowledge imposed by their dispositions. Only fragments of all the available knowledge is possessed by an individual. Hayek sees no reason to believe that members of government would gather more information in order to be able to interfere into individual plans.

17. Hayek distinguishes between ‘law’ and ‘legislation’. His emphasis lies on the ‘law’ that refers to the rules of just conduct which developed mostly spontaneously and therefore to primary dispositions. ‘Legislation’ describes the rules that governments lays down with a certain goal in mind and comprises the rules that government needs to work as an organization.

18. Hayek (1988, p. 69), referring to Popper (1971) emphasizes that piecemeal improvement of moral traditions is based on immanent criticism, “that is, by analyzing the compatibility and consistency of their parts, and tinkering with the system accordingly”.

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MAKING SENSE OUT OF *THE SENSORY ORDER*

Daniel J. D'Amico and Peter J. Boettke

ABSTRACT

Purpose – To comment on how The Sensory Order by F. A. Hayek is understood within the context of Hayek's broader research program.

Methodology/approach – Earlier and current perspectives on The Sensory Order are surveyed, quoted extensively, and commented upon. An alternative framework for understanding The Sensory Order is offered and compared to the existing perspectives. Some textual and archival evidence are combined with insights from the history of thought literature to present how Hayek himself may have viewed the role of The Sensory Order in his broader research project.

Findings – Earlier and current perspectives on The Sensory Order are found wanting. The available alternative hypothesis – that Hayek's economics is foundational to his theory of mind – is presented as a more fruitful approach to motivate modern Austrian economics as a progressive research program.

Research limitations/implications – There is limited archival and source material available on this topic and apparently competing versions circulating. Such a discussion has a relatively small and narrow field of interest among scholars intimately familiar with one another's work.

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Originality/value of paper – *If correct, this chapter offers a unique and original perspective on how to perceive the insights from Hayek's The Sensory Order. It also reaffirms the role of methodological pluralism in progressing contemporary political economy.*

INTRODUCTION: THE NEW NEURO-HAYEKIANS

Aside from being one of the most influential economists of the twentieth century, F. A. Hayek was a popularizer of ideas, a knowledgeable historian, a precise etymologist, a legal philosopher, and an innovative theorist of psychology and cognitive neuroscience. A renaissance man, a true scholar, Hayek's research interests stretched throughout the entire corpus of academia and he approached all topics with an intense rigor and thoughtfulness. In no uncertain terms, *The Sensory Order* is a testament to this fact. In his review, R. G. Grenell (1954) wrote "[*The Sensory Order* could] only be successfully carried out by a well trained scientific mind extremely well informed and flexible enough to see clearly across the 'boundaries' between different fields of thought" (p. 409). This chapter offers a comment on recent writings from Hayekian scholars who pay particular attention to Hayek's theories of psychology and neuroscience found within *The Sensory Order*. While we applaud these writers in their calling for more attention to be paid to Hayek, particularly Hayek's less-recognized works, we feel compelled to comment on how Hayek's psychology has been characterized within this literature. Yes, it is important to read Hayek in full in order to digest the complete message of his work (Boettke, 2005; Boettke, Coyne, & Beaulier, 2005), and yes we see a consistent parallel between *The Sensory Order* and the rest of Hayek's writings (Boettke & Subrick, 2002), but it is an overstatement to herald *The Sensory Order* as an essential foundation for Hayek's economics. We do not think that Hayek's own thinking was developed in this manner, nor is such a foundational arrangement essential to understanding Hayek's general theory.

In contrast to earlier theorists such as Machlup (1974), Runde (1988), and Tomlison (1990) who viewed *The Sensory Order* as "separate and different from the rest of Hayek's work," a handful of contemporary scholars have returned to Hayek's psychology and consider it a work of great significance. Gray (1983), Miller (1979), Nadeau (1987), De Vries (1993), Vanberg (1994), Witt (1989), Rizzello (1993), and Cubeddu (1996) all "think the text contains the most coherent, complete and systematic description of all the

methodological aspects used by Hayek” (Rizzello, 1997, p. 24). They emphasize the similarities between *The Sensory Order* and Hayek’s economics, philosophy, and politics rather than the differences. Most recently another generation of Hayek scholars has attempted to flesh out explicit links between Hayek’s cognitive psychology and Austrian capital theory (Mulligan, forthcoming; Horwitz, 2008). Drawing on insights from *The Sensory Order*, Mulligan (2010) characterizes epistemology as an evolutionary adaptation, Horwitz (2010) applies insights from *The Sensory Order* to inform management and organizational theory, and Koppl, Kurzban, and Kobilinsky (2008) have applied Hayekian-inspired epistemic reasoning to analyze forensic procedures.

We prefer the more recent interpretations of *The Sensory Order* (Gray, 1983; Miller, 1979; Nadeau, 1987; De Vries, 1993; Vanberg, 1994; Witt, 1989; Rizzello, 1993; Cubeddu, 1996; Buto & Koppl, 1996; Mulligan, forthcoming, 2010; Horwitz, 2000, 2008, 2010) over the earlier interpretations (Machlup, 1974; Runde, 1988; Tomlison, 1990). *The Sensory Order* is indeed a valuable text in the collection of Hayek’s writings. We believe that there are more similarities between *The Sensory Order* and Hayek’s other writings than there are differences, but we contend that, this is because *The Sensory Order* is a consistent application of (rather than a foundation to) his general theory – institutional rules engender patterned outcomes. As a matter of history of thought, we argue and offer evidence that Hayek returned to his original manuscript – drafted as a graduate student (Hayek, 1952a, p. v) – on psychology only after he had worked out his vision of the economic problem of society. For Hayek, the central economic problem of society is one of complex coordination of the dispersed and often divergent plans of actors. In short, the analogy of the market as a communication network that enables complex coordination, rather than a unified plan that assures coordination, was worked through first by Hayek in order to explain the economy and then this similar theoretical framework was applied to construct his theory of mind. This framework as to the flow of influence amidst Hayek’s work is preferable to motivate current Hayekians pursuing a progressive research agenda.

We also agree with the new neuro-Hayekian’s that *The Sensory Order* is consistent with and complementary to Hayek’s broader research agenda – philosophy, politics, and economics. If one puts on a pair of Hayekian eyeglasses one will see a great amount of similarities between the structural nature of the economy and the structural nature of the mind – and rightly so. But we fear that some of the phraseology used in the new neuro-Hayekian literature confusingly heralds *The Sensory Order* as a sort of key to unlocking

Hayek's vision of the economy.¹ We invite our readers to entertain the alternative position – it is Hayek's vision of the working economy that is the key to understanding his theoretical psychology.

Our claim can be summarized with a simple thought experiment of constrained optimization. Given the limited opportunity to read either Hayek's psychology or Hayek's economics, which text will offer the motivated reader a clearer picture of the other? Will a Hayekian economist be able to infer the basic gist of Hayekian psychology better (having not read the other) than a Hayekian psychologist will be able to infer the gist of Hayekian economics? We would bet good money on the economist over the psychologist.

Hayek's major contribution to economics elaborated the role of tacit and incomplete knowledge to coordinate the production and distribution of goods and services throughout the economy. Prices serve as road maps, constantly informing actors, and thus promoting coordination along an otherwise unnavigable and infinitely complex route. As Hayek's mentor, Ludwig von Mises argued relative price adjustments in the market economy enable actors to negotiate the "bewildering throng of economic possibilities" and to coordinate the production plans of some with the consumption demands of others (Mises, 1922, p. 101). Without the aid of the relative prices for goods and services established on the market, Mises argued that production decisions within the economy would be reduced to little more than stabs in the dark. The knowledge of what to produce, how to produce it, who should do the producing, as well as who would value the product most highly, all must be generated by exchange then discovered, disseminated, and acted upon within the market process for a complex social order to be realized. Knowledge in a modern economy is overwhelming. We are confronted with numerous possibilities and thus we need to economize on knowledge. Knowledge is often suppressed, dispersed, and incomplete, thus in need of being mobilized in order to coordinate economic plans in a manner that realizes the mutual gains from exchange. Such economizing and mobilization of knowledge creates wealth.

Hayek's description of neuroscience is a parallel application of his general theoretical insight to the applied topic of sensory perception – the functioning of the central nervous system of a human body.² The unifying aspect of Hayek's research is the spontaneous, generative, communicative, and finally coordinative properties of complex adaptive systems. The brain must digest information in a way to simplify data and stimuli that would otherwise be infinitely complex and confusing – in much the same way as prices make calculative decisions digestible in the economy.

In our favor, the parallel interpretation of Hayek's approach is not limited only to economics and neuroscience. Hayek's work on legal systems, culture, history, and ideology all allude to a similar relationship between the theory of institutions, their structural influence, and its many applications. Institutions such as common law, proverbial wisdom, language, and the ideological preference for liberty serve as crude but accurate estimators of "good" decision making within complicated and uncertain contexts.

There is a common pattern to Hayek's ideas. Given that Hayek's theories themselves allude to and explain structural patterns in the mind and the social world, there seems to be a pattern of patterns. But it is the science of economics that is the most thoroughly fleshed out application of Hayek's thought.³ It is economics that offers the clearest application of Hayek's works, and therefore we argue that it is Hayek's economics that should inform his psychology rather than the other way around.

The remainder of this chapter is organized as follows. The next section explains that the interpretation of Hayek's *The Sensory Order* as a critical birthplace to Hayek's economics is plausible but not necessarily true. The third section offers the alternative hypothesis that Hayek's psychology shares a similar methodology to his other areas of applied research. Structural frameworks, guiding principles, and institutional rules bear influence upon the outcomes of the processes that they exist within. The fourth section responds to Butois and Koppl (1996). "Does *The Sensory Order* have a useful economic future?" Not necessarily and certainly not by necessity. Finally, some concluding remarks are presented. Economics is an objective science in which the social theorist must recognize the subjective characteristics of the individuals he studies.

IS *THE SENSORY ORDER* THE KEY TO UNDERSTANDING HAYEK?

Early interpretations viewed *The Sensory Order* as an aside, ancillary to the rest of Hayek's work on philosophy, politics, and economics (Machlup, 1974; Runde, 1988; Tomlison, 1990). For economists first engaging *The Sensory Order*, its contributions were mostly methodological. Vanberg (2004) argues that Hayek's research program, when paying proper attention to the insights of *The Sensory Order*, can be termed "naturalistic subjective" and thus compatible with empiricism. Vanberg uses this point to distinguish Hayek from others within the Austrian School. Vanberg's position is

different from earlier writers on *The Sensory Order* in so far as his is an attempt to recognize the methodological context of Hayek's psychology within his larger research program of human and social science. Horwitz (2000, p. 25) and Buto and Koppl (1996, p. 30) also explain this unique quality of Hayek's work as bridging a methodological gap between Kantian a priorism and Humeian empirical evolution. This recent interpretation of *The Sensory Order* as part and parcel of Hayek's broader research agenda is a preferable portrayal compared to the earlier understandings. *The Sensory Order* is not a fluke digression.

Recent interpretations see *The Sensory Order* as a consistent and important portion of Hayek's corpus. Hayek was foremost an Austrian economist (Caldwell, 2004b), a tradition of social science similar to that elaborated by Max Weber (Boettke & Storr, 2002; Boettke, 1998). To understand human behaviors and social phenomena requires a recognition as to the *meanings* that lay behind actions. Before one can analyze the success or failure, efficiency or wastefulness of an action, he must fit that behavior into a structure of means and ends. Subjectivism plays a major role in Austrian social science because it is a constant reminder as to the cognitive dissonance (1) between the social scientist and the agents in his model, (2) between the agents within the model and other agents within the model, and (3) between policy planners and individuals in society. "Why did he do that?" No one knows for certain, but certainly there must be some reason, some motivation, some purpose. In this tradition there is a welcome seat for Hayek's psychology to fill. Buto and Koppl (1996) are right to point out that *The Sensory Order* is compatible with subjectivism and that it succeeds in demonstrating how there is a consistent structure among individuals' perceptions of reality. How do actors interpret means to relate to ends? How do they judge the success or failure of their actions? *The Sensory Order* shows that diverse individuals share a common structure of perception. "We can understand each other because we have more or less the same mental make up. Hayek's cognitive theory supports an argument against 'polylogism,' the doctrine that different cultures have different logics" (Buto & Koppl, 1996, p. 36).

We see in Hayek's *Sensory Order* a similarly patterned relationship between theory and application as we have seen in his legal philosophy and economics. The mind is a simplifying tool to digest an otherwise infinitely complex reality just as institutions such as the law are coping mechanisms to mitigate risk, uncertainty, and transaction costs. Prices are local and clear yet informed by dispersed and tacitly formed subjective evaluations. For Hayek, the mind perceives incomplete and limited bits of information,

it then fits such information onto a memorial “map” and in doing so the mind maintains a functional and well-operating human body and life.

Our concern is that the most recent revival of Hayek’s neuroscience overuses bold terms to describe the role that Hayek’s theory of psychology plays in relation to the rest of his work. This could be interpreted in two (not necessarily exclusive ways): (1) Hayek perceived *The Sensory Order* as foundational to the rest of his work, or (2) current scholars would be right to view *The Sensory Order* as foundational to the rest of Hayek’s work. We disagree with both and to the extent that the second may rest upon the first critiquing the first weakens the case for the latter. As a result (or perhaps part and parcel) of these perspectives the prominence of *The Sensory Order* within the Hayekian research program we fear has been at times overstated. For example, Vanberg (2004) writes, “these internal representations or models provide *the clue to our understanding* of adaptive or purposive behavior. The fact that these internal models allow the organism to anticipate the likely consequences from different kinds of behavior is... *the essential ingredient of purposive, problem-solving behavior*” (italics are ours, p. 26).

Steve Horwitz (2000) begins his applications of *The Sensory Order* insight with statements that seem compatible to our framework: “Hayek’s arguments for a constitutionally constrained government *are consistent with... his work in theoretical psychology*” (italics are ours, p. 23). Also, he writes – and we agree – “Hayek’s theory of mind *fits into* his economic and social thought. Ultimately, Hayek’s conception of the human mind is that it is a spontaneous order much like the various social and economic phenomena he has explored in other works” (italics are ours, p. 24). Again, in his most recent papers concerning *The Sensory Order*, Horwitz (2008) writes, “It is striking how similar the Austrian theory of capital is to Hayek’s work on cognition... They are... analogous models of complexity... [Hayek’s] work on capital got him thinking about ideas and approaches that led to his later work in the mind and complexity more generally” (p. 145). And Horwitz (2010), “some have argued that [Hayek’s] vision of the mind as spontaneous order was perhaps in some ways always present as a guiding idea for his work in economics... [t]here is another way... [r]ather than mind being central, the concept of spontaneous order was always there” (p. 1).

But in other places, Horwitz appears to award *The Sensory Order* a sort of foundational role to the entire stretch of Hayek’s thinking. Beyond *The Sensory Order* being “consistent with” Hayek’s notion of constitutionally constrained government, his (2000) abstract reads, “Hayek’s arguments for

a constitutionally constrained government are consistent with, *and to some extent rest upon*, his work in theoretical psychology... The Austrian view of the microeconomic coordination *is a logical outgrowth of Hayek's theory of mind*," (italics are ours, p. 23) and further "*The Sensory Order, is crucial to understanding both his economics and his politics*" (italics are ours, *ibid.*). Horwitz (2000) retains this language – "crucial" and "resting-upon" – throughout his article, and their instances outnumber the compatible phrases referenced above. He continues, "the linkage between that theory [of mind] and Hayek's economic and social theories is what constitutes his *true contribution*," (p. 26) and later in the same section, "[t]he limits of *explicit human knowledge form the basis for Hayek's economic and social thought and are the crucial difference* between his approach and that of both socialism and modern neoclassical economics" (italics are ours, p. 27). Later, Horwitz describes Hayek's economic and social research as applications of *The Sensory Order*, "Hayek's emphasis in the 1937 paper [*Economics and Knowledge*] on the interaction between expectations, the market process, and human learning *is the framework of The Sensory Order applied to actors in the economy*" (italics are ours, p. 28). Horwitz concludes that section by stating that Hayek, "*needed to make the case for his epistemological perspective (and its psychological foundation) in order to show where his evolving and developing perspective on political economy was derived from*" (italics are ours, p. 29), though he footnotes Hayek (1952a) as we perceive to be the opposite effect, "it was concern with the logical character of social theory which forced me to reexamine systematically my ideas on theoretical psychology" (p. v). In some places, for Horwitz, Hayek's economics stem from his psychology and also, "Hayek's philosophy of law appears to be another application of his theory of mind," (p. 35) and later "for understanding the argument in *The Road to Serfdom*, *it is not only necessary to understand Hayek's Austrian economics, but also his theory of mind*" (italics are ours, p. 37). Horwitz closes boldly asserting that a progressive Hayekian research program today, "*must begin with a better understanding of the human mind*" (italics are ours, p. 38).

Such a research program has been investigated by Horwitz (2008, 2010) himself. In each piece he attempts to make analogies devoid of implying an influential magnitude from mind to management (forthcoming) or mind to capital structure (2008). "So the question of which set of idea might have been the inspiration for the other, as it seems possible it could go either of both ways" (p. 24). In short we argue that this question is best answered in the opposite direction. The analogy that the mind works like a capital structure or like a complex organizational structure instead of a single

autonomous entity is a more fruitful tool of political economy than the reverse analogies. Perhaps coincidentally, Horwitz (2008) phrases them the other way, “the capital structure as an analogy to the mind” (p. 160). And later, “[m]uch as the biological processes of the brain lead to the emergent phenomena of the mind and consciousness, so do the individual capital-using plans of entrepreneurs get translated via the market into the emergent phenomenon of a more or less integrated capital structure” (p. 162).

In Butos and Koppl (1996), there are both phrasings that we find compatible and conflicting with our framework, but the latter seem to outnumber the former. Compatible: they explain, “Hayek’s view of the mind as taxonomic order *follows from* the motivating insight of his theory” (italics are ours, p. 27) and later, “the cognitive problem Hayek sets out to resolve *is identical to* the social problem he has addressed over the years: how do complex phenomena like the mind and markets resolve inherent limitations on knowledge?” (italics are ours, p. 39). But other passages lead us to presume that Butos and Koppl view *The Sensory Order* as foundational to the remainder of Hayek’s work. They state clearly that, “economists should let *The Sensory Order* inform their thinking” (italics are ours, p. 20). For Butos and Koppl, “Hayek’s theory [of mind] *establishes the cognitive base* for tacit knowledge” (italics are ours, p. 29). They use the same phrasing – *cognitive basis* – again later (p. 37), and refer to *The Sensory Order* as “*the foundation*” of “strong linkages [that] exist between [Hayek’s] methodology and his interest in complexly organized adaptive phenomena” (p. 39). Butos and Koppl argue that common misinterpretations of *The Sensory Order* act as a *pons asinorum* – a fool’s bridge that “[u]ntil you have crossed... you have not properly entered the field and you cannot yet form opinions on the subject,” (p. 29) and state clearly, “[w]e believe that this lacuna (or possibly neglect) within the social sciences is surprising and unwarranted” (p. 25).

Lastly, some passages give hint as to what precisely Butos and Koppl (1996) attribute as the essentially unique contribution of *The Sensory Order*. “While Hayek’s treatment of the knowledge problem in the catallactic domain *clearly emphasized the discovery* and use of decentralized knowledge, his treatment in *his cognitive work* [*The Sensory Order*] should be seen as an account of its *generation*” (italics are ours, p. 42). This would seem to imply that Butos and Koppl’s appreciation for *The Sensory Order* stems from their interpretation that Hayek’s economics and social theory lacked particular attention to the *generation* of knowledge, though they admit an ambiguity here later in their text, “it is not quite obvious why we need this theory to understand market level path-dependence or, more generally, how the

cognitive theory specifically generates a coherent theory of social institutions” (p. 44). They clearly argue for the applicable power of *The Sensory Order* to explain economic and social phenomena, “any theory of the market process *requires* some kind of articulated theory of learning... it seems difficult if not *ad hoc*, to make claims about the conditions under which markets will tend or not tend toward the coordination of individual plans” (italics are ours, p. 45). We do not share in this interpretation that the Hayekian paradigm without *The Sensory Order* lacks attention to knowledge generation, nor are we convinced that Butos and Koppl have adequately identified a uniquely applicable feature of *The Sensory Order* unfulfilled by Hayekian economics, but we will address this more in later sections.

One explanation for why such overemphasis has been placed upon the role of *The Sensory Order* in influencing Hayek’s work stems from the history of thought surrounding the writing process, publication, reception, and recollection of the book. Did *The Sensory Order* inform Hayek as current neuro-Hayekians hope it to inform their own work? Horwitz (2000) points out that there was a great change in Hayek’s writings from neoclassicism toward more innovative, knowledge-based economics that centered around the early 1930s (*ibid.*, p. 27 and Caldwell, 1988, 2004b). We share Horwitz’s (2008) later take – Hayek’s epistemic turn and the development of his core theory was most essentially influenced and developed within the context of the socialist calculation debates (Boettke, Schaeffer, & Snow, forthcoming) over the perspective implied from the quotations we selected above.

It is well known, that Hayek wrote a portion of *The Sensory Order* very early in his career, arguably earlier than much of his groundbreaking economics. Hayek says “thirty years earlier” according to the preface of *The Sensory Order* (1952a, p. v). Given that the final book was published in its first edition in 1952 – by this account that places Hayek’s writing of the original manuscript around the year 1922 – while Hayek was still in graduate school. Thus, it appears that the dates form a sort of linear progression. If we first admit that Hayek’s framework – structure and its influence on ordered outcomes – runs parallel throughout his economics, social theory, legal philosophy, and neuroscience; and second, we admit that his neuroscience was investigated, theorized, and developed before the other topics, then it would be reasonable to conclude that a flow of influence or theoretical foundation exists stemming from *The Sensory Order* onto Hayek’s later works. We deny this second point and instead argue that (1) there is insufficient evidence to argue that Hayek fleshed out (what we

consider to be) the essential portion of *The Sensory Order* at this early drafting date and (2) the available evidence points to a flow of influence in the opposite direction – Hayek returned to his underdeveloped manuscript on the topic of psychology after he had developed his framework of complex social phenomena through the lens of rational-choice and rational-action-based economics.

Hayekian biographer and historian of thought Bruce Caldwell was kind enough to supply us a translated copy of the original manuscript that Hayek refers to drafting long before the publication of *The Sensory Order*. It bore the title: “Contributions to a Theory of How Consciousness Develops,” and is dated 1920 – a 36-page, double-spaced Word document in 12-point font with significant page margins and white space. Caldwell (2004b, p. 249) refers to this archival document as “less mature” than the final publication – 232 pages in its current-University of Chicago Press – paperback edition. If so much of *The Sensory Order* was envisioned by Hayek in 1920 then why did its rewriting and revision require such significant time and effort? Also worth noting is the fact that Hayek completed his most formal and technical work on economics, *Prices and Production* for finalized publication in 1931, well before the final manuscript of *The Sensory Order* was ready in 1952.

One could argue that comparing the length of the 1920 manuscript with its final publication is insignificant so long as the core insight of *The Sensory Order* was included in that original manuscript. Hayek seems to imply as much in the final version’s preface, “it [his student manuscript] contains the whole principle of the theory I am now putting forward” (p. 5). On this point we disagree with Hayek and share Caldwell’s perception; the manuscript is significantly “less mature” than the published version. Had the published version more closely resembled the older document, then we would agree more with *The Sensory Order*-as-ancillary perspective (Machlup, 1974; Runde, 1988; Tomlison, 1990). The final publication of *The Sensory Order* contains a more-nuanced version of Hayek’s theory of mind. This final version of the theory contains a crucial element not contained (as best we can decipher) in the original 1920 manuscript.

What is the crucial difference between the 1920 unpublished manuscript and the 1952 version of *The Sensory Order*? In the final publication, Hayek fleshes out the implications of neuroscience when one recognizes the scarcity and constraints of perception. In other words, Hayek applies an economic argument to the topic of neuroscience.⁴

Butos and Koppl (1996) write, “[p]erhaps the key insight of Hayek’s approach [in *The Sensory Order*] is that the set of connections [formed by the central-nervous system] creates a classification over sensory inputs”

(p. 10). Perception is not a mirror of reality. They (p. 13) quote Hayek's (1952a) final publication "we do not first have sensations which are then preserved by memory, but it is as a result of physiological memory that the physiological impulses are converted into sensations" (p. 53). Memory precedes perception. We identify and label this version – perhaps we should say portion – of Hayek's theory of mind as underdeveloped. In some form or another, it is found in both the 1920 and 1952 manuscripts. The closest portrayal Hayek (1920) achieves in the original manuscript (as best we can decipher – the document is dense yet poorly communicated) is as follows: "[w]e are therefore justified in saying that becoming conscious means integrating an impression into a preexisting nexus of meanings, a system of qualities, and it is this process that we shall designate as uptake into consciousness" (p. 4), and later, "[p]hysiological memory, which is based on sensory experiences, is in any case a precursor of this sensation. We have seen how sensation was created by memory... We do not have sensations, which are then stored in our memory; instead, it is only because of memory that physiological stimuli become sensations or other attributes of consciousness" (p. 8). Or as Hayek quotes in 1952 from the 1920 manuscript, "we do not first have sensations which are then preserved by memory, but it is as a result of physiological memory that the physiological impulses are converted into sensations. The connexions between the physiological elements are thus the primary phenomenon which creates the mental phenomena" (Hayek, 1952a, p. 53). In its simplest terms: memory precedes perception.

Several points are worth mentioning that strike against upholding the underdeveloped version of Hayek's theory of mind as the foundation or influential force upon the rest of his research agenda. First, Hayek appears to accredit this insight to others by citing informally in the document, "Wertung or Auffassung" (p. 4), meaning Hayek himself did not take credit for this supposedly unique insight.⁵ Second, Hayek appears to precisely choose the word "meaning," in the above quotation. We interpret this word choice as his attempt to emphasize the primacy of choice over neurological or biological processes. Again this puts Hayek firmly in the Austrian tradition akin to Max Weber and other economists within the Austrian camp (Boettke & Storr, 2002; Boettke, 1998). Third, Hayek's motivations for this paper are clearly stated and narrowly defined. He does not mention that the theory of mind could or should be cross-applied to garner otherwise inaccessible insight into other realms of social science.⁶ Instead, Hayek very specifically targets what he considers to be false psychological perspectives popular at the time. Hayek (1920) writes, "[t]he author's views in this

matter are in the sharpest opposition to the prevalent ‘dogmatic-atomistic concept of sensations’ and to any concept relying on a primal, clear-cut association of consciousness event and brain process” (p. 1). Hayek wanted to avoid treating perception like a black box of psychic processes, a view he attributes to W. Jerusalem’s (1907) textbook, “[w]hen the excitation penetrates as far as the cerebrum, it is converted there, inexplicably into a psychic state, or, in the case of a simple stimulus, into a sensation” (p. 18) (Hayek, 1920, p. 3, fn.1). These are more modest intentions behind Hayek’s theory of the mind compared to those found in the later version of *The Sensory Order*. Hayek’s student essay was predominantly reactionary. If the central point of the 1920 manuscript is memory precedes perception, we argue that this is a weak point compared to its fuller version in 1952.

What is the fuller-developed insight of *The Sensory Order*? Compare the quotations above and Butos and Koppl’s accurate summary with the following from the 1952 published version of *The Sensory Order*:

The more this process leads us away from the immediately given sensory qualities, and the more the elements described in terms of these qualities are replaced by new elements defined in terms of consciously experienced relations, the greater becomes the part of our knowledge which is embodied in the definitions of the elements, and which therefore is necessarily true. At the same time the part of our knowledge which is subject to control by experiences becomes correspondingly smaller. (Hayek, 1952a, p. 170)

Hayek is explaining how different magnitudes of different pieces of cognitive information cause different perceptions and therefore actions. Hayek is explaining the scarcities and abundances – dare we say the relative prices – dare we say the economics of cognition and perception. Butos and Koppl (1996) are right to point out that Hayek’s fuller theory of the mind (1952a) emphasizes the “scarcity” conditions under which the mind processes knowledge. The physical logistics of mental processes act as a form of “constraint” upon the amount and type of knowledge accessible to any individual person. Our claim is merely that this is a larger point than memory precedes perception, and that this is an essentially unique point made in the final published version of *The Sensory Order* and not found in its original 1920 manuscript. Second, this is an essentially economic argument – scarcity induces constraints and implies particular strategies are efficient at accomplishing specifically defined goals and objectives. Structure induces particular forms of order. Without this economic core, Hayek’s theory of the mind is not cross-applicable to resolve other social issues and it does not seem evident that this core is somehow uniquely dependent upon

neuroscience as its genetic birthplace, certainly not the neuroscience found within Hayek's 1920 student paper.

One is tempted to point out Hayek's comment that he almost became a psychologist rather than an economist (1952a, p. v) as evidence of *The Sensory Order's* foundational role. But Hayek considered pursuing multiple advanced degrees, he obtained a joint degree in law and economics, but did not complete a Ph.D. in psychology, nor did he publish in the field of psychology ever after the publication of *The Sensory Order*. He in fact became an economist and introduces himself as such in the very first sentence of the preface to *The Sensory Order* (Hayek, 1952a, p. v). Furthermore, as Caldwell (2004b) rightly points out, "[h]e would later say that the resulting volume, *The Sensory Order*, was extremely important for understanding his later work. But he never said how or why, and, for that matter, subsequent references to *The Sensory Order* were not particularly prominent" (p. 7) – a peculiar trend for Hayek since he regularly cross-references his research to his other relevant pieces.

Boettke (2005) argues that Hayek's motivation behind *The Sensory Order* was in part the result of the popular success from *The Road to Serfdom*. Such popularizing was and still is looked down upon in the academic profession – unscholarly, unsophisticated, and unscientific (a particular pet peeve for Hayek). Seeking to reestablish his academic identity and reputation, Hayek dedicated himself to *The Sensory Order* as his most intensely theoretical work. Caldwell's (2004a) take is compatible with our interpretation that Hayek sought to attack the proponents of behavioralism and physicalism on their own turf – the physical sciences (pp. 246–248). Again, Hayek's (1952a) preface seems to support this view,

... one of the basic problems of psychology were the result of the prevalence during this period of an all too exclusively empirical approach and of an excessive contempt for 'speculation'. It seems almost as if 'speculation' (which, be it remembered, is merely another word for thinking) had become so discredited among psychologists that it has to be done by outsiders who have no professional reputation to lose. (p. vi)

Thus, one could argue that *The Sensory Order* was a strategic publication for both career and argumentative purposes. But this argument has a particular implication; if *The Sensory Order* is a strategic publication to advance Hayek's career and research program then one must also admit that Hayek possessed a vision of his analytical framework at the time of this strategic action. In other words, Hayek's vision of the spontaneously ordered economy and society predated and sits as foundational to his theory of the mind rather than the other way around.

Despite his own recollection and insistence that the book was one of his most important pieces (*ibid.*, pp. 248–250), by Hayek’s own admission he is a bad source of reliable information on his own intellectual history. Caldwell (2004b) refers to Popper biographer Malachi Hacohen (2000) who draws a similar conclusion, “autobiographical accounts are inaccurate,” and proceeds to footnote that “Hayek’s own psychological work on memory [presumably a reference to *The Sensory Order*] leads us to expect this kind of problem” (p. 6). Just as the mind tends to lump stimuli and information into coherent clusters and categories, one’s mind can be prone to fuse memories together. What in reality were separate occurrences – in his honest recollection – sometimes seem more like the same event (Hayek, 1994).

Hayek wrote a significant portion of *The Sensory Order* in the early 1920s, he then saw a strategic opportunity for its elaboration and publication to advance his career and ideas years later after 1944. He then revisited his original manuscript, rewrote it and finally published it in 1952. After its publication he was forced to defend his thesis by reexplaining it against those who failed to grasp its meaning. And lastly, he was asked to reminisce about its value and importance after he had won the Nobel Prize (Hayek, 1994). We argue it was only natural for both him and later scholars to overemphasize its theoretical value given the publication’s long and dramatic life cycle.

The overstatement of the link between *The Sensory Order* and the remainder of Hayek’s work is a perfectly reasonable conclusion and in fact can be thought of as one such phenomenon like those that Hayek refers to within *The Sensory Order*. Hayek’s work is dense and complex, where differences and varieties exist, the interpreters’ mind may naturally lump ideas together or construct narratives and connections in order to make sense and award meaning out of the apparent complexity. In some sense, recent Hayek scholars have attributed a false “anthropomorphism” to *The Sensory Order*, much like Hayek warns of in *Law, Legislation and Liberty* (1973, pp. 26–29). It is almost a natural instinct for us to draw stronger connections than what actually exists, potential connections where none exist at all, or infer causal relationships where there are really only correlations. Just as the mind is not a mirror of reality, recent portrayals of *The Sensory Order* are not a mirror of its reality nor the reality of Hayek’s intentions for it. We feel compelled to comment on such matters because such overemphasis on the physiological components of Hayek’s theory run the risk of portraying Hayek (incorrectly) as scientistic (Hayek, 1952b) and/or functionalist (Gauss, 2006) – the latter Horwitz (2008, p. 146 and p. 151) seems to accept, while Butos and Koppl (1996) walk delicately close to the former.

And lastly we call upon Hayek to offer his own counter-claim at the time of *The Sensory Order*'s publication. On what drove him to write book, Hayek (1952a) penned: "it was concern with the logical character of *social theory which forced me to re-examine systematically my ideas on theoretical psychology*" (italics are ours, p. v). Again, Hayek's social theory informed his psychology rather than the other way around.

AN ALTERNATIVE HYPOTHESIS

If the claims (1) that *The Sensory Order* is a foundational text to Hayek's economics and (2) that reading and understanding *The Sensory Order* sheds some otherwise unknown light upon Hayekian economics – are to be strong and valid they must be preferable to a pair of reasonably framed alternative hypotheses. If simpler and more plausible interpretations of *The Sensory Order* are as reasonable or more so than the two listed above then they are in effect weakened. We submit two alternative hypothesis that are in their essence anti-thetical to *The Sensory Order*-as-foundational perspective. (1) Hayek's understanding of economics is foundational to his remaining theories in philosophy, politics, and psychology, and (2) reading Hayek's economics better informs a reader as to the general outline of his theoretical insight than does reading *The Sensory Order*. We believe these alternative and opposing hypotheses to be more reasonable and therefore more likely an accurate portrayal of the theoretical content of *The Sensory Order* and its potential role to guide progressive research.

Hayek's economics, politics, philosophy, and psychology are all consistent applications of his general theory. The outcomes of processes are predictable in so far as the processes contain elements of a universal structure. When first reviewed in the early 1950s, *The Sensory Order* did not receive much attention from economists or social scientists. It was instead reviewed by individuals within its own fields of specialty, philosophy, biology, psychology, etc., but still rarely so. Several reviewers acclaimed its thoroughness and insight, but they often had complaints that it lacked sufficient examples and applications (Spratt, 1954; Boring, 1953; Chisholm, 1954; Kneale, 1954; Grenell, 1954). This could be in part explained if Hayek did not view *The Sensory Order* as a key to unlocking the rest of his work's examples. Instead, if Hayek viewed *The Sensory Order* as we do: another consistent application of the relationship between structure and outcomes. In this light, it would be almost tedious for *The Sensory Order* to contain elaborate empirical support and applied evidence. From this perspective, the

entire text and theoretical exercise of *The Sensory Order* was and is itself an empirical case study that serves as an example to support his more general theory and larger body of research.

Our second counterhypothesis is best argued for by use of a simple thought experiment in marginal value theory. What is the marginal value of reading *The Sensory Order*? We admit that this question presupposes an answer to the question, “value for what?” We define such value as the amount of understanding awarded to the reader by means of a careful analysis. While the marginal value of reading *The Sensory Order* may in fact be relatively high – relative to reading mainstream economics, given the under-recognition of Hayek’s work in general – it seems obvious that the marginal value of reading Hayek’s works on economics are at least greater than the marginal returns of reading *The Sensory Order*. We find it difficult to accept a perspective of Hayek that would imply a teacher or instructor to have his students read *The Sensory Order* before, for example, *Prices and Production*, *Individualism and Economic Order*, *The Constitution of Liberty*, or *Law, Legislation and Liberty*. While none of the neuro-Hayekians have or presumably would make this explicit claim, we feel the message is worth communicating to current and future Hayek scholars.

DOES *THE SENSORY ORDER* HAVE A USEFUL ECONOMIC FUTURE?

Butos and Koppl (1996) have argued that *The Sensory Order* has had the misfortune to be both under investigated and often misunderstood. In doing so, we argue that they and Horwitz (2000) have overstated the magnitude of influence that Hayek scholars should garner from *The Sensory Order* and/or award to it. In so far as one values truth for truth’s sake, overstatements are problematic because they are misrepresentations of reality, that is, not truth. As Butos and Koppl (1996, p. 34) refer to Khalil (2002) who writes, “the mind is not a mirror of reality” (p. 334). In much the same tone, we argue that the *Sensory Order*-as-foundational presentation of Hayek is also not a symmetric mirroring as to the reality of Hayek’s work. We are skeptical that Hayek meant *The Sensory Order* to be what it has been implied it to be, and we are skeptical of the evidence for the argument that one grows a significantly greater understanding of Hayek by reading it.

Surely every piece of scholarship cannot be the foundational or essential element. With the awarding of first-place prizes comes the necessity for

second-place losers. If we are to believe that *The Sensory Order* is the foundation of Hayekian thought, and we are wrong, then we are not only overestimating *The Sensory Order* but we are also underestimating whatever actually is the foundational theory of Hayek's vision.

Now to return to the earlier topic broached by Buto and Koppl (1996), "it is not quite obvious why we need this theory to understand market level path-dependence or, more generally, how the cognitive theory specifically generates a coherent theory of social institutions" (p. 44). We argue that the cognitive limitations between individuals on the one hand and the widespread social order on the other – Hayek's general unique contribution – are driven predominantly and more essentially not by the neurological nature of the mind but by the inherent complexity of the subject matter that they attempt to explain – the social world around us. With Hayek's theory of mind in hand, one can recognize Hayek's major contribution – knowledge is scarce and society is complex – but it is not the only way to get there nor does the nature of neuroscience appear to be the dominant source of cognitive dissonance in the social world. In other words, the degree of complexity that Hayek attributes to the mind in forming sensory perceptions is proportionate to the degree of simplicity suffered by an individual in relation to his amount of knowledge in society. But this amount of complexity is dwarfed by the degree of complexity that Hayek ascribes to the physical and social worlds – the very subject matters that sensory perception processes are trying to make sense of. Thus, we are left to conclude that the heavy lifting in promoting social order is predominantly done by emergent social institutions compared to complex neural processes.

Furthermore, it seems obvious that Hayek's (1937, 1945) writings on the economics of knowledge were clear that the process of exchange does in fact *generate* unique points of knowledge. In fact this was the essential point made by Hayek in the infamous socialist calculation debates. Hayek's (1945) point was framed directly against market-socialists Oskar Lange (1936, 1937) and Wassely Leontief (1951, 1986), who argued that with the right calculative ability production and distribution in a socialist economy could be achieved. Hayek's contribution was to specifically point out that without individual actors engaged in buying and selling at market prices, the necessary datum to plug into such calculations does not exist. Hence markets generate uniquely necessary knowledge within economic systems. It is not only Hayek's work in neuroscience that addresses the generative aspects of knowledge despite Buto and Koppl's (1996, pp. 41–42) contrary claim. Does *The Sensory Order* have a useful economic future? – not

necessarily, certainly not by necessity and perhaps not without sacrificing an understanding that may be more true to form and more powerfully explanatory.

CONCLUSIONS

Economics is an objective science that attempts to deal with the infinitely troublesome nature of subjective agents. But regardless as to the role of intersubjectivity and sensory perception there are truths of economic law. Demand curves slope downward and the division of labor is limited by the extent of the market. Our position in the debate on the role of *The Sensory Order* is simply one that while acknowledging Hayek's great depth and breadth as a thinker, argues that the relative weight ought to be on his contributions to economics and political economy.

All we are asking our fellow Austrians is to consider the possibility that many economists might not find the conversation in neuroscience – no matter how interesting it is for psychology – to improve the explanatory power of economics or advance our understanding of the market process. Is Ariel Rubenstein (2008) completely off track in his critique of the field with respect to the mainstream of economic thinking? He argues that neuroeconomics hopes to understand “all brains” yet falls short of its goals. Different patterns of choice are found among agents who make decisions in different amounts of time. It is unclear as to how such patterns map from the laboratory into real market behavior and thus it is unclear what amount or type of influence neuroeconomics may have upon the broader field of economic theory. In conclusion, we offer a set of brief comments and remarks.

First, Austrians such as Mises and Hayek already rejected the homo economicus description of man in economics and instead focused on a notion of homo agents (or *Homo sapiens*). Hayek's description of *The Sensory Order*, obviously fleshes out an important aspect of this rejection – agents are ill-informed by their own nature and cognitive limitations in addition to the quality or accuracy of the signals to which they perceive. So the Austrian vision of economics is less devastated by the sort of “behavioralist” critiques of “rationality” currently *en vogue* than are the traditional mainstream visions of economics. The same can be said for laissez faire conclusions supported on Austrian grounds.

Second, in the argument for an economic understanding of a free society, is not the heavy lifting of the explanation being done by institutional context, rather than by behavioral assumptions? With minimal behavioral

assumptions operating within an institutional context, we see a variety of complex and profound spontaneous orders in result. Rather than studying the mind per se, would we not learn more as economists and political economists by concentrating our efforts on history, politics, law, sociology, and economics? How is it that individuals act and react to institutions grown and imposed, new and old, successful and failed?

Third, we want to make it clear that we are not denying that *the mind matters*, but we think the mind matters mainly because of man's cognitive limitations, not because of things like emotion or chemical reactions, etc. We think that Lachmann's response to Nozick on methodological reductionism in the mid-1970s is still relevant (Lavoie, 1978) – we focus on the individual and his choices rather than the brain and its activities because it is only at the level of the individual that we can attribute *meaning* to action. It is only in reference to the meaning of action that we can begin to explain why an action did or did not occur, and what range of opportunity costs and benefits an individual faced. Austrian economics is an economics that strives to account for human meaning as an interpretive social science. To know the meaning of action is to make sense out of the social world.

Fourth, we agree that complexity, neuroscience, string theory, experimental techniques, and computer simulations are all fascinating fields of research and many have illuminating power with respect to economic principles. But the core of economic theory has been unaffected by these developments. To us, we can learn as much if not more in terms of illuminating exercises from explorations in history, anthropology, ethnography, and even literature as we do from a serious study of these more scientific exercises. We know that this runs contrary to our cultural bias, which is for the perceived “scientific” discipline to infiltrate the “less scientific” one – Mathematical economics compared to an economic approach to politics or law. We realize that in the debate over neuroscience and Austrian economics, we are pushing for a “softer” economics – an economic core which is used to do narrative history, an economic core which is discovered in the fictional discussions of great literature, and an economic core which is applied to understand politics, law, and society, etc. Where do we get the core economic theory? Not from complexity and neuroscience, we would contend, but from a close study of economic doctrine – Adam Smith (1776) to F. A. Hayek (1945), tested by the pure logic of choice. The claim that demand curves slope downward is true for Aboriginal cultures as it is for modern man.

We are all for letting every flower bloom intellectually, but we must also confess that we do not think we are heading to a state of intellectual

oblivion because neuroscience is not the thing we pay most attention to as *economists*. Others have alternative intellectual preferences. To put it another way, for the questions that we are interested in as economists and political economists we learn a lot from Andrei Shleifer (see Shleifer & Hay, 1998; Shleifer, Hay, & Vishny, 1996; Shleifer, La Porta, Lopez-de-Silanes, & Vishny, 1999), not as much from Colin Camerer (see Camerer, Loewenstein, & Prelec, 2005; Camerer & Fehr, 2006; Ming, Bhatt, Adolphs, Tranel, & Camerer, 2005). Not a knock on Camerer, but a judgment of relative importance is always a function of what topics most interest you for your research and teaching interests.

Loosely speaking Shleifer is pursuing the Hayekian program from *The Constitution of Liberty*, Camerer is exploring the Hayekian program from *The Sensory Order*. There is value added to bringing the Hayekian limits of knowledge argument to both tables. We merely claim that an economist or a political economist should be more focused on *The Constitution of Liberty* research agenda than *The Sensory Order* one.

NOTES

1. If our representation of these authors is not inline with their intended meaning then this paper merely serves as a clarification on such matters. We read them as we represent them here, and presume that others would garner similar interpretations as we have. In some cases, those we critique have phrased their arguments in terms that are at times both compatible and incompatible with our framework for understanding Hayek. In such cases, we have tried our best to quote both perspectives and clearly identify them.

2. Horwitz (2008, 2010) offers accessible summaries of *The Sensory Order*, as do McQuade and Butos (2005), Steele (2002), and Butos and Koppl (1993).

3. We mean to imply something analogous to how Hayek's mentor Ludwig von Mises described his own research agenda. Economics is recognized as the "best-developed branch" of praxeology – the broader study and science of human action (Mises, 1949, p. 885; 1957, p. 309).

4. One insightful referee points out the following: "[B]y 1952, Shackle (1949a, 1949b, later culminated in 1952, 1953) had already published his theory of choice under uncertainty which includes an economic view of attention via its ascendancy function, set out in terms of indifference curves... in Shackle's case it is very clear that the economic thinking came before the theory of attention."

5. It is also interesting to point out that Adam Smith's body of work is similar to Hayek's in its wide reach of topics. Thus, many scholars have been caught up on a "problem" in Smith's work (Viner, 1991). How could the scholar so convinced by self-interest in *The Wealth of Nations* (1776) be the same who is so concerned with sympathy in *Theory of Moral Sentiments* (1759)? More recently, some have arguably resolved the paradox (Smith, 1998) by showing the consistent general theory

throughout Smith's work. Most relevant here is the fact that in Smith's work on these other non-economics topics he comes strikingly close to Hayek's supposedly unique contribution that memory precedes perception. Smith's hypothetical construct of the impartial spectator admits to a gap between the appropriateness of one's own behaviors perceived by one's self and perceived by society (Smith, 1759, p. 110). Smith (1980, p. 38) also explains the classificatory role of perception and goes on to elaborate the primary role of memory to the practice of scientific investigation (p. 124).

6. At this point, the debate becomes very focused around the quality and or accuracy of archival evidence of which there appears to be different versions of (though the difference is subtle). On Hayek's motivation behind *The Sensory Order*, Koppl has reported that his translation of Hayek's 1920 manuscript reads,

[h]ere the author is making every effort to base his explanation exclusively on the physiological processes that underlie association processes and to avoid relying on any special psychic perceptive capacity of any special physiological hypothesis (such as cell memory). He hopes thereby to limit the explanation to a single recognized physiological law and to *integrate it into the world view of the natural sciences* (italics are ours, p. 1).

Our version contains a difference in translation to the closing sentence. "He hopes thereby to limit the explanation to a single recognized physiological law *compatible with the perspective of the natural sciences*" (italics are ours). We leave this matter for better-trained historians of thought and German language translators to resolve.

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I AM NOT A “NEURO-HAYEKIAN,” I’M A SUBJECTIVIST

Steven Horwitz

ABSTRACT

Purpose – To respond to a paper by D’Amico and Boettke arguing that certain scholars, including myself, whom they label “Neuro-Hayekians” have both oversold the importance of Hayek’s The Sensory Order for understanding his economics and misunderstood the importance of institutions as opposed to brains/minds in generating social order.

Methodology/approach – I offer a different interpretation of my own work, particularly my use of the word “foundational” to describe the role of The Sensory Order in Hayek’s system as well as a criticism of D’Amico and Boettke’s apparent dualism.

Finding – On a more careful reading of my own work, as well as that of Hayek himself, I argue that I am not guilty of holding the view that D’Amico and Boettke attribute to me.

Research limitations/implications – The major implication of this exchange is that there is much more agreement than D’Amico and Boettke seem to think.

Originality/value of paper – The value of this paper is found in its attempt to make clear that those scholars arguing for the importance of Hayek’s cognitive theory in understanding his work are not arguing that it

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is a necessary or sufficient condition for understanding his system. Rather, it is valuable for grasping the interconnectedness of his theories of the mind and the market and the relationship between them.

I have been called many things in my career but a “Neuro-Hayekian” is a first. For reasons I still cannot quite understand, D’Amico and Boettke (2010) have worked themselves up into a tizzy over a set of issues that seems extremely narrow and whose stakes seem very small. In their zeal to make their case, I think that they have not sufficiently exercised what the great Austrian economist Ludwig Lachmann called the “Principle of Charitable Interpretation” in reading the ideas of those with whom they disagree. Though I will admit to some occasional lack of clarity, the bottom line result is that they have misread and misinterpreted my position on the relationship between *The Sensory Order* and Hayek’s economics.

In this reply, I will both respond to some of their specific claims and attempt to articulate my own position on the issues they raise as clearly as I can. Let me start the latter process by referring to their thought experiment (pp. 4–5). I completely agree that someone familiar with Hayek’s economics only will be in a better position to understand his cognitive theory than someone familiar with his cognitive theory only would be in trying to understand his economics. Moreover, being familiar with his economics is not just helpful, but also, I would argue, *necessary* to understanding his defense of classical liberalism more broadly. At places, D’Amico and Boettke seem to think that what I and others are arguing is that knowing *The Sensory Order* is both *necessary and sufficient* to grasp fully Hayek’s entire system. I hold neither view. I know there are those who hold it necessary, and the authors seem to think there are those who think it is sufficient. Perhaps. But that is not my position, nor is there any reading of my work that would suggest otherwise. If D’Amico and Boettke are indeed imagining such a position, they should have made that clear and specify who holds it. My view is this: *The Sensory Order* is very helpful in understanding Hayek’s entire system for reasons I’ll argue below. But it is neither the most important book for doing so, nor the first that one should read in attempting to grasp Hayek.

Furthermore, I agree completely that “complex adaptive systems” is the right over-arching metaphor for Hayek’s thought. In fact, I say so in Horwitz (2008). From the mind to the market to the Great Society as a whole, each can be understood as a complex adaptive system, and each can be understood as nesting within each of the others. (One could add biological evolution as yet another such complex adaptive system overlaying

the entire social world.) D'Amico and Boettke claim that I believe that *The Sensory Order* is "foundational" for Hayek's system, suggesting something in contrast to what I say above. Below, I also attempt to clarify what "foundational" might or might not mean in this context. I conclude with a response to their claim that the "Neuro-Hayekians" are ignoring institutions for a preferred focus on brains.

NEURO-HAYEKIAN? NOT ME

My objection to the label of "Neuro-Hayekian" is quite straightforward: nowhere in my published work, including that cited by D'Amico and Boettke, do I make any references to neuroeconomics or argue that neuroscience should somehow become part of economics properly understood. I'm not especially interested in using MRIs to do economics. The "neuro" prefix rhetorically connects me to those who do, and for those who do not read the work cited by D'Amico and Boettke, it might leave the impression that I have such a connection. I do not. In agreement with Hayek, I do not believe we can gather more than an "explanation of the principle" of how brains operate. Brain scan technology might be useful in understanding said principles, but when it comes to actually doing economics, we have to work at the level of meaning and intentionality in human actors. That, however, does not mean we can afford to completely ignore the role of the physical brain in producing mind and ultimately choice, especially if the more we learn about the brain, the more we realize that Austrian subjectivism has a basis in science.

As Koppl (2010) notes, D'Amico and Boettke have an odd sentence that contrasts "meaning" with "neuroscience" and "biology." I agree that minds make meaning and that in our attempts to render intelligible human action, we must operate at the level of meaning. But the mind remains an emergent outcome of what is happening in the brain, so if economics is to analyze choice, we cannot afford to ignore the fact that choice is the outcome of a physical process to which an understanding of the biology of the brain might contribute. It is almost as if D'Amico and Boettke wish to treat "choice" as something utterly distinct from physical or material processes, suggesting some sort of dualist rejection of the material basis of human choice for reasons left unargued.

But this runs against the grain of both Mises's and Hayek's rejection of these forms of dualism. For both, the claim was that we can never understand those physical processes completely enough to substitute them

for the belief that humans are “free choosers.” But neither denied that choice and other mental events were “ultimately” the product of biological processes in the brain. The pragmatic belief that choice is free is just that – pragmatic. We cannot get behind it, so we start our analysis of humans from the assumption of choice and the meanings people attach to their choices. In this sense, I would describe Hayek’s position as “non-reductive materialism.” The human mind is an emergent phenomenon from the material brain, but it cannot be “reduced” to those underlying physical processes, because it is beyond our ability to understand totalistically the human mind with the human mind, leaving us with “explanations of the principles” at best. To draw the strict line between “meaning” and “biology” that D’Amico and Boettke would seem to put them in the dualist world of “mind as spirit” or the like. However, as Mises (1966, p. 46) said:

Individual man is the product of a long line of zoological evolution which has shaped his physiological inheritance The innate and inherited biological qualities and all that life has worked upon him make a man what he is at any instant of his pilgrimage. They are his fate and destiny. His will is not “free” in the metaphysical sense of the term. It is determined by his background and all the influences to which he himself and his ancestors were exposed.

The meanings our minds make cannot be *reduced* to neurons or biology, but they are nonetheless their emergent outcome.

FROM MIND TO MARKET

D’Amico and Boettke present evidence from my work that suggests that I have argued both their position (that Hayek’s work on the mind is derived from his economics and therefore secondary) and the position they wish to criticize (that Hayek’s cognitive theory is “foundational” for understanding his economics and social theory). They kindly offer both sets of evidence as a way of illustrating their comfort with the former and concern with the latter. I think the confusion is about words, such as “foundational.” To say that A is foundational for understanding B could mean one of three things. First, it could mean that A is temporally prior to B in the development of the theorist’s ideas. Second, it could mean more narrowly that one cannot understand B without understanding A. Finally, it could mean that theory A helps us understand the part of theory B that is the beginning of what theory B attempts to do or the story it attempts to tell.

To clarify, I do not believe that Hayek’s cognitive theory came first before his economics, nor do I believe that one cannot understand his economics

without understanding his cognitive theory. I concur with D'Amico and Boettke's reading of the insufficiency of Hayek's 1920 essay as a complete proxy for the full-bodied argument of *The Sensory Order*. As D'Amico and Boettke note, I offer explicit evidence that Hayek thought the economics came first in Horwitz (2008) and the footnote in Horwitz (2000) they discuss (p. 11) has me quoting Hayek approvingly that it was his concern about social theory that led him to reexamine his work on theoretical psychology. Thus, the first view of foundational is out. I also reject the second view. Many people have a very sound grasp of Hayek's economics without having read his cognitive theory. Again, I think the cognitive theory is neither necessary nor sufficient in this regard.

What I do mean by "foundational" or similar words starts from the observation that Hayek's (and Mises's) economics can be best understood as taking us from "the mind to the market." We start with the human mind and its purposes and plans and from there we trace out how human interactions in various institutional contexts do or do not produce patterns of outcomes and order. If this is what we do, then it would not seem unreasonable to say that attempting to understand how the mind works is "foundational" in the sense that our explanations of social order *begin* with the human mind. D'Amico and Boettke might object in general to probing how minds work beyond taking plans and intentions as given. If so, then they need to say so more explicitly. *The Sensory Order* is "foundational" to economics in the same sense that minds are "foundational" to markets.

There is yet one more reason to spend time on Hayek's cognitive theory. As Bruce Caldwell (1994) has argued, if Hayek is largely right about how the mind works, then we have a scientific underpinning for tacit knowledge and subjectivism, two ideas at the core of Austrian economics and Hayekian social theory. Critics of Hayek might claim that his views on knowledge are just a convenient assumption, so that he can then make the sociopolitical arguments that follow from them. In other words, the Hayekian view of knowledge could be dismissed as ideologically driven. What *The Sensory Order* and sympathetic work in neuroscience can do for Hayekian social science is to knock down that objection. Hayekian cognitive theory is important, if not "foundational," precisely because *it provides a compelling rationale for the subjectivist view of knowledge and the phenomenological starting point of social theory*. But neuroscience is not social science, nor does a good social scientist need to produce neuroscience. We should, however, be cautious consumers of it.

One of the practical conclusions of *The Sensory Order* is that we, of necessity, live in a world of expectations. The mind is at base a collection of

expectations about our interactions with the world. If one is to do social science in a phenomenological way à la Schutz, then one will have to start with the concept of “ideal types,” which are similarly collections of expectations. If our understanding of social action in the phenomenological tradition is to start with human meaning and plans that are based upon ideal typical constructions that inform our expectations of the world, then does it not seem reasonable to emphasize a cognitive theory with the major conclusion that the very nature of the brain and mind means that we perceive the world through expectations? *The Sensory Order* helps us understand the structural reasons for why a phenomenological approach to social science is both necessary and valuable.

MINDS OR INSTITUTIONS?

Toward the end of the paper, D’Amico and Boettke (p. 26) argue that the “heavy lifting” of generating social order is predominantly due to “social institutions” rather than “complex neural processes.” On page 28 they say “is not the heavy lifting of the explanation [of social order] being done by institutional context, rather than by behavioral assumptions?” As with other parts of their paper, it is simply not clear what Hayekian believes otherwise on either count. It is noteworthy that no author is explicitly cited as holding these views, which suggests it is either a straw man or simply reflects a concern the authors have about what others “might” believe. If it’s the latter, then this should be far more clear, as the paper strongly implies that the “Neuro-Hayekians” named in the paper hold this sort of view, yet no textual evidence is provided to support it. This leaves the accusation hanging, and creates the impression that I might hold the view they criticize in this section.

So to be clear once again: I do not hold the view that complex neural processes or behavioral assumptions are more important than institutions in explaining the emergence of social order or social disorder. Institutions do indeed do the heavy lifting. In fact, institutions are so important in the role they play that perhaps it is worthwhile to probe deeper into the reasons *why* we require institutions to generate social order. As the authors note, it is a consequence of our ignorance of the details of an overly complex social world. Well, why are we not able to grasp that complexity? Perhaps it has something to do with our cognitive limits. Perhaps it has something to do with the environments in which our brains evolved. Perhaps cognitive theory can be of some help in understanding why it is that the heavy lifting of social order is done by institutions. And perhaps in that way, not temporally, nor

as a necessary or sufficient condition, cognitive theory is “foundational” to the social sciences.

Under no circumstances would I tell, say, a graduate student in economics that she should spend time reading *The Sensory Order* rather than *The Constitution of Liberty* or *Individualism and Economic Order* or pretty much anything else of Hayek’s economics if she wanted to understand Hayek’s economics and social theory. Hayek’s cognitive theory is really important to understand his thought, but one needs to grasp his economics and social theory to even realize why a theory of cognition might be so valuable. But once one *does* grasp his social thought, reading his cognitive theory enriches one’s understanding of his whole system and provides a scientific foundation for his subjectivism and his work on the division of knowledge. If all D’Amico and Boettke are trying to say is that it’s more important to read Hayek’s economics than his cognitive theory to understand his economics, who has said otherwise? I still cannot shake the feeling that my work has been used as a proxy and convenient target for some other unnamed set of people who genuinely are “Neuro-Hayekians” in a way that matches the inaccurate picture of my work that they sometimes draw. If all they are trying to say is that we should not “oversell” the importance of *The Sensory Order* for understanding Hayek’s social thought, then it will come down to a judgment of what is “overselling.” I hope that my own position is now clear and that they realize that it is, I believe, not the one their paper appears to associate with my work.

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CONFESSIONS OF A NEURO-HAYEKIAN

Roger Koppl

ABSTRACT

Purpose – To defend my past work against the charge that it has exaggerated the importance of The Sensory Order or mistaken its proper role in Hayek’s scholarship.

Methodology/approach – I review the evidence given by D’Amico and Boettke and show that such evidence does not imply exaggeration or distortion regarding the nature and importance of The Sensory Order in Hayek’s oeuvre.

Findings – Hayek’s The Sensory Order was very important, but you can understand his economics without having read it.

Research limitations/implications – The paper only discusses some criticisms of my work on Hayek and a few related points. It does not provide a detailed or extensive account of either The Sensory Order or the role of that work in Hayek’s economics.

Originality/value of paper – The work is “original” because it defends my past work against a criticism that previously did not exist. My defense is valuable to the extent that the criticism of D’Amico and Boettke is considered relevant and interesting.

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D'Amico and Boettke do Butos and me the kindness of discussing our 2006 paper at some length. Knowing that criticism is kind of praise, I appreciate their attention even though I am not persuaded that we exaggerated the importance of *The Sensory Order* or mistook its proper role in Hayek's scholarship. Horwitz (2000) and ourselves (Butos & Koppl, 2006), seem to be the leading examples of "neuro-Hayekians." We are charged with exaggerating the role and importance of Hayek's psychology for understanding Hayek's economics. We are admonished, "it is an overstatement to herald *The Sensory Order* as an essential foundation for Hayek's economics." D'Amico and Boettke want neuro-Hayekians to tone it down. I will try not give too detailed a reply, but I will address the question of whether Butos and I suggest that *The Sensory Order* is "an essential foundation for Hayek's economics." I will also address the "thought experiment" D'Amico and Boettke propose. Then I will comment on the importance of *The Sensory Order* for economics before addressing what I think might be an unstated underlying issue. This unstated issue will give me an excuse to say a few words about the divide between the "two cultures" of literature and science.

D'Amico and Boettke say, certain passages of Butos and Koppl (2006) "lead us to presume that Butos and Koppl view *The Sensory Order* as foundational to the remainder of Hayek's work." I am not sure how clear a meaning "foundational" has in this quote, but I think it means something like the following: "You cannot really understand Hayek's economics if you do not understand his psychology." I do not think Butos and I made any such claim. We say, "economists should let *The Sensory Order* inform their thinking" (2006, p. 20), which might be true even if the work is not "foundational." We say, "Hayek's theory [of mind] establishes the cognitive base for tacit knowledge" (2006, p. 29). We do not say that without such a cognitive base you cannot recognize that some knowledge is tacit or that tacit knowledge is important in economics. I would not make either claim. Finally, we say, "it is also clear that strong linkages exist between his methodology (and its foundation within *The Sensory Order*) and his interest in complexly organized adaptive phenomena" (p. 39). It is possible that this remark overstates the importance of *The Sensory Order* for Hayek's methodology, but not for his economics. I think it does not overstate the importance of *The Sensory Order* for Hayek's methodology, however, in part because we say Hayek's other book of 1952, *The Counter-Revolution of Science* (1952b), is "perhaps his most important work" in methodology (Butos & Koppl, 2006, p. 37). Butos and I think *The Sensory Order* is important, but I do not think we represented it as "foundational" in the way D'Amico and Boettke seem to say.

D'Amico and Boettke propose a thought experiment. "Will a Hayekian economist be able to infer the basic gist of Hayekian psychology better (having not read the other) than a Hayekian psychologist will be able to infer the gist of Hayekian economics? We would bet good money on the economist over the psychologist." I disagree, given that Hayek says the central insight of his psychology is that memory precedes sensation. Although Hayek says the "central thesis" (1952a, p. 53) of his model is that memory precedes sensation, D'Amico and Boettke seem to doubt it, at least for Hayek's 1920 manuscript. Without this thesis, however, Hayek's 1920 explanation of consciousness as a physiological process falls apart. It is this "central thesis" that decides the outcome of D'Amico and Boettke's thought experiment.

What happens in D'Amico and Boettke's thought experiment if the insight Hayek calls the "central thesis" of his psychology is considered to be the central thesis of his psychology? A Hayekian psychologist would recognize that Hayek's model of the mind is quite close to that of Hebb (1949), as Hayek himself notes (1952a, p. viii). He might therefore be reminded of John Holland's representation of the Hebb model as a complex adaptive system (Holland, 1992, pp. 58–65). He might know that Holland uses the same basic framework to describe both market economies and Hebb's model of mind. Thus, it would not be that difficult for our imaginary Hayekian psychologist to make the connection between minds and markets, especially considering what Lavoie, Baetjer, and Tulloh (2004) rightly call Holland's "economic model for the problem of credit assignment" in complex adaptative systems (p. 114). It is even possible that our Hayekian psychologist could infer something of Hayek's doubts about prediction in economics from Hayek's discussion of "explanation in principle" in the final chapter of *The Sensory Order*. Conceivably, therefore, he could guess that Hayek generally favors "free markets" in some sense and looks askance upon government attempts to direct the economy.

Our imaginary Hayekian economist, by contrast, would not be able to guess the central thesis of Hayek's psychology. Perhaps he could guess that Hayek viewed the mind as a spontaneous order, although such a guess seems a stretch to me. But he would not be able to guess that Hayek's central thesis is that memory precedes sensation. Nothing in economics points to such an insight. D'Amico and Boettke would lose their bet.

While I think D'Amico and Boettke have not quite hit the target when criticizing Buto and Koppl (2006), it does seem fair to say that we rank Hayek's psychology higher than they do. Certainly, I personally think *The Sensory Order* is a very important work. I do not think one is somehow

unable to understand Hayek's economics without first immersing oneself in Hayek's psychology. That would indeed be an exaggeration. But I think it is very helpful in understanding his methodology, especially as developed in Hayek (1952b) and, relatedly, his theory of complex phenomena (Hayek, 1967).

Most importantly, Hayek's psychology can be of great assistance to any economist who takes knowledge issues seriously. If, as Hayek argued in 1937, "the empirical element in economic theory ... consists of propositions about the acquisition of knowledge" (1937, p. 33), then it seems fair to say that economics needs a theory of learning and that any such theory will benefit from a model of human cognition.

The most fundamental models in economics are robust to the particulars of human cognition. We can identify what is seen and what is not seen and why socialism will not "work" without a modern theory of cognitive psychology. As Vernon Smith (2008, pp. 61–68) has emphasized, the epistemic properties of some institutions such as double auctions are invariant across rival models of human cognition. Nevertheless, as Smith also emphasizes, a good economic model may sometimes require a good model of cognition (2008, pp. 61–68). We cannot understand observed outcomes in the trust game, for example, without appeal to human psychology, as the pioneering fMRI study of McCabe, Houser, Ryan, Smith, and Trouard (2001) illustrates.

D'Amico and Boettke quote Hayek's (1920) manuscript, "We are therefore justified in saying that becoming conscious means integrating an impression into a pre-existing nexus of meanings, a system of qualities, and it is this process that we shall designate as uptake into consciousness." Referring (I think) to this passage they later say, "Hayek appears to precisely choose the word 'meaning,' in the above quotation. We interpret this word choice as his attempt to emphasize the primacy of choice over neurological or biological processes." D'Amico and Boettke seem to view choice as something distinct from all "neurological or biological processes" and to input such a view to Hayek's (1920) manuscript. I do not know how to square such an interpretation with Hayek's explicit rejection (in 1920) of the "dualist–parallelist doctrine" according to which, "there is no equivalence between psychic association and physiological activation processes, only an unexplained parallelism." Hayek opens his 1920 manuscript with the sentence, "This study is an attempt to create a basis for explaining consciousness phenomena physiologically by investigating the simplest consciousness responses, particularly sensory experience, and by deducing them from the operation of well-known physiological laws."

In the first paragraph of the manuscript he says that he wishes to “integrate” his explanation of consciousness “into the worldview of the natural science.” He want to “limit the explanation to a single recognized physiological law” (Hayek, 1920).

Hayek considers the argument of his 1920 manuscript to vindicate Henri Bergson’s model of consciousness. (Presumably, however, Hayek did not mean to endorse Bergson’s metaphysics.) He says, “All that we have just explained speaks against this conception of sensory experience as a constant element of consciousness processes It is worth noting [th]at H. Bergson reached very similar results by a very different approach and also rejected this atomistic conception most vigorously.”

D’Amico and Boettke are right to say that Hayek is “firmly in the Austrian tradition akin to Max Weber.” And yet he “naturalized” consciousness by reducing it to physiological processes. As I have argued elsewhere, Hayek brought the epistemology of the *lebensphilosophie* of Bergson and Dilthey fully within the orbit of natural science (Koppl, 2009). His 1920 manuscript attempts to do just that. He says *both* that Bergson “reached very similar results” to his *and* that he wishes to “integrate” his explanation of consciousness “into the worldview of the natural science.” In my opinion he succeeded.

The notion that we can bring “the worldview of the natural science” and *lebenphilosophie* together seems bizarre. Hayek’s bizarre position cannot be sustained, I think, without the argument we find in the final “philosophical” chapter of *The Sensory Order*. He uses a diagonal argument of the sort pioneered by the great mathematician Georg Cantor to show that there is a sense in which the mind cannot fully explain itself. His physiological explanation of consciousness is an “explanation of the principle.”

Hayek’s position seems bizarre in large part because we still suffer from C. P. Snow’s great divide between the “two cultures.” Snow’s motivating observation is likely to have been accurate for his time and seems applicable to ours, as the Sokal hoax suggests (Sokal, 1996, p. 343). Snow said, “I believe the intellectual life of the whole of western society is increasingly being split into two polar groups.” On the one side, we have “the literary intellectuals” and on the other side we have “scientists, and as the most representative, the physical scientists. Between the two a gulf of mutual incomprehension – sometimes (particularly among the young) hostility and dislike, but most of all lack of understanding” (Snow, 1990 [1959], p. 169). Dilthey, Bergson, and Max Weber belong to “the literary intellectuals.” Georg Cantor, John Holland, and Hal Varian belong to the “scientists.” Hayek was personally comfortable and at home in both groups.

More importantly, his scholarship was rooted equally in both groups, bridging the great divide. Hayek brought the “literary” view of man found in Bergson and Dilthey “into the worldview of the natural science” through his theoretical psychology and theory of complex phenomena. Snow’s divide is so strong, however, that this achievement is generally not recognized. Instead, we see Hayek’s admirers divided into the “literary” group that emphasizes subjectivism (D’Amico & Boettke, 2010; Boettke, 1990; Burczak, 1994; Madison, 1989) and the “scientific” group that emphasizes science and mathematical complexity theory (Caldwell, 2004; Vanberg, 2004). Hayek gave rigorously scientific foundations to a literary vision of man. I pine for the day when scholars will recognize how completely Hayek transcended the two cultures.

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FROM NEURO-HAYEKIANS TO SUBJECTIVIST HAYEKIANS: A REPLY TO HORWITZ AND KOPPL

Daniel J. D'Amico and Peter J. Boettke

ABSTRACT

Purpose – To recognize the comments made by Horwitz (2010) and Koppl (2010) in their attempts to reply to D'Amico and Boettke (forthcoming), “Making Sense out of The Sensory Order.” Furthermore, this paper hopes to explain what role D'Amico and Boettke do see for cognitive neuroscience in the study of Austrian economics.

Methodology/approach – Some brief summary comments are presented about Horwitz (2010) and Koppl (2010). Then a general framework of individual learning and its effects upon social institutions and economic processes is described by referring to Cowan and Rizzo (1996) and Denzau and North (1994).

Findings – Hayek was a political economist first and foremost. Whatever the status of his research in theoretical psychology attains, it does not change the fact that we as economists would do well (especially young economists) to focus on his substantive contributions to economics and political economy.

Research limitations/implications – Though space and time constraints did not afford this at present, further research would benefit from an

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intensive survey of the empirical findings available in the neuroscience and neuroeconomics literatures. How do such findings map onto the proposed frameworks of Hayekian economics provided by Koppl compared to D'Amico and Boettke.

Originality/value of paper – *This paper takes notice of the historical linkage between Cowan and Rizzo's (1996) cognitive model of individual learning within the broader tradition of subjectivist/Hayekian/Austrian economics.*

Rather than specifically responding to all the points made by Horwitz (2010) and Koppl (2010) in their replies to our critique of Neuro-Hayekianism, we would rather take advantage of this opportunity to make explicit what role we *do* see for neuroscience and cognitive psychology in pursuing Austrian economics. While this may seem like evasion on our part, we feel justified in our original endeavor – commenting on how Hayek's *The Sensory Order* should be interpreted by Austrian economists – if only because our original essay succeeded in generating oppositional perspectives from Horwitz and Koppl themselves. Secondly, we feel justified in this follow up to explain our vision as to the role of neuroscience in Austrian economics because we hope to inspire (if only mildly) a forward momentum for understanding the links between cognition and Austrian economics. We are not critics of neuro and cognitive sciences, let alone efforts to learn from these disciplines for the purposes of improving our understanding of human choice and social interaction. Boettke, for example, is one of the editors of *Cambridge Studies in Economics, Cognition and Society* (along with Timur Kuran) and has attempted to relate the work on philosophy of mind to the underlying philosophy of the market found in Hayek (Boettke & Subrick, 2002). What we are trying to emphasize is that Hayek was a political economist first and foremost, and that whatever the status of his research in theoretical psychology attains, it does not change the fact that we as economists would do well (especially young economists) to focus on his substantive contributions to economics and political economy. We deny that *The Sensory Order* is the key to unlocking Hayek's contributions; we do not deny that *The Sensory Order* is a compelling work, and important for Hayek's development.

Now we will proceed to explain particularly what we see as the role of neuroscience and cognitive psychology in advancing Austrian economics. Rather than create an entirely original framework to explain the processes

of perception – we admit to the benefits of a division of labor – all we hope to do is point to those conceptual frameworks that we see as being most compatible with the Austrian research program pursued by Hayek. Conveniently enough, these selections are also particularly attuned to the functional properties of spontaneous orders. More specifically, they fit within a body of research that pays particular attention to the role of subjectivism in generating spontaneous orders. Cowan and Rizzo's (1996) "The Genetic-Causal Tradition and Modern Economic Theory," and Denzau and North's (1994) "Shared Mental Models: Ideologies and Institutions," well explain cognitive and perceptual learning processes at the individual and social levels.

Biographer Bruce Caldwell (2004) has explained Hayek's research project as a rejection of, and a progression away from, physiological theories of economics – where values, actions, exchange rates, and economic growth were explained by referencing the physical and objective qualities of capital resources and individual's skill sets. Hayek's work emphasized human choosers who are sometimes smart, other times dull, most often bumbling and erring. Yet his work also emphasized the institutions within which choice and exchange takes place and it is the institutional environment that either directs behavior in a direction to realize the mutual gains from cooperation, or steers human actors into situations mired in conflict and frustration. Hayekian scholars have fleshed out a variety of implications stemming from this emphasis on the subjective and humane foundations to the patterns of economic and social phenomena. This list certainly does not need to be repeated for scholars like Koppl and Horwitz as they are, no doubt, thoroughly familiar with its insights. The following comments are offered for the interested reader and to make a minor point about history of thought.

Lachmann (1956) applied the notion of radical subjectivism and recognized the heterogeneous and particularistic qualities of the capital structure. Physiological goods and services throughout the economy take on the shapes and distributive patterns that they do because they are continuously being *purposefully* reshaped according to the tastes and preferences of sovereign consumers. Capital goods are interchangeable with one another to the extent that they are acceptable substitutes from the subjective perspectives of human agents attempting to accomplish specific plans and expectations. O'Driscoll and Rizzo (1985) offer a framework for rational decision making. Theirs is a contextual environment that relaxes the unrealistic assumptions of perfect information and instantaneous choice often found in neoclassical models. Instead, they favor more realistic presumptions of individual ignorance and the passage of time. For example,

individuals cannot be expected to quantify probability estimates for events if they are completely ignorant that the event has any possibility of occurring. Much like an American colonialist in the late 1700s was ignorant that physical resources could be arranged in such a way to produce cellular telephone technologies, individuals in everyday economic affairs do not know what they do not know (Boettke, 2002). Perhaps most significantly, Kirzner (1973) emphasizes the nonphysiological when he distinguished between the role of owning capital resources and the purely knowledge driven and discovery role of the entrepreneur. We argue that Cowan and Rizzo (1996) should be recognized as a useful and accurate extension of this Austrian/Hayekian research project, hence they write “[p]urposive behavior, then is behavior caused by *desires and beliefs*, and economics is about the individual and social implications of such behavior” (p. 3, italics are ours).

Cowan and Rizzo explain human decision making as an evolutionary process. Agents in their model learn over time, but rather than characterizing this as a neural process, Cowan and Rizzo emphasize the inevitable accumulation of information *ex post action*. With the benefit of more information, cause and effect relationships are easier for individuals to recognize. Denzau and North (1994) similarly avoid discussing the neurological component of learning. Instead, they focus upon the similarities and differences of beliefs across groups. It is no surprise to admit that different people in different times and places view the world differently. While this may relate to certain neurological differences across these groups the causations versus correlations between the two remains ambiguous. For the purposes of economics – understanding the patterns of exchange, production, and distribution throughout society – the physiological processes of neurocognition are not essential. Economists begin by making simplifying presumptions as to the meaning, intentionality and beliefs that motivate human actions. This methodological practice leads to understanding the division of labor and the spontaneous emergence of functional social institutions.

While we admit that cognitive science can contribute to understanding these social processes, we do not think that neurological processes are the driving force behind the bulk of human cooperation. In fact, we are inclined to believe the opposite. To repeat, though Hayek offered an explanation for cognitive learning in *The Sensory Order*, the remainder of his research in political economy was particularly unique and successful because he operated under the presumption that his agents were ill informed, ignorant, and erring. Institutions function where human perceptions fail. They have been recognized to proxy for trust, reputation, risk, uncertainty, etc. It is perhaps important to note that in Hayek’s philosophical anthropology,

we have reason because we followed rules, not that we used our reason to design the appropriate rules. Hayek's (1945) major contribution is worth repeating here, economic knowledge conveyed through social institutions, such as market prices, is more crucial to maintaining an advanced system of exchange and production than is the accurate digestions of stimuli by any individual human mind. This is what we have wanted to say in a nutshell for good or bad. Hayek was a political economist who had a fascination with the workings of the human mind, not a neuroscientist who had an interest in the economy and public policy. For the vast majority of work-a-day economists and political economists who want to use Hayek's ideas in developing their own arguments in economics, politics, and history, it would be more productive to place relatively more weight on his contributions to those areas of research than to his efforts in philosophical/theoretical psychology (however, profound and important they may be).

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